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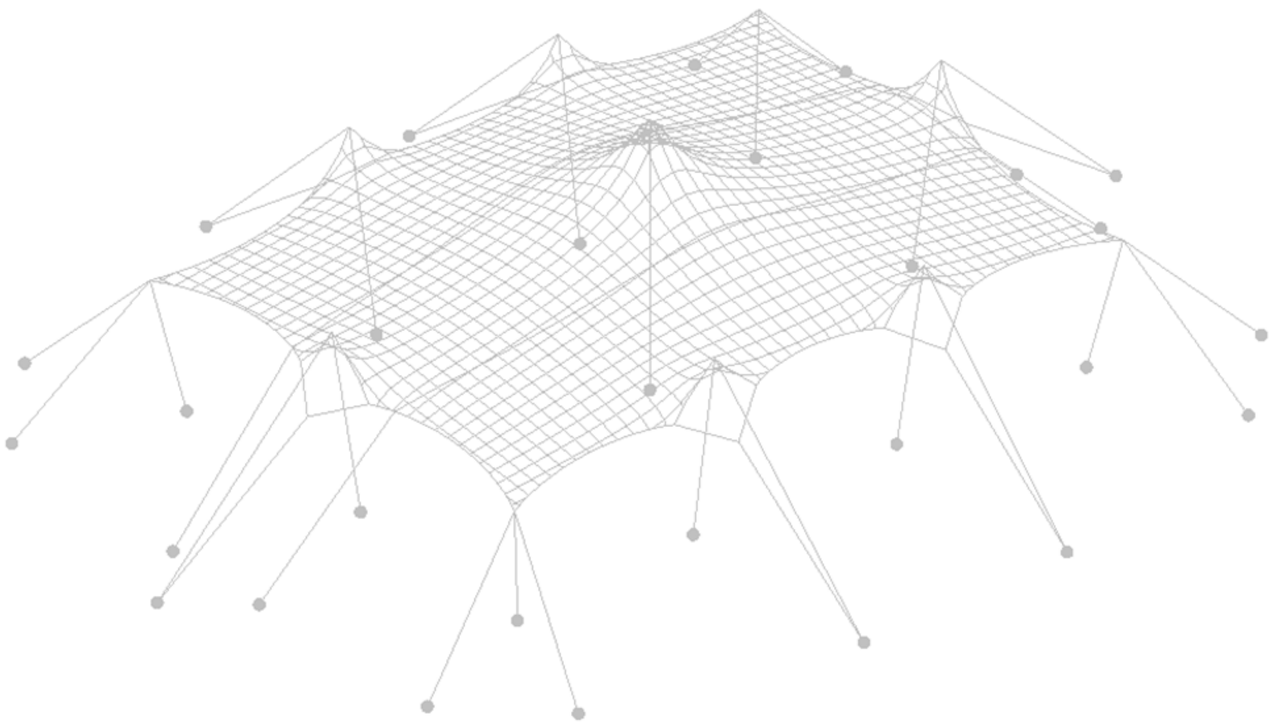

Inhoud:	**Tentboek (conform NEN-EN 8020-41:2012)**
Object:	**7.5m stretchtent en alternatieven**
Eigenaar object:	**Eurostretch B.V.**
Documentcode:	**26.03.00033.2**
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A. Inleiding

Eurostretch B.V. heeft Tentech B.V. opdracht gegeven om een statische analyse uit te voeren van een stretchtent. Eurostretch B.V. is een Nederlands bedrijf dat gespecialiseerd is in de verhuur en verkoop van tentconstructies van een rekbaar membraan, zogenaamde Stretch Tents of Bedouin tents. De tenten zijn gemaakt van rekbaar membraan, hetgeen een vrije vorm mogelijk maakt. Afhankelijk van de locatie kan gevarieerd worden met het aantal, de lengte en de plaatsing van de masten en afspanningen. Dit resulteert in een op maat gemaakte overkapping.

De vormvrijheid wordt mogelijk gemaakt door een rekbaar materiaal, aangezien de gewenste vorm uit een rechthoekig stuk materiaal wordt gerekt. Het nadeel van deze vormflexibiliteit is dat het moeilijk is om alle verschillende mogelijkheden te onderzoeken en deze te generaliseren voor een statische analyse.

Dit rapport bevat de statische analyse van een 10m stretchtent serie, bestaande uit verschillende afmetingen: **7.5x10, 7.5x15, en 6.5x10m**. Het rapport toont de statische analyse van de representatieve afmetingen: 7.5x10 m en 7.5x15 m. De statische analyse van de tent van 7.5x10 m is geldig voor de kleinere afmeting van 6.5x10m dankzij de toepassing van het **schaalprincipe**. Alle modellen en varianten worden onderzocht in een 'floating" opstelling – de hoeken en zijkanten van het membraan zijn boven de grond verbonden, voornamelijk met palen.

Schaalprincipe:
Een tent met een vergelijkbare opstelling van palen en afspanningen, maar die kleiner is of een gunstigere vorm heeft, wordt automatisch goedgekeurd zodra de berekening van de minder gunstige of grotere tent is goedgekeurd, op voorwaarde dat de afstand tussen de palen en afspanningen van de geschaalde tent gelijk is aan of kleiner is dan die van de oorspronkelijke tent. Bij het bouwen van kleinere tenten met verschillende vormen is het van belang om vergelijkbare (of gunstigere) opstellingen van palen en afspanningen te gebruiken op basis van een verhouding van afstanden (om de juiste afstand tussen palen en afspanningen te garanderen).

In dit document worden de benodigde gegevens voor een tentboek volgens de NEN 8020-41:2012 gebundeld en gepresenteerd voor de 7.5m stretchtent serie van Eurostretch B.V.

Dit tentboek bevat:
- Eigendom gegevens;
- Tekeningen van de verschillende varianten van de tent inclusief maatvoering onderdelenaanduiding en benodigde verankering en ballast;
- Toelaatbare nuttige belasting;
- Maximaal toelaatbare windsnelheden (conform NEN-EN 1991-1-4:2005);
- Constructieve berekening (conform NEN-EN 13782:2015);
- Materiaalcertificaten (sterkte-eigenschappen en brandeigenschappen).

Utrecht, 10.08.2026, S. Banga

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C. Normen

- NEN-EN 1990 Eurocode, Grondslag van het constructief ontwerp
- NEN-EN 1991 Belastingen op constructies; Deel 1-4: Algemene belastingen – Windbelasting
- NEN-EN 1995 Ontwerp en berekening van houtconstructies; Deel 1-1: Gemeenschappelijke regels en regels voor gebouwen
- NEN-EN 13782 Tijdelijke constructies – Tenten – Veiligheid
- NEN-EN 12195-2 Banden gemaakt van synthetische vezels
- ISO 1141 Vezeltouwen – polyester
- CTS Franse norm - Chapiteaux, Tentes, Structures

D. Samenvatting

Fabrikant:	Eurostretch B.V. Akkermansbeekweg 15A 7061 ZA Terborg Nederland info@eurostretch.com

Dit document geldt voor drie tentafmetingen: 7.5x10 m, 6.5x10 m en 7.5x15 m. Hoewel er geen afzonderlijke berekening wordt gemaakt voor de afmeting 6.5x10 m, kunnen alle elementen, afmetingen en verankeringskrachten worden gebaseerd op het 7.5x10 m model. Voor gebruik in Nederland zijn twee opbouwopties mogelijk – met en zonder stormbanden.

D.1. Samenvatting - opbouw met stormbanden

Stormbanden zijn altijd nodig.

Algemene informatie:			
Hoofd-afmetingen:	Breedte: Lengte: Zijhoogte van het membraan: Max. hoogte:	**7.5x10m / 6.5x10m** 7.5m / 6.5m 10m 3m 4m	**7.5x15m** 7.5m 15m 3m 4m
	Hoofdmasten: *	4.0m Ø95 mm [Spruce,C18]	4.0m Ø95 mm [Spruce, C18]
	Zijmasten: *	3.0m Ø 75 mm [Spruce, C18]	3.0m Ø 75 mm [Spruce, C18]
	Hoekmasten: *	2.5m Ø 70 mm [Spruce, C18]	2.5m Ø 70 mm [Spruce, C18]
	Stormbanden:	50mm PES [EN 12195-2], LC=1000 daN	50mm PES [EN 12195-2], LC=1000 daN
	Touwen:	Ø 8mm polyester, breeksterkte ≥ 900 daN **	Ø 8mm polyester, breeksterkte ≥ 900 daN **
	Lange zijde – gecombineerd:	3 sneden	3 sneden
	Lange zijde - V-touwen:	2 sneden	2 sneden
	Korte zijde – gecombineerd:	3 sneden	3 sneden
	Korte zijde - V-touwen:	2 sneden	2 sneden
	Banden:	50mm PES [EN 12195-2], LC=1000 daN	50mm PES [EN 12195-2], LC=1000 daN
	Verbindings-middelen:	Carbine 180x14 M16 & M12 Oogbout Ratchet, BL = 3.5 kN	Carbine 180x14 M16 & M12 Oogbout Ratchet, BL = 3.5 kN

	Basisplaat voor zij- en hoekmast Basisplaat voor hoofdmast	25x25cm 30x30cm	25x25cm 30x30cm
Gebruiks-belasting:	Het is toegestaan om max. 25 kg per hoofdmast aan decoratie, geluid- of lichtinstallaties toe te passen. De belasting moet centrisch aangebracht worden.		
Sneeuw-belasting:	Er is rekening gehouden met een sneeuwbelasting van 0.1 kN/m² (4cm) overeenkomstig met de Franse CTS norm.		
Wind-belasting:	**Conform NEN-EN 13782** De volgende stuwdrukwaarden zijn gehanteerd: h < 5m: 0.50kN/m² Deze waarde komt overeen met windkracht 8 beaufort. Boven deze windkracht (bij aanvang windkracht 9) is de veiligheid van de constructie niet meer gewaarborgd en dient deze ontruimd te worden.		

* De gemiddelde diameter, als minimumvereiste in het midden van de paal.

** De vereiste breeksterkte (BL_{tot}) kan worden bereikt door meerdere touwsecties te gebruiken (n): $BL_{rope} = BL_{tot}/n$

Conform NEN-EN 13782	Bebouwd*1 NL cat: III	Onbebouwd*2 NL cat: II	Kust*3 NL cat: 0
A. Beaufort (indicatief)	< 10 BFT	< 9 BFT	< 8 BFT
B. 10 min. gemiddelde windsnelheid	< 25.1 m/s	< 23.6 m/s	< 17.8 m/s
C. Piekwindsnelheid (windvlaag)	< 113 km/u	< 113 km/u	< 113 km/u

Tabel 1. Toelaatbare windsnelheden

1. Met een bebouwde omgeving wordt bedoeld: binnen de bebouwde kom met voldoende beschutting in Nederland.
2. Met een onbebouwde omgeving wordt bedoeld: buiten de bebouwde kom, of binnen de bebouwde kom met weinig beschutting in Nederland.
3. Met kust wordt bedoeld: de kortste afstand van de constructie tot aan het water bedraagt minder dan 50m en het betreffende water is 2km lang vrij van obstakels.

Veiligheid tegen verschuiven, opwaaien en omwaaien:			
Ankers:	Aannames: hoek van 45 graden De volgende ontwerpweerstand van de verankeringskrachten is vereist: (Zie meer over ankertesten volgens EN 13782 onder statische analyse)		
		7.5x10m / 6.5x10m	**7.5x15m**
	Afspanning lange zijde Afspanning korte zijde Afspanning hoek Stormbanden:	5.00 kN (γ = 1.2) 4.09 kN (γ = 1.2) 4.32 kN (γ = 1.2) 5.10 kN (γ = 1.2)	5.53 kN (γ = 1.2) 4.93 kN (γ = 1.2) 4.86 kN (γ = 1.2) 4.67 kN (γ = 1.2)
Vereiste ankers:	Bij dichte, niet-cohesieve grond (bijv. zandgrond). Waneer ankers **Ø30mm x 1000mm** (effectieve lengte) worden gebruikt:		
		7.5x10m / 6.5x10m	**7.5x15m**
	Afspanning lange zijde Afspanning korte zijde Afspanning hoek	1 anker 1 anker 1 anker 1 anker	2 ankers 1 anker 1 anker 1 anker

	Stormbanden:		
Vereiste ballast:	De ballastberekening wordt uitgevoerd voor 2 verschillende situaties: - Beton op beton – er worden betonnen ballasten op het beton geplaatst - Beton op hout of rubber – de betonnen ballasten worden op hout geplaatst, of er wordt een antislipmat onder de ballast gelegd		
	7.5x10m / 6.5x10m		
		Beton op beton	Beton op hout of rubber
	Afspanning lange zijde Afspanning korte zijde Afspanning hoek Stormbanden:	1060 kg 870 kg 920 kg 1080 kg	940 kg 770 kg 820 kg 960 kg
	7.5x15m		
		Beton op beton	Beton op hout of rubber
	Afspanning lange zijde Afspanning korte zijde Afspanning hoek Stormbanden:	1170 kg 1050 kg 1030 kg 990 kg	1040 kg 930 kg 920 kg 880 kg

D.2. Samenvatting - opbouw zonder stormbanden

Algemene informatie:			
Hoofd-afmetingen:		**7.5x10m / 6.5x10m**	**7.5x15m**
	Breedte: Lengte: Zijhoogte van het membraan: Max. hoogte:	7.5m / 6.5m 10m 3m 4m	7.5m 15m 3m 4m
	Hoofdmasten: *	4.0m Ø95 mm [Spruce,C18]	4.0m Ø95 mm [Spruce, C18]
	Zijmasten: *	3.0m Ø 75 mm [Spruce, C18]	3.0m Ø 75 mm [Spruce, C18]
	Hoekmasten: *	2.5m Ø 70 mm [Spruce, C18]	2.5m Ø 70 mm [Spruce, C18]
	Touwen:	Ø 8mm polyester, breeksterkte ≥ 900 daN **	Ø 8mm polyester, breeksterkte ≥ 900 daN **
	Lange zijde – gecombineerd:	3 sneden	3 sneden
	Lange zijde - V-touwen:	2 sneden	2 sneden
	Korte zijde – gecombineerd:	2 sneden	2 sneden
	Korte zijde - V-touwen:	2 sneden	2 sneden
	Banden:	50mm PES [EN 12195-2], LC=1000 daN	50mm PES [EN 12195-2], LC=1000 daN
	Verbindings-middelen:	Carbine 180x14 M16 & M12 Oogbout Ratchet, BL = 3.5 kN	Carbine 180x14 M16 & M12 Oogbout Ratchet, BL = 3.5 kN
	Basisplaat voor zij- en hoekmast	25x25cm	25x25cm
	Basisplaat voor hoofdmast	30x30cm	30x30cm
Gebruiks-belasting:	Het is toegestaan om max. 25 kg per hoofdmast aan decoratie, geluid- of lichtinstallaties toe te passen. De belasting moet centrisch aangebracht worden.		
Sneeuw-belasting:	Er is rekening gehouden met een sneeuwbelasting van 0.1 kN/m^2 (4cm) overeenkomstig met de Franse CTS norm.		
Wind-belasting:	**Conform NEN-EN 13782** De volgende stuwdrukwaarden zijn gehanteerd: h < 5m: 0.25kN/m²		

* De gemiddelde diameter, als minimumvereiste in het midden van de paal.

** De vereiste breeksterkte (BL_{tot}) kan worden bereikt door meerdere touwsecties te gebruiken (n): $BL_{rope} = BL_{tot}/n$

	Bebouwd*1 NL cat: III	Onbebouwd*2 NL cat: II	Kust*3 NL cat: 0
A. Beaufort (indicatief)	< 8 BFT	< 7 BFT	< 6 BFT
B. 10 min. gemiddelde windsnelheid	< 17.8 m/s	< 16.7 m/s	< 12.6 m/s
C. Piekwindsnelheid (windvlaag)	< 80 km/u	< 80 km/u	< 80 km/u

Tabel 2. Toelaatbare windsnelheden

1. Met een bebouwde omgeving wordt bedoeld: binnen de bebouwde kom met voldoende beschutting in Nederland.
2. Met een onbebouwde omgeving wordt bedoeld: buiten de bebouwde kom, of binnen de bebouwde kom met weinig beschutting in Nederland.
3. Met kust wordt bedoeld: de kortste afstand van de constructie tot aan het water bedraagt minder dan 50m en het betreffende water is 2km lang vrij van obstakels.

Veiligheid tegen verschuiven, opwaaien en omwaaien:

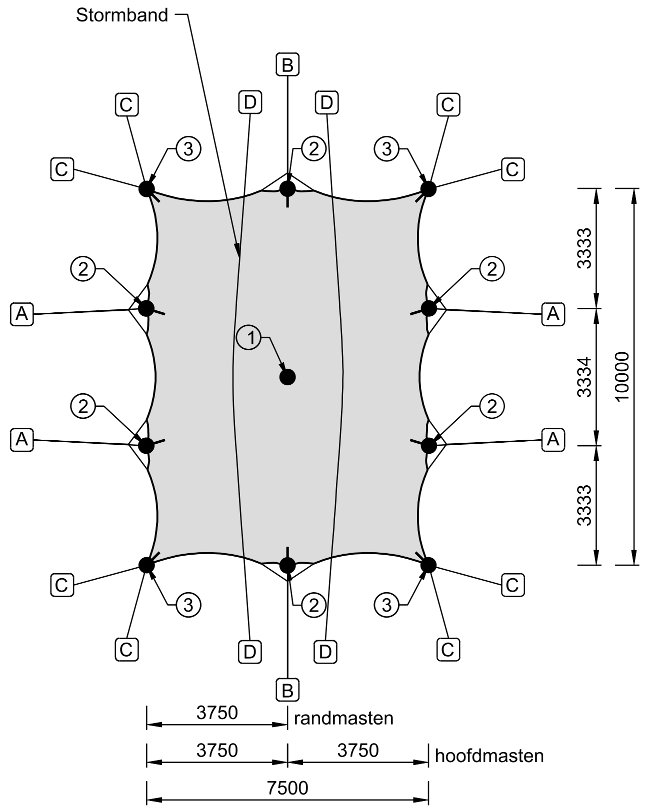
Ankers:	Aannames: hoek van 45 graden De volgende ontwerpweerstand van de verankeringskrachten is vereist: (Zie meer over ankertesten volgens EN 13782 onder statische analyse)		
		7.5x10m / 6.5x10m	**7.5x15m**
	Afspanning lange zijde Afspanning korte zijde Afspanning hoek	4.12 kN (γ = 1.2) 3.97 kN (γ = 1.2) 3.76 kN (γ = 1.2)	5.02 kN (γ = 1.2) 4.62 kN (γ = 1.2) 3.86 kN (γ = 1.2)
Vereiste ankers:	Bij dichte, niet-cohesieve grond (bijv. zandgrond). Waneer ankers **Ø30mm x 1000mm** (effectieve lengte) worden gebruikt:		
		7.5x10m / 6.5x10m	**7.5x15m**
	Afspanning lange zijde Afspanning korte zijde Afspanning hoek	1 anker 1 anker 1 anker	1 anker 1 anker 1 anker
Vereiste ballast:	De ballastberekening wordt uitgevoerd voor 2 verschillende situaties: - Beton op beton – er worden betonnen ballasten op het beton geplaatst - Beton op hout of rubber – de betonnen ballasten worden op hout geplaatst, of er wordt een antislipmat onder de ballast gelegd		
	7.5x10m / 6.5x10m		
		Beton op beton	Beton op hout of rubber
	Afspanning lange zijde Afspanning korte zijde Afspanning hoek	870 kg 840 kg 800 kg	780 kg 750 kg 710 kg
	7.5x15m		
		Beton op beton	Beton op hout of rubber

	Afspanning lange zijde Afspanning korte zijde Afspanning hoek	1060 kg 980 kg 820 kg	950 kg 870 kg 730 kg

E. Tekeningen: hoofdmaatvoering en verankering

In dit hoofdstuk worden tekeningen voor de volgende afmetingen getoond:

- 7.5x10m model
- 7.5x15m model
- 6.5x10m model (gebaseerd op 7.5x10m model)



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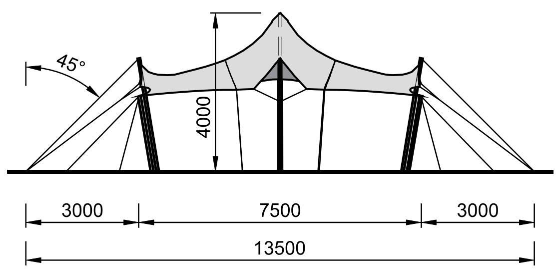

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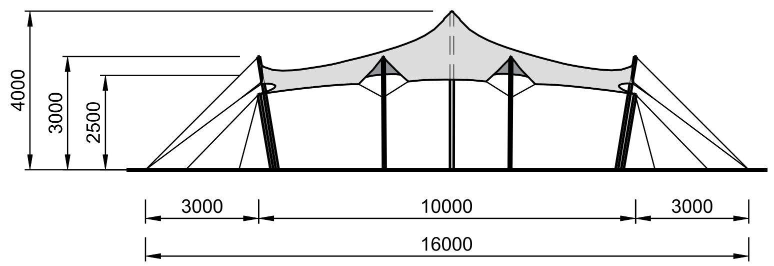

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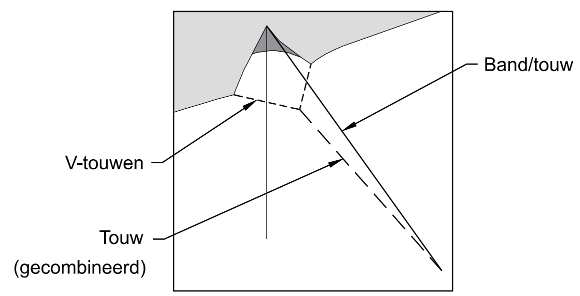

Afspanningsschema
[1:100]

Veiligheid tegen verschuiven, opwaaien en omwaaien

Voor dichte, niet samenhangende grond.

Bij opbouw met stormbanden:	Ankers (Ø30x1000mm)	Ballast (beton op beton)	Ballast (beton op hout of rubber)
A Afspanning - lange zijde	1x ankers	1060 kg	940 kg
B Afspanning - korte zijde	1x ankers	870 kg	770 kg
C Afspanning - hoek	1x ankers	920 kg	820 kg
D Stormband *(Per uiteinde)*	1x ankers	1080 kg	960 kg

Bij opbouw zonder stormbanden:	Ankers (Ø30x1000mm)	Ballast (beton op beton)	Ballast (beton op hout of rubber)
A Afspanning - lange zijde	1x ankers	870 kg	780 kg
B Afspanning - korte zijde	1x ankers	840 kg	750 kg
C Afspanning - hoek	1x ankers	800 kg	710 kg

Alle spanbanden/touwen en stormbelts van het membraan moeten onder een hoek van ≥ 45° aan de ankers worden bevestigd.

Masten

Hoofdmasten 4.0m:
1 - Ø95mm [C18, Spruce]
Randmasten 3.0m:
2 - Ø75mm [C18, Spruce]
Hoekmasten 2.5m:
3 - Ø70mm [C18, Spruce]

Afspanningen

Stormband:	[PES] BL 3000 daN = LC 1000 daN
Afspanning band:	[PES] BL 3000 daN = LC 1000 daN
Afspanning touw:	Touw: Ø 8mm polyester, breeksterkte ≥ 900 daN

Gecombineerd en v-touwen:	Met stormbanden Touw	Zonder stormbanden Touw
n - aantal secties	n	n
Afspanning - lange zijde (gecombineerd)	3x	3x
Afspanning - lange zijde (v-touwen)	2x	2x
Afspanning - korte zijde (gecombineerd)	3x	2x
Afspanning - korte zijde (v-touwen)	2x	2x

Banden (alternatief touw):	Band	Touw
n - aantal secties	(aantal)	n
Afspanning - lange zijde	1x	1x
Afspanning - korte zijde	1x	1x
Afspanning - hoek	1x	3x

Alternatieve tentmaten

7.5 x 10 m 7.5 x 15 m ** Basis-afmetingen*	6.5 x 10 m

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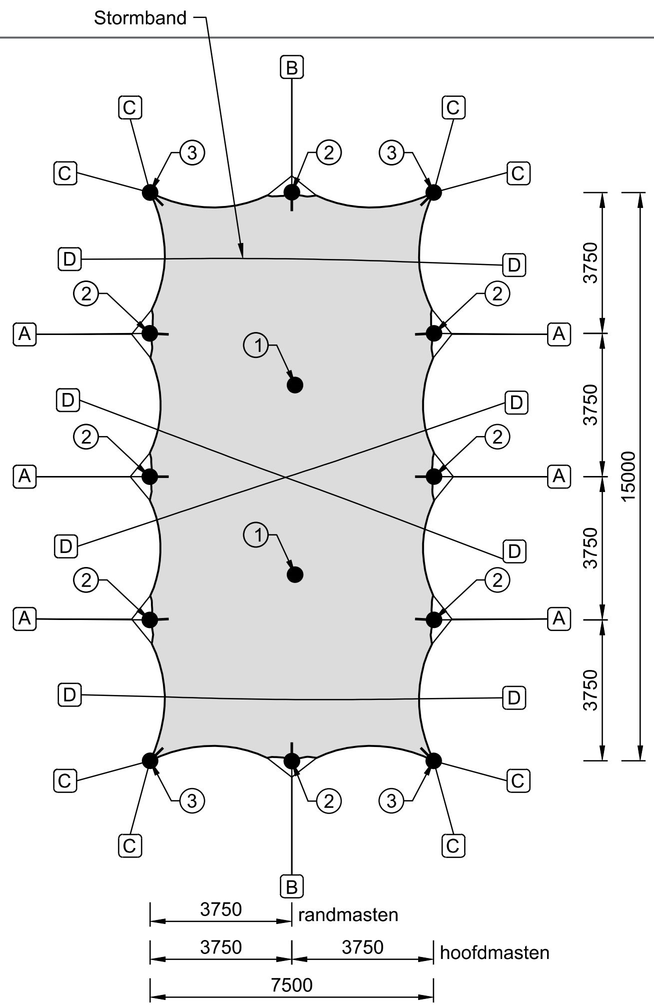

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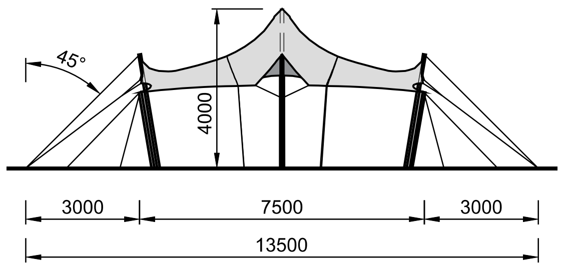

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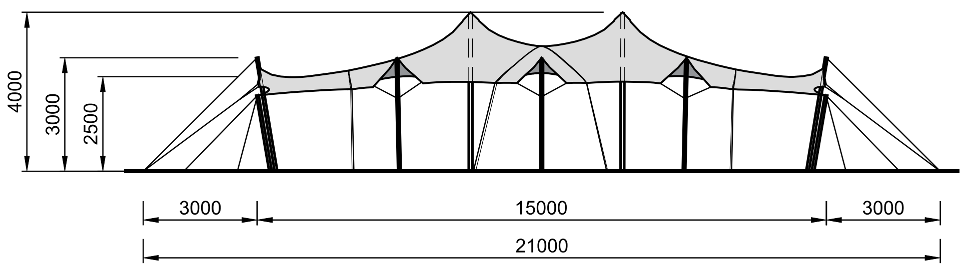

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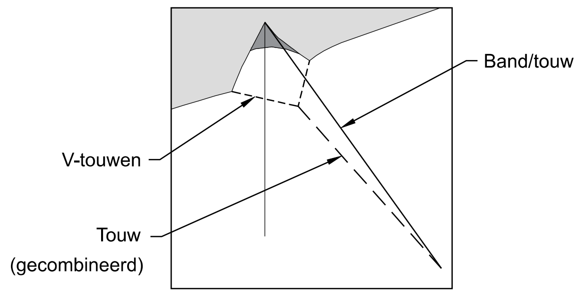

Afspanningsschema
[1:100]

Veiligheid tegen verschuiven, opwaaien en omwaaien

Voor dichte, niet samenhangende grond.

Bij opbouw met stormbanden:	Ankers (Ø30x1000mm)	Ballast (beton op beton)	Ballast (beton op hout of rubber)
A Afspanning - lange zijde	2x ankers	1170 kg	1040 kg
B Afspanning - korte zijde	1x ankers	1050 kg	930 kg
C Afspanning - hoek	1x ankers	1030 kg	920 kg
D Stormband *(Per uiteinde)*	1x ankers	990 kg	880 kg

Bij opbouw zonder stormbanden:	Ankers (Ø30x1000mm)	Ballast (beton op beton)	Ballast (beton op hout of rubber)
A Afspanning - lange zijde	1x ankers	1060 kg	950 kg
B Afspanning - korte zijde	1x ankers	980 kg	870 kg
C Afspanning - hoek	1x ankers	820 kg	730 kg

Alle spanbanden/touwen en stormbelts van het membraan moeten onder een hoek van ≥ 45° aan de ankers worden bevestigd.

Masten

Hoofdmasten 4.0m:
1 - Ø95mm [C18, Spruce]
Randmasten 3.0m:
2 - Ø75mm [C18, Spruce]
Hoekmasten 2.5m:
3 - Ø70mm [C18, Spruce]

Afspanningen

Stormband: [PES] BL 3000 daN = LC 1000 daN

Afspanning band: [PES] BL 3000 daN = LC 1000 daN
Afspanning touw: Touw: Ø 8mm polyester, breeksterkte ≥ 900 daN

Gecombineerd en v-touwen:	Met stormbanden	Zonder stormbanden
	Touw	Touw
n - aantal secties	n	n
Afspanning - lange zijde (gecombineerd)	3x	3x
Afspanning - lange zijde (v-touwen)	2x	2x
Afspanning - korte zijde (gecombineerd)	3x	2x
Afspanning - korte zijde (v-touwen)	2x	2x

Banden (alternatief touw):	Band	Touw
n - aantal secties	(aantal)	n
Afspanning - lange zijde	1x	1x
Afspanning - korte zijde	1x	1x
Afspanning - hoek	1x	3x

Alternatieve tentmaten

7.5 x 10 m 7.5 x 15 m ** Basis-afmetingen*	6.5 x 10 m

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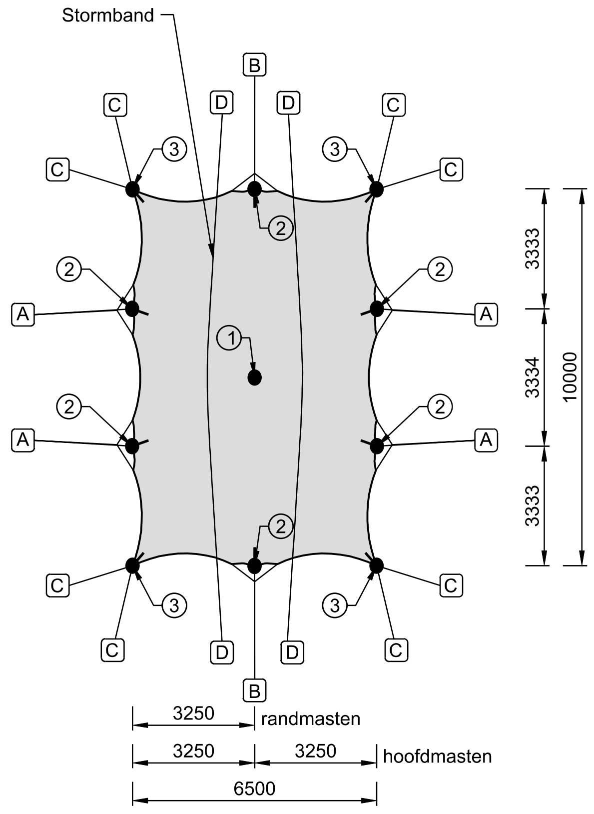

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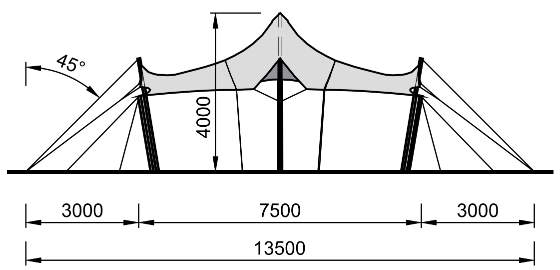

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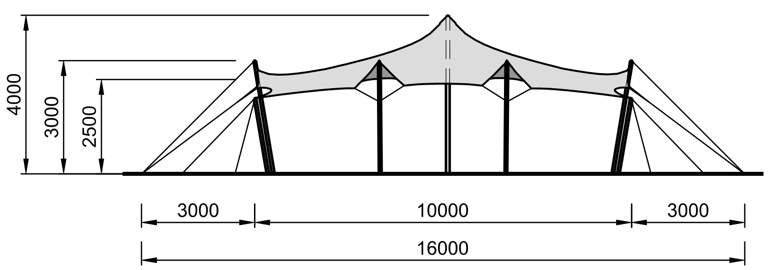

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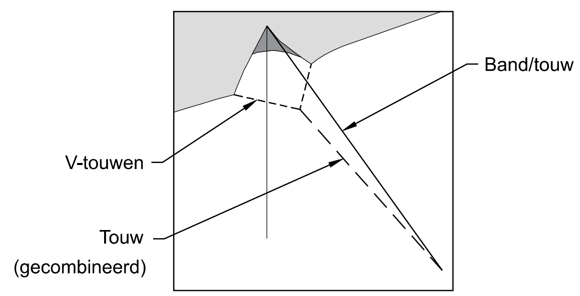

Afspanningsschema
[1:100]

Veiligheid tegen verschuiven, opwaaien en omwaaien

Voor dichte, niet samenhangende grond.

Bij opbouw met stormbanden:	Ankers (Ø30x1000mm)	Ballast (beton op beton)	Ballast (beton op hout of rubber)
A Afspanning - lange zijde	1x ankers	1060 kg	940 kg
B Afspanning - korte zijde	1x ankers	870 kg	770 kg
C Afspanning - hoek	1x ankers	920 kg	820 kg
D Stormband *(Per uiteinde)*	1x ankers	1080 kg	960 kg

Bij opbouw zonder stormbanden:	Ankers (Ø30x1000mm)	Ballast (beton op beton)	Ballast (beton op hout of rubber)
A Afspanning - lange zijde	1x ankers	870 kg	780 kg
B Afspanning - korte zijde	1x ankers	840 kg	750 kg
C Afspanning - hoek	1x ankers	800 kg	710 kg

Alle spanbanden/touwen en stormbelts van het membraan moeten onder een hoek van ≥ 45° aan de ankers worden bevestigd.

Masten

Hoofdmasten 4.0m:
1 - Ø95mm [C18, Spruce]
Randmasten 3.0m:
2 - Ø75mm [C18, Spruce]
Hoekmasten 2.5m:
3 - Ø70mm [C18, Spruce]

Afspanningen

Stormband: [PES] BL 3000 daN = LC 1000 daN

Afspanning band: [PES] BL 3000 daN = LC 1000 daN
Afspanning touw: Touw: Ø 8mm polyester, breeksterkte ≥ 900 daN

Gecombineerd en v-touwen:	Met stormbanden Touw	Zonder stormbanden Touw
n - aantal secties	n	n
Afspanning - lange zijde (gecombineerd)	3x	3x
Afspanning - lange zijde (v-touwen)	2x	2x
Afspanning - korte zijde (gecombineerd)	3x	2x
Afspanning - korte zijde (v-touwen)	2x	2x

Banden (alternatief touw):	Band	Touw
n - aantal secties	(aantal)	n
Afspanning - lange zijde	1x	1x
Afspanning - korte zijde	1x	1x
Afspanning - hoek	1x	3x

Alternatieve tentmaten

7.5 x 10 m 7.5 x 15 m * *Basis-afmetingen*	6.5 x 10 m

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F. Uitgangspunten en randvoorwaarden

Dit document geldt voor de gebouwde constructie wanneer de volgende uitgangspunten en randvoorwaarden zijn aangehouden:

- De gebruikte materialen, onderdelen en doorsneden (membraan, masten, afspanningen, stormbanden, verankering) zijn in overeenstemming met dit document.
- De afmetingen van de opgebouwde constructie komen overeen met de maten in dit document.
- Er mogen geen onderdelen (masten, afspanningen, ankers) verwijderd worden.
- **Waterophopingen dient onmiddellijk te worden verwijderd.**
- Obstakels dienen op ten minste 0.5m afstand van het membraan geplaatst te worden (gemeten loodrecht op het doek).
- Bij sterke wind (zie samenvatting, onderdeel windbelasting) wordt de tent gesloten voor publieke toegang.
- Er mag maximaal 25 kg per mast aan decoratie, geluid- of lichtinstallaties bevestigd worden aan de hoofdmasten. De belasting dient centrisch aangebracht te worden.
- Alle masten moeten droog worden opgeslagen. Als ze nat worden, moet er een mogelijkheid zijn om ze te drogen.
- De verankering is gebaseerd op dichte, niet-samenhangende grond. Wanneer op andere ondergrond wordt gebouwd, moet mogelijk aanvullende verankering worden voorzien of ankertesten worden uitgevoerd.
- Er wordt uitgegaan van een standaardbelasting van 0,1 kN/m² volgens EN 13782, wat overeenkomt met de vereiste sneeuwbelasting (4 cm) volgens de Franse CTS.

G. Toelaatbare windsnelheden

G.1. Toelaatbare windsnelheden

Wind kan op verschillende manieren uit gedrukt worden:

- 10 minuten gemiddelde windsnelheid – een gemiddelde windsnelheid over 10minuten gemeten op 10m hoogte in een open terrein (NEN-EN 1991-1-4 NB NL terreincategorie II).
- Piekwindsnelheid – een kortstondige maximale windstoot met een bepaalde snelheid, afhankelijk van de hoogte. Vaak gegeven in km/u.

De stuwdruk waarop een tent berekend is, is bepalend voor de sterkte van de tent. Het gaat er dus om dat op de juiste manier wordt vastgesteld welke windsnelheid moet worden aangehouden om te kunnen bepalen of de stuwdruk overschreden wordt.

G.1.1 Opbouw met stormbanden

In de constructieve berekening is een stuwdruk van **0.50 kN/m² tot 5m** hoogte aangehouden, deze stuwdruk is omgerekend naar windsnelheden. Boven onderstaande windsnelheden dient de structuur ontruimd en gesloten te worden. Boven de gegeven windsnelheid is de constructie niet meer gewaarborgd betreffende sterkte en/of stabiliteit.

Conform NEN-EN 13782	Bebouwd*1 NL cat: III	Onbebouwd*2 NL cat: II	Kust*3 NL cat: 0
A. Beaufort (indicatief)	< 10 BFT	< 9 BFT	< 8 BFT
B. 10 min. gemiddelde windsnelheid	< 25.1 m/s	< 23.6 m/s	< 17.8 m/s
C. Piekwindsnelheid (windvlaag)	< 113 km/u	< 113 km/u	< 113 km/u

Tabel 3. Toelaatbare windsnelheden

1. Met een bebouwde omgeving wordt bedoeld: binnen de bebouwde kom met voldoende beschutting in Nederland.
2. Met een onbebouwde omgeving wordt bedoeld: buiten de bebouwde kom, of binnen de bebouwde kom met weinig beschutting in Nederland.
3. Met kust wordt bedoeld: de kortste afstand van de constructie tot aan het water bedraagt minder dan 50m en het betreffende water is 2km lang vrij van obstakels.

Wanneer de hierboven genoemde windsnelheden bereikt worden dient de structuur ontruimd en voor publiek gesloten te worden.

Bovenstaande waardes (A, B en C) kunnen op verschillende manieren gemeten worden en kunnen alle drie onafhankelijk van elkaar gebruikt worden:

A. Dit is een indicatieve Beaufort schaal welke hoort bij de 10 minuten gemiddelde windsnelheid, deze waarde moet afkomstig zijn van het dichtstbijzijnde meteostation.
B. 10 min gemiddelde windsnelheid op 10 meter hoogte in een open terrein, deze waarde moet afkomstig zijn van het dichtstbijzijnde meteostation.
C. Windvlaagsnelheid op 10 meter hoogte in een open terrein, deze waarde moet afkomstig zijn van het dichtstbijzijnde meteostation.

G.1.2 Opbouw zonder stormbanden

In de constructieve berekening is een stuwdruk van **0.25 kN/m^2 tot 5m** hoogte aangehouden, deze stuwdruk is omgerekend naar windsnelheden. Boven onderstaande windsnelheden dient de structuur ontruimd en gesloten te worden. Boven de gegeven windsnelheid is de constructie niet meer gewaarborgd over sterkte en/of stabiliteit.

	Bebouwd*1 NL cat: III	Onbebouwd*2 NL cat: II	Kust*3 NL cat: 0
A. Beaufort (indicatief)	< 8 BFT	< 7 BFT	< 6 BFT
B. 10 min. gemiddelde windsnelheid	17.8 m/s	16.7 m/s	12.6 m/s
C. Piekwindsnelheid (windvlaag)	80 km/u	80 km/u	80 km/u

Tabel 4. Toelaatbare windsnelheden

1. Met een bebouwde omgeving wordt bedoeld: binnen de bebouwde kom met voldoende beschutting in Nederland.
2. Met een onbebouwde omgeving wordt bedoeld: buiten de bebouwde kom, of binnen de bebouwde kom met weinig beschutting in Nederland.
3. Met kust wordt bedoeld: de kortste afstand van de constructie tot aan het water bedraagt minder dan 50m en het betreffende water is 2km lang vrij van obstakels.

Wanneer de hierboven genoemde windsnelheden bereikt worden dient de structuur ontruimd en voor publiek gesloten te worden.

Bovenstaande waardes (A, B en C) kunnen op verschillende manieren gemeten worden en kunnen alle drie onafhankelijk van elkaar gebruikt worden:

A. Dit is een indicatieve Beaufort schaal welke hoort bij de 10 minuten gemiddelde windsnelheid, deze waarde moet afkomstig zijn van het dichtstbijzijnde meteostation.
B. 10 min gemiddelde windsnelheid op 10 meter hoogte in een open terrein, deze waarde moet afkomstig zijn van het dichtstbijzijnde meteostation.
C. Windvlaagsnelheid op 10 meter hoogte in een open terrein, deze waarde moet afkomstig zijn van het dichtstbijzijnde meteostation.

G.2. Windsnelheid berekening

G.2.1 Opbouw met stormbanden

De maximale stuwdruk wordt middels de NEN-EN 1991-1-4 teruggerekend naar een basiswindsnelheid voor het kustgebied, een bebouwde omgeving en een onbebouwde omgeving. (uitgaande van terreinruwheid volgens de Nederlandse NB).

Piekwindsnelheid op 10m hoogte

$605 \times {}^{1}/_{2} \times \rho \times v^2 = {}^{1}/_{2} \times 1.25 \times v^2 \rightarrow v = 31.1\ {}^{m}/_{s} \rightarrow 113\ {}^{km}/_{u}$	**Verg. 4.10 NEN-EN 1991-1-4** **Basisstuwdruk**

Windsnelheid kust gebied op 10m hoogte

$K_r = 0.19 \times \left(\frac{z_0}{0.05}\right)^{0.07} = 0.19 \times \left(\frac{0.005}{0.05}\right)^{0.07} = 0.16$	**Verg. 4.5 NEN-EN 1991-1-4** **Terreinfactor**
$C_r = K_r \times \ln\left(\frac{z}{z_0}\right) = 0.16 \times ln\left(\frac{4}{0.005}\right) = 1.08$	**Verg. 4.4 NEN-EN 1991-1-4** **Ruwheidsfactor**
$V_m = C_r \times V_b = 1.08 \times V_b$	**Verg. 4.3 NEN-EN 1991-1-4** **Gemiddelde windsnelheid op hoogte**
$\sigma_v = K_r \times V_b = 0.16 \times V_b$	**Verg. 4.6 NEN-EN 1991-1-4** **Standaardafwijking van de turbulentie**
$L_v = \frac{\sigma_v}{V_m} = \frac{0.16 \times V_b}{1.08 \times V_b} = 0.15$	**Verg. 4.7 NEN-EN 1991-1-4** **Turbulentie intensiteit**
$Q_p = (1 + 7 \times L_v) \times {}^{1}/_{2} \times \rho \times {V_m}^2 = 1.5 \times {V_b}^2$	**Verg. 4.8 NEN-EN 1991-1-4** **Extreme stuwdruk**
$500 = 1.5 \times {V_b}^2 \rightarrow V_b = 18.3\ {}^{m}/_{s}$	**Karakteristieke windsnelheid**

Windsnelheid onbebouwd gebied op 10m hoogte

$K_r = 0.19 \times \left(\frac{z_0}{0.05}\right)^{0.07} = 0.19 \times \left(\frac{0.2}{0.05}\right)^{0.07} = 0.21$	**Verg. 4.5 NEN-EN 1991-1-4** **Terreinfactor**
$C_r = K_r \times \ln\left(\frac{z}{z_0}\right) = 0.21 \times ln\left(\frac{4}{0.2}\right) = 0.63$	**Verg. 4.4 NEN-EN 1991-1-4** **Ruwheidsfactor**
$V_m = C_r \times V_b = 0.63 \times V_b$	**Verg. 4.3 NEN-EN 1991-1-4** **Gemiddelde windsnelheid op hoogte**
$\sigma_v = K_r \times V_b = 0.21 \times V_b$	**Verg. 4.6 NEN-EN 1991-1-4** **Standaardafwijking van de turbulentie**
$L_v = \frac{\sigma_v}{V_m} = \frac{0.21 \times V_b}{0.63 \times V_b} = 0.33$	**Verg. 4.7 NEN-EN 1991-1-4** **Turbulentie intensiteit**
$Q_p = (1 + 7 \times L_v) \times {}^{1}/_{2} \times \rho \times {V_m}^2 = 0.82 \times {V_b}^2$	**Verg. 4.8 NEN-EN 1991-1-4** **Extreme stuwdruk**
$500 = 0.82 \times {V_b}^2 \rightarrow V_b = 24.7\ {}^{m}/_{s}$	**Karakteristieke windsnelheid**

Windsnelheid bebouwd gebied op 10m hoogte

$K_r = 0.19 \times \left(\frac{z_0}{0.05}\right)^{0.07} = 0.19 \times \left(\frac{0.5}{0.05}\right)^{0.07} = 0.22$	**Verg. 4.5 NEN-EN 1991-1-4 Terreinfactor**
$C_r = K_r \times \ln\left(\frac{z}{z_0}\right) = 0.22 \times ln\left(\frac{7}{0.5}\right) = 0.59$	**Verg. 4.4 NEN-EN 1991-1-4 Ruwheidsfactor**
$V_m = C_r \times V_b = 0.59 \times V_b$	**Verg. 4.3 NEN-EN 1991-1-4 Gemiddelde windsnelheid op hoogte**
$\sigma_v = K_r \times V_b = 0.22 \times V_b$	**Verg. 4.6 NEN-EN 1991-1-4 Standaardafwijking van de turbulentie**
$L_v = \frac{\sigma_v}{V_m} = \frac{0.22 \times V_b}{0.59 \times V_b} = 0.38$	**Verg. 4.7 NEN-EN 1991-1-4 Turbulentie intensiteit**
$Q_p = (1 + 7 \times L_v) \times {}^{1}/_{2} \times \rho \times {V_m}^2 = 0.79 \times {V_b}^2$	**Verg. 4.8 NEN-EN 1991-1-4 Extreme stuwdruk**
$500 = 0.79 \times {V_b}^2 \rightarrow V_b = 25.1\ {}^{m}/_{s}$	**Karakteristieke windsnelheid**

G.2.2 Opbouw zonder stormbanden

De maximale stuwdruk wordt middels de NEN-EN 1991-1-4 teruggerekend naar een basiswindsnelheid voor het kustgebied, een bebouwde omgeving en een onbebouwde omgeving. (uitgaande van terreinruwheid volgens de Nederlandse NB).

Piekwindsnelheid op 10m hoogte

$303 \times {}^{1}/_{2} \times \rho \times v^2 = {}^{1}/_{2} \times 1.25 \times v^2 \rightarrow v = 22.0\ {}^{m}/_{s} \rightarrow 80\ {}^{km}/_{u}$	**Verg. 4.10 NEN-EN 1991-1-4 Basisstuwdruk**

Windsnelheid kust gebied op 10m hoogte

$K_r = 0.19 \times \left(\frac{z_0}{0.05}\right)^{0.07} = 0.19 \times \left(\frac{0.005}{0.05}\right)^{0.07} = 0.16$	**Verg. 4.5 NEN-EN 1991-1-4 Terreinfactor**
$C_r = K_r \times \ln\left(\frac{z}{z_0}\right) = 0.16 \times ln\left(\frac{4}{0.005}\right) = 1.08$	**Verg. 4.4 NEN-EN 1991-1-4 Ruwheidsfactor**
$V_m = C_r \times V_b = 1.08 \times V_b$	**Verg. 4.3 NEN-EN 1991-1-4 Gemiddelde windsnelheid op hoogte**
$\sigma_v = K_r \times V_b = 0.16 \times V_b$	**Verg. 4.6 NEN-EN 1991-1-4 Standaardafwijking van de turbulentie**
$L_v = \frac{\sigma_v}{V_m} = \frac{0.16 \times V_b}{1.08 \times V_b} = 0.15$	**Verg. 4.7 NEN-EN 1991-1-4 Turbulentie intensiteit**
$Q_p = (1 + 7 \times L_v) \times {}^{1}/_{2} \times \rho \times {V_m}^2 = 1.5 \times {V_b}^2$	**Verg. 4.8 NEN-EN 1991-1-4 Extreme stuwdruk**
$250 = 1.5 \times {V_b}^2 \rightarrow V_b = 12.62\ {}^{m}/_{s}$	**Karakteristieke windsnelheid**

Windsnelheid onbebouwd gebied op 10m hoogte

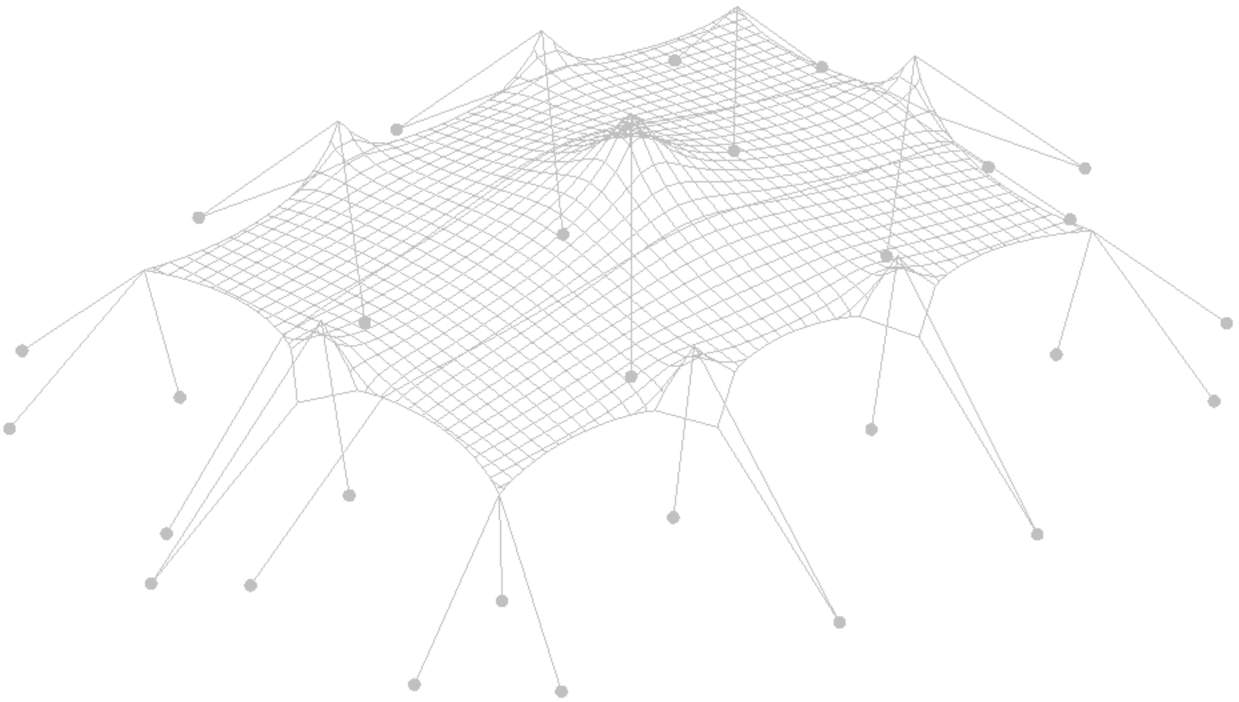
$K_r = 0.19 \times \left(\frac{z_0}{0.05}\right)^{0.07} = 0.19 \times \left(\frac{0.2}{0.05}\right)^{0.07} = 0.21$	**Verg. 4.5 NEN-EN 1991-1-4 Terreinfactor**

$C_r = K_r \times \ln\left(\frac{z}{z_0}\right) = 0.21 \times ln\left(\frac{4}{0.2}\right) = 0.63$	**Verg. 4.4 NEN-EN 1991-1-4 Ruwheidsfactor**
$V_m = C_r \times V_b = 0.63 \times V_b$	**Verg. 4.3 NEN-EN 1991-1-4 Gemiddelde windsnelheid op hoogte**
$\sigma_v = K_r \times V_b = 0.21 \times V_b$	**Verg. 4.6 NEN-EN 1991-1-4 Standaardafwijking van de turbulentie**
$L_v = \frac{\sigma_v}{V_m} = \frac{0.21 \times V_b}{0.63 \times V_b} = 0.33$	**Verg. 4.7 NEN-EN 1991-1-4 Turbulentie intensiteit**
$Q_p = (1 + 7 \times L_v) \times {}^1/_2 \times \rho \times {V_m}^2 = 0.82 \times {V_b}^2$	**Verg. 4.8 NEN-EN 1991-1-4 Extreme stuwdruk**
$250 = 0.82 \times {V_b}^2 \rightarrow V_b = 16.66\ {}^m/_s$	**Karakteristieke windsnelheid**

Windsnelheid bebouwd gebied op 10m hoogte

$K_r = 0.19 \times \left(\frac{z_0}{0.05}\right)^{0.07} = 0.19 \times \left(\frac{0.5}{0.05}\right)^{0.07} = 0.22$	**Verg. 4.5 NEN-EN 1991-1-4 Terreinfactor**
$C_r = K_r \times \ln\left(\frac{z}{z_0}\right) = 0.22 \times ln\left(\frac{7}{0.5}\right) = 0.59$	**Verg. 4.4 NEN-EN 1991-1-4 Ruwheidsfactor**
$V_m = C_r \times V_b = 0.59 \times V_b$	**Verg. 4.3 NEN-EN 1991-1-4 Gemiddelde windsnelheid op hoogte**
$\sigma_v = K_r \times V_b = 0.22 \times V_b$	**Verg. 4.6 NEN-EN 1991-1-4 Standaardafwijking van de turbulentie**
$L_v = \frac{\sigma_v}{V_m} = \frac{0.22 \times V_b}{0.59 \times V_b} = 0.38$	**Verg. 4.7 NEN-EN 1991-1-4 Turbulentie intensiteit**
$Q_p = (1 + 7 \times L_v) \times {}^1/_2 \times \rho \times {V_m}^2 = 0.79 \times {V_b}^2$	**Verg. 4.8 NEN-EN 1991-1-4 Extreme stuwdruk**
$250 = 0.79 \times {V_b}^2 \rightarrow V_b = 17.76\ {}^m/_s$	**Karakteristieke windsnelheid**

DESIGN AND ENGINEERING OF LIGHTWEIGHT STRUCTURES



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Owner object: **Eurostretch B.V.**

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A. Introduction

Eurostretch B.V. has commissioned Tentech B.V. to perform a static analysis of a stretch tent. Eurostretch B.V. is a Dutch company that specializes in rental and selling of tent structures made of a stretchable membrane, so called Stretch Tents or Bedouin tents. The stretchable membrane enables a freedom of form as there is not a pre-described shape necessary. Depending on the location, the number of poles, length of the poles, and the placement of the poles, number and type of ties can be varied. This results in a custom canopy at each location.

The freedom of form is enabled by a stretchable fabric, as the desired form is stretched from a rectangular fabric. The drawback of this flexibility in shape is the difficulty of investigating all the different possibilities and generalizing them for a static analysis.

This report presents the static analysis of a 7.5m stretch tent series which consists of different sizes – **7.5x10, 7.5x15, and 6.5x10m**. The report shows the static analysis of the decisive sizes – 7.5x10m and 7.5x15m. The static analysis of tent 7.5x10m is valid for a smaller size of 6.5x10m due to the application of the **scaling principle**. All models and variants are investigated in a “floating” set-up – membrane corners and sides are connected above ground, primarily to poles.

Scaling principle: A tent with a similar arrangement of poles and ties but smaller in size or having a more favourable shape is automatically verified when the calculation of the less favourable or bigger tent has been approved, provided that the spacing between the poles and ties of the scaled tent is similar or smaller than the original tent. While constructing smaller tent sizes of different shapes, it is significant to use similar (or more favourable) arrangements of poles and ties based on ratio of distances (to ensure correct pole and tie spacing).

This document contains the data required for a tent book, according to EN 13782, for the 7.5m tent series of Eurostretch B.V., including:

- Ownership data;
- Drawings of the different variants of the tent, including dimensions, indications of elements and required anchoring/ballast;
- Permitted live load;
- Maximum wind speeds (according to EN 1991-1-4:2005);
- Structural analysis (according to EN 13782:2015);
- Material certificates (strength properties and fire properties).

Utrecht, S. Banga, 10.08.2026

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C. Codes and standards

- EN 1990 Eurocode, Basis of structural Design
- EN 1991 Eurocode 1, Part 1-4: General actions - wind actions.
- EN 1995 Eurocode 5, Design of wooden structures
- EN 13782 Temporary Structures – Tents - Safety
- EN 12195-2 Belts made from synthetic fibres
- ISO 1141 Fibre ropes – Polyester
- CTS French standard - Chapiteaux, Tentes, Structures

D. Summary

Manufacturer:	Eurostretch B.V. Akkermansbeekweg 15A 7061 ZA Terborg Nederland info@eurostretch.com	

This document is valid for three tent sizes: 7.5x10m, 7.5x15m and 6.5x10m. Although no separate calculation is made for 6.5x10m size, all elements, sizes and anchoring forces can be based on 7.5x10m model.

General information:

Main dimensions:		**7.5x10m / 6.5x10m**	**7.5x15m**
	Width:	7.5/6.5m	7.5m
	Length:	10m	15m
	Side height of membrane:	3m	3m
	Max. height:	4m	4m
	Main poles: *	4.0m Ø 95 mm [Spruce, C18]	4.0m Ø 95 mm [Spruce, C18]
	Side poles: *	3.0m Ø 75 mm [Spruce, C18]	3.0m Ø 75 mm [Spruce, C18]
	Corner poles: *	2.5m Ø 70 mm [Spruce, C18]	2.5m Ø 70 mm [Spruce, C18]
	Stormbelts:	50mm PES [EN 12195-2], LC=1000 daN	50mm PES [EN 12195-2], LC=1000 daN
	Guy ropes:	Ø 8mm polyester, breaking strength ≥ 900 daN **	Ø 8mm polyester, breaking strength ≥ 900 daN **
	Long side – combined:	3 sections	3 sections
	Long side - V-ropes:	2 sections	2 sections
	Short side – combined:	3 sections	3 sections
	Short side - V-ropes:	2 sections	2 sections
	Belts:	50mm PES [EN 12195-2], LC=1000 daN	50mm PES [EN 12195-2], LC=1000 daN
	Fasteners:	Carabine 180x14	

		M16 & M12 Eye bolt Ratchet, BL = 3.5 kN	Carabine 180x14 M16 & M12 Eye bolt Ratchet, BL = 3.5 kN
	Side & corner pole base plate	25x25cm	25x25cm
	Main pole base plate	30x30cm	30x30cm
User load:	A maximum of 25 kg per main mast may be used for decorations, sound or lighting installations. The load must be applied centrally.		
Snow load:	A snow load of 0.1 kN/m² (4 cm) has been considered in accordance with the French CTS standard.		
Wind load:	**According to EN 13782** The following pressure values have been used: h < 5m: 0.50kN/m² This value corresponds to wind force 8 on the Beaufort scale. Above this wind force (starting at wind force 9), the safety of the structure can no longer be guaranteed, and it must be evacuated.		

* The average diameter, as a minimum required at the middle of the pole.

** The required breaking load (BL_{tot}) may be achieved by using multiple rope sections (n): $BL_{rope} = BL_{tot} / n$, see page 47

Wind speeds for default European terrain categories (not Country specific)

	Out of service (1)				
	Coast	Flattened, open area	Rural area	Village	City
10 min. average wind speed (2)	> 17.94 m/s > 64.57 km/h	> 18.89 m/s > 67.99 km/h	> 21.08 m/s > 75.88 km/h	> 24.99 m/s > 89.97 km/h	> 26.08 m/s > 93.89 km/h
Beaufort (3)	> 7 BFT	> 7 BFT	> 8 BFT	> 9 BFT	> 9 BFT
Peak wind speed (4)	> 113 km/h	> 113 km/h	> 113 km/h	> 113 km/h	> 113 km/h

(1) *'Out of service' means: above the given wind speed the structure is no longer guaranteed regarding strength and/or stability.*
(2) *10min average wind speed at 10m height measured at the nearest weather stations.*
(3) *wind data in Beaufort (BFT) are indicative values.*
(4) *3 second peak wind speed measured on site at 10m height.*

Safety against sliding, overturning and uplifting:

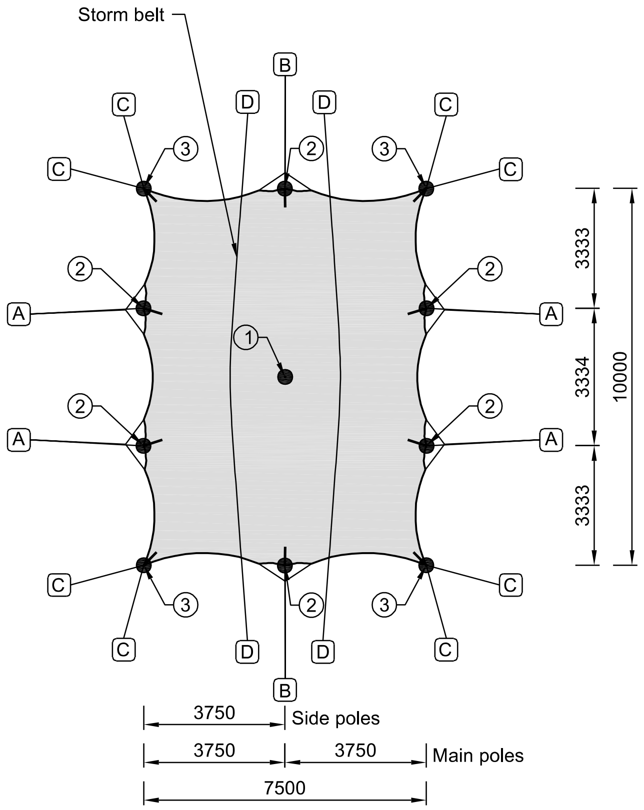
Anchor forces:	Assumptions: angle of 45 degrees The following design resistance of the anchor forces is required: (See H.8.3, page 56 for Anchor tests according to EN 13782)		
		7.5x10m / 6.5x10m	**7.5x15m**
	Long side guy ropes: Short side guy ropes: Corner ties: Stormbelt:	5.00 kN (γ = 1.2) 4.09 kN (γ = 1.2) 4.32 kN (γ = 1.2) 5.10 kN (γ = 1.2)	5.53 kN (γ = 1.2) 4.93 kN (γ = 1.2) 4.86 kN (γ = 1.2) 4.67 kN (γ = 1.2)
Required anchors:	Based on dense, non-cohesive soil (e.g. sandy soils). In the case anchors **Ø30mm x 1000mm** (effective length) are being used:		
		7.5x10m / 6.5x10m	**7.5x15m**
	Long side guy ropes: Short side guy ropes: Corner ties: Stormbelt:	1 anchor 1 anchor 1 anchor 1 anchor	2 anchors 1 anchor 1 anchor 1 anchor
Required ballast:	Ballast calculation is performed for 2 different situations: - Concrete on concrete – concrete ballasts are placed on concrete - Concrete on wood or rubber – concrete ballasts are placed on wood, or an anti-slip mat is placed underneath the ballast		
	7.5x10m / 6.5x10m		
		Concrete on concrete	Concrete on wood or rubber
	Long side guy ropes: Short side guy ropes: Corner ties: Stormbelt:	1060 kg 870 kg 920 kg 1080 kg	940 kg 770 kg 820 kg 960 kg
	7.5x15m		
		Concrete on concrete	Concrete on wood or rubber
	Long side guy ropes: Short side guy ropes: Corner ties: Stormbelt:	1170 kg 1050 kg 1030 kg 990 kg	1040 kg 930 kg 920 kg 880 kg

E. **Drawings: main dimensions and anchoring**

Drawings for the following sizes are presented in this chapter:

- 7.5x10m model
- 7.5x15m model
- 6.5x10m model (based on 7.5x10m model)

TEN TECH



TOP VIEW
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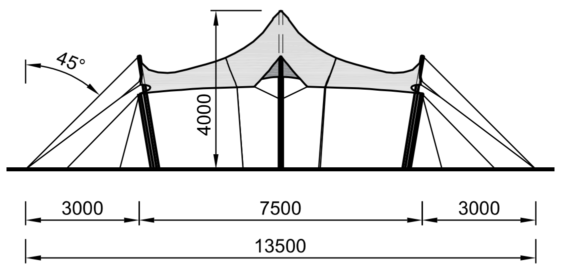

FRONT VIEW
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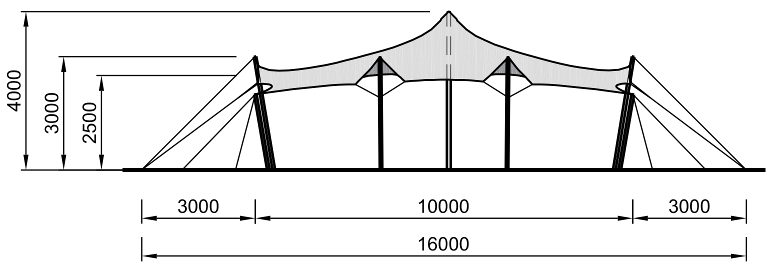

SIDE VIEW
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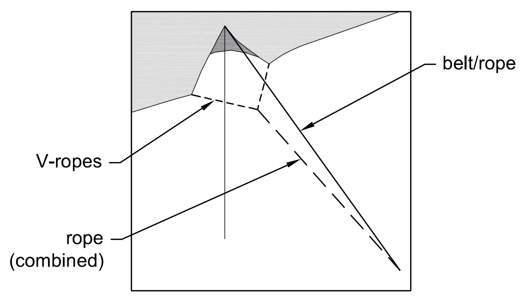

GUYING DIAGRAM
[1:100]

Safety against sliding, overturning and uplifting

Based on dense, non-cohesive soil (e.g. sandy soils).

	Anchors (Ø30x1000mm)	Ballast (concrete on concrete)	Ballast (concrete on wood or rubber)
A Guying - long side	1x anchors	1060 kg	940 kg
B Guying - short side	1x anchors	870 kg	770 kg
C Guying - corner	1x anchors	920 kg	820 kg
D Storm belt *(per end)*	1x anchors	1080 kg	960 kg

All belts/ropes and storm belts of the membrane must be placed under a 45° angle of ≥ 45° be attached to the anchors.

Poles

Main poles 4.0m:

1 - Ø95mm [C18, Spruce]

Side poles 3.0m:

2 - Ø75mm [C18, Spruce]

Corner poles 2.5m:

3 - Ø70mm [C18, Spruce]

Guying

Storm belt: [PES] BL 3000 daN = LC 1000 daN

Guying belt: [PES] BL 3000 daN = LC 1000 daN

Guying rope: rope: Ø 8mm polyester, breaking strength ≥ 900 daN

Combined & v-ropes:	Rope
n - number of sections	n
Guying - long side (combined)	3x
Guying - long side (v-ropes)	2x
Guying - short side (combined)	3x
Guying - short side (v-ropes)	2x

Belts (alternative rope):	Belt	Rope
n - number of sections	(number)	n
Guying - long side	1x	1x
Guying - short side	1x	1x
Guying - corner	1x	3x

Alternative configuration

7.5 x 10 m 7.5 x 15 m * *Basic configurations*	6.5 x 10 m

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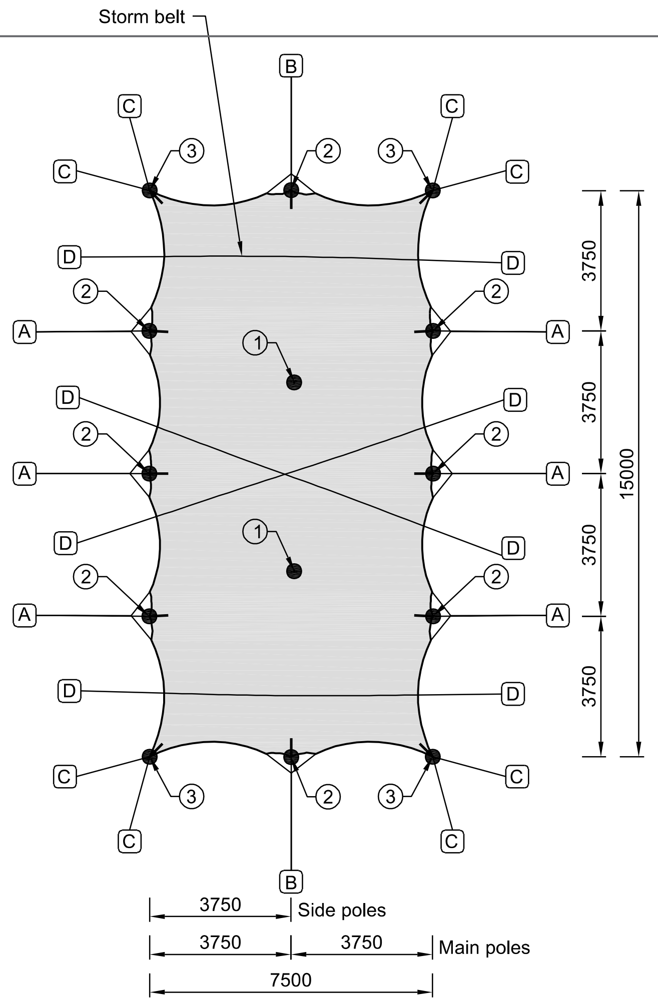
TEN TECH



TOP VIEW
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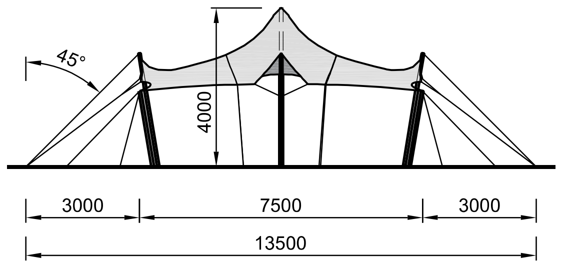

FRONT VIEW
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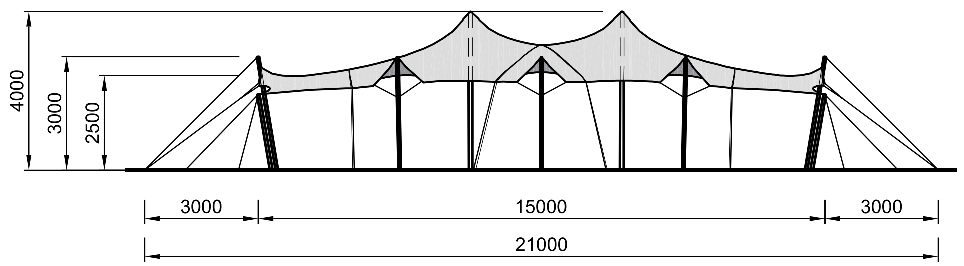

SIDE VIEW
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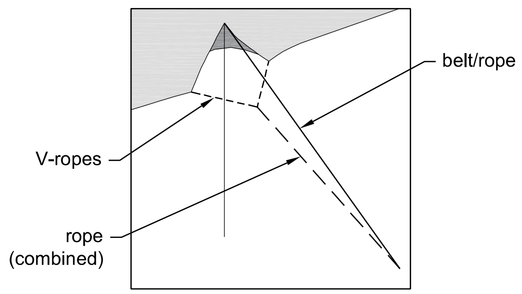

GUYING DIAGRAM
[1:100]

Safety against sliding, overturning and uplifting

Based on dense, non-cohesive soil (e.g. sandy soils).

	Anchors (Ø30x1000mm)	Ballast (concrete on concrete)	Ballast (concrete on wood or rubber)
A Guying - long side	2x anchors	1170 kg	1040 kg
B Guying - short side	1x anchors	1050 kg	930 kg
C Guying - corner	1x anchors	1030 kg	920 kg
D Storm belt *(per end)*	1x anchors	990 kg	880 kg

All belts/ropes and storm belts of the membrane must be placed under a 45° angle of ≥ 45° be attached to the anchors.

Poles

Main poles 4.0m:
1 - Ø95mm [C18, Spruce]
Side poles 3.0m:
2 - Ø75mm [C18, Spruce]
Corner poles 2.5m:
3 - Ø70mm [C18, Spruce]

Guying

Storm belt: [PES] BL 3000 daN = LC 1000 daN

Guying belt: [PES] BL 3000 daN = LC 1000 daN
Guying rope: rope: Ø 8mm polyester, breaking strength ≥ 900 daN

Combined & v-ropes:	Rope
n - number of sections	n
Guying - long side (combined)	3x
Guying - long side (v-ropes)	2x
Guying - short side (combined)	3x
Guying - short side (v-ropes)	2x

Belts (alternative rope):	Belt	Rope
n - number of sections	(number)	n
Guying - long side	1x	1x
Guying - short side	1x	1x
Guying - corner	1x	3x

Alternative configuration

7.5 x 10 m 7.5 x 15 m * *Basic configurations*	6.5 x 10 m

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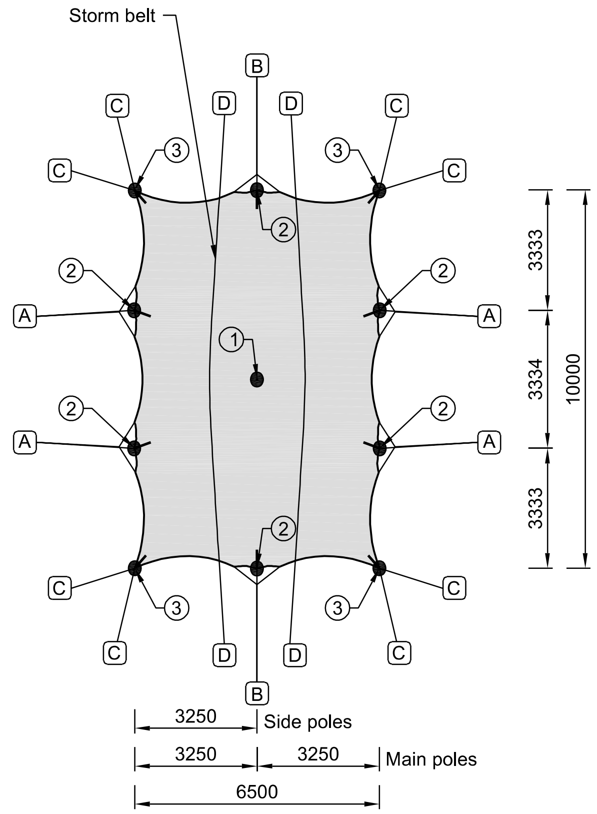

TOP VIEW
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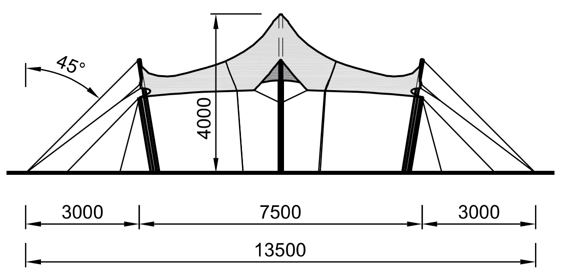

FRONT VIEW
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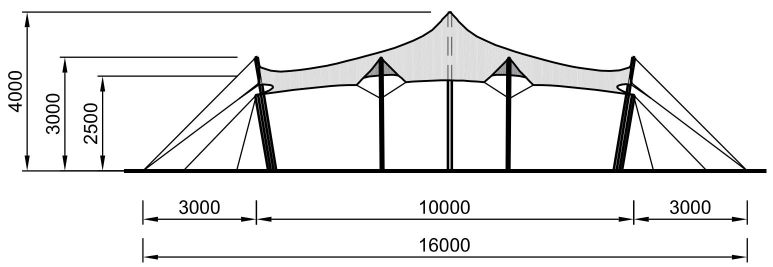
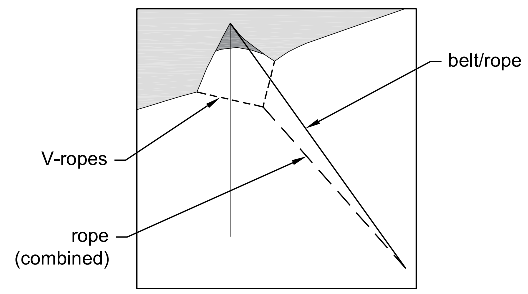


SIDE VIEW
[1:200]



GUYING DIAGRAM
[1:100]

Safety against sliding, overturning and uplifting

Based on dense, non-cohesive soil (e.g. sandy soils).

	Anchors (Ø30x1000mm)	Ballast (concrete on concrete)	Ballast (concrete on wood or rubber)
A Guying - long side	1x anchors	1060 kg	940 kg
B Guying - short side	1x anchors	870 kg	770 kg
C Guying - corner	1x anchors	920 kg	820 kg
D Storm belt *(per end)*	1x anchors	1080 kg	960 kg

All belts/ropes and storm belts of the membrane must be placed under a 45° angle of ≥ 45° be attached to the anchors.

Poles

Main poles 4.0m:
1 - Ø95mm [C18, Spruce]
Side poles 3.0m:
2 - Ø75mm [C18, Spruce]
Corner poles 2.5m:
3 - Ø70mm [C18, Spruce]

Guying

Storm belt: [PES] BL 3000 daN = LC 1000 daN

Guying belt: [PES] BL 3000 daN = LC 1000 daN
Guying rope: rope: Ø 8mm polyester, breaking strength ≥ 900 daN

Combined & v-ropes:	Rope
n - number of sections	n
Guying - long side (combined)	3x
Guying - long side (v-ropes)	2x
Guying - short side (combined)	3x
Guying - short side (v-ropes)	2x

Belts (alternative rope):	Belt	Rope
n - number of sections	(number)	n
Guying - long side	1x	1x
Guying - short side	1x	1x
Guying - corner	1x	3x

Alternative configuration

7.5 x 10 m 7.5 x 15 m * *Basic configurations*	6.5 x 10 m

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F. Important terms and conditions

This document applies to the built structure if the following principles and conditions are met:

- The used materials, parts and sections (membrane, poles, ties, anchoring) are in accordance with this document.
- The dimensions of the built structure match the dimensions stated in this document.
- Parts (poles, ties, anchors) may not be removed.
- Obstacles should be placed at least 0.5m from the membrane (measured perpendicular to the fabric); The fabric needs a certain freedom to deform in all directions to prevent damages caused by collision with objects located closely to the fabric (see also EN 13782, article 8.7).
- Above the maximum allowable wind speeds (see D) the structure should be evacuated and access for the public must be denied.
- Only decorations, music- and light installations of less than 25 kg per pole, can be attached to the structure. They must be attached centrically.
- A conventional load of 0.1 kN/m^2 is considered according to EN 13782, which corresponds with the required snow load (4cm) according to the French CTS.
- Anchoring is based on dense, non-cohesive soil. When soil differs, additional anchoring might be necessary, or anchor tests need to be performed.
- Poles must be stored dry. If they become wet, there needs to be a possibility for them to dry.
- All water accumulation must be removed as soon as possible.

G. Allowed wind speeds

The maximum wind speed is converted into a basic wind speed for a coastal area, flattened/open area, rural area, village and city according to EN 1991-1-4. Terrain roughness is taken according to the recommended general values for the different terrain categories for Europe. Illustrations of these terrain categories are presented on page 16.

	Full wind load	
Wind pressure p_w	500 N/m²	At **5m** height
Corresponding peak wind pressure $p_{w,peak}$	605 N/m²	At **10m** height

Peak wind speed at 10m height

Equation:

$$605 = {}^{1}/_{2} \times \rho \times v^2 = {}^{1}/_{2} \times 1.25 \times v^2 \rightarrow v = 31.1\ m/s \rightarrow \pm 113\ km/h$$

Eq. 4.10 EN 1991-1-4
Basic wind pressure

Wind speed coastal area at 10m height

$$K_r = 0.19 \times \left(\frac{z_0}{0.05}\right)^{0.07} = 0.19 \times \left(\frac{0.003}{0.05}\right)^{0.07} = 0.156$$

Eq. 4.5 EN 1991-1-4
Terrain factor for coastal area

$$C_r = K_r \times \ln\left(\frac{z}{z_0}\right) = 0.156 \times \ln\left(\frac{4.0}{0.003}\right) = 1.123$$

Eq. 4.4 EN 1991-1-4
Roughness factor
Z = 4.0 > Z_{min} = 1

$$V_m = C_r \times V_b = 1.123 \times V_b$$

Eq. 4.3 EN 1991-1-4
Average wind speed at height

$$\sigma_v = K_r \times V_b = 0.141 \times V_b$$

Eq. 4.6 EN 1991-1-4
Standard deviation of turbulence

$$L_v = \frac{\sigma_v}{V_m} = \frac{0.156 \times V_b}{1.158 \times V_b} = 0.139$$

Eq. 4.7 EN 1991-1-4
Turbulence intensity

$$Q_p = (1 + 7 \times L_v) \times {}^{1}/_{2} \times \rho \times {V_m}^2 = 1.554 \times {V_b}^2$$

Eq. 4.8 EN 1991-1-4
Extreme wind pressure

Equation:

$$500 = 1.554 \times {V_b}^2 \rightarrow solving\ gives \rightarrow V_b = 17.94\ m/s$$

Characteristic wind speed

Wind speed flattened, open area at 10m height

$$K_r = 0.19 \times \left(\frac{z_0}{0.05}\right)^{0.07} = 0.19 \times \left(\frac{0.01}{0.05}\right)^{0.07} = 0.170$$

Eq. 4.5 EN 1991-1-4
Terrain factor for coastal area

$$C_r = K_r \times \ln\left(\frac{z}{z_0}\right) = 0.170 \times \ln\left(\frac{4.0}{0.01}\right) = 1.017$$

Eq. 4.4 EN 1991-1-4
Roughness factor at 3.5m height
Z = 4.0 > Z_{min} = 1

$$V_m = C_r \times V_b = 1.017 \times V_b$$

Eq. 4.3 EN 1991-1-4
Average wind speed at height

$$\sigma_v = K_r \times V_b = 0.170 \times V_b$$

Eq. 4.6 EN 1991-1-4
Standard deviation of turbulence

$$L_v = \frac{\sigma_v}{V_m} = \frac{0.170 \times V_b}{1.017 \times V_b} = 0.167$$

Eq. 4.7 EN 1991-1-4
Turbulence intensity

$$Q_p = (1 + 7 \times L_v) \times {}^{1}/_{2} \times \rho \times {V_m}^2 = 1.402 \times {V_b}^2$$

Eq. 4.8 EN 1991-1-4
Extreme wind pressure

Equation:

$$500 = 1.402 \times {V_b}^2 \rightarrow solving\ gives \rightarrow V_b = 18.89\ m/s$$

Characteristic wind speed

Wind speed rural area at 10m height

$$K_r = 0.19 \times \left(\frac{z_0}{0.05}\right)^{0.07} = 0.19 \times \left(\frac{0.05}{0.05}\right)^{0.07} = 0.190$$

Eq. 4.5 EN 1991-1-4
Terrain factor for unbuilt area

$$C_r = K_r \times \ln\left(\frac{z}{z_0}\right) = 0.190 \times \ln\left(\frac{4.0}{0.05}\right) = 0.833$$

Eq. 4.4 EN 1991-1-4
Roughness factor at 4m height
Z = 4.0 > Z_{min} = 2

$$V_m = C_r \times V_b = 0.833 \times V_b$$

Eq. 4.3 EN 1991-1-4
Average wind speed at height

$$\sigma_v = K_r \times V_b = 0.190 \times V_b$$

Eq. 4.6 EN 1991-1-4
Standard deviation of turbulence

$$L_v = \frac{\sigma_v}{V_m} = \frac{0.190 \times V_b}{0.833 \times V_b} = 0.228$$

Eq. 4.7 EN 1991-1-4
Turbulence intensity

$$Q_p = (1 + 7 \times L_v) \times {}^1\!/_2 \times \rho \times {V_m}^2 = 1.125 \times {V_b}^2$$

Eq. 4.8 EN 1991-1-4
Extreme wind pressure

Equation:

$$500 = 1.125 \times {V_b}^2 \rightarrow solving\ gives \rightarrow V_b = 21.08\ m/s$$

Characteristic wind speed

Wind speed village at 10m height

$$K_r = 0.19 \times \left(\frac{z_0}{0.05}\right)^{0.07} = 0.19 \times \left(\frac{0.3}{0.05}\right)^{0.07} = 0.215$$

Eq. 4.5 EN 1991-1-4
Terrain factor for unbuilt area

$$C_r = K_r \times \ln\left(\frac{z}{z_0}\right) = 0.215 \times \ln\left(\frac{5}{0.3}\right) = 0.606$$

Eq. 4.4 EN 1991-1-4
Roughness factor at 7m height
Z = Z_{min} = 5

$$V_m = C_r \times V_b = 0.606 \times V_b$$

Eq. 4.3 EN 1991-1-4
Average wind speed at height

$$\sigma_v = K_r \times V_b = 0.215 \times V_b$$

Eq. 4.6 EN 1991-1-4
Standard deviation of turbulence

$$L_v = \frac{\sigma_v}{V_m} = \frac{0.215 \times V_b}{0.606 \times V_b} = 0.355$$

Eq. 4.7 EN 1991-1-4
Turbulence intensity

$$Q_p = (1 + 7 \times L_v) \times {}^1\!/_2 \times \rho \times {V_m}^2 = 0.801 \times {V_b}^2$$

Eq. 4.8 EN 1991-1-4
Extreme wind pressure

Equation:

$$500 = 0.801 \times {V_b}^2 \rightarrow solving\ gives \rightarrow V_b = 24.99\ m/s$$

Characteristic wind speed

Wind speed city at 10m height

$$K_r = 0.19 \times \left(\frac{z_0}{0.05}\right)^{0.07} = 0.19 \times \left(\frac{1}{0.05}\right)^{0.07} = 0.234$$

Eq. 4.5 EN 1991-1-4
Terrain factor for unbuilt area

$$C_r = K_r \times \ln\left(\frac{z}{z_0}\right) = 0.234 \times \ln\left(\frac{10}{1}\right) = 0.540$$

Eq. 4.4 EN 1991-1-4
Roughness factor at 7m height
Z = Z_{min} =10

$$V_m = C_r \times V_b = 0.540 \times V_b$$

Eq. 4.3 EN 1991-1-4
Average wind speed at height

$$\sigma_v = K_r \times V_b = 0.234 \times V_b$$

Eq. 4.6 EN 1991-1-4
Standard deviation of turbulence

$$L_v = \frac{\sigma_v}{V_m} = \frac{0.234 \times V_b}{0.540 \times V_b} = 0.434$$

Eq. 4.7 EN 1991-1-4
Turbulence intensity

$$Q_p = (1 + 7 \times L_v) \times {}^1\!/_2 \times \rho \times {V_m}^2 = 0.735 \times {V_b}^2$$

Eq. 4.8 EN 1991-1-4
Extreme wind pressure

Equation:

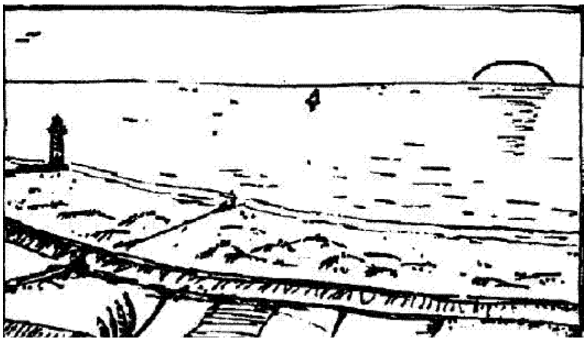
$$500 = 0.735 \times {V_b}^2 \rightarrow solving\ gives \rightarrow V_b = 26.08\ m/s$$

Characteristic wind speed

Terrain categories

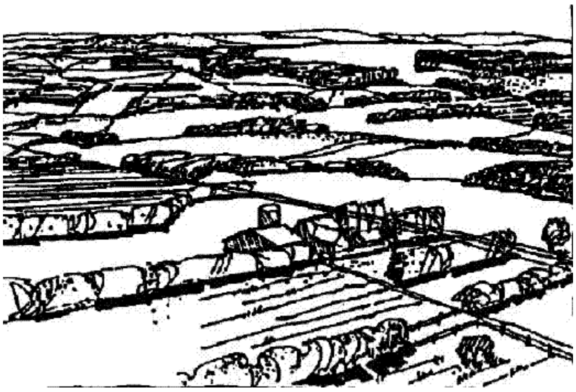
Default European terrain categories (not Country specific)

0: Coastal area:

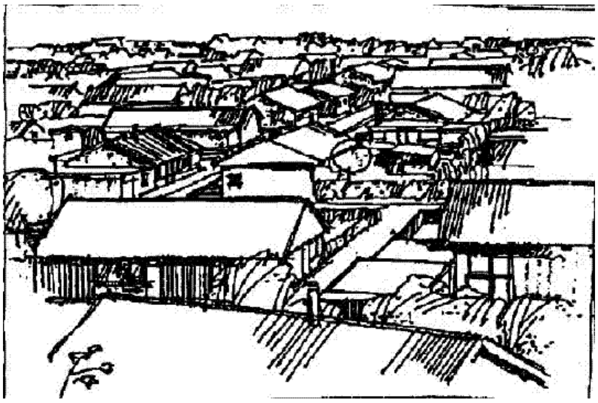


I: Flattened, open area:

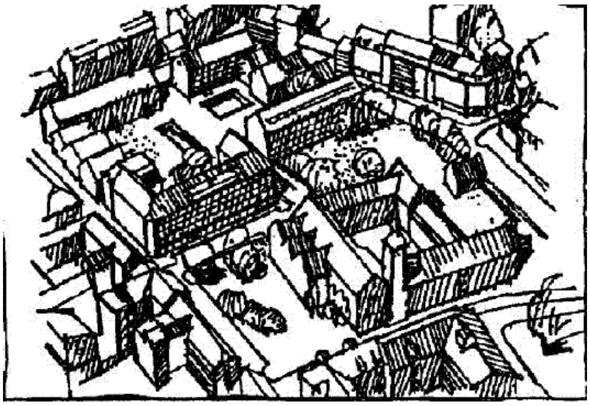
II: Rural area



III: Village



IV: City



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H.1. Project description

The principle of a stretchtent is based on a rectangular piece of stretchable membrane with a reinforcement along the boundary. The membrane is supported by poles, both at the edge and under the membrane. The side poles are stabilized by guy ropes and belts. The poles do not require a fixed position which ensures a freedom of shape.

Tentech B.V. has performed a static analysis of two tents – 7.5x10m and 7.5x15m always assumed to have stormbelts. Further chapters include material and cross-section description, calculation method, load cases and combinations, results, element checks, anchor and ballast calculations. These chapters present calculation results valid for Europe according to EN 13782.

As an exception, two additions to the static analysis are made (see I and J on pages 59 and 68 respectively):

1. For use only in Netherlands – element check for reduced wind speeds.

2. For use in France or countries following French CTS – element check for combined wind and snow loads

For use in other countries than Netherlands or France, the statical analysis must be considered without its additions. When relevant information is mentioned for these specific cases either [NL] or [FR] immediately follows this information. This information is not considered while making the base calculations. Addition J on page 68 must always be considered for use in France.

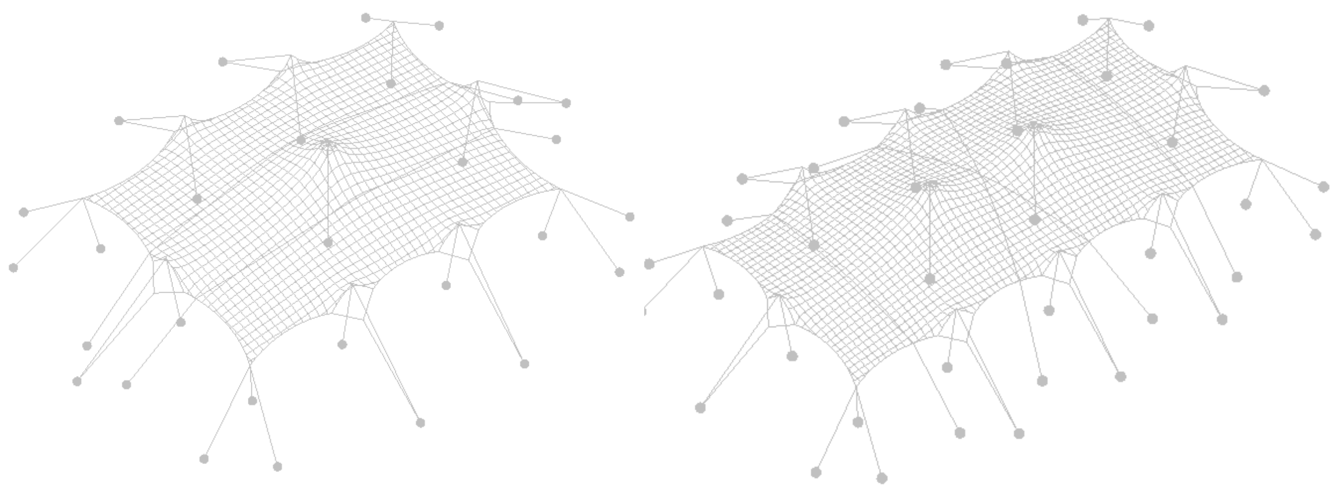


Figure 1. 7.5x10m and 7.5x15m models

H.2. Materials

H.2.1 Fabric

Design tensile strength	f_d	f_{tk} / γ_m	art 8.6 EN 13782
Characteristic tensile strength (warp)	$f_{tk,\ warp}$		
Characteristic tensile strength (weft)	$f_{tk,\ weft}$		
Material factor – global, permanent load	γ_m	2.5	tbl 4. EN13782
Material factor – global, short duration load	γ_m	2.0	tbl 4. EN13782

Table 1. Used symbols, codes and standard for fabric materials

Material	Type	Weight	$f_{rd;warp;perm}$	$f_{rd;\ weft;perm}$	$f_{rd;\ warp;short}$	$f_{rd:\ weft;short}$
Sioen RP5639	-	≈ 730 gr/m^2	9.6 kN/m	7.2 kN/m	12.0 kN/m	9.0 kN/m

Table 2. Used fabrics

H.2.2 Guy ropes

Design resistance	F_{rd}	R_m / γ_{m1}	art 10.2. EN13782
Characteristic tensile strength	R_m		art 10.2. EN13782
Material factor	$\gamma_{m1\ <\ 12mm}$ $\gamma_{m1\ >\ 12mm}$	4.0 3.3	art 10.3. EN13782 art 10.3. EN13782

Table 3. Used symbols, codes and standard for rope materials

Material	Cross section	Breaking strength	Rm	F_{rd}
8mm Polyester Braid rope	Ø 8 mm	≥ 900 daN	9.0 kN	2.25 kN

Table 4. Used guy ropes

H.2.3 Belts

Design resistance	F_{rd}	R_m / γ_{m1}	art 10.2. EN13782
Characteristic breaking strength	R_m	LC x γ_{m2}	art 10.2. EN13782
Lashing capacity	LC		Conform EN 12195-2
Material factor	γ_{m1}	2.0	art 10.2. EN13782
Material factor	γ_{m2}	3.0	EN1492-1

Table 5. Used symbols, codes and standard for belt materials

Material	LC	Rm	F_{rd}
Belt [PES] EN 12195-2, 50mm	1000 daN 10 kN	3000 daN 30 kN	15 kN

Table 6. Used belts

H.2.4 Wood

Material	γ_{m1}	1.3	tbl. 2.3. EN 1995-1-1

Table 7. Used material factors

Material	Weight	$E_{0.05}$	F_{c0k}	F_{mk}
Wood, Spruce, C18	320 kg/m^3	6.0 kN/m^2	18 N/mm^2	18 N/mm^2

Table 8. Used wooden materials

H.3. Cross sections

Profile	Material	Ø mm	G kg/m	A mm^2	I_y mm^4	$W_{el;y}$ mm^3
Ø ≈ 95 mm *	Wood C18	95	2.26	7088	3998198	84173
Ø ≈ 75 mm *	Wood C18	75	1.41	4418	1553156	41417
Ø ≈ 70 mm *	Wood C18	70	1.23	3848	1178588	33674

Table 9. Used cross sections

* the average diameter, as a minimum required at the middle of the pole.

H.4. Calculation method

H.4.1 Modelling

The analysis of the structure is performed with the software package EASY FCS supplied by TECHNET GmbH, Berlin. This software is specially developed for structures with large deformability, such as membrane structures. The performed analysis is a full non-linear second order analysis.

The membrane structure is modelled in 3D. The membrane is modelled as a cable net structure and is supported by poles. These centre poles will be stabilized by the tensioned membrane. The side poles are stabilized and tied down by tension belts or ropes, which are attached to ground anchors or ballast.

The membrane edges are reinforced. They consist of multiple layers of fabric about 10cm wide with additional patches where the lugs are placed. Storm belts are required constantly and placed in the valleys between the centre poles.

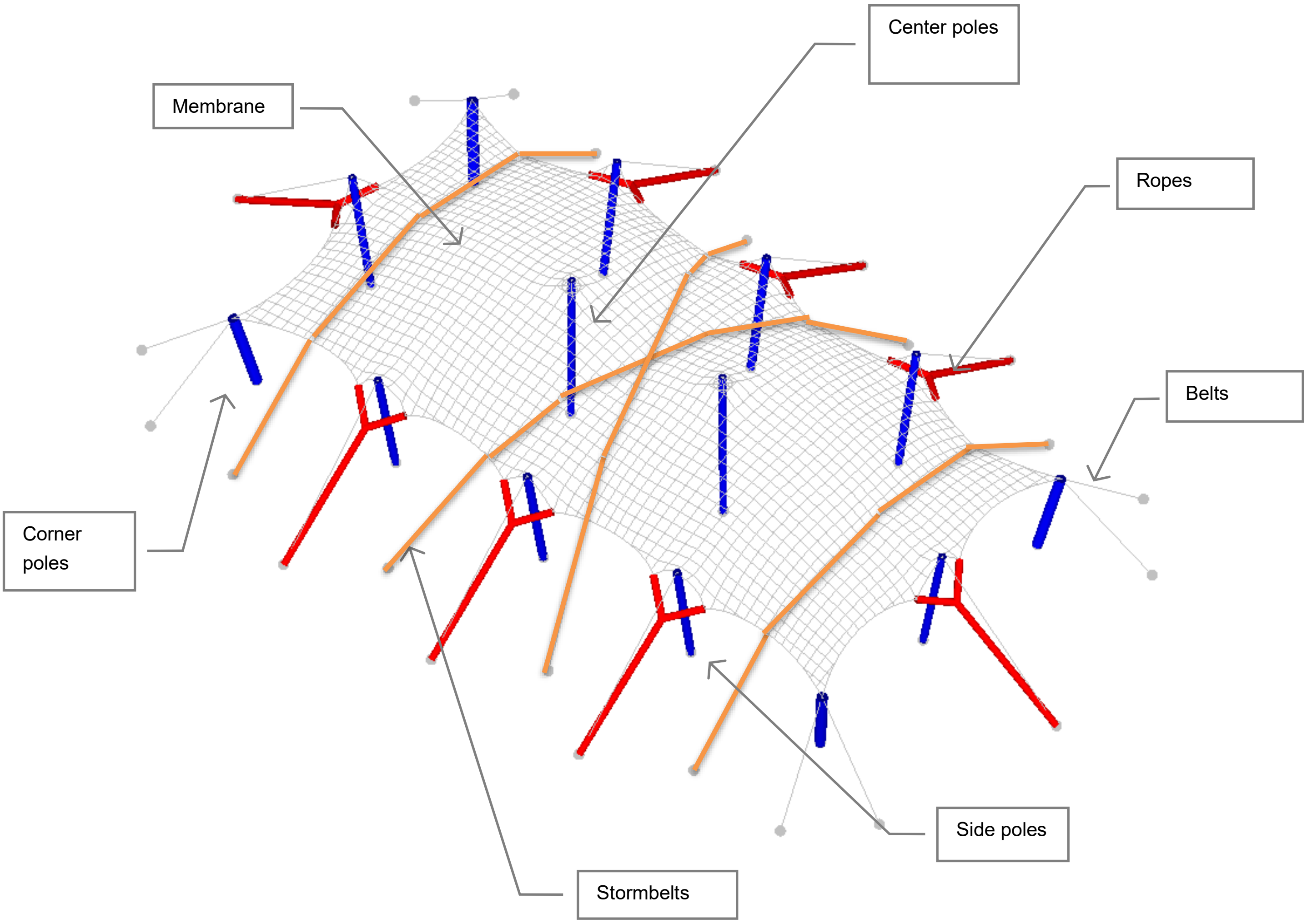


Figure 2. Calculation model

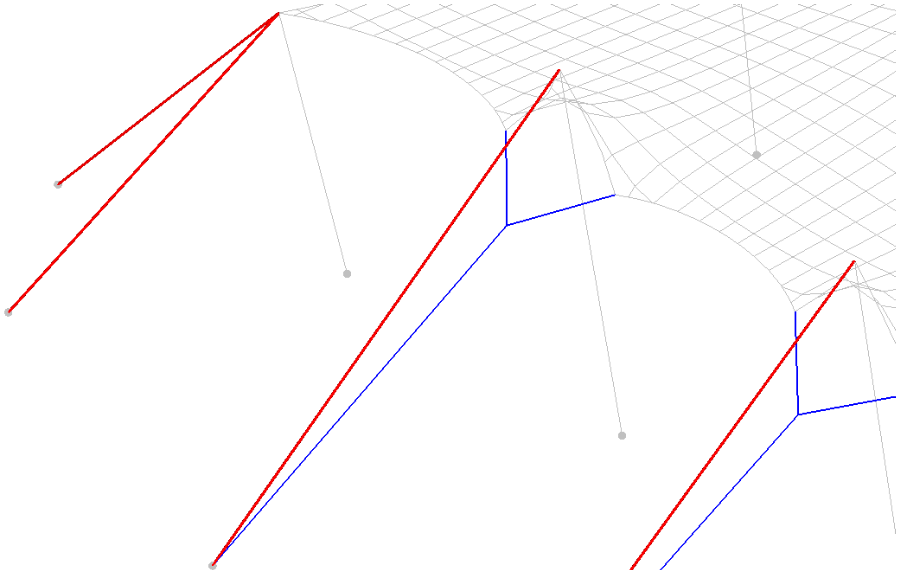


Figure 3. Principle of ties

Ties stabilize the tent and create the shape of the stretchable membrane. Guy ropes (see blue above) transfer the forces to the ground when wind suction is present. Belts (see red above) help transfer the pressure forces.

H.4.2 Structural behaviour of membrane structures

Since the fabric is a highly deformable material, it is only possible to calculate stresses and deformations with a non-linear method. FEM-software EASY is used to perform these calculations. Because of the non-linearity of the calculations the partial safety factors are not applied beforehand, since the deformations will be greater due to these safety factors, resulting in lower stresses in the fabric. See figure below.

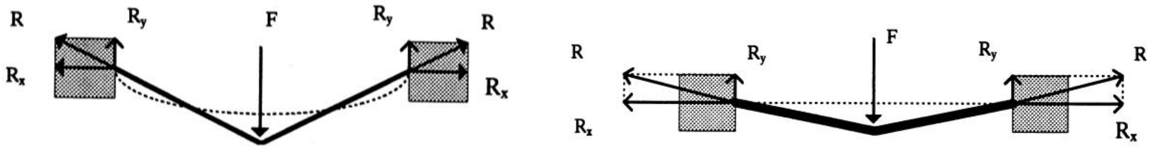


Figure 4. membrane behavior

As a membrane structure is a form-active structure, article 6.3 (4) b) of the EN-1990:2002 applies:

> *When the action effect increases less than the action, the partial factor γ_F should be applied to the action effect of the representative value of the action.*

This means that no load factors are applied on the load beforehand, but afterwards.

H.4.3 Load combinations

H.4.3.1 Fundamental - Ultimate limit state

For determination of strength and to check elements and connections ultimate limit state is used.

	One variable load	Multiple variable loads
Unfavorable permanent load	$1.35 \times G + 1.5 \times Q$	$1.35 \times G + \sum 1.35 \times Q_i$
Favorable permanent load	$1.0 \times G + 1.5 \times Q$	$1.0 \times G + \sum 1.35 \times Q_i$

Table 10. Load combinations according to EN 13782

This means that the following load combinations will be checked/calculated:

1. 1.35 x Own weight + 1.5 x Wind load
2. 1.35 x Own weight + 1.5 x Snow load
3. 1.35 x Own weight + 1.35 x Wind load + 1.35 x Snow load [FR]

H.4.3.2 Safety against overturning, sliding and uplifting - Ultimate limit state

For determination and check of the needed ballast weight and/or anchors ultimate limit state is used

	One or multiple variable loads
Unfavorable permanent load	$1.1 \times G + 1.2 \times Q_{wind} + \sum 1.3 \times Q_i$
Favorable permanent load	$1.0 \times G + 1.2 \times Q_{wind} + \sum 1.3 \times Q_i$

Table 11. Load combinations according to EN 13782

This means the following load combinations will be checked/calculated:

1. 1.0 x Own weight + 1.2 x Wind load
2. 1.0 x Own weight + 1.2 x Wind load + 1.3 x Snow load [FR]

H.5. Load cases

H.5.1 Own weight
The own weight of the fabric is 730 g/m² = 0.0073 kN/m² and is considered automatically within EASY software.

H.5.2 Pretension
The structure will be pretensioned with guy ropes and belts. With a ratchet it is possible to achieve 2-3 kN force in corner belts. Approximately 2kN force is achieved in side ropes. The tensile force in ropes and belts results in a membrane stress of approximately 0.50 kN/m². Higher membrane forces can be present near connection points due to the stretchable nature of the fabric.

H.5.3 Conventional / snow load
Conventional load according to EN 13782: The stability shall be checked with a conventional vertical load of 0.1 kN/m². This can be seen as a snow load of 0.1 kN/m² (4cm) according to the French CTS, where snow load must be combined with wind load. This condition is tackled in J. Addition 2 – element check for combined wind and snow load on page 68.

H.5.4 User load
A user defined load (for light, sound and/or decoration purposes) is 25 kg per pole and is added after the analysis while performing element checks.

H.5.5 Wind load

H.5.5.1 Wind pressure

Wind load according to EN 13782, 7.4.2.2:
q_p (z_e) of 0.50 kN/m² if the height of the structure does not exceed 5m.

Wind load can be reduced for use in the Netherlands. A reduction factor α=0.5 is applied, which leads to **q_p (z_e) of 0.25 kN/m² [NL],** this is tackled in I. Addition 1 – element check for reduced wind speeds [NL]

H.5.5.2 Wind shape values (Cp-factors)
Two different wind situations are reviewed for the membrane:

1. The whole tent is subjected to wind suction (Cp values given in EN 13782)
2. The whole tent is subjected to wind pressure (Cp values given in EN 13782)

<u>Wind suction</u>

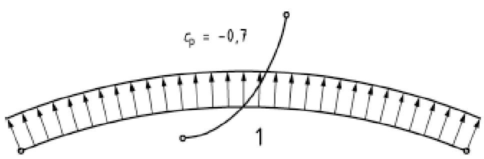


Figure 5. C_p values for rectangle structures EN 13782

P_w	$p_{w,rep}$	
500 N/m^2	-0.7 x 0.500 = **-0.350 kN/m^2**	
250 N/m^2	-0.7 x 0.250 = **-0.125 kN/m^2**	[NL]

Wind pressure

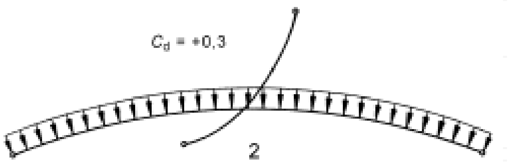


Figure 6. C_p values for rectangle structures EN 13782

P_w	$p_{w,rep}$
500 N/m^2	0.3 x 0.500 = **0.150 kN/m^2**

H.6. Calculation results

H.6.1 Listing of calculated load combinations

LC1 = Pretension
LC2 = Own weight
LC3 = Snow load
LC4 = Wind pressure
LC5 = Wind suction
LC6 = Wind suction reduced

The following load combinations are considered:
partial safety factors are added after the static analysis (see H.4.2).

	LC1	LC2	LC3	LC4	LC5	LC6 [NL]
CO1	1 x	1 x				
CO2	1 x	1 x	1 x			
CO3	1 x	1 x		1 x		
CO4	1 x	1 x			1 x	
CO5 [NL]	1 x	1 x				1 x
CO6 [FR]	1 x	1 x	1 x	1 x		

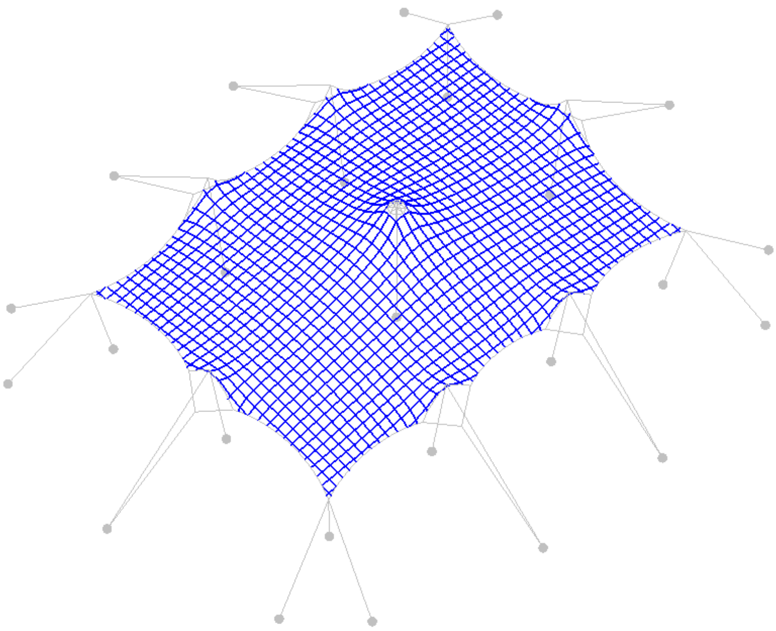
Table 12. Combinations (CO)

The results for load combinations CO5 and CO6 are mentioned in the upcoming chapter but considered in calculations only in chapters I and J on pages 59 and 68 respectively.

H.6.2 Overview: Global results of static analysis

H.6.2.1 Membrane

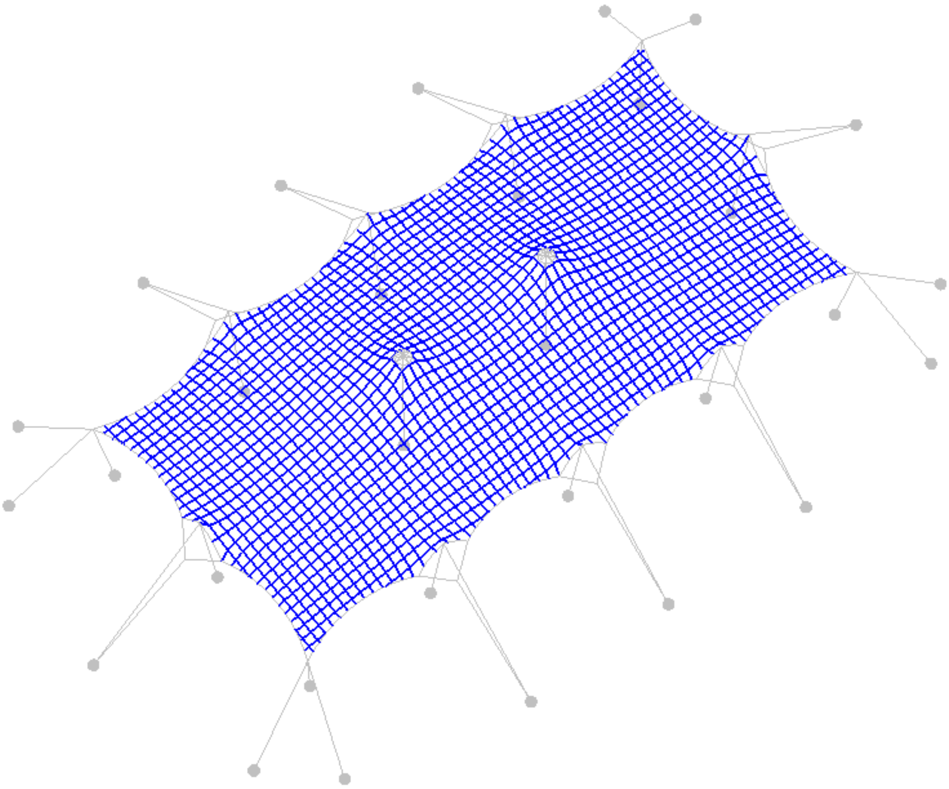
H.6.2.1.1. 7.5x10m model



		Load combination	F_{rep}	Pag.
Warp		CO1. Own weight + pretension	0.96 kN/m	101
		CO2. Own weight + pretension + snow	1.93 kN/m	105
		CO3. Own weight + pretension + wind pressure	2.37 kN/m	109
	Max	CO4. Own weight + pretension + wind suction	2.98 kN/m	113
		CO5. Own weight + pretension + wind suction reduced [NL]	2.06 kN/m	117
		CO6. Own weight + pretension + wind pressure + snow [FR]	3.46 kN/m	121
Weft		CO1. Own weight + pretension	1.02 kN/m	101
		CO2. Own weight + pretension + conventional / snow	2.55 kN/m	105
	Max	CO3. Own weight + pretension + wind pressure	3.33 kN/m	109
		CO4. Own weight + pretension + wind suction	0.98 kN/m	113
		CO5. Own weight + pretension + wind suction reduced [NL]	1.11 kN/m	117
		CO6. Own weight + pretension + wind pressure + snow [FR]	4.54 kN/m	121
Edge		CO1. Own weight + pretension	2.04 kN	101
		CO2. Own weight + pretension + conventional / snow	2.30 kN	105
		CO3. Own weight + pretension + wind pressure	2.55 kN	109
	Max	CO4. Own weight + pretension + wind suction	3.26 kN	113
		CO5. Own weight + pretension + wind suction reduced [NL]	2.45 kN	117
		CO6. Own weight + pretension + wind pressure + snow [FR]	2.92 kN	121

Table 13. Leading forces in membrane – 7.5x10m

H.6.2.1.2. 7.5x15m model

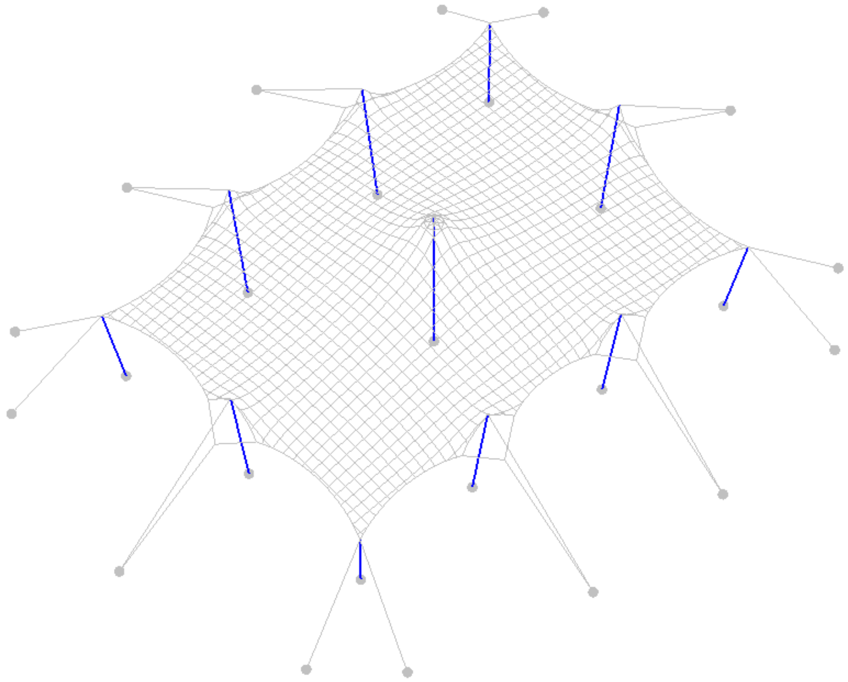


		Load combination	F_{rep}	Pag.
Warp		CO1. Own weight + pretension	1.22 kN/m	125
		CO2. Own weight + pretension + snow	2.56 kN/m	129
	Max	CO3. Own weight + pretension + wind pressure	3.38 kN/m	133
		CO4. Own weight + pretension + wind suction	2.23 kN/m	137
		CO5. Own weight + pretension + wind suction reduced [NL]	1.24 kN/m	141
		CO6. Own weight + pretension + wind pressure + snow [FR]	4.72 kN/m	145
Weft		CO1. Own weight + pretension	1.05 kN/m	125
		CO2. Own weight + pretension + conventional / snow	1.84 kN/m	129
		CO3. Own weight + pretension + wind pressure	2.43 kN/m	133
	Max	CO4. Own weight + pretension + wind suction	2.52 kN/m	137
		CO5. Own weight + pretension + wind suction reduced [NL]	2.14 kN/m	141
		CO6. Own weight + pretension + wind pressure + snow [FR]	3.39 kN/m	145
Edge		CO1. Own weight + pretension	1.96 kN	125
		CO2. Own weight + pretension + conventional / snow	2.29 kN	129
		CO3. Own weight + pretension + wind pressure	2.62 kN	133
	Max	CO4. Own weight + pretension + wind suction	3.36 kN	137
		CO5. Own weight + pretension + wind suction reduced [NL]	2.60 kN	141
		CO6. Own weight + pretension + wind pressure + snow [FR]	3.04 kN	145

Table 14. Leading forces in membrane – 7.5x15m

H.6.2.2 Poles

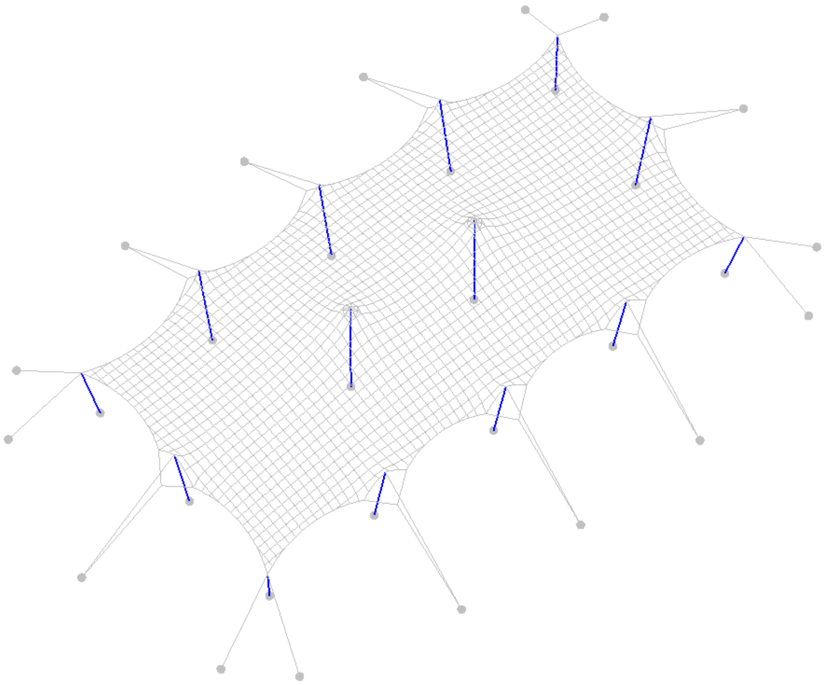
H.6.2.2.1. 7.5x10m model



		Load combination	F_{rep}	Pag.
Main poles		CO1. Own weight + pretension	-2.19 kN	101
		CO2. Own weight + pretension + snow	-4.72 kN	105
	Max	CO3. Own weight + pretension + wind pressure	-5.91 kN	109
		CO4. Own weight + pretension + wind suction	-0.06 kN	113
		CO5. Own weight + pretension + wind suction reduced [NL]	-0.22 kN	117
		CO6. Own weight + pretension + wind pressure + snow [FR]	-8.57 kN	121
Side poles		CO1. Own weight + pretension	-1.50 kN	101
		CO2. Own weight + pretension + conventional / snow	-2.61 kN	105
	Max	CO3. Own weight + pretension + wind pressure	-3.32 kN	109
		CO4. Own weight + pretension + wind suction	-2.29 kN	113
		CO5. Own weight + pretension + wind suction reduced [NL]	-0.82 kN	117
		CO6. Own weight + pretension + wind pressure + snow [FR]	-4.49 kN	121
Corner poles		CO1. Own weight + pretension	-2.69 kN	101
		CO2. Own weight + pretension + conventional / snow	-3.26 kN	105
	Max	CO3. Own weight + pretension + wind pressure	-3.79 kN	109
		CO4. Own weight + pretension + wind suction	-3.92 kN	113
		CO5. Own weight + pretension + wind suction reduced [NL]	-3.15 kN	117
		CO6. Own weight + pretension + wind pressure + snow [FR]	-4.52 kN	121

Table 15. Leading forces in poles – 7.5x10m model

H.6.2.2.2. 7.5x15m model

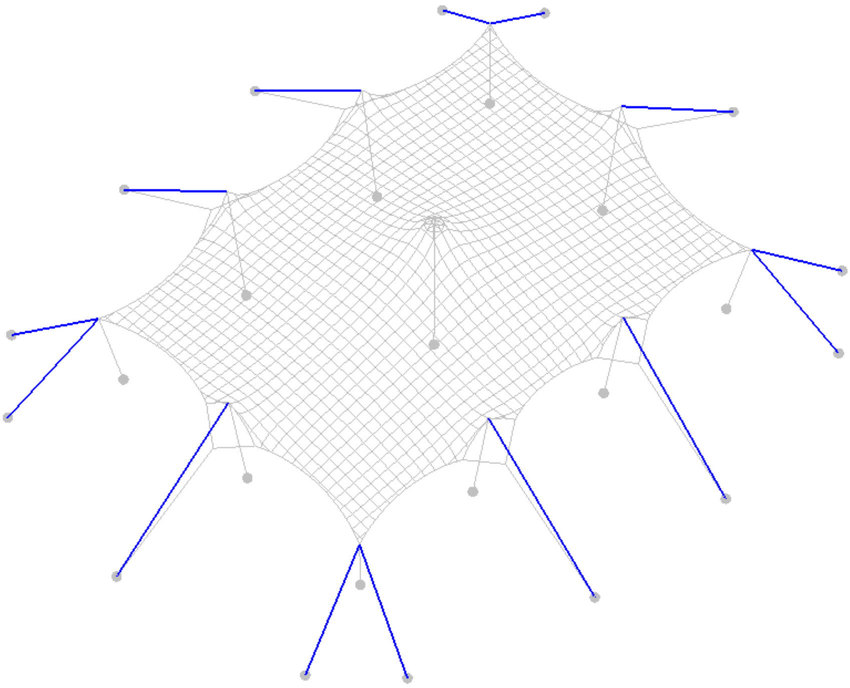


		Load combination	F_{rep}	Pag.
Main poles		CO1. Own weight + pretension	-1.94 kN	125
		CO2. Own weight + pretension + snow	-4.40 kN	129
	Max	CO3. Own weight + pretension + wind pressure	-5.60 kN	133
		CO4. Own weight + pretension + wind suction	-0.01 kN	137
		CO5. Own weight + pretension + wind suction reduced [NL]	-0.10 kN	141
		CO6. Own weight + pretension + wind pressure + snow [FR]	-8.20 kN	145
Side poles		CO1. Own weight + pretension	-1.55 kN	125
		CO2. Own weight + pretension + conventional / snow	-2.83 kN	129
	Max	CO3. Own weight + pretension + wind pressure	-3.69 kN	133
		CO4. Own weight + pretension + wind suction	-1.89 kN	137
		CO5. Own weight + pretension + wind suction reduced [NL]	-0.78 kN	141
		CO6. Own weight + pretension + wind pressure + snow [FR]	-5.04 kN	145
Corner poles		CO1. Own weight + pretension	-2.80 kN	125
		CO2. Own weight + pretension + conventional / snow	-3.45 kN	129
		CO3. Own weight + pretension + wind pressure	-4.08 kN	133
	Max	CO4. Own weight + pretension + wind suction	-4.44 kN	137
		CO5. Own weight + pretension + wind suction reduced [NL]	-3.27 kN	141
		CO6. Own weight + pretension + wind pressure + snow [FR]	-4.90 kN	145

Table 16. Leading forces in poles – 7.5x15m model

H.6.2.3 Belts

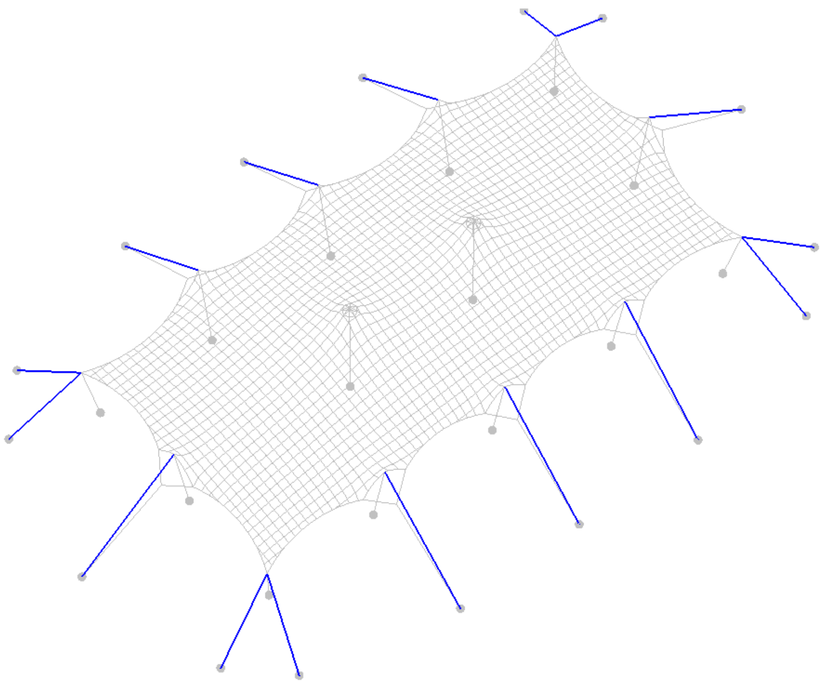
H.6.2.3.1. 7.5x10m model



		Load combination	F_{rep}	Pag.
Corner belts		CO1. Own weight + pretension	2.21 kN	101
		CO2. Own weight + pretension + snow	2.48 kN	105
		CO3. Own weight + pretension + wind pressure	2.82 kN	109
	Max	CO4. Own weight + pretension + wind suction	3.60 kN	113
		CO5. Own weight + pretension + wind suction reduced [NL]	3.13 kN	117
		CO6. Own weight + pretension + wind pressure + snow [FR]	3.26 kN	121
Belts – long side		CO1. Own weight + pretension	0.12 kN	101
		CO2. Own weight + pretension + conventional / snow	0.35 kN	105
	Max	CO3. Own weight + pretension + wind pressure	0.51 kN	109
		CO4. Own weight + pretension + wind suction	0.12 kN	113
		CO5. Own weight + pretension + wind suction reduced [NL]	0.15 kN	117
		CO6. Own weight + pretension + wind pressure + snow [FR]	0.82 kN	121
Belts – short side		CO1. Own weight + pretension	0.24 kN	101
		CO2. Own weight + pretension + conventional / snow	0.72 kN	105
	Max	CO3. Own weight + pretension + wind pressure	1.01 kN	109
		CO4. Own weight + pretension + wind suction	0.07 kN	113
		CO5. Own weight + pretension + wind suction reduced [NL]	0.01 kN	117
		CO6. Own weight + pretension + wind pressure + snow [FR]	1.47 kN	121

Table 17. Leading forces in belts – 7.5x10m model

H.6.2.3.2. 7.5x15m model

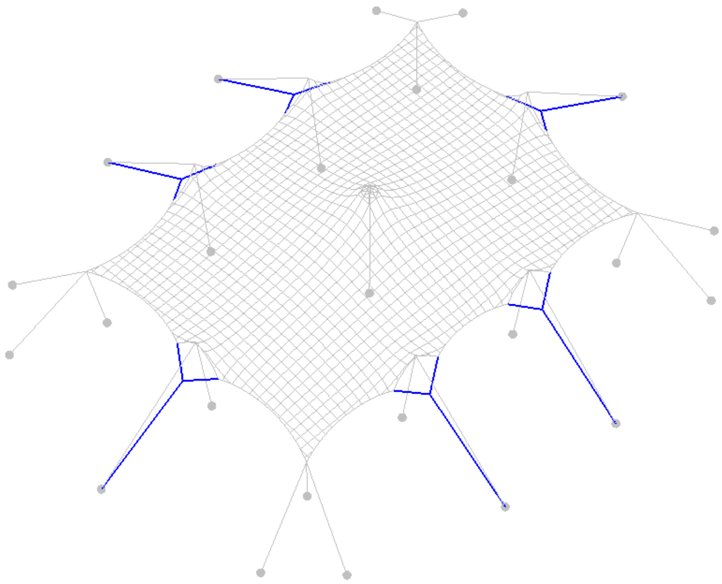


		Load combination	F_{rep}	Pag.
Corner belts		CO1. Own weight + pretension	2.24 kN	125
		CO2. Own weight + pretension + snow	2.55 kN	129
		CO3. Own weight + pretension + wind pressure	2.95 kN	133
	Max	CO4. Own weight + pretension + wind suction	4.05 kN	137
		CO5. Own weight + pretension + wind suction reduced [NL]	3.22 kN	141
		CO6. Own weight + pretension + wind pressure + snow [FR]	3.47 kN	145
Belts – long side		CO1. Own weight + pretension	0.17 kN	125
		CO2. Own weight + pretension + conventional / snow	0.42 kN	129
	Max	CO3. Own weight + pretension + wind pressure	0.61 kN	133
		CO4. Own weight + pretension + wind suction	0.34 kN	137
		CO5. Own weight + pretension + wind suction reduced [NL]	0.18 kN	141
		CO6. Own weight + pretension + wind pressure + snow [FR]	0.93 kN	145
Belts – short side		CO1. Own weight + pretension	0.25 kN	125
		CO2. Own weight + pretension + conventional / snow	0.83 kN	129
	Max	CO3. Own weight + pretension + wind pressure	1.23 kN	133
		CO4. Own weight + pretension + wind suction	0.14 kN	137
		CO5. Own weight + pretension + wind suction reduced [NL]	0.01 kN	141
		CO6. Own weight + pretension + wind pressure + snow [FR]	1.85 kN	145

Table 18. Leading forces in belts – 7.5x15m model

H.6.2.4 Guy ropes

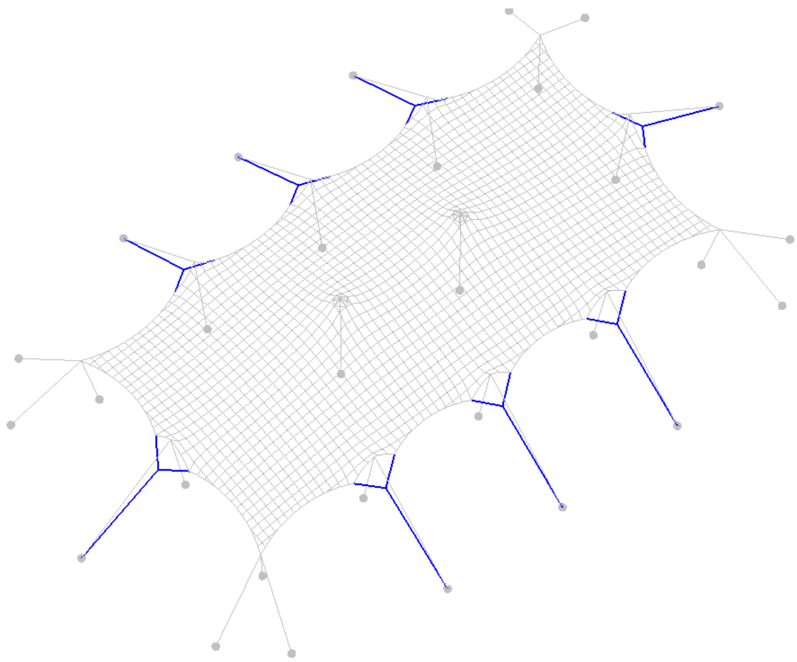
H.6.2.4.1. 7.5x10m model



		Load combination	F_{rep}	Pag.
Combined rope – long side		CO1. Own weight + pretension	1.83 kN	101
		CO2. Own weight + pretension + snow	1.94 kN	105
		CO3. Own weight + pretension + wind pressure	2.16 kN	109
	Max	CO4. Own weight + pretension + wind suction	4.05 kN	113
		CO5. Own weight + pretension + wind suction reduced [NL]	3.28 kN	117
		CO6. Own weight + pretension + wind pressure + snow [FR]	2.43 kN	121
V-ropes – long side		CO1. Own weight + pretension	1.38 kN	101
		CO2. Own weight + pretension + conventional / snow	1.48 kN	105
		CO3. Own weight + pretension + wind pressure	1.64 kN	109
	Max	CO4. Own weight + pretension + wind suction	2.97 kN	113
		CO5. Own weight + pretension + wind suction reduced [NL]	2.43 kN	117
		CO6. Own weight + pretension + wind pressure + snow [FR]	1.83 kN	121
Combined rope – short side		CO1. Own weight + pretension	1.96 kN	101
		CO2. Own weight + pretension + conventional / snow	2.08 kN	105
		CO3. Own weight + pretension + wind pressure	2.30 kN	109
	Max	CO4. Own weight + pretension + wind suction	3.36 kN	113
		CO5. Own weight + pretension + wind suction reduced [NL]	2.92 kN	117
		CO6. Own weight + pretension + wind pressure + snow [FR]	2.56 kN	121
V-ropes – short side		CO1. Own weight + pretension	1.43 kN	101
		CO2. Own weight + pretension + conventional / snow	1.47 kN	105
		CO3. Own weight + pretension + wind pressure	1.60 kN	109
	Max	CO4. Own weight + pretension + wind suction	2.56 kN	113
		CO5. Own weight + pretension + wind suction reduced [NL]	2.19 kN	117
		CO6. Own weight + pretension + wind pressure + snow [FR]	1.75 kN	121

Table 19. Leading forces in ropes – 7.5x10m model

H.6.2.4.2. 7.5x15m model

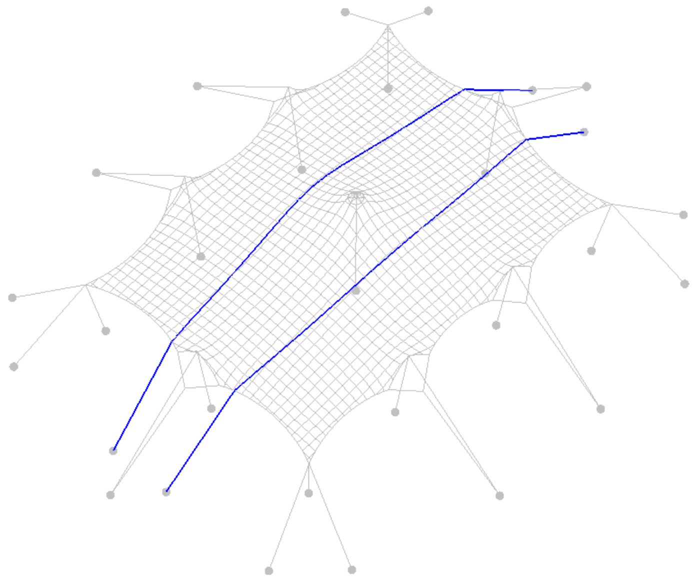


		Load combination	F_{rep}	Pag.
Combined rope – long side		CO1. Own weight + pretension	2.11 kN	125
		CO2. Own weight + pretension + snow	2.18 kN	129
		CO3. Own weight + pretension + wind pressure	2.47 kN	133
	Max	CO4. Own weight + pretension + wind suction	4.29 kN	137
		CO5. Own weight + pretension + wind suction reduced [NL]	4.01 kN	141
		CO6. Own weight + pretension + wind pressure + snow [FR]	2.75 kN	145
V-ropes – long side		CO1. Own weight + pretension	1.47 kN	125
		CO2. Own weight + pretension + conventional / snow	1.59 kN	129
		CO3. Own weight + pretension + wind pressure	1.80 kN	133
	Max	CO4. Own weight + pretension + wind suction	2.98 kN	137
		CO5. Own weight + pretension + wind suction reduced [NL]	2.74 kN	141
		CO6. Own weight + pretension + wind pressure + snow [FR]	2.00 kN	145
Combined rope – short side		CO1. Own weight + pretension	1.96 kN	125
		CO2. Own weight + pretension + conventional / snow	2.28 kN	129
		CO3. Own weight + pretension + wind pressure	2.64 kN	133
	Max	CO4. Own weight + pretension + wind suction	3.98 kN	137
		CO5. Own weight + pretension + wind suction reduced [NL]	2.99 kN	141
		CO6. Own weight + pretension + wind pressure + snow [FR]	3.04 kN	145
V-ropes – short side		CO1. Own weight + pretension	1.42 kN	125
		CO2. Own weight + pretension + conventional / snow	1.59 kN	129
		CO3. Own weight + pretension + wind pressure	1.81 kN	133
	Max	CO4. Own weight + pretension + wind suction	2.87 kN	137
		CO5. Own weight + pretension + wind suction reduced [NL]	2.26 kN	141
		CO6. Own weight + pretension + wind pressure + snow [FR]	2.05 kN	145

Table 20. Leading forces in ropes – 7.5x15m model

H.6.2.5 Stormbelts

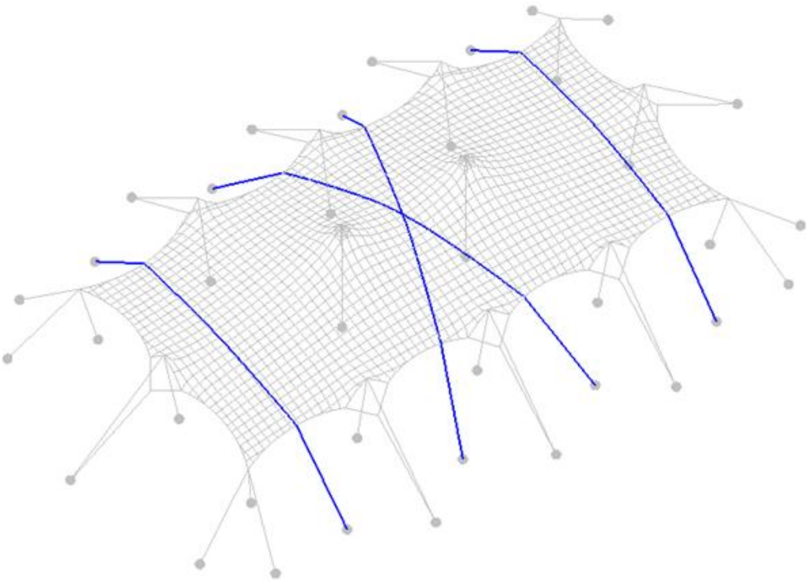
H.6.2.5.1. 7.5x10m model



		Load combination	F_{rep}	Pag.
Stormbelts		CO1. Own weight + pretension	0.00 kN	101
		CO2. Own weight + pretension + snow	0.00 kN	105
		CO3. Own weight + pretension + wind pressure	0.00 kN	109
	Max	CO4. Own weight + pretension + wind suction	4.25 kN	113
		CO5. Own weight + pretension + wind suction reduced [NL]	0.00 kN	117
		CO6. Own weight + pretension + wind pressure + snow [FR]	0.00 kN	121

Table 21. Leading forces in stormbelts – 7.5x10m model

H.6.2.5.2. 7.5x15m model

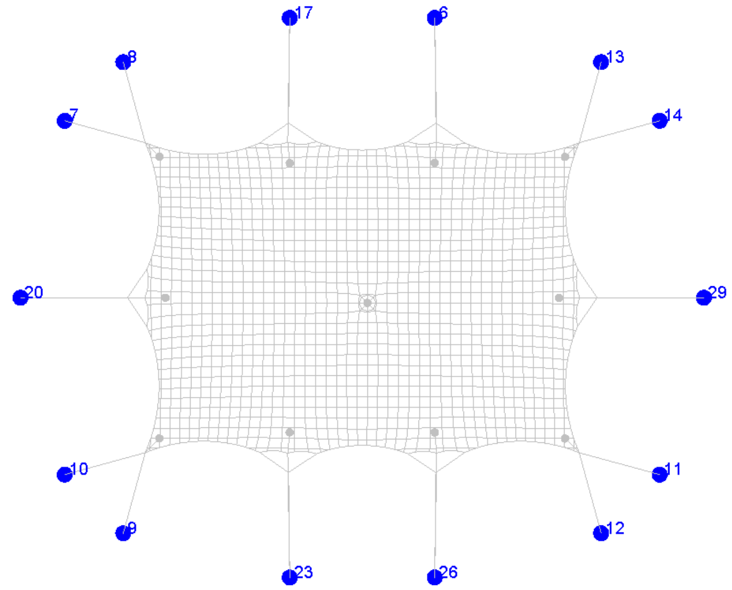


		Load combination	F_{rep}	Pag.
Stormbelts		CO1. Own weight + pretension	0.00 kN	125
		CO2. Own weight + pretension + snow	0.00 kN	129
		CO3. Own weight + pretension + wind pressure	0.00 kN	133
	Max	CO4. Own weight + pretension + wind suction	3.89 kN	137
		CO5. Own weight + pretension + wind suction reduced [NL]	0.00 kN	141
		CO6. Own weight + pretension + wind pressure + snow [FR]	0.00 kN	145

Table 22. Leading forces in stormbelts – 7.5x15m model

H.6.2.6 Reaction forces

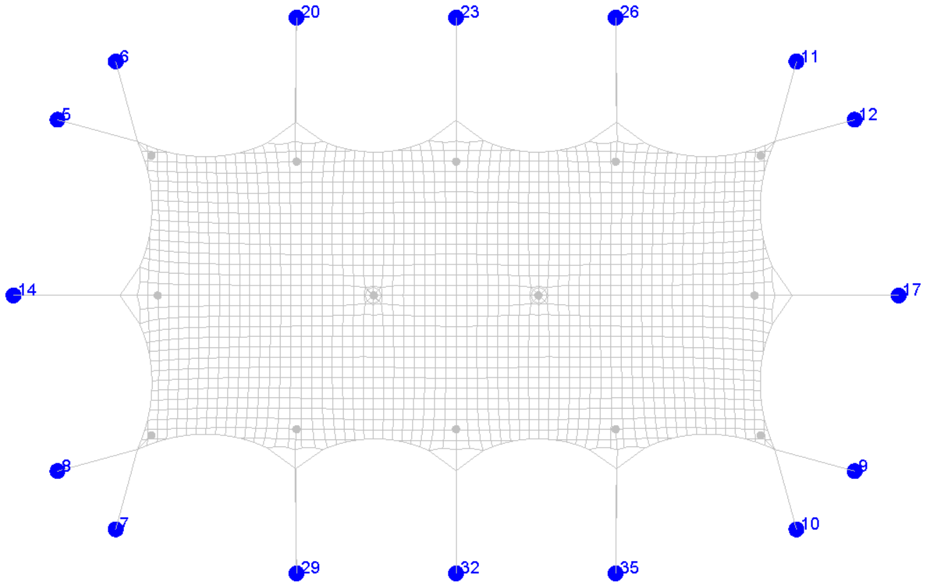
H.6.2.6.1. 7.5x10m model



		Load combination	F_{rep}	Pag.
Long side		CO1. Own weight + pretension	1.95 kN	101
		CO2. Own weight + pretension + snow	2.29 kN	105
		CO3. Own weight + pretension + wind pressure	2.67 kN	109
	Max	CO4. Own weight + pretension + wind suction	4.17 kN	113
		CO5. Own weight + pretension + wind suction reduced [NL]	3.43 kN	117
		CO6. Own weight + pretension + wind pressure + snow [FR]	3.29 kN	121
Short side		CO1. Own weight + pretension	2.19 kN	101
		CO2. Own weight + pretension + conventional / snow	2.79 kN	105
		CO3. Own weight + pretension + wind pressure	3.31 kN	109
	Max	CO4. Own weight + pretension + wind suction	3.41 kN	113
		CO5. Own weight + pretension + wind suction reduced [NL]	2.92 kN	117
		CO6. Own weight + pretension + wind pressure + snow [FR]	4.07 kN	121
Corner		CO1. Own weight + pretension	2.21 kN	101
		CO2. Own weight + pretension + conventional / snow	2.48 kN	105
		CO3. Own weight + pretension + wind pressure	2.82 kN	109
	Max	CO4. Own weight + pretension + wind suction	3.80 kN	113
		CO5. Own weight + pretension + wind suction reduced [NL]	3.13 kN	117
		CO6. Own weight + pretension + wind pressure + snow [FR]	3.30 kN	121

Table 23. Leading reaction forces – 7.5x10m model

H.6.2.6.2. 7.5x15m model



		Load combination	F_{rep}	Pag.
Long side		CO1. Own weight + pretension	2.21 kN	125
		CO2. Own weight + pretension + snow	2.59 kN	129
		CO3. Own weight + pretension + wind pressure	3.07 kN	133
	Max	CO4. Own weight + pretension + wind suction	4.61 kN	137
		CO5. Own weight + pretension + wind suction reduced [NL]	4.18 kN	141
		CO6. Own weight + pretension + wind pressure + snow [FR]	3.72 kN	145
Short side		CO1. Own weight + pretension	2.21 kN	125
		CO2. Own weight + pretension + conventional / snow	3.10 kN	129
		CO3. Own weight + pretension + wind pressure	3.85 kN	133
	Max	CO4. Own weight + pretension + wind suction	4.11 kN	137
		CO5. Own weight + pretension + wind suction reduced [NL]	2.99 kN	141
		CO6. Own weight + pretension + wind pressure + snow [FR]	4.93 kN	145
Corner		CO1. Own weight + pretension	2.24 kN	125
		CO2. Own weight + pretension + conventional / snow	2.55 kN	129
		CO3. Own weight + pretension + wind pressure	2.95 kN	133
	Max	CO4. Own weight + pretension + wind suction	4.05 kN	137
		CO5. Own weight + pretension + wind suction reduced [NL]	3.22 kN	141
		CO6. Own weight + pretension + wind pressure + snow [FR]	3.51 kN	145

Table 24. Leading reaction forces – 7.5x15m model

H.6.3 Summary of forces

H.6.3.1 7.5x10m model

kN or kN/m	CO1	CO2	CO3	CO4	CO5 [NL]	CO6 [FR]
Warp	0.96	1.93	2.37	2.98	2.06	3.46
Weft	1.02	2.55	3.33	0.98	1.11	4.54
Edge	2.04	2.30	2.55	3.26	2.45	2.92
Main pole	-2.19	-4.72	-5.91	-0.06	-0.22	-8.57
Side pole	-1.50	-2.61	-3.32	-2.29	-0.82	-4.49
Corner pole	-2.69	-3.26	-3.79	-3.92	-3.15	-4.52
Corner belt	2.21	2.48	2.82	3.60	3.13	3.26
Belts - long side	0.12	0.35	0.51	0.12	0.15	0.82
Belts - short side	0.24	0.72	1.01	0.07	0.00	1.47
Stormbelts	0.00	0.00	0.00	4.25	0.00	0.00
Combined rope - long side	1.83	1.94	2.16	4.05	3.28	2.43
V-ropes - long side	1.38	1.48	1.64	2.97	2.43	1.83
Combined rope - short side	1.96	2.08	2.30	3.36	2.92	2.56
V-ropes short side	1.43	1.47	1.60	2.56	2.19	1.75
Reaction - long side	1.95	2.29	2.67	4.17	3.43	3.29
Reaction - short side	2.19	2.79	3.31	3.41	2.92	4.07
Reaction - corner	2.21	2.48	2.82	3.60	3.13	3.30

Table 25. Summary of forces for 7.5x10m model

H.6.3.2 7.5x15m model

kN or kN/m	CO1	CO2	CO3	CO4	CO5 [NL]	CO6 [FR]
Warp	1.22	2.56	3.38	2.23	1.24	4.72
Weft	1.05	1.84	2.43	2.52	2.14	3.39
Edge	1.96	2.29	2.62	3.36	2.60	3.04
Main pole	-1.94	-4.40	-5.60	-0.01	-0.10	-8.20
Side pole	-1.55	-2.83	-3.69	-1.89	-0.78	-5.04
Corner pole	-2.80	-3.45	-4.08	-4.44	-3.27	-4.90
Corner belt	2.24	2.55	2.95	4.05	3.22	3.47
Belts - long side	0.17	0.42	0.61	0.34	0.18	0.93
Belts - short side	0.25	0.83	1.23	0.14	0.00	1.85
Stormbelts	0.00	0.00	0.00	3.89	0.00	0.00
Combined rope - long side	2.11	2.18	2.47	4.29	4.01	2.75
V-ropes - long side	1.47	1.59	1.80	2.98	2.74	2.00
Combined rope - short side	1.96	2.28	2.64	3.98	2.99	3.04
V-ropes short side	1.42	1.59	1.81	2.87	2.26	2.05
Reaction - long side	2.21	2.59	3.07	4.61	4.18	3.72
Reaction - short side	2.21	3.10	3.85	4.11	2.99	4.93
Reaction - corner	2.24	2.55	2.95	4.05	3.22	3.51

Table 26. Summary of forces for 7.5x15m model

H.7. Element check

H.7.1 Membrane

	Load combination	Element	Representative stress	Design value stress	Pag.
7.5x15m	CO1. Own weight + pretension	Long term warp	1.22 kN/m	1.65 kN/m (γ = 1.35)	41
7.5x15m	CO1. Own weight + pretension	Long term weft	1.05 kN/m	1.42 kN/m (γ = 1.35)	41
7.5x15m	CO3. Own weight + pretension + wind pressure	Short term warp	3.38 kN/m	5.07 kN/m (γ = 1.5)	41
7.5x15m	CO4. Own weight + pretension + wind suction	Short term weft	2.52 kN/m	3.78 kN/m (γ = 1.5)	41

Sioen RP5639 fabric is being used.

UC.1a	Long term warp	$S_{Ed} / S_{rd} < 1$	1.65 / 9.6 = 0.17 < 1	OK
UC.1b	Long term weft	$S_{Ed} / S_{rd} < 1$	1.42 / 7.2 = 0.20 < 1	OK
UC.1c	Short term warp	$S_{Ed} / S_{rd} < 1$	5.07 / 12.0 = 0.42 < 1	OK
UC.1d	Short term weft	$S_{Ed} / S_{rd} < 1$	3.78 / 9.0 = 0.42 < 1	OK

For capacity of membrane see H.2.1, page 21

H.7.2 Main poles

	Load combinations	Element	Representative force	Design value force	Pag.
7.5x10m	CO3. Own weight + pretension + wind pressure	Main pole 4.0m	-5.91 kN	-8.87 kN (γ = 1.5)	40

User load of max. 25 kg is applied, loaded centrically.

4.0m pole Ned = -5.91 x 1.50 + (-0.25 x 1.35) = -9.20 kN

H.7.2.1 4.0m pole - Wood, Spruce C18

Profile	=	Pole, Ø ≈ 95 mm
		average diameter, as a minimum required at the middle of the pole
Length	=	max. 4.00m
Quality	=	C18

The poles are considered as hinged poles; the buckling length is equivalent to the pole length.

UC.2	4.0m	$\sigma_{c,0,d}$ / ($f_{c,0,d}$ x k_{cy}) = 1.30 / (12.46 x 0.109) = 0.96 ≤ 1	OK

See Annex E.1 Main pole 4.0m – Wood, on page 149 for the elaborate check

H.7.3 Side poles

	Load combinations	Element	Representative force	Design value force	Pag.
7.5x15m	CO3. Own weight + pretension + wind pressure	Side pole 3.0m	-3.69 kN	-5.54 kN (γ = 1.5)	41

User load of max. 25 kg is applied, loaded centrically.

3m pole Ned = -3.69 x 1.50 + (-0.25 x 1.35) = -5.87 kN

H.7.3.1 3m pole - Wood, Spruce C18

Profile = Pole, Ø ≈ 75 mm
average diameter, as a minimum required at the middle of the pole

Length = max. 3.0m

Quality = C18

The poles are considered as hinged poles; the buckling length is equivalent to the pole length.

UC.3	3m	$\sigma_{c,0,d}$ / ($f_{c,0,d}$ x k_{cy}) = 1.33 / (12.46 x 0.12) = 0.89 < 1	OK

See Annex E.2 Side pole 3m – Wood, Spruce C18, on page 150 for the elaborate check

H.7.4 Corner poles

	Load combinations	Element	Representative force	Design value force	Pag.
7.5x15m	CO4. Own weight + pretension + wind suction	Corner pole 2.5m	-4.44 kN	-6.66 kN (γ = 1.5)	41

User load of max. 25 kg is applied, loaded centrically.

2.5m corner pole Ned = -4.44 x 1.50 + (-0.25 x 1.35) = -7.00 kN

H.7.4.1 2.5m pole - Wood, Spruce C18

Profile = Pole, Ø ≈ 70 mm
average diameter, as a minimum required at the middle of the pole

Length = max. 2.5m

Quality = C18

The poles are considered as hinged poles; the buckling length is equivalent to the pole length.

UC.4	2.5m	$\sigma_{c,0,d}$ / ($f_{c,0,d}$ x k_{cy}) = 1.82 / (12.46 x 0.149) = 0.98 < 1	OK

See Annex E.3 Corner pole 2.5m - Wood, C18, on page 151 for the elaborate check

H.7.5 Guy ropes

	Load combination	Element	Representa tive force	Design value force	Pag.
7.5x15m	CO4. Own weight + pretension + wind suction	long side - combined	4.29 kN	6.44 kN (γ = 1.5)	41
7.5x10m	CO4. Own weight + pretension + wind suction	long side - combined	4.05 kN	6.08 kN (γ = 1.5)	40
7.5x15m	CO4. Own weight + pretension + wind suction	long side – V-ropes	2.98 kN	4.47 kN (γ = 1.5)	41
7.5x10m	CO4. Own weight + pretension + wind suction	long side – V-ropes	2.97 kN	4.46 kN (γ = 1.5)	40
7.5x15m	CO4. Own weight + pretension + wind suction	short side - combined	3.98 kN	5.97 kN (γ = 1.5)	41
7.5x10m	CO4. Own weight + pretension + wind suction	short side - combined	3.36 kN	5.04 kN (γ = 1.5)	40
7.5x15m	CO4. Own weight + pretension + wind suction	short side – V-ropes	2.87 kN	4.31 kN (γ = 1.5)	41
7.5x10m	CO4. Own weight + pretension + wind suction	short side – V-ropes	2.56 kN	3.84 kN (γ = 1.5)	40

Synthetic rope Ø8mm, BL 900 daN. Is being used → $F_{rd,8mm}$ = 2.25 kN.

In a case of using individual guy ropes made of multiple sections, the ropes need to be tied back, creating multiple rope sections that together will take the load.

7.5x10m model

UC.5a	long side - combined	1 individual rope made of 3 sections Ø8mm	F_d / F_{rd} < 1	6.08 / (1 x 3 x 2.25) = 0.92 < 1	OK
UC.5b	long side – V-ropes	1 individual rope made of 2 sections Ø8mm	F_d / F_{rd} < 1	4.46 / (1 x 2 x 2.25) = 0.99 < 1	OK
UC.5c	short side - combined	1 individual rope made of 3 sections Ø8mm	F_d / F_{rd} < 1	5.04 / (1 x 3 x 2.25) = 0.78 < 1	OK
UC.5d	short side – V-ropes	1 individual rope made of 2 sections Ø8mm	F_d / F_{rd} < 1	3.84 / (1 x 2 x 2.25) = 0.87 < 1	OK

For capacity of ropes see H.2.2, page 21

7.5x15m model

UC.5e	long side - combined	1 individual rope made of 3 sections Ø8mm	F_d / F_{rd} < 1	6.44 / (1 x 3 x 2.25) = 0.95 < 1	OK
UC.5f	long side – V-ropes	1 individual rope made of 2 sections Ø8mm	F_d / F_{rd} < 1	4.47 / (1 x 2 x 2.25) = 0.99 < 1	OK
UC.5g	short side - combined	1 individual rope made of 3 sections Ø8mm	F_d / F_{rd} < 1	5.97 / (1 x 3 x 2.25) = 0.88 < 1	OK

UC.5h	short side – V-ropes	1 individual rope made of 2 sections Ø8mm	$F_d / F_{rd} < 1$	4.31 / (1 x 2 x 2.25) = 0.96 < 1	OK

For capacity of ropes see H.2.2, page 21

It is also possible to use alternative **ropes** with a higher breaking strength (BL_{rope}) and less rope sections (n), as long as $\mathbf{BL_{rope} \times n_{sections} \times n_{ropes} \geq BL_{required}}$

Required breaking loads for ropes:

<u>7.5x10m model</u>

Long side - combined: $\mathbf{BL_{rope} \times n_{sections} \times n_{ropes} \geq 2432}$ **kg**
Long side – V-ropes: $\mathbf{BL_{rope} \times n_{sections} \times n_{ropes} \geq 1784}$ **kg**
Short side - combined: $\mathbf{BL_{rope} \times n_{sections} \times n_{ropes} \geq 2016}$ **kg**
Short side – V-ropes: $\mathbf{BL_{rope} \times n_{sections} \times n_{ropes} \geq 1536}$ **kg**

<u>7.5x15m model</u>

Long side - combined: $\mathbf{BL_{rope} \times n_{sections} \times n_{ropes} \geq 2576}$ **kg**
Long side – V-ropes: $\mathbf{BL_{rope} \times n_{sections} \times n_{ropes} \geq 1788}$ **kg**
Short side - combined: $\mathbf{BL_{rope} \times n_{sections} \times n_{ropes} \geq 2388}$ **kg**
Short side - V-ropes: $\mathbf{BL_{rope} \times n_{sections} \times n_{ropes} \geq 1724}$ **kg**

H.7.6 Belts

	Load combination	Element	Representative force	Design value force	Pag.
7.5x15m	CO4. Own weight + pretension + wind suction	Corner belt	4.05 kN	6.08 kN (γ = 1.5)	41
7.5x15m	CO3. Own weight + pretension + wind pressure	Belt – long side	0.61 kN	0.92 kN (γ = 1.5)	41
7.5x15m	CO3. Own weight + pretension + wind pressure	Belt – short side	1.23 kN	1.85 kN (γ = 1.5)	41

H.7.6.1 Belt capacity check

PES belt with a breaking strength of 3000kg is being used.

UC.6a	Corner belt	1 PES belt with minimum BL = 3000 kg	F_d / F_{rd} < 1	6.08 / (1 x 15) = 0.41 < 1	OK
UC.6b	Belt – long side	1 PES belt with minimum BL = 3000 kg	F_d / F_{rd} < 1	0.92 / (1 x 15) = 0.06 < 1	OK
UC.6c	Belt – short side	1 PES belt with minimum BL = 3000 kg	F_d / F_{rd} < 1	1.85 / (1 x 15) = 0.12 < 1	OK

For capacity of belts see H.2.3, page 21

H.7.6.2 Alternative – rope capacity check

It is possible to use either a rope or a belt. Synthetic rope Ø8mm, BL 900 daN. Is being used → $F_{rd,8mm}$ = 2.25 kN.

In a case of using individual guy ropes, the ropes need to be tied back, creating multiple rope sections that together will take the load.

UC.6d	Corner rope	1 individual rope made of <u>3 sections</u> Ø8mm	F_d / F_{rd} < 1	6.08 / (1 x 3 x 2.25) = 0.90 < 1	OK
UC.6e	Rope – long side	1 individual rope made of <u>1 section</u> Ø8mm	F_d / F_{rd} < 1	0.92 / (1 x 1 x 2.25) = 0.41 < 1	OK
UC.6f	Rope – short side	1 individual rope made of <u>1 section</u> Ø8mm	F_d / F_{rd} < 1	1.85 / (1 x 1 x 2.25) = 0.82 < 1	OK

For capacity of ropes see H.2.2, page 21

It is also possible to use alternative **ropes** with a higher breaking strength (BL_{rope}) and less rope sections (n), as long as $\mathbf{BL_{rope} \times n_{sections} \times n_{ropes} \geq BL_{required}}$
Required breaking loads for ropes:
Corner rope: $\mathbf{BL_{rope} \times n_{sections} \times n_{ropes} \geq 2432}$ **kg**
Rope – long side: $\mathbf{BL_{rope} \times n_{sections} \times n_{ropes} \geq 368}$ **kg**
Rope – long side: $\mathbf{BL_{rope} \times n_{sections} \times n_{ropes} \geq 740}$ **kg**

H.7.7 Stormbelts

	Load combination	Element	Representative force	Design value force	Pag.
7.5x10m	CO4. Own weight + pretension + wind suction	Stormbelt	4.25 kN	6.38 kN (γ = 1.5)	41

PES belt with a breaking strength of 3000kg is being used.

UC.7	Storm belt	1 PES belt with minimum BL = 3000 kg	$F_d / F_{rd} < 1$	6.38 / (1 x 15) = 0.43 < 1.0	OK

For capacity of belts see H.2.3, page 21

H.7.8 Fasteners

Force acting on fasteners is equal to the force in belts or ropes. See these values in chapters H.7.6 and H.7.5.

H.7.8.1 Corner belt connection

H.7.8.1.1. M16 Eye bolt, DIN580

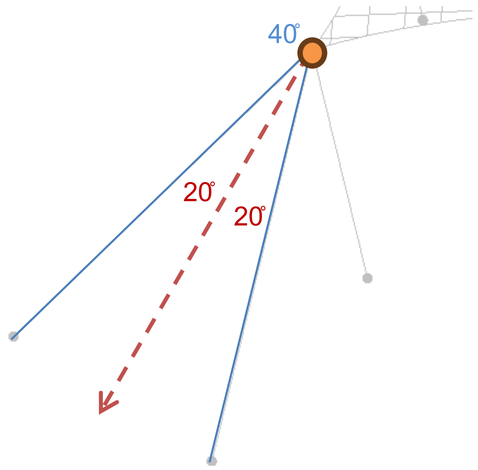
For ≤45°:

M16, WLL 700kg → BL = 6 x 7 kN = 42 kN → Frd = BL / γ_m = 42 / 2 =21 kN

$F_d = 2 \times F_{corner\ belt} \times \cos 20° = 2 \times 6.08 \times \cos 20° = 11.56\ kN$

UC.8a	Corner eye bolt	F_d / F_{rd} < 1	11.56 / 21 = 0.55 < 1.0	OK

For capacity of eye bolts see K.3 Capacity of Eye bolts on page 84.



H.7.8.1.2. Corner belt connection

Similar connection piece with a capacity of at least 11.5 kN can be used.

Orange Pin D-sluiting, WLL 1.0 t → BL = 6 x 10 kN = 60 kN → Frd = BL / ɣm = 60 kN / 2 = 30 kN

UC.8b	D-sluiting	F_d / F_{rd} < 1	11.56 / 11.5 = 0.39 < 1.0	OK

For capacity of D-sluiting see K.4 Capacity of Orange Pin 1 Ton D-sluiting, DIN EN 13889 on page 84.

H.7.8.1.3. Rope ratchet – for alternative corner connection with ropes

Ratchet, BL = 3.5 kN → Frd = BL / ɣm = 3.5 kN / 1.5 = 2.33 kN

F_d=6.08 kN (Instead of corner belts corner ropes are used.)

UC.8c	Corner ropes (alternative)	**F_d / F_{rd} < 1**	**6.08 / 3*x2.33= 0.87 < 1**	**OK**

For capacity of rope ratchet see K.6 Rope ratchet, page 85

* Number of cross sections making a rope. Ratchet is applied only on one cross-section.

H.7.8.2 Side belt connection

H.7.8.2.1. M12 Eye bolt, DIN580

For ≤45°:

M12, WLL 340kg → BL = 6 x 3.4 kN = 20.4 kN → Frd = BL / γ_m = 20.4 / 2 = 10.2 kN

F_d=1.85 kN

UC.8d	Corner eye bolt	F_d / F_{rd} < 1	1.85 / 10.2 = 0.18 < 1.0	OK

For capacity of eye bolts see K.3 Capacity of Eye bolts on page 84.

H.7.8.2.2. Carabine, DIN5299C

Carabine 180x14, WLL 580kg → BL = 2 x 5.8 kN = 11.6 kN → Frd = BL / ɣm = 11.6 kN / 2 = 5.8 kN

UC.8e	Corner eye bolt	F_d / F_{rd} < 1	1.85 / 5.8 = 0.32 < 1.0	OK

For capacity of carabines see K.5 Carabine, DIN 5299C, page 85

H.7.8.3 Side rope connection

H.7.8.3.1. Carabine, DIN5299C

Carabine 180x14, WLL 580kg → BL = 2 x 5.8 kN = 11.6 kN → Frd = BL / ɣm = 11.6 kN / 2 = 5.8 kN

F_d=4.47 kN

UC.8f	**F_d / F_{rd} < 1**	**4.47 / 5.80 = 0.77 < 1**	**OK**

For capacity of carabines see K.5 Carabine, DIN 5299C, page 85

H.7.8.3.2. Rope ratchet

Ratchet, BL = 3.5 kN → Frd = BL / ɣm = 3.5 kN / 1.5 = 2.33 kN

7.5x15m	UC.8g	Long side – combined rope	**F_d / F_{rd} < 1**	**6.44 / 3*x2.33= 0.92 < 1**	**OK**
7.5x10m	UC.8h	Long side – combined rope	**F_d / F_{rd} < 1**	**6.08 / 3*x2.33= 0.87 < 1**	**OK**
7.5x15m	UC.8i	Short side – combined rope	**F_d / F_{rd} < 1**	**5.97 / 3*x2.33= 0.85 < 1**	**OK**
7.5x10m	UC.8j	Short side – combined rope	**F_d / F_{rd} < 1**	**5.04 / 3*x2.33= 0.72 < 1**	**OK**

For capacity of rope ratchet see K.6 Rope ratchet, page 85

* Number of cross sections making a rope. Ratchet is applied only on one cross-section.

H.7.8.4 Base plate

H.7.8.4.1. Side pole & corner pole

See H.6.3 for the highest design pressure force. Poles are subjected only to compressive force.

The worst case is assumed – pole is located on soil:

1. Soil (base plate size 25x25 cm, p_{soil} = 125 kN/m^2)

UC.8k	Soil	$\frac{F_d}{A}/p \leq 1$	**7.00 / (0.25*0.25) / 125 = 0.90 < 1**	**OK**

H.7.8.4.2. Main pole

See H.6.3 for the highest design pressure force. Poles are subjected only to compressive force.

The worst case is assumed – pole is located on soil:

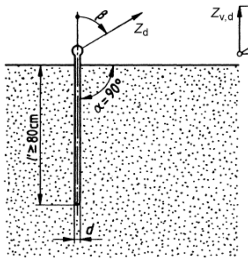
1. Soil (base plate size 30x30 cm, p_{soil} = 150 kN/m^2)

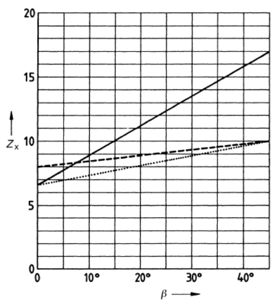
UC.8l	Soil	$\frac{F_d}{A}/p \leq 1$	**8.74 / (0.30*0.30) / 150 = 0.65 < 1**	**OK**

H.8. Safety against overturning, sliding and uplifting

The calculations in this chapter provide a guideline for dense cohesionless soil, in case no anchor tests have been performed. Anchor tests on location can show that a different number of anchors or different size of anchors have sufficient capacity for the specific soil conditions. (see paragraph H.8.3)

H.8.1 Anchor capacity





Angle of pull	Load bearing capacity N
$\beta = 0$	Z_d = 6,5 dl' for stiff cohesive and for dense cohesion less soils
$\beta = 0$	Z_d = 8 dl' for very stiff cohesive soils
$\beta \geq 45$	Z_d = 10 dl' for cohesive soils of at least medium to stiff consistency
$\beta \geq 45$	Z_d = 17 dl' for dense cohesion less soils
$0 < \beta < 45$	The load bearing capacity for the soil types shall be determined by interpolation

Z_d is the anchor service load (service load), in N;

$Z_{h,d}$ is the horizontal anchor service load, in N;

$Z_{v,d}$ is the vertical anchor service load, in N;

d is the anchor diameter, in cm;

l' is the depth of penetration in cm;

α is the angle of penetration;

ß is the angle of acting tensile force to the vertical

figure 7. Taken from EN 13782: Figures 4 & 5, table 5

Anchors **of Ø30mm** are used, considering an **effective length of at least 1000 mm.** If multiple anchors are placed at the same location, the anchors need to be at least 5xØ apart to use the full capacity of the anchors.

<u>Anchor capacity for anchors based in dense cohesionless soil (sandy soil)</u>

		Ø30 x 1000 mm
Angle	β	≥ 45
Effective length anchor	l’	100 cm
Diameter anchor	d	3.0 cm
Anchor capacity*	Z_d	5.10 kN

H.8.2 Required anchor pins

	Load combination	Element	Representative force	Design value force	Pag.
7.5x10m	CO4. Own weight + pretension + wind suction	Long side anchor point	4.17 kN	5.00 kN (γ = 1.2)	40
7.5x15m	CO4. Own weight + pretension + wind suction	Long side anchor point	4.61 kN	5.53 kN (γ = 1.2)	41
7.5x10m	CO4. Own weight + pretension + wind suction	Short side anchor point	3.41 kN	4.09 kN (γ = 1.2)	40
7.5x15m	CO4. Own weight + pretension + wind suction	Short side anchor point	4.11 kN	4.93 kN (γ = 1.2)	41
7.5x10m	CO4. Own weight + pretension + wind suction	Corner anchor point	3.60 kN	4.32 kN (γ = 1.2)	40
7.5x15m	CO4. Own weight + pretension + wind suction	Corner anchor point	4.05 kN	4.86 kN (γ = 1.2)	41
7.5x10m	CO4. Own weight + pretension + wind suction	Stormbelt	4.25 kN	5.1 kN (γ = 1.2)	40
7.5x15m	CO4. Own weight + pretension + wind suction	Stormbelt	3.89 kN	4.67 kN (γ = 1.2)	41

Required anchor amount for dense cohesionless soil – **7.5x10m model**

Long side guy ropes	1x	Ø30 x 1000 mm	F_d / F_{rd} = 5.00 / (1 x 5.10) = 0.98 ≤ 1	OK
Short side guy ropes	1x	Ø30 x 1000 mm	F_d / F_{rd} = 4.09 / (1 x 5.10) = 0.80 ≤ 1	OK
Corner ties	1x	Ø30 x 1000 mm	F_d / F_{rd} = 4.56 / (1 x 5.10) = 0.85 ≤ 1	OK
Stormbelt	1x	Ø30 x 1000 mm	F_d / F_{rd} = 5.54 / (2 x 5.10) = 1.00 ≤ 1	OK

Required anchor amount for dense cohesionless soil – **7.5x15m model**

Long side guy ropes	2x	Ø30 x 1000 mm	F_d / F_{rd} = 5.53 / (2 x 5.10) = 0.54 ≤ 1	OK
Short side guy ropes	1x	Ø30 x 1000 mm	F_d / F_{rd} = 4.93 / (1 x 5.10) = 0.97 ≤ 1	OK
Corner ties	1x	Ø30 x 1000 mm	F_d / F_{rd} = 4.86 / (1 x 5.10) = 0.95 ≤ 1	OK

Stormbelt	1x	Ø30 x 1000 mm	F_d / F_{rd} = 4.67 / (1 x 5.10) = 0.92 ≤ 1	OK

H.8.3 Anchor tests according to BS-EN 13782

It is advised to perform anchor test on location when there is a reason to doubt the “pull-out force” of the anchors, which could be when ground conditions differ from dense, non-cohesive soil.

Anchor tests should be carried out according to the following procedure:
Three anchors spread throughout the terrain should be put perpendicular into the ground. The anchors should be pulled out with the aid of a spring balance in the direction of the force acting on the anchor. The lowest of the three measured values should be used.

A safety factor of v = 1.6 regarding ultimate limit load is to apply for the lowest test value to determine the load bearing capacity in subsequent calculation. The load bearing capacity determined in this manner shall not result in anchor movement which would result in stresses, deformations or instability inadmissible for the structure.
If the foundation conditions are comparable, test loadings carried out in another location may be adduced for substantiation purposes.

For example:
Force in belts: Frep = 3.60 kN
Fsd_{belt} = 1.2 x Frep = 1.2 x 3.60 = 4.32 kN

The partial safety factor γ = 1.6 is applied on the ultimate limit load:
$Z_{u,d,test}$ > 1.6 x Fsd = 1.6 x 4.32 = 6.91 kN

If for example the anchor test point has a minimal anchor capacity of 5 kN (500 kg), then 2 anchors are needed: 2x5 = 10 kN > $Z_{u,d,test}$

H.8.4 Required ballast

The safety against sliding shall be calculated from:
$\sum \mu * \gamma * N_k \geq \sum \gamma * H_k$ (eq. 5. NEN-EN 13782)
where

μ the coefficient of friction according to table 3
γ The safety factor according to Table 2, γ = 1.2 (unfavorable wind load)
N the vertical load components (characteristic value), corresponds to the required ballast
H the horizontal load components (characteristic value).

The safety against lifting shall be calculated from:
$\sum \gamma * N_{Rk,stb} \geq \sum \gamma * N_{Ek,dst}$ (eq. 7. NEN-EN 13782)
where

$N_{Rk,stb}$ are the vertical stabilizing load components (characteristic value);
$N_{Ek,dst}$ are the vertical lifting destabilizing load components (characteristic value).

For load calculations against sliding and lifting, Frep is split into the horizontal (Fh) and vertical (Fv) forces. If the separate reaction force factors are not known:
$F_h = F_{rep} \times \cos\alpha$ and $F_v = F_{rep} \times \sin\alpha$
Where α is equal to the angle between F_{rep} and the ground plane.

It follows from the above equation:
Required load (against sliding): $B_h = (\gamma F_h) / \mu$
Required load (against lifting): $B_v = \gamma F_v$

$B_{tot} = B_h + B_v$
B_{tot} = total required load
B_h = required load due to horizontal forces
B_v = required load due to vertical forces

7.5x10m model

Location	F_{rep} (γ = 1.2) (kN)	F_H (kN)	F_V (kN)	α °	Ballast for μ = 0.5 (concrete on concrete) (KG)	Ballast for μ = 0.6 (concrete on rubber or wood) (KG)
Long side guy ropes	5.00	3.54	3.54	*45*	*1060*	*940*
Short side guy ropes	4.09	2.89	2.89	*45*	*870*	*770*
Corner ties	3.60	3.05	3.05	*45*	*920*	*820*
Stormbelts	5.10	3.61	3.61	45	*1080*	960

7.5x15m model

Location	F_{rep} (γ = 1.2) (kN)	F_H (kN)	F_V (kN)	α °	Ballast for μ = 0.5 (concrete on concrete) (KG)	Ballast for μ = 0.6 (concrete on rubber or wood) (KG)
Long side guy ropes	5.53	3.91	3.91	*45*	*1170*	*1040*
Short side guy ropes	4.93	3.49	3.49	*45*	*1050*	*930*
Corner ties	4.86	3.44	3.44	*45*	*1030*	*920*
Stormbelts	4.67	3.30	3.30	45	*990*	*880*

I. Addition 1 – element check for reduced wind speeds [NL]

For stretchtent use in the Netherlands, a reduction of maximal allowable wind speeds is specified. As a result, the stretchtent is calculated for two wind scenarios – full wind and reduced wind. Maximal wind speeds are specified and if the structure exceeds them, the tent must be evacuated and access must be denied. For these wind speeds no stormbelts are necessary.

A reduction factor α=0.5 is applied, which leads to **q_p (z_e) of 0.25 kN/m^2 [NL]. No stormbelts are required.**

I.1. Summary of results – low wind in Netherlands

This document is valid for three tent sizes: 7.5x10m, 7.5x15m and 6.5x10m. Although no separate calculation is made for 6.5x10m size, all elements, sizes and anchoring forces can be based on 7.5x10m model.

Changes that come from lower wind and no stormbelts are highlighted in orange.

General information:			
Main dimensions:		**7.5x10m / 6.5x10m**	**7.5x15m**
	Width:	7.5m	7.5m
	Length:	10m	15m
	Side height of membrane:	3m	3m
	Max. height:	4m	4m
	Main poles: *	4.0m Ø 95 mm [Spruce, C18]	4.0m Ø 95 mm [Spruce, C18]
	Side poles: *	3.0m Ø 75 mm [Spruce, C18]	3.0m Ø 75 mm [Spruce, C18]
	Corner poles: *	2.5m Ø 70 mm [Spruce, C18]	2.5m Ø 70 mm [Spruce, C18]
	Guy ropes:	Ø 8mm polyester, breaking strength ≥ 900 daN **	Ø 8mm polyester, breaking strength ≥ 900 daN **
	Long side – combined:	3 sections	3 sections
	Long side - V-ropes:	2 sections	2 sections
	Short side – combined:	**2 sections**	**2 sections**
	Short side - V-ropes:	2 sections	2 sections
	Belts:	50mm PES	50mm PES

	Fasteners:	[EN 12195-2], LC=1000 daN Carabine 180x14 M16 & M12 Eye bolt Ratchet, BL = 3.5 kN	[EN 12195-2], LC=1000 daN Carabine 180x14 M16 & M12 Eye bolt Ratchet, BL = 3.5 kN
	Side & corner pole base plate	25x25cm	25x25cm
	Main pole base plate	30x30cm	30x30cm
User load:	A maximum of 25 kg per main mast may be used for decorations, sound or lighting installations. The load must be applied centrally.		
Snow load:	A snow load of 0.1 kN/m² (4 cm) has been considered in accordance with the French CTS standard.		
Wind load:	**According to EN 13782** The following pressure values have been used: h < 5m: 0.35kN/m²		

* The average diameter, as a minimum required at the middle of the pole.

** The required breaking load (BL_{tot}) may be achieved by using multiple rope sections (n): $BL_{rope} = BL_{tot} / n$, see page 63

Safety against sliding, overturning and uplifting:			
Anchor forces:	Assumptions: angle of 45 degrees The following design resistance of the anchor forces is required: (See H.8.3, page 56 for Anchor tests according to EN 13782)		
		7.5x10m / 6.5x10m	**7.5x15m**
	Long side guy ropes: Short side guy ropes: Corner ties:	**4.12 kN** (γ = 1.2) **3.97 kN** (γ = 1.2) **3.76 kN** (γ = 1.2)	**5.02 kN** (γ = 1.2) **4.62 kN** (γ = 1.2) **3.86 kN** (γ = 1.2)
Required anchors:	Based on dense, non-cohesive soil (e.g. sandy soils). In the case anchors **Ø30mm x 1000mm** (effective length) are being used:		
		7.5x10m / 6.5x10m	**7.5x15m**
	Long side guy ropes: Short side guy ropes: Corner ties:	1 anchor 1 anchor 1 anchor	**1 anchor** 1 anchor 1 anchor
Required ballast:	Ballast calculation is performed for 2 different situations: - Concrete on concrete – concrete ballasts are placed on concrete - Concrete on wood or rubber – concrete ballasts are placed on wood, or an anti-slip mat is placed underneath the ballast		
	7.5x10m / 6.5x10m		

	Concrete on concrete	Concrete on wood or rubber
Long side guy ropes: Short side guy ropes: Corner ties:	**870 kg** **840 kg** **800 kg**	**780 kg** **750 kg** **710 kg**
	7.5x15m	
	Concrete on concrete	Concrete on wood or rubber
Long side guy ropes: Short side guy ropes: Corner ties:	**1060 kg** **980 kg** **820 kg**	**950 kg** **870 kg** **730 kg**

I.2. Summary of forces

Further chapters include the calculation only for highlighted elements in the CO5 [NL] column if they show higher values than calculated in the previous chapters or if recalculating them show a clear difference in results. Other element checks can be looked up in H.7.

I.2.1 7.5x10m model

kN or kN/m	CO1	CO2	CO3	CO4	CO5 [NL]	CO6 [FR]
Warp	0.96	1.93	2.37	2.98	2.06	3.46
Weft	1.02	2.55	3.33	0.98	1.11	4.54
Edge	2.04	2.30	2.55	3.26	2.45	2.92
Main pole	-2.19	-4.72	-5.91	-0.06	-0.22	-8.57
Side pole	-1.50	-2.61	-3.32	-2.29	-0.82	-4.49
Corner pole	-2.69	-3.26	-3.79	-3.92	-3.15	-4.52
Corner belt	2.21	2.48	2.82	3.60	3.13	3.26
Belts - long side	0.12	0.35	0.51	0.12	0.15	0.82
Belts - short side	0.24	0.72	1.01	0.07	0.00	1.47
Stormbelts	0.00	0.00	0.00	4.25	0.00	0.00
Combined rope - long side	1.83	1.94	2.16	4.05	3.28	2.43
V-ropes - long side	1.38	1.48	1.64	2.97	2.43	1.83
Combined rope - short side	1.96	2.08	2.30	3.36	2.92	2.56
V-ropes short side	1.43	1.47	1.60	2.56	2.19	1.75
Reaction - long side	1.95	2.29	2.67	4.17	3.43	3.29
Reaction - short side	2.19	2.79	3.31	3.41	2.92	4.07
Reaction - corner	2.21	2.48	2.82	3.60	3.13	3.30

Table 27. Summary of forces for 7.5x10m model

I.2.2 7.5x15m model

kN or kN/m	CO1	CO2	CO3	CO4	CO5 [NL]	CO6 [FR]
Warp	1.22	2.56	3.38	2.23	1.24	4.72
Weft	1.05	1.84	2.43	2.52	2.14	3.39
Edge	1.96	2.29	2.62	3.36	2.60	3.04
Main pole	-1.94	-4.40	-5.60	-0.01	-0.10	-8.20
Side pole	-1.55	-2.83	-3.69	-1.89	-0.78	-5.04
Corner pole	-2.80	-3.45	-4.08	-4.44	-3.27	-4.90
Corner belt	2.24	2.55	2.95	4.05	3.22	3.47
Belts - long side	0.17	0.42	0.61	0.34	0.18	0.93
Belts - short side	0.25	0.83	1.23	0.14	0.00	1.85
Stormbelts	0.00	0.00	0.00	3.89	0.00	0.00
Combined rope - long side	2.11	2.18	2.47	4.29	4.01	2.75
V-ropes - long side	1.47	1.59	1.80	2.98	2.74	2.00
Combined rope - short side	1.96	2.28	2.64	3.98	2.99	3.04
V-ropes short side	1.42	1.59	1.81	2.87	2.26	2.05
Reaction - long side	2.21	2.59	3.07	4.61	4.18	3.72
Reaction - short side	2.21	3.10	3.85	4.11	2.99	4.93
Reaction - corner	2.24	2.55	2.95	4.05	3.22	3.51

Table 28. Summary of forces for 7.5x15m model

I.3. Element check - low wind in Netherlands

I.3.1 Guy ropes

	Load combination	Element	Representative force	Design value force	Pag.
7.5x15m	CO5. Own weight + pretension + wind suction reduced	long side - combined	4.01 kN	6.02 kN (γ = 1.5)	62
7.5x10m	CO5. Own weight + pretension + wind suction reduced	long side - combined	3.28 kN	4.92 kN (γ = 1.5)	61
7.5x15m	CO5. Own weight + pretension + wind suction reduced	long side – V-ropes	2.74 kN	4.11 kN (γ = 1.5)	62
7.5x10m	CO5. Own weight + pretension + wind suction reduced	long side – V-ropes	2.43 kN	3.65 kN (γ = 1.5)	61
7.5x15m	CO5. Own weight + pretension + wind suction reduced	short side - combined	2.99 kN	4.49 kN (γ = 1.5)	62
7.5x10m	CO5. Own weight + pretension + wind suction reduced	short side - combined	2.92 kN	4.38 kN (γ = 1.5)	61
7.5x15m	CO5. Own weight + pretension + wind suction reduced	short side – V-ropes	2.26 kN	3.39 kN (γ = 1.5)	62
7.5x10m	CO5. Own weight + pretension + wind suction reduced	short side – V-ropes	2.19 kN	3.29 kN (γ = 1.5)	61

Synthetic rope Ø8mm, BL 900 daN. Is being used → $F_{rd,8mm}$ = 2.25 kN.

In a case of using individual guy ropes made of multiple sections, the ropes need to be tied back, creating multiple rope sections that together will take the load.

7.5x10m model

UC.1a	long side - combined	1 individual rope made of 3 sections Ø8mm	F_d / F_{rd} < 1	4.92 / (1 x 3 x 2.25) = 0.73 < 1	OK
UC.1b	long side – V-ropes	1 individual rope made of 2 sections Ø8mm	F_d / F_{rd} < 1	3.65 / (1 x 2 x 2.25) = 0.81 ≤ 1	OK
UC.1c	short side - combined	1 individual rope made of 2 sections Ø8mm	F_d / F_{rd} < 1	4.38 / (1 x 2 x 2.25) = 0.97 < 1	OK
UC.1d	short side – V-ropes	1 individual rope made of 2 sections Ø8mm	F_d / F_{rd} < 1	3.29 / (1 x 2 x 2.25) = 0.73 < 1	OK

For capacity of ropes see H.2.2, page 21

7.5x15m model

UC.1e	long side - combined	1 individual rope made of 3 sections Ø8mm	F_d / F_{rd} < 1	6.02 / (1 x 3 x 2.25) = 0.89 < 1	OK
UC.1f	long side – V-ropes	1 individual rope made of 2 sections Ø8mm	F_d / F_{rd} < 1	4.11 / (1 x 2 x 2.25) = 0.91 < 1	OK
UC.1g	short side - combined	1 individual rope made of 2 sections Ø8mm	F_d / F_{rd} < 1	4.49 / (1 x 2 x 2.25) = 1.00 ≤ 1	OK
UC.1h	short side – V-ropes	1 individual rope made of 2 sections Ø8mm	F_d / F_{rd} < 1	3.39 / (1 x 2 x 2.25) = 0.75 < 1	OK

For capacity of ropes see H.2.2, page 21

It is also possible to use alternative **ropes** with a higher breaking strength ($BL_{rope)}$ and less rope sections (n), as long as **BL_{rope} x $n_{sections}$ x n_{ropes} ≥ $BL_{required}$**

Required breaking loads for ropes:

7.5x10m model

Long side - combined: **BL_{rope} x $n_{sections}$ x n_{ropes} ≥ 1968 kg**
Long side – V-ropes: **BL_{rope} x $n_{sections}$ x n_{ropes} ≥ 1458 kg**
Short side - combined: **BL_{rope} x $n_{sections}$ x n_{ropes} ≥ 1752 kg**
Short side – V-ropes: **BL_{rope} x $n_{sections}$ x n_{ropes} ≥ 1314 kg**

7.5x15m model

Long side - combined: **BL_{rope} x $n_{sections}$ x n_{ropes} ≥ 2406 kg**
Long side – V-ropes: **BL_{rope} x $n_{sections}$ x n_{ropes} ≥ 1644 kg**
Short side - combined: **BL_{rope} x $n_{sections}$ x n_{ropes} ≥ 1794 kg**
Short side – V-ropes: **BL_{rope} x $n_{sections}$ x n_{ropes} ≥ 1356 kg**

I.3.2 Fasteners

Force acting on fasteners is equal to the force in belts or ropes. See these values in chapter I.3.1.

I.3.2.1 Side rope connection

I.3.2.1.1. Carabine, DIN5299C

Carabine 180x14, WLL 580kg → BL = 2 x 5.8 kN = 11.6 kN → Frd = BL / ɣm = 11.6 kN / 2 = 5.8 kN
F_d=5.21 kN

UC.2a	**F_d / F_{rd} < 1**	**5.21 / 5.8 = 0.90 ≤ 1**	**OK**

For capacity of carabines see K.5 Carabine, DIN 5299C, page 85

I.3.2.1.2. Rope ratchet

Ratchet, BL = 3.5 kN → Frd = BL / ɣm = 3.5 kN / 1.5 = 2.33 kN

7.5x15m	**UC.2b**	Long side – combined rope	**F_d / F_{rd} < 1**	**6.02 / 3*x2.33= 0.86 < 1**	**OK**
7.5x10m	**UC.2c**	Long side – combined rope	**F_d / F_{rd} < 1**	**4.92 / 3*x2.33= 0.70 < 1**	**OK**
7.5x15m	**UC.2d**	Short side – combined rope	**F_d / F_{rd} < 1**	**4.49 / 2*x2.33= 0.96 < 1**	**OK**
7.5x10m	**UC.2e**	Short side – combined rope	**F_d / F_{rd} < 1**	**4.38 / 2*x2.33= 0.94 < 1**	**OK**

For capacity of rope ratchet see K.6 Rope ratchet, page 85

* Number of cross sections making a rope. Ratchet is applied only on one cross-section.

I.4. Safety against overturning, sliding and uplifting - low wind in Netherlands

The calculation in this chapter provides a guideline for dense cohesion less soil, in case no anchor tests have been performed. Anchor tests on location can show that a different number of anchors or different size of anchors have sufficient capacity for the specific soil conditions. (see paragraph H.8.3)

Assumed anchor capacity Frd=5.1 kN (see H.8.1)

I.4.1 Required anchor pins

	Load combination	Element	Representative force	Design value force	Pag.
7.5x10m	CO5. Own weight + pretension + wind suction reduced	Long side anchor point	3.43 kN	4.12 kN (γ = 1.2)	61
7.5x15m	CO5. Own weight + pretension + wind suction reduced	Long side anchor point	4.18 kN	5.02 kN (γ = 1.2)	62
7.5x10m	CO3. Own weight + pretension + wind pressure	Short side anchor point	3.31 kN	3.97 kN (γ = 1.2)	61
7.5x15m	CO3. Own weight + pretension + wind pressure	Short side anchor point	3.85 kN	4.62 kN (γ = 1.2)	62
7.5x10m	CO5. Own weight + pretension + wind suction reduced	Corner anchor point	3.13 kN	3.76 kN (γ = 1.2)	61
7.5x15m	CO5. Own weight + pretension + wind suction reduced	Corner anchor point	3.22 kN	3.86 kN (γ = 1.2)	62

Required anchor amount for dense cohesionless soil – **7.5x10m model**

Long side guy ropes	1x	Ø30 x 1000 mm	F_d / F_{rd} = 4.12 / (1 x 5.10) = 0.81 ≤ 1	OK
Short side guy ropes	1x	Ø30 x 1000 mm	F_d / F_{rd} = 3.97 / (1 x 5.10) = 0.78 < 1	OK
Corner ties	1x	Ø30 x 1000 mm	F_d / F_{rd} = 3.76 / (1 x 5.10) = 0.74 < 1	OK

Required anchor amount for dense cohesionless soil – **7.5x15m model**

Long side guy ropes	1x	Ø30 x 1000 mm	F_d / F_{rd} = 5.02 / (1 x 5.10) = 0.98 < 1	OK
Short side guy ropes	1x	Ø30 x 1000 mm	F_d / F_{rd} = 4.62 / (1 x 5.10) = 0.91 < 1	OK

Corner ties	1x	Ø30 x 1000 mm	F_d / F_{rd} = 3.86 / (1 x 5.10) = 0.76 < 1	OK

I.4.2 Required ballast

For calculation method see H.8.4.

7.5x10m model

Location	F_{rep} (γ = 1.2) (kN)	F_H (kN)	F_V (kN)	α °	Ballast for μ = 0.5 (concrete on concrete) (KG)	Ballast for μ = 0.6 (concrete on rubber or wood) (KG)
Long side guy ropes	4.12	2.91	2.91	*45*	*870*	*780*
Short side guy ropes	3.97	2.81	2.81	*45*	*840*	*750*
Corner ties	3.76	2.66	2.66	*45*	*800*	*710*

7.5x15m model

Location	F_{rep} (γ = 1.2) (kN)	F_H (kN)	F_V (kN)	α °	Ballast for μ = 0.5 (concrete on concrete) (KG)	Ballast for μ = 0.6 (concrete on rubber or wood) (KG)
Long side guy ropes	5.02	3.55	3.55	*45*	*1060*	*950*
Short side guy ropes	4.62	3.27	3.27	*45*	*980*	*870*
Corner ties	3.86	2.73	2.73	*45*	*820*	*730*

J. Addition 2 – element check for combined wind and snow load [FR]

For stretchtent use in France, the static analysis must include a load combination including snow and wind loads, corresponding to article 7 of CTS. This addition represents changes in element checks and anchor and ballast determination if this load combination is considered. Stormbelts are always required.

J.1. Summary of results – snow+wind in France

This document is valid for three tent sizes: 7.5x10m, 7.5x15m and 6.5x10m. Although no separate calculation is made for 6.5x10m size, all elements, sizes and anchoring forces can be based on 7.5x10m model.

Changes that come from snow and wind combination are highlighted in orange.

General information:			
Main dimensions:		**7.5x10m / 6.5x10m**	**7.5x15m**
	Width:	7.5m	7.5m
	Length:	10m	15m
	Side height of membrane:	3m	3m
	Max. height:	4m	4m
	Main poles: *	4.0m **Ø 100 mm [Spruce, C18]**	4.0m **Ø 100 mm [Spruce, C18]**
	Side poles: *	3.0m **Ø 80 mm [Spruce, C18]**	3.0m **Ø 80 mm [Spruce, C18]**
	Corner poles: *	2.5m Ø 70 mm [Spruce, C18]	2.5m Ø 70 mm [Spruce, C18]
	Stormbelts:	50mm PES [EN 12195-2], LC=1000 daN	50mm PES [EN 12195-2], LC=1000 daN
	Guy ropes:	Ø 8mm polyester, breaking strength ≥ 900 daN **	Ø 8mm polyester, breaking strength ≥ 900 daN **
	Long side – combined:	3 sections	3 sections
	Long side - V-ropes:	2 sections	2 sections
	Short side – combined:	3 sections	3 sections
	Short side - V-ropes:	2 sections	2 sections
	Belts:	50mm PES [EN 12195-2],	50mm PES [EN 12195-2],

	Fasteners:	LC=1000 daN Carabine 180x14 M16 & M12 Eye bolt Ratchet, BL = 3.5 kN	LC=1000 daN Carabine 180x14 M16 & M12 Eye bolt Ratchet, BL = 3.5 kN
	Side & corner pole base plate	25x25cm	25x25cm
	Main pole base plate	30x30cm	30x30cm
User load:	A maximum of 25 kg per main mast may be used for decorations, sound or lighting installations. The load must be applied centrally.		
Snow load:	A snow load of 0.1 kN/m² (4 cm) has been considered in accordance with the French CTS standard.		
Wind load:	**According to EN 13782** The following pressure values have been used: h < 5m: 0.50kN/m² This value corresponds to wind force 8 on the Beaufort scale. Above this wind force (starting at wind force 9), the safety of the structure can no longer be guaranteed, and it must be evacuated.		

* The average diameter, as a minimum required at the middle of the pole.

** The required breaking load (BL_{tot}) may be achieved by using multiple rope sections (n): $BL_{rope} = BL_{tot} / n$, see page 47

Safety against sliding, overturning and uplifting:

Anchor forces:	Assumptions: angle of 45 degrees The following design resistance of the anchor forces is required: (See H.8.3, page 56 for Anchor tests according to EN 13782)		
		7.5x10m / 6.5x10m	**7.5x15m**
	Long side guy ropes:	5.08 kN (γ = 1.2)	6.14 kN (γ = 1.2)
	Short side guy ropes:	**4.88 kN** (γ = 1.2)	**5.92 kN** (γ = 1.2)
	Corner ties:	4.56 kN (γ = 1.2)	4.92 kN (γ = 1.2)
	Stormbelt:	5.54 kN (γ = 1.2)	4.51 kN (γ = 1.2)
Required anchors:	Based on dense, non-cohesive soil (e.g. sandy soils). In the case anchors **Ø30mm x 1000mm** (effective length) are being used:		
		7.5x10m / 6.5x10m	**7.5x15m**
	Long side guy ropes:	1 anchor	2 anchors
	Short side guy ropes:	1 anchor	**2 anchors**
	Corner ties:	1 anchor	1 anchor
	Stormbelt:	1 anchor	1 anchor
Required ballast:	Ballast calculation is performed for 2 different situations: - Concrete on concrete – concrete ballasts are placed on concrete - Concrete on wood or rubber – concrete ballasts are placed on wood, or an anti-slip mat is placed underneath the ballast		

7.5x10m / 6.5x10m		
	Concrete on concrete	Concrete on wood or rubber
Long side guy ropes: Short side guy ropes: Corner ties: Stormbelt:	1080 kg **1040 kg** 970 kg 1180 kg	960 kg **920 kg** 860 kg 1050 kg
7.5x15m		
	Concrete on concrete	Concrete on wood or rubber
Long side guy ropes: Short side guy ropes: Corner ties: Stormbelt:	1300 kg **1260 kg** 1045 kg 960 kg	1160 kg **1120 kg** 930 kg 850 kg

J.2. Cross sections

Profile	Material	Ø mm	G kg/m	A mm^2	I_y mm^4	$W_{el;y}$ mm^3
Ø ≈ 100 mm *	Wood C18	100	2.51	7854	4908738	98175
Ø ≈ 80 mm *	Wood C18	80	1.61	5026	2010619	50265
Ø ≈ 70 mm *	Wood C18	70	1.23	3848	1178588	33674

Table 29. Used cross sections

* The average diameter, as a minimum required at the middle of the pole.

J.3. Summary of forces

Further chapters include the calculation only for highlighted elements in the CO6 [FR] column if they show higher values than calculated in the previous chapters or if recalculating them show a clear difference in results. Other element checks can be looked up in H.7.

J.3.1 7.5x10m model

kN or kN/m	CO1	CO2	CO3	CO4	CO5 [NL]	CO6 [FR]
Warp	0.96	1.93	2.37	2.98	2.06	3.46
Weft	1.02	2.55	3.33	0.98	1.11	4.54
Edge	2.04	2.30	2.55	3.26	2.45	2.92
Main pole	-2.19	-4.72	-5.91	-0.06	-0.22	-8.57
Side pole	-1.50	-2.61	-3.32	-2.29	-0.82	-4.49
Corner pole	-2.69	-3.26	-3.79	-3.92	-3.15	-4.52
Corner belt	2.21	2.48	2.82	3.60	3.13	3.26
Belts - long side	0.12	0.35	0.51	0.12	0.15	0.82
Belts - short side	0.24	0.72	1.01	0.07	0.00	1.47
Stormbelts	0.00	0.00	0.00	4.25	0.00	0.00
Combined rope - long side	1.83	1.94	2.16	4.05	3.28	2.43

V-ropes - long side	1.38	1.48	1.64	2.97	2.43	1.83
Combined rope - short side	1.96	2.08	2.30	3.36	2.92	2.56
V-ropes short side	1.43	1.47	1.60	2.56	2.19	1.75
Reaction - long side	1.95	2.29	2.67	4.17	3.43	3.29
Reaction - short side	2.19	2.79	3.31	3.41	2.92	4.07
Reaction - corner	2.21	2.48	2.82	3.60	3.13	3.30

Table 30. Summary of forces for 7.5x10m model

J.3.2 7.5x15m model

kN or kN/m	CO1	CO2	CO3	CO4	CO5 [NL]	CO6 [FR]
Warp	1.22	2.56	3.38	2.23	1.24	4.72
Weft	1.05	1.84	2.43	2.52	2.14	3.39
Edge	1.96	2.29	2.62	3.36	2.60	3.04
Main pole	-1.94	-4.40	-5.60	-0.01	-0.10	-8.20
Side pole	-1.55	-2.83	-3.69	-1.89	-0.78	-5.04
Corner pole	-2.80	-3.45	-4.08	-4.44	-3.27	-4.90
Corner belt	2.24	2.55	2.95	4.05	3.22	3.47
Belts - long side	0.17	0.42	0.61	0.34	0.18	0.93
Belts - short side	0.25	0.83	1.23	0.14	0.00	1.85
Stormbelts	0.00	0.00	0.00	3.89	0.00	0.00
Combined rope - long side	2.11	2.18	2.47	4.29	4.01	2.75
V-ropes - long side	1.47	1.59	1.80	2.98	2.74	2.00
Combined rope - short side	1.96	2.28	2.64	3.98	2.99	3.04
V-ropes short side	1.42	1.59	1.81	2.87	2.26	2.05
Reaction - long side	2.21	2.59	3.07	4.61	4.18	3.72
Reaction - short side	2.21	3.10	3.85	4.11	2.99	4.93
Reaction - corner	2.24	2.55	2.95	4.05	3.22	3.51

Table 31. Summary of forces for 7.5x15m model

J.4. Element check – snow + wind in France

J.4.1 Membrane

	Load combination	Element	Representative stress	Design value stress	Pag.
7.5x15m	CO1. Own weight + pretension	Long term warp	1.22 kN/m	1.65 kN/m (γ = 1.35)	71
7.5x15m	CO1. Own weight + pretension	Long term weft	1.05 kN/m	1.42 kN/m (γ = 1.35)	70
7.5x15m	CO6. Own weight + pretension + wind pressure + snow	Short term warp	4.72 kN/m	6.37 kN/m (γ = 1.35)	71
7.5x15m	CO6. Own weight + pretension + wind pressure + snow	Short term weft	3.39 kN/m	4.58 kN/m (γ = 1.35)	71

Sioen RP5639 fabric is being used.

UC.1a	Long term warp	$S_{Ed} / S_{rd} < 1$	1.65 / 9.6 = 0.17 < 1	OK
UC.1b	Long term weft	$S_{Ed} / S_{rd} < 1$	1.42 / 7.2 = 0.20 < 1	OK
UC.1c	Short term warp	$S_{Ed} / S_{rd} < 1$	6.37 / 12.0 = 0.53 < 1	OK
UC.1d	Short term weft	$S_{Ed} / S_{rd} < 1$	4.58 / 9.0 = 0.51 < 1	OK

For capacity of membrane see H.2.1, page 21

J.4.2 Main poles

	Load combinations	Element	Representative force	Design value force	Pag.
7.5x10m	CO6. Own weight + pretension + wind pressure + snow	Main pole 4.0m	-8.57 kN	-11.57 kN (γ = 1.35)	70

User load of max. 25 kg is applied, loaded centrically.

4.0m pole Ned = -8.57 x 1.35 + (-0.25 x 1.35) = -11.91 kN

J.4.2.1 4.0m pole - Wood, Spruce C18

Profile = Pole, Ø ≈ 100 mm
average diameter, as a minimum required at the middle of the pole

Length = max. 4.00m

Quality = C18

The poles are considered as hinged poles; the buckling length is equivalent to the pole length.

UC.2	4.0m	$\sigma_{c,0,d}$ / ($f_{c,0,d}$ x k_{cy}) = 1.52 / (12.46 x 0.120) = 1.02 ≤ 1	OK

See Annex E.4 Main pole 4.0m – Wood, Spruce C18 – for snow and wind combination [FR], on page 152 for the elaborate check

J.4.3 Side poles

	Load combinations	Element	Representative force	Design value force	Pag.
7.5x15m	CO6. Own weight + pretension + wind pressure + snow	Side pole 3.0m	-5.04 kN	-6.80 kN (γ = 1.35)	71

User load of max. 25 kg is applied, loaded centrically.

3m pole Ned = -5.04 x 1.35 + (-0.25 x 1.35) = -7.14 kN

J.4.3.1 3m pole - Wood, Spruce C18

Profile = Pole, Ø ≈ 80 mm

average diameter, as a minimum required at the middle of the pole

Length = max. 3.0m

Quality = C18

The poles are considered as hinged poles; the buckling length is equivalent to the pole length.

UC.3	3m	$\sigma_{c,0,d}$ / ($f_{c,0,d}$ x k_{cy}) = 1.42 / (12.46 x 0.136) = 0.84 < 1	OK

See Annex E.5 Side pole 3.0m – Wood, Spruce C18 – for snow and wind combination [FR], on page 153 for the elaborate check

J.4.4 Corner poles

	Load combinations	Element	Representative force	Design value force	Pag.
7.5x15m	CO6. Own weight + pretension + wind pressure + snow	Corner pole 2.5m	-4.90 kN	-6.62 kN (γ = 1.35)	71

User load of max. 25 kg is applied, loaded centrically.

2.5m corner pole Ned = -4.90 x 1.35 + (-0.25 x 1.35) = -6.95 kN

J.4.4.1 2.5m pole - Wood, Spruce C18

Profile = Pole, Ø ≈ 70 mm
average diameter, as a minimum required at the middle of the pole

Length = max. 2.5m

Quality = C18

The poles are considered as hinged poles; the buckling length is equivalent to the pole length.

UC.4	2.5m	$\sigma_{c,0,d}$ / ($f_{c,0,d}$ x k_{cy}) = 1.81 / (12.46 x 0.149) = 0.97 < 1	OK

See Annex E.6 Corner pole 2.5m – Wood, Spruce C18 – for snow and wind combination [FR], on page 154 for the elaborate check

J.4.5 Belts

	Load combination	Element	Representative force	Design value force	Pag.
7.5x15m	CO4. Own weight + pretension + wind suction	Corner belt	4.05 kN	6.08 kN (γ = 1.5)	71
7.5x15m	CO6. Own weight + pretension + wind pressure + snow	Belt – long side	0.93 kN	1.26 kN (γ = 1.35)	71
7.5x15m	CO6. Own weight + pretension + wind pressure + snow	Belt – short side	1.85 kN	2.50 kN (γ = 1.35)	71

J.4.5.1 Belt capacity check

PES belt with a breaking strength of 3000kg is being used.

UC.5a	Corner belt	1 PES belt with minimum BL = 3000 kg	$F_d / F_{rd} < 1$	6.08 / (1 x 15) = 0.41 < 1	OK
UC.5b	Belt – long side	1 PES belt with minimum BL = 3000 kg	$F_d / F_{rd} < 1$	1.26 / (1 x 15) = 0.08 < 1	OK
UC.5c	Belt – short side	1 PES belt with minimum BL = 3000 kg	$F_d / F_{rd} < 1$	2.50 / (1 x 15) = 0.17 < 1	OK

For capacity of belts see H.2.3, page 21

J.4.5.2 Alternative – rope capacity check

It is possible to use either a rope or a belt. Synthetic rope Ø8mm, BL 900 daN. Is being used → $F_{rd,8mm}$ = 2.25 kN.

In a case of using individual guy ropes, the ropes need to be tied back, creating multiple rope sections that together will take the load.

UC.5d	Corner rope	1 individual rope made of <u>3 sections</u> Ø8mm	$F_d / F_{rd} < 1$	6.08 / (1 x 3 x 2.25) = 0.90 < 1	OK
UC.5e	Rope – long side	1 individual rope made of <u>1 section</u> Ø8mm	$F_d / F_{rd} < 1$	1.26 / (1 x 1 x 2.25) = 0.56 < 1	OK
UC.5f	Rope – short side	1 individual rope made of <u>2 sections</u> Ø8mm	$F_d / F_{rd} < 1$	2.50 / (1 x 2 x 2.25) = 0.56 < 1	OK

For capacity of ropes see H.2.2, page 21

It is also possible to use alternative **ropes** with a higher breaking strength (BL_{rope}) and less rope sections (n), as long as $\mathbf{BL_{rope} \times n_{sections} \times n_{ropes} \geq BL_{required}}$

Required breaking loads for ropes:

Corner rope: $\mathbf{BL_{rope} \times n_{sections} \times n_{ropes} \geq 2432\ kg}$

Rope – long side: $\mathbf{BL_{rope} \times n_{sections} \times n_{ropes} \geq 315\ kg}$

Rope – long side: $\mathbf{BL_{rope} \times n_{sections} \times n_{ropes} \geq 1000\ kg}$

J.4.6 Fasteners

Force acting on fasteners is equal to the force in belts or ropes. See these values in chapters H.7.6 and H.7.5.

J.4.6.1 Side belt connection

J.4.6.1.1. M12 Eye bolt, DIN580

For ≤45°:

M12, WLL 340kg → BL = 6 x 3.4 kN = 20.4 kN → Frd = BL / γ_m = 20.4 / 2 = 10.2 kN

F_d=2.50 kN

UC.6a	Corner eye bolt	F_d / F_{rd} < 1	2.50 / 10.2 = 0.25 < 1.0	OK

For capacity of eye bolts see K.3 Capacity of Eye bolts on page 84.

J.4.6.1.2. Carabine, DIN EN 5299C

Carabine 180x14, WLL 580kg → BL = 2 x 5.8 kN = 11.6 kN → Frd = BL / γm = 11.6 kN / 2 = 5.8 kN

UC.6b	Corner eye bolt	F_d / F_{rd} < 1	2.50 / 5.8 = 0.43 < 1.0	OK

For capacity of carabines see K.5 Carabine, DIN 5299C, page 85

J.4.6.2 Base plate

J.4.6.2.1. Side pole & corner pole

See J.3 for the highest design pressure force. Poles are subjected only to compressive force.

The worst case is assumed – pole is located on soil:

- Soil (base plate size 25x25 cm, p_{soil} = 125 kN/m^2)

UC.6c	Soil	$\frac{F_d}{A}/p \leq 1$	**7.14 / (0.25*0.25) / 125 = 0.91 < 1**	**OK**

J.4.6.2.2. Main pole

See J.3 for the highest design pressure force. Poles are subjected only to compressive force.

The worst case is assumed – pole is located on soil:

- Soil (base plate size 30x30 cm, p_{soil} = 150 kN/m^2)

UC.6d	Soil	$\frac{F_d}{A}/p \leq 1$	**11.41 / (0.30*0.30) / 150 = 0.85 < 1**	**OK**

J.5. Safety against overturning, sliding and uplifting – snow + wind in France

The calculation in this chapter provides a guideline for dense cohesion less soil, in case no anchor tests have been performed. Anchor tests on location can show that a different number of anchors or different size of anchors have sufficient capacity for the specific soil conditions. (see paragraph H.8.3)

Assumed anchor capacity Frd=5.1 kN (see H.8.1)

J.5.1 Required anchor pins

	Load combination	Element	Representative force	Design value force	Pag.
7.5x10m	CO5. Own weight + pretension + wind pressure + snow	Short side anchor point	4.07 kN	4.88 kN (γ = 1.2)	70
7.5x15m	CO5. Own weight + pretension + wind pressure + snow	Short side anchor point	4.93 kN	5.92 kN (γ = 1.2)	71

Required anchor amount for dense cohesionless soil – **7.5x10m model**

Short side guy ropes	1x	Ø30 x 1000 mm	F_d / F_{rd} = 4.88 / (1 x 5.10) = 0.96 < 1	OK

Required anchor amount for dense cohesionless soil – **7.5x15m model**

Short side guy ropes	2x	Ø30 x 1000 mm	F_d / F_{rd} = 5.92 / (2 x 5.10) = 0.58 < 1	OK

J.5.2 Required ballast

For calculation method see H.8.4.

7.5x10m model

Location	F_{rep} (γ = 1.2) (kN)	F_H (kN)	F_V (kN)	α °	Ballast for μ = 0.5 (concrete on concrete) (KG)	Ballast for μ = 0.6 (concrete on rubber or wood) (KG)
Short side guy ropes	4.88	3.45	3.45	*45*	*1040*	*920*

7.5x15m model

Location	F_{rep} (γ = 1.2) (kN)	F_H (kN)	F_V (kN)	α °	Ballast for μ = 0.5 (concrete on concrete) (KG)	Ballast for μ = 0.6 (concrete on rubber or wood) (KG)
Short side guy ropes	5.92	4.18	4.18	*45*	*1260*	*1120*

K. Material certificates

K.1. Fabric specifications

SIOEN FABRICS

Z.I. du Blanc Ballot, Avenue Urbino 6, B-7700 Mouscron - Tel +32(0)56 85 68 80 - Fax +32(0)56 34 61 31
sioenfabrics@sioen.be - http://www.sioen.com

Technical Data Sheet

RP5639

FLEXITENT SINGLE M2 (fiche provisoire) - Width 153 cm

Total weight / Poids total	730 [+/-10%] g/m²	ISO 3801/5 1977
Weight of fabric [PES] / Poids tissu [PES]	280 [+/-10] g/m²	ISO 3801/5 1977
Weight of coating [PU/PVC] / Poids enduction [PU/PVC]	450 [+/-10%] g/m²	ISO 3801/5 1977
Breaking strength warp Force de rupture chaîne	1200 N/5cm	ISO 13934 1999
Breaking strength weft Force de rupture trame	900 N/5cm	ISO 13934 1999
Breaking extension warp Allongement à la rupture chaîne	110 %	ISO 13934 1999
Breaking extension weft Allongement à la rupture trame	110 %	ISO 13934 1999
Tear strenght warp Résistance à la déchirure chaîne	100 N	ISO 4674A1 1977
Tear strenght weft Résistance à la déchirure trame	180 N	ISO 4674A1 1977
Adhesion warp / Adhérence chaîne	23 N/5cm	D41 1015
Adhesion weft / Adhérence trame	23 N/5cm	D41 1015
Waterproofness before washing Imperméabilité avant lavage	OK:2000 G M D mm	ISO 811 1981
Waterproofness after washing Imperméabilité après lavage	OK:2000 G M D mm	ISO 811 1981
Flammability / Inflammabilité	OK:M2 -	NF P 92-507 (2004)

*May be cleaned with mild, PH neutral, non-ionogenic detergents. Certainly avoid solvent containing cleaning agents (alcohols, polar solvents,etc.). Testing is always advised before using cleaning agents.
After surface cleaning, please let air dry.*

All our technical characteristics are indicative
Rev. 10/2012

K.2. Fabric fire certificate

FCBA
INSTITUT TECHNOLOGIQUE

INSTITUT TECHNOLOGIQUE
FCBA
Laboratoire d'essais feu
10 rue Galilée - 77420 Champs-sur-Marne

Laboratoire agréé par le Ministère de l'Intérieur et de l'Aménagement du Territoire
JO du 10 février 2007
PROCES-VERBAL DE CLASSEMENT DE REACTION AU FEU
D'UN MATERIAU PREVU A L'ARTICLE 5
DE L'ARRETE DU 21 NOVEMBRE 2002

Page 1

N° CM - 21 - P - 096

et 1 annexe de 4 pages.
Valable 5 ans à compter du mardi 13 juillet 2021

Matériau présenté par : **SIOEN FABRICS SA**
Avenue Urbino 6

7700 MOUSCRON

Marque commerciale : **F5639**

Description sommaire:

Type de textile enduit tricot polyester enduit de PVC ignifugé
Composition support textile/ fabric 100% polyester (280 g/m²) + enduction/ coating PVC ignifugé (450 g/m²)
Masse surfacique 730 g/m²
Epaisseur 0,75 mm
Coloris tous coloris/ all colors

Nature de l'essai : Essai au brûleur électrique
Essai de persistance de flamme
Essai de gouttes

Classement : **M2**

Observations : /

Durabilité du classement (l'annexe 22) : Non limitée a priori

Usage : Revêtement de murs tendus, voilages, rideaux,... **Conditions d'entretien :** selon préconisations du client

Compte tenu des critères résultant des essais annexés décrits dans le(s) rapport(s) d'essais N° : *FE 21-0216-1182*

Ce procès-verbal atteste uniquement des caractéristiques de l'échantillon soumis aux essais et ne préjuge pas des caractéristiques de produits similaires. Il ne constitue donc pas une certification de produits au sens de l'article L.115-27 du code de la consommation et de la loi du 3 juin 1994.
NOTA : Seules sont autorisées les reproductions intégrales et par photocopies du présent procès-verbal de classement ou de l'ensemble du procès-verbal de classement et rapport(s) d'essais annexé(s).Ces conclusions ne portent que sur les performances de résistance / réaction au feu de l'élément objet du présent procès verbal de classement. Elles ne préjugent, en aucun cas, des autres performances liées à son incorporation à un ouvrage"

"Non valable pour toute application couverte par l'article AM 18 (Arrêté du 6 mars 2006) concernant les sièges rembourrés"

Champs-sur-Marne, le vendredi 23 juillet 2021

Technicien d'essais du Laboratoire Feu
Bilel HAMILA

Responsable Technique du laboratoire Feu du Pôle Ameublement
Clotilde ETIEMVRE

INSTITUT TECHNOLOGIQUE FCBA CHAMPS-SUR-MARNE LABORATOIRE ESSAIS ET MESURES

Siège social
10 rue Galilée
77420 Champs-sur-Marne
Tél + 33 (0)1 72 84 97 84
www.fcba.fr
Siret 775 680 903 00132
APE 7219 Z
Code TVA CEE : FR 14 775 680 903

Institut technologique FCBA : Forêt, Cellulose, Bois - construction, Ameublement

INSTITUT TECHNOLOGIQUE
FCBA
Fire Laboratory

10 rue Galilée - 77420 Champs-sur-Marne

Laboratory approved by the Ministry for the Interior
JO of February 10, 2007
OFFICIAL REPORT OF CLASSIFICATION OF FIRE PERFORMANCE
Of A MATERIAL ENVISAGED IN ARTICLE 5
DECREE OF November 21, 2002

N° CM - 21 - P - 096

et 1 annexe de 4 pages.

Valid 5 years as from mardi 13 juillet 2021

Material presents by : **SIOEN FABRICS SA**
Avenue Urbino 6
7700 MOUSCRON

Trade mark: **F5639**

Summary description

Kind of coating fabric	tricot polyester enduit de PVC ignifugé
Composition	support textile/ fabric 100% polyester (280 g/m²) + enduction/ coating PVC ignifugé (450 g/m²)
Mass per unit area	730 g/m²
Tickness	0,75 mm
Color	tous coloris/ all colors

Nature of the test : Electrical burner test
Flame persistence
Dripping test

Classification : **M2**

Observations : /

Durability of the classification (Article 5 de l'annexe 2) Not limited

Usual term Walls hung covering, sheers, curtains,… **Cleaning conditions** Recommendation of Customer

Taking into account the criteria resulting from the tests annexed described in the test report N° *FE 21-0216-1182*

This official report attests only characteristics of the sample submitted for testing and does not prejudge similar characteristics of products. It thus does not constitute a certification of products within the meaning of the L.115-27 article of the code of the consumption and the law of June 3rd, 1994.
NOTE: The integral reproductions and by photocopy of this official report of classification are only authorized or the official report unit of classification and report of annexed test. These conclusions relate only to the resistance / reaction to fire performance of the element that is the subject of the present classification report. They do not prejudge, in any case, other performances related to its incorporation to a work "

"Not legally acceptable for any use under the pretence of the article AM 18 (Decree of March 6th, 2006) for upholstered seat"

Champs-sur-Marne, le vendredi 23 juillet 2021

Technicien d'essais du Laboratoire Feu
Bilel HAMILA

Responsable Technique du laboratoire Feu du Pôle Ameublement
Clotilde ETIEMVRE

INSTITUT TECHNOLOGIQUE FCBA LABORATOIRE ESSAIS ET MESURES CHAMPS-SUR-MARNE

Only the french version remains valid

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Siret 775 680 903 00132
APE 7219 Z
Code TVA CEE : FR 14 775 680 903

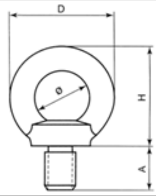
Institut technologique FCBA : Forêt, Cellulose, Bois - construction, Ameublement

K.3. Capacity of Eye bolts

Karta techniczna Nr 2GM/KAR

1. Nazwa dostawcy: **„GÓRALMET" M. i J. Góral Sp. J.** Adres dostawcy: **ul. Krakowska 68, 32-860 Czchów, tel/fax 014 6635260**

2. Nazwa wyrobu: **GM580 Śruba z uchem**



3. Opis i przeznaczenie: Śruba stalowa kuta z uchem, z gwintem metrycznym zwykłym prawym, do wykorzystania jako punkt mocujący do podnoszenia przedmiotów, materiałów i towarów, a także do innych zastosowań, w zakresie parametrów technicznych podanych w punkcie 5 i wymagań Dyrektywy 2006/42/WE
4. Dopuszczalne obciążenia robocze w kolumnie 11 i punkcie 6 podano dla normalnych warunków użytkowania. Przy zastosowaniu wyrobu w wyjątkowo niebezpiecznych warunkach, zakres zastosowania i redukcję dopuszczalnego obciążenia roboczego należy uzgodnić z dostawcą. Przechowywanie i montaż - zgodnie z instrukcją montażu i stosowania elementów osprzętu do podnoszenia.

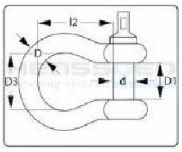
5. Parametry techniczne wyrobu:

LP.	Rozmiar	Wymiary nominalne [mm]							Masa [kg/szt]	WLL Obciążenie robocze [t]*	MPF Siła próby [kN]	BF Siła niszcząca [min. kN]	Materiał	Powłoki
		D	ø	H	A	-	-	-						
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	**M6**	36	20	36	13	-	-	-	0,052	0,09	1,77	5,3	C15	Ocynk elektrolityczny
2	**M8**	36	20	36	13	-	-	-	0,054	0,14	2,75	8,2		
3	**M10**	45	25	45	17	-	-	-	0,099	0,23	4,51	13,5		
4	**M12**	54	30	53	20,5	-	-	-	0,164	0,34	6,67	20,0		
5	**M14**	54	30	53	20,5	-	-	-	0,171	0,50	9,81	29,4		
6	**M16**	63	35	62	27	-	-	-	0,288	0,70	13,73	41,2		
7	**M18**	63	35	62	27	-	-	-	0,300	0,93	18,24	54,7		
8	**M20**	72	40	71	30	-	-	-	0,457	1,20	23,54	70,6		
9	**M22**	72	40	71	30	-	-	-	0,463	1,50	29,42	88,3		
10	**M24**	90	50	90	36	-	-	-	0,874	1,80	35,30	105,9		
11	**M27**	90	50	90	36	-	-	-	0,936	2,50	49,03	147,1		
12	**M30**	108	60	109	45	-	-	-	1,626	3,60	70,61	211,8		

6. Informacje dodatkowe:

- Współczynnik bezpieczeństwa – 6 (*dla obciążenia wzdłuż osi gwintu). Dopuszczalne obciążanie siłą równoległą do płaszczyzny ucha pod kątem do 45° od osi gwintu pod warunkiem redukcji WLL o 30%, lub pod kątem 45° do 90° przy redukcji WLL o 50%. Zabrania się obciążania śruby prostopadle do płaszczyzny ucha. Podane obciążenia są ważne tylko dla prawidłowo zamontowanego wyrobu, tj. gdy gwint śruby jest całkowicie wkręcony w otwór montażowy, a kołnierz śruby spoczywa na powierzchni nośnej
- MPF=2,0 x g x WLL, BF =6,0 x g x WLL gdzie g - przyspieszenie ziemskie w m/s^2 (tj. 9,80665)
- Kod identyfikacyjny producenta i produktu: GM580
- Cechowanie: symbol producenta – GMG; CE ; kod materiału – C15; rozmiar; WLL....T
- Informacje dodatkowe – etykieta: CE ; nazwa wyrobu; rozmiar; ilość szt.; numer dostawy; dane producenta i adres strony internetowej.

K.4. Capacity of Orange Pin 1 Ton D-sluiting, DIN EN 13889



DIN EN 13889
U.S. Fed. Spec. RR-C-271
Quality class = 6

	name	Ident Nr.	materiaal	kleur	oppervlakte	d in mm	D in mm	D inch	D1 in mm	D3 in mm	l2 in mm	SWL /WLL daN	gewicht g	VE			
	50001-00,50	933729	HS	orange pin	fe	8	6,5	1/4	11	19	28	500	48	25	1.00	stuks	aanvragen
	50001-00,75	933730	HS	orange pin	fe	10	8	5/16	12	21	31	750	78	25	1.00	stuks	aanvragen
	50001-01,00	933731	HS	orange pin	fe	11	10	3/8	17	26	35	1000	136	1	1.00	stuks	aanvragen
	50001-01,50	933732	HS	orange pin	fe	12	13	7/16	18	30	42	1500	178	1	1.00	stuks	aanvragen
	50001-02,00	933733	HS	orange pin	fe	16	13,5	1/2	20	32	47	2000	300	1	1.00	stuks	aanvragen

K.5. Carabine, DIN 5299C

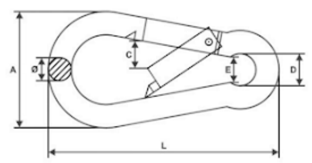
Karta Techniczna Nr 10GM/KAR

1. Nazwa dostawcy: **„GÓRALMET" M. i J. Góral Sp. J.** — Adres dostawcy: **ul. Krakowska 68, 32-860 Czchów, tel/fax 014 6635260**

2. Nazwa wyrobu: **Karabińczyk do lin 5299C**

3. Opis i przeznaczenie: Zatrzaśniki stalowe, służące jako elementy łączące w zakończeniach lin, łańcuchów gospodarczych, oraz do innych zastosowań ogólnych. Do stosowania w zakresie parametrów technicznych podanych w punkcie 5.

4. Karabińczyki do lin nie mogą być użyte w zastosowaniach posiadających inne - uregulowane prawem czy normami wymagania bezpieczeństwa, np. do podnoszenia ładunków, jako środek do ochrony indywidualnej przed upadkiem z wysokości, lub osprzęt alpinistyczny. Nie przeciążać ponad wartość WLL (kG) podaną w kolumnie 13.



5. Parametry techniczne wyrobu:

LP.	Rozmiar [mm]	Wymiary nominalne [mm]									Masa [kg/szt]	WLL Obciążenie robocze [kG]	BF Siła niszcząca [min. kG]	Materiał	Powłoki
		A [mm]	C min [mm]	D [mm]	E min [mm]	Ø [mm]	L [mm]	---	---	---					
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	**4**	19,5	4,5	6	4,5	3,9	40	---	---	---	0,008	70	140	C10 ~ C15	Ocynk elektrolityczny
2	**5**	24	5,5	7,5	5,5	4,8	50	---	---	---	0,015	100	200		
3	**6**	29	8	8,5	6	5,5	60	---	---	---	0,023	120	240		
4	**7**	34	9	10,5	8,5	6,7	70	---	---	---	0,040	180	360		
5	**8**	39	10	11,5	9	7,4	80	---	---	---	0,056	230	460		
6	**9**	44	11	13,5	10	8,4	90	---	---	---	0,083	250	500		
7	**10**	48	11,5	14,5	11	9,4	100	---	---	---	0,115	350	700		
8	**11**	57	17	17,5	13	10,7	120	---	---	---	0,179	400	800		
9	**12**	67	22	20	15	11,5	140	---	---	---	0,246	450	900		
10	**13**	80	26	21	17	12,6	160	---	---	---	0,343	530	1060		
11	**14**	87	28	24	20	13,6	180	---	---	---	0,443	580	1160		
12	**15**	98	32	24	20	14,7	200	---	---	---	0,563	700	1400		

6. Informacje dodatkowe:

- Współczynnik bezpieczeństwa - 2
- BF =2,0 x WLL
- Cechowanie: symbol producenta – GM lub GMG; kod rozmiaru
- Informacje dodatkowe – etykieta: nazwa wyrobu; rozmiar; ilość szt.; numer dostawy; dane producenta i adres strony internetowej.

K.6. Rope ratchet

3/8" Rope Ratchet

Carolina North Manufacturing

10022

$10.89 ~~$12.89~~

Maximum load limit 250 lbs. Outer casings are made from #6-33% glass filled nylon with interior mechanisms of die cast zinc. Perfect for large jobs:

- heavy loads for transport
- hoisting items
- straightening items
- securing objects under tension
- product weight 0.16 lbs

Product is compatible with our 3/8" S-hooks and 3/8" solid braided rope.

Bulk purchasing options available.

ADD TO CART | BUY IT NOW

Share:

Collections: Bulk Rope Ratchets

Type: Bulk Ratchet

Annex

Annex A. Software input (load combination) – 7.5x10m model 87

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Annex A. Software input (load combination) – 7.5x10m model

Annex A.1. CO1. Own weight + pretension

```
EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)
ORDERED BY LOADGROUPS
-------------------------------------------------------------------------------------------------------------------
     LOADGROUP        LOADMODE      FACTOR  VOLUME/AREA(P.U)          SUM_X          SUM_Y          SUM_Z   VOLUME/AREA
-------------------------------------------------------------------------------------------------------------------
        STRUTS      SELFWEIGHT        1.00         78.500000         0.0000         0.0000        -0.6020      0.007669
        CABLES      SELFWEIGHT        1.00         78.500000         0.0000         0.0000        -0.0252      0.000320
 MEM-LINKS1000      SELFWEIGHT        1.00          0.005000         0.0000         0.0000        -0.3575     71.495529
 MEM-LINKS2000      SELFWEIGHT        1.00          0.005000         0.0000         0.0000        -0.3555     71.094423
-------------------------------------------------------------------------------------------------------------------
           SUM                                                       0.0000         0.0000        -1.3401
-------------------------------------------------------------------------------------------------------------------


EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADGROUPS
-------------------------------------------------------------------------------------------------------
 LOADGROUP        LOADMODE      LOAD    FACTOR          SUM_X          SUM_Y          SUM_Z  LOADED AREA
-------------------------------------------------------------------------------------------------------
-------------------------------------------------------------------------------------------------------
       SUM                                             0.0000         0.0000         0.0000
-------------------------------------------------------------------------------------------------------


EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADMODES
------------------------------------------------------------------------------
                 LOADMODE                      SUM_X          SUM_Y          SUM_Z
------------------------------------------------------------------------------
------------------------------------------------------------------------------
       SUM      AREA-LOADS                    0.0000         0.0000         0.0000
------------------------------------------------------------------------------


EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS
==============================================================================
                                               SUM_X          SUM_Y          SUM_Z
==============================================================================
                                              0.0000         0.0000        -1.3401
==============================================================================
```

```
Inner Energy=            1.342832


StatikD:  Sum of external loads:       0.000000       0.000000      -1.340042
StatikD:  Sum of support forces:       0.000062      -0.000000       1.340657
StatikD:  P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek1\7.5x10\CO1_OW+VO
StatikD: Time of calculation:       0.25s
StatikD: Version 2025.38 done.
```

Annex A.2. CO2. Own weight + pretension + conventional / snow

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	FACTOR	VOLUME/AREA(P.U)	SUM_X	SUM_Y	SUM_Z	VOLUME/AREA
STRUTS	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.6020	0.007669
CABLES	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.0252	0.000320
MEM-LINKS1000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.3575	71.495529
MEM-LINKS2000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.3555	71.094423
SUM				0.0000	0.0000	-1.3401	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	LOAD	FACTOR	SUM_X	SUM_Y	SUM_Z	LOADED AREA
1	SCHNEE	0.1000	1.00	0.0000	0.0000	-6.8360	68.36
SUM				0.0000	0.0000	-6.8360	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADMODES

	LOADMODE	SUM_X	SUM_Y	SUM_Z
SUM	SCHNEE	0.0000	0.0000	-6.8360
SUM	AREA-LOADS	0.0000	0.0000	-6.8360

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

SUM_X	SUM_Y	SUM_Z
0.0000	0.0000	-8.1761

```
Inner Energy=            1.919365


StatikD:  Sum of external loads:       0.000000       0.000000      -8.176114
StatikD:  Sum of support forces:       0.000093      -0.000001       8.175749
StatikD:  P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek1\7.5x10\CO2_OW+VO+SNOW
StatikD: Time of calculation:       0.30s
StatikD: Version 2025.38 done.
```

Annex A.3. CO3. Own weight + pretension + wind pressure

```
EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)
ORDERED BY LOADGROUPS
-------------------------------------------------------------------------------------------------------------------
     LOADGROUP        LOADMODE      FACTOR  VOLUME/AREA(P.U)         SUM_X          SUM_Y          SUM_Z    VOLUME/AREA
-------------------------------------------------------------------------------------------------------------------
        STRUTS      SELFWEIGHT        1.00         78.500000        0.0000         0.0000        -0.6020       0.007669
        CABLES      SELFWEIGHT        1.00         78.500000        0.0000         0.0000        -0.0252       0.000320
 MEM-LINKS1000      SELFWEIGHT        1.00          0.005000        0.0000         0.0000        -0.3575      71.495529
 MEM-LINKS2000      SELFWEIGHT        1.00          0.005000        0.0000         0.0000        -0.3555      71.094423
-------------------------------------------------------------------------------------------------------------------
           SUM                                                      0.0000         0.0000        -1.3401
-------------------------------------------------------------------------------------------------------------------


EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADGROUPS
------------------------------------------------------------------------------------------------------
 LOADGROUP       LOADMODE       LOAD     FACTOR           SUM_X          SUM_Y          SUM_Z   LOADED AREA
------------------------------------------------------------------------------------------------------
         1           WIND    -0.1500       1.00         -0.0000         0.0000       -10.2540         72.08
------------------------------------------------------------------------------------------------------
       SUM                                              -0.0000         0.0000       -10.2540
------------------------------------------------------------------------------------------------------


EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADMODES
-------------------------------------------------------------------------------
                LOADMODE                        SUM_X          SUM_Y          SUM_Z
-------------------------------------------------------------------------------
       SUM         WIND                       -0.0000         0.0000       -10.2540
-------------------------------------------------------------------------------
-------------------------------------------------------------------------------
       SUM     AREA-LOADS                     -0.0000         0.0000       -10.2540
-------------------------------------------------------------------------------


EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS
===============================================================================
                                                SUM_X          SUM_Y          SUM_Z
===============================================================================
                                              -0.0000         0.0000       -11.5941
===============================================================================
```

```
Inner Energy=          2.514336


StatikD:  Sum of external loads:      -0.000008      -0.000005     -11.594129
StatikD:  Sum of support forces:       0.000107       0.000001      11.594147
StatikD:  P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek1\7.5x10\CO3_OW+VO+WD
StatikD: Time of calculation:       0.31s
StatikD: Version 2025.38 done.
```

Annex A.4. CO4. Own weight + pretension + wind suction

```
EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)
ORDERED BY LOADGROUPS
---------------------------------------------------------------------------------------------------------------------
      LOADGROUP        LOADMODE     FACTOR  VOLUME/AREA(P.U)          SUM_X          SUM_Y          SUM_Z     VOLUME/AREA
---------------------------------------------------------------------------------------------------------------------
         STRUTS      SELFWEIGHT       1.00         78.500000         0.0000         0.0000        -0.6020        0.007669
         CABLES      SELFWEIGHT       1.00         78.500000         0.0000         0.0000        -0.0345        0.000440
  MEM-LINKS1000      SELFWEIGHT       1.00          0.005000         0.0000         0.0000        -0.3609       72.185733
  MEM-LINKS2000      SELFWEIGHT       1.00          0.005000         0.0000         0.0000        -0.3588       71.761810
---------------------------------------------------------------------------------------------------------------------
            SUM                                                      0.0000         0.0000        -1.3563
---------------------------------------------------------------------------------------------------------------------


EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADGROUPS
--------------------------------------------------------------------------------------------------------
 LOADGROUP        LOADMODE      LOAD    FACTOR          SUM_X          SUM_Y          SUM_Z    LOADED AREA
--------------------------------------------------------------------------------------------------------
         1            WIND    0.3500      1.00        -0.0002         0.0003        24.1239          72.73
--------------------------------------------------------------------------------------------------------
       SUM                                            -0.0002         0.0003        24.1239
--------------------------------------------------------------------------------------------------------


EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADMODES
------------------------------------------------------------------------------
                LOADMODE                          SUM_X          SUM_Y          SUM_Z
------------------------------------------------------------------------------
       SUM            WIND                      -0.0002         0.0003        24.1239
------------------------------------------------------------------------------
------------------------------------------------------------------------------
       SUM      AREA-LOADS                      -0.0002         0.0003        24.1239
------------------------------------------------------------------------------


EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS
==============================================================================
                                                  SUM_X          SUM_Y          SUM_Z
==============================================================================
                                                -0.0002         0.0003        22.7677
==============================================================================
```

```
Inner Energy=            4.301083


StatikD:  Sum of external loads:      -0.000205       0.000300      22.767683
StatikD:  Sum of support forces:       0.000658      -0.000148     -22.767530
StatikD:  P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek1\7.5x10\CO4_OW+VO+WU
StatikD: Time of calculation:       1.11s
StatikD: Version 2025.38 done.
```

Annex A.5. CO5. Own weight + pretension + wind suction reduced [NL]

```
EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)
ORDERED BY LOADGROUPS
-----------------------------------------------------------------------------------------------
     LOADGROUP       LOADMODE    FACTOR  VOLUME/AREA(P.U)         SUM_X       SUM_Y       SUM_Z    VOLUME/AREA
-----------------------------------------------------------------------------------------------
        STRUTS     SELFWEIGHT      1.00        78.500000         0.0000      0.0000     -0.6020       0.007669
        CABLES     SELFWEIGHT      1.00        78.500000         0.0000      0.0000     -0.0252       0.000320
 MEM-LINKS1000     SELFWEIGHT      1.00         0.005000         0.0000      0.0000     -0.3575      71.495529
 MEM-LINKS2000     SELFWEIGHT      1.00         0.005000         0.0000      0.0000     -0.3555      71.094423
-----------------------------------------------------------------------------------------------
           SUM                                                   0.0000      0.0000     -1.3401
-----------------------------------------------------------------------------------------------


EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADGROUPS
-------------------------------------------------------------------------------------
 LOADGROUP       LOADMODE     LOAD    FACTOR          SUM_X        SUM_Y        SUM_Z   LOADED AREA
-------------------------------------------------------------------------------------
         1           WIND   0.1750      1.00         0.0000      -0.0000      11.9630         72.08
-------------------------------------------------------------------------------------
       SUM                                           0.0000      -0.0000      11.9630
-------------------------------------------------------------------------------------


EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADMODES
-----------------------------------------------------------------
                 LOADMODE                 SUM_X        SUM_Y        SUM_Z
-----------------------------------------------------------------
       SUM           WIND                0.0000      -0.0000      11.9630
-----------------------------------------------------------------
-----------------------------------------------------------------
       SUM     AREA-LOADS                0.0000      -0.0000      11.9630
-----------------------------------------------------------------


EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS
=================================================================
                                          SUM_X        SUM_Y        SUM_Z
=================================================================
                                         0.0000      -0.0000      10.6229
=================================================================
```

```
Inner Energy=           2.730369


StatikD:  Sum of external loads:       0.000003       0.000003      10.622948
StatikD:  Sum of support forces:       0.000016      -0.000042     -10.621775
StatikD:  P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek1\7.5x10\CO5_OW+VO+WU_red
StatikD: Time of calculation:       0.46s
StatikD: Version 2025.38 done.
```

Annex A.6. CO6. Own weight + pretension + wind pressure + snow [FR]

Annex A.6.1. For element check

```
EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)
ORDERED BY LOADGROUPS
-------------------------------------------------------------------------------------------------------------------
      LOADGROUP        LOADMODE     FACTOR  VOLUME/AREA(P.U)          SUM_X          SUM_Y          SUM_Z     VOLUME/AREA
-------------------------------------------------------------------------------------------------------------------
         STRUTS      SELFWEIGHT       1.00         78.500000         0.0000         0.0000        -0.6020        0.007669
         CABLES      SELFWEIGHT       1.00         78.500000         0.0000         0.0000        -0.0252        0.000320
   MEM-LINKS1000     SELFWEIGHT       1.00          0.005000         0.0000         0.0000        -0.3575       71.495529
   MEM-LINKS2000     SELFWEIGHT       1.00          0.005000         0.0000         0.0000        -0.3555       71.094423
-------------------------------------------------------------------------------------------------------------------
            SUM                                                      0.0000         0.0000        -1.3401
-------------------------------------------------------------------------------------------------------------------


EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADGROUPS
------------------------------------------------------------------------------------------------------
 LOADGROUP        LOADMODE        LOAD     FACTOR           SUM_X          SUM_Y          SUM_Z   LOADED AREA
------------------------------------------------------------------------------------------------------
         1          SCHNEE      0.1000       1.00          0.0000         0.0000        -6.8360         68.36
         1            WIND     -0.1500       1.00         -0.0000         0.0000       -10.2540         72.08
------------------------------------------------------------------------------------------------------
       SUM                                                -0.0000         0.0000       -17.0900
------------------------------------------------------------------------------------------------------


EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADMODES
------------------------------------------------------------------------------
                LOADMODE                          SUM_X          SUM_Y          SUM_Z
------------------------------------------------------------------------------
       SUM        SCHNEE                         0.0000         0.0000        -6.8360
------------------------------------------------------------------------------
       SUM          WIND                        -0.0000         0.0000       -10.2540
------------------------------------------------------------------------------
------------------------------------------------------------------------------
       SUM    AREA-LOADS                        -0.0000         0.0000       -17.0900
------------------------------------------------------------------------------


EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS
==============================================================================
                                                  SUM_X          SUM_Y          SUM_Z
==============================================================================
                                                -0.0000         0.0000       -18.4301
==============================================================================
```

```
Inner Energy=           3.654297


StatikD:  Sum of external loads:      -0.000008      -0.000005     -18.430164
StatikD:  Sum of support forces:       0.000129       0.000002      18.430528
StatikD:  P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek1\7.5x10\CO6_OW+VO+WD+SNOW
StatikD: Time of calculation:       0.30s
StatikD: Version 2025.38 done.
```

Annex A.6.2. For anchor and ballast calculation

```
EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)
ORDERED BY LOADGROUPS
--------------------------------------------------------------------------------------------------------------------
     LOADGROUP        LOADMODE    FACTOR  VOLUME/AREA(P.U)          SUM_X          SUM_Y          SUM_Z   VOLUME/AREA
--------------------------------------------------------------------------------------------------------------------
        STRUTS      SELFWEIGHT      1.00         78.500000         0.0000         0.0000        -0.6020      0.007669
        CABLES      SELFWEIGHT      1.00         78.500000         0.0000         0.0000        -0.0252      0.000320
 MEM-LINKS1000      SELFWEIGHT      1.00          0.005000         0.0000         0.0000        -0.3575     71.495529
 MEM-LINKS2000      SELFWEIGHT      1.00          0.005000         0.0000         0.0000        -0.3555     71.094423
--------------------------------------------------------------------------------------------------------------------
           SUM                                                     0.0000         0.0000        -1.3401
--------------------------------------------------------------------------------------------------------------------


EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADGROUPS
--------------------------------------------------------------------------------------------------------
 LOADGROUP       LOADMODE      LOAD    FACTOR           SUM_X          SUM_Y          SUM_Z  LOADED AREA
--------------------------------------------------------------------------------------------------------
         1         SCHNEE    0.1083      1.00          0.0000         0.0000        -7.4055        68.36
         1           WIND   -0.1500      1.00         -0.0000         0.0000       -10.2540        72.08
--------------------------------------------------------------------------------------------------------
       SUM                                            -0.0000         0.0000       -17.6595
--------------------------------------------------------------------------------------------------------


EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADMODES
--------------------------------------------------------------------------------
                 LOADMODE                     SUM_X          SUM_Y          SUM_Z
--------------------------------------------------------------------------------
       SUM         SCHNEE                    0.0000         0.0000        -7.4055
--------------------------------------------------------------------------------
       SUM           WIND                   -0.0000         0.0000       -10.2540
--------------------------------------------------------------------------------
--------------------------------------------------------------------------------
       SUM     AREA-LOADS                   -0.0000         0.0000       -17.6595
--------------------------------------------------------------------------------


EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS
================================================================================
                                              SUM_X          SUM_Y          SUM_Z
================================================================================
                                            -0.0000         0.0000       -18.9996
================================================================================
```

```
Inner Energy=           3.756092


StatikD:  Sum of external loads:       -0.000008       -0.000005      -18.999588
StatikD:  Sum of support forces:        0.000130        0.000003       19.000350
StatikD:  P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek1\7.5x10\C06a_OW+VO+WD+SNOW_anchors
StatikD: Time of calculation:       0.27s
StatikD: Version 2025.38 done.
```

Annex B. Software input (load combination) – 7.5x15m model

Annex B.1. CO1. Own weight + pretension

```
EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)
ORDERED BY LOADGROUPS
---------------------------------------------------------------------------------------------------------------------
     LOADGROUP        LOADMODE     FACTOR  VOLUME/AREA(P.U)          SUM_X          SUM_Y          SUM_Z    VOLUME/AREA
---------------------------------------------------------------------------------------------------------------------
        STRUTS      SELFWEIGHT       1.00         78.500000         0.0000         0.0000        -0.8365       0.010656
        CABLES      SELFWEIGHT       1.00         78.500000         0.0000         0.0000        -0.0353       0.000449
 MEM-LINKS1000      SELFWEIGHT       1.00          0.005000         0.0000         0.0000        -0.5402     108.042120
 MEM-LINKS2000      SELFWEIGHT       1.00          0.005000         0.0000         0.0000        -0.5370     107.393517
---------------------------------------------------------------------------------------------------------------------
           SUM                                                      0.0000         0.0000        -1.9490
---------------------------------------------------------------------------------------------------------------------


EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADGROUPS
--------------------------------------------------------------------------------------------------------
 LOADGROUP        LOADMODE       LOAD     FACTOR           SUM_X          SUM_Y          SUM_Z   LOADED AREA
--------------------------------------------------------------------------------------------------------
--------------------------------------------------------------------------------------------------------
       SUM                                                 0.0000         0.0000         0.0000
--------------------------------------------------------------------------------------------------------


EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADMODES
-------------------------------------------------------------------------------
                  LOADMODE                        SUM_X          SUM_Y          SUM_Z
-------------------------------------------------------------------------------
-------------------------------------------------------------------------------
       SUM      AREA-LOADS                       0.0000         0.0000         0.0000
-------------------------------------------------------------------------------


EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS
===============================================================================
                                                  SUM_X          SUM_Y          SUM_Z
===============================================================================
                                                 0.0000         0.0000        -1.9490
===============================================================================
```

```
Inner Energy=            1.864821


StatikD:  Sum of external loads:       0.000000       0.000000      -1.948950
StatikD:  Sum of support forces:      -0.000516       0.000859       1.948154
StatikD:  P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek1\7.5x15\CO1_OW+VO
StatikD: Time of calculation:       0.55s
StatikD: Version 2025.38 done.
```

Annex B.2. CO2. Own weight + pretension + conventional / snow

EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	FACTOR	VOLUME/AREA(P.U)	SUM_X	SUM_Y	SUM_Z	VOLUME/AREA
STRUTS	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.8365	0.010656
CABLES	SELFWEIGHT	1.00	78.500000	0.0000	0.0000	-0.0353	0.000449
MEM-LINKS1000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.5402	108.042120
MEM-LINKS2000	SELFWEIGHT	1.00	0.005000	0.0000	0.0000	-0.5370	107.393517
SUM				0.0000	0.0000	-1.9490	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADGROUPS

LOADGROUP	LOADMODE	LOAD	FACTOR	SUM_X	SUM_Y	SUM_Z	LOADED AREA
1	SCHNEE	0.1000	1.00	0.0000	0.0000	-10.3249	103.25
SUM				0.0000	0.0000	-10.3249	

EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADMODES

	LOADMODE	SUM_X	SUM_Y	SUM_Z
SUM	SCHNEE	0.0000	0.0000	-10.3249
SUM	AREA-LOADS	0.0000	0.0000	-10.3249

EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS

SUM_X	SUM_Y	SUM_Z
0.0000	0.0000	-12.2739

```
Inner Energy=            2.729394


StatikD:  Sum of external loads:       0.000000       0.000000     -12.273883
StatikD:  Sum of support forces:       0.000099       0.000429      12.274127
StatikD:  P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek1\7.5x15\CO2_OW+VO+SNOW
StatikD: Time of calculation:       1.03s
StatikD: Version 2025.38 done.
```

Annex B.3. CO3. Own weight + pretension + wind pressure

```
EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)
ORDERED BY LOADGROUPS
-----------------------------------------------------------------------------------------------------------------------
     LOADGROUP        LOADMODE     FACTOR  VOLUME/AREA(P.U)          SUM_X          SUM_Y          SUM_Z    VOLUME/AREA
-----------------------------------------------------------------------------------------------------------------------
        STRUTS      SELFWEIGHT      1.00         78.500000         0.0000         0.0000        -0.8365       0.010656
        CABLES      SELFWEIGHT      1.00         78.500000         0.0000         0.0000        -0.0353       0.000449
 MEM-LINKS1000      SELFWEIGHT      1.00          0.005000         0.0000         0.0000        -0.5402     108.042120
 MEM-LINKS2000      SELFWEIGHT      1.00          0.005000         0.0000         0.0000        -0.5370     107.393517
-----------------------------------------------------------------------------------------------------------------------
           SUM                                                     0.0000         0.0000        -1.9490
-----------------------------------------------------------------------------------------------------------------------


EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADGROUPS
-----------------------------------------------------------------------------------------------------------
 LOADGROUP        LOADMODE       LOAD     FACTOR          SUM_X          SUM_Y          SUM_Z   LOADED AREA
-----------------------------------------------------------------------------------------------------------
         1            WIND    -0.1500       1.00        -0.0148         0.0059       -15.4874        108.86
-----------------------------------------------------------------------------------------------------------
       SUM                                              -0.0148         0.0059       -15.4874
-----------------------------------------------------------------------------------------------------------


EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADMODES
--------------------------------------------------------------------------------
                  LOADMODE                    SUM_X          SUM_Y          SUM_Z
--------------------------------------------------------------------------------
       SUM            WIND                  -0.0148         0.0059       -15.4874
--------------------------------------------------------------------------------
--------------------------------------------------------------------------------
       SUM      AREA-LOADS                  -0.0148         0.0059       -15.4874
--------------------------------------------------------------------------------


EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS
================================================================================
                                              SUM_X          SUM_Y          SUM_Z
================================================================================
                                            -0.0148         0.0059       -17.4363
================================================================================
```

```
Inner Energy=            3.657800


StatikD:  Sum of external loads:       -0.014794       0.005925      -17.436344
StatikD:  Sum of support forces:        0.015695      -0.006346       17.435585
StatikD:  P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek1\7.5x15\CO3_OW+VO+WD
StatikD: Time of calculation:       0.98s
StatikD: Version 2025.38 done.
```

Annex B.4. CO4. Own weight + pretension + wind suction

```
EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)
ORDERED BY LOADGROUPS
-----------------------------------------------------------------------------------------------------------------------
     LOADGROUP       LOADMODE     FACTOR  VOLUME/AREA(P.U)          SUM_X          SUM_Y          SUM_Z     VOLUME/AREA
-----------------------------------------------------------------------------------------------------------------------
        STRUTS     SELFWEIGHT      1.00         78.500000         0.0000         0.0000        -0.8365        0.010656
        CABLES     SELFWEIGHT      1.00         78.500000         0.0000         0.0000        -0.0510        0.000650
 MEM-LINKS1000     SELFWEIGHT      1.00          0.005000         0.0000         0.0000        -0.5402      108.042120
 MEM-LINKS2000     SELFWEIGHT      1.00          0.005000         0.0000         0.0000        -0.5370      107.393517
-----------------------------------------------------------------------------------------------------------------------
           SUM                                                    0.0000         0.0000        -1.9647
-----------------------------------------------------------------------------------------------------------------------


EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADGROUPS
----------------------------------------------------------------------------------------------------------
 LOADGROUP       LOADMODE       LOAD   FACTOR           SUM_X          SUM_Y          SUM_Z   LOADED AREA
----------------------------------------------------------------------------------------------------------
         1           WIND     0.3500     1.00          0.0392        -0.0134        36.0574        108.83
----------------------------------------------------------------------------------------------------------
       SUM                                             0.0392        -0.0134        36.0574
----------------------------------------------------------------------------------------------------------


EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADMODES
--------------------------------------------------------------------------------
                 LOADMODE                     SUM_X          SUM_Y          SUM_Z
--------------------------------------------------------------------------------
       SUM           WIND                    0.0392        -0.0134        36.0574
--------------------------------------------------------------------------------
--------------------------------------------------------------------------------
       SUM     AREA-LOADS                    0.0392        -0.0134        36.0574
--------------------------------------------------------------------------------


EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS
================================================================================
                                              SUM_X          SUM_Y          SUM_Z
================================================================================
                                             0.0392        -0.0134        34.0927
================================================================================
```

```
Inner Energy=           6.465364


StatikD:  Sum of external loads:       0.039256      -0.013340      34.092686
StatikD:  Sum of support forces:      -0.037114       0.012729     -34.123287
StatikD:  P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek1\7.5x15\CO4_OW+VO+WU
StatikD: Time of calculation:       1.51s
StatikD: Version 2025.38 done.
```

Annex B.5. CO5. Own weight + pretension + wind suction reduced [NL]

```
EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)
ORDERED BY LOADGROUPS
----------------------------------------------------------------------------------------------------------------------------
     LOADGROUP        LOADMODE    FACTOR  VOLUME/AREA(P.U)          SUM_X          SUM_Y          SUM_Z      VOLUME/AREA
----------------------------------------------------------------------------------------------------------------------------
        STRUTS      SELFWEIGHT     1.00         78.500000          0.0000         0.0000        -0.8365         0.010656
        CABLES      SELFWEIGHT     1.00         78.500000          0.0000         0.0000        -0.0353         0.000449
 MEM-LINKS1000      SELFWEIGHT     1.00          0.005000          0.0000         0.0000        -0.5402       108.042120
 MEM-LINKS2000      SELFWEIGHT     1.00          0.005000          0.0000         0.0000        -0.5370       107.393517
----------------------------------------------------------------------------------------------------------------------------
           SUM                                                    0.0000         0.0000        -1.9490
----------------------------------------------------------------------------------------------------------------------------


EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADGROUPS
--------------------------------------------------------------------------------------------------------------
 LOADGROUP        LOADMODE       LOAD     FACTOR          SUM_X          SUM_Y          SUM_Z    LOADED AREA
--------------------------------------------------------------------------------------------------------------
         1            WIND     0.1750      1.00          0.0173        -0.0069        18.0686         108.86
--------------------------------------------------------------------------------------------------------------
       SUM                                               0.0173        -0.0069        18.0686
--------------------------------------------------------------------------------------------------------------


EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADMODES
-------------------------------------------------------------------------------
                LOADMODE                           SUM_X          SUM_Y          SUM_Z
-------------------------------------------------------------------------------
       SUM          WIND                          0.0173        -0.0069        18.0686
-------------------------------------------------------------------------------
-------------------------------------------------------------------------------
       SUM      AREA-LOADS                        0.0173        -0.0069        18.0686
-------------------------------------------------------------------------------


EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS
===============================================================================
                                                   SUM_X          SUM_Y          SUM_Z
===============================================================================
                                                  0.0173        -0.0069        16.1196
===============================================================================
```

```
Inner Energy=            4.271551


StatikD:  Sum of external loads:       0.017233      -0.006919      16.119644
StatikD:  Sum of support forces:      -0.016079       0.003593     -16.122554
StatikD:  P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek1\7.5x15\CO5_OW+VO+WU_red
StatikD: Time of calculation:       1.26s
StatikD: Version 2025.38 done.
```

Annex B.6. CO6. Own weight + pretension + wind pressure + snow [FR]

Annex B.6.1. For element check

```
EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)
ORDERED BY LOADGROUPS
------------------------------------------------------------------------------------------------------------------------
      LOADGROUP        LOADMODE     FACTOR  VOLUME/AREA(P.U)         SUM_X          SUM_Y          SUM_Z     VOLUME/AREA
------------------------------------------------------------------------------------------------------------------------
         STRUTS      SELFWEIGHT       1.00         78.500000        0.0000         0.0000        -0.8365        0.010656
         CABLES      SELFWEIGHT       1.00         78.500000        0.0000         0.0000        -0.0353        0.000449
 MEM-LINKS1000       SELFWEIGHT       1.00          0.005000        0.0000         0.0000        -0.5402      108.042120
 MEM-LINKS2000       SELFWEIGHT       1.00          0.005000        0.0000         0.0000        -0.5370      107.393517
------------------------------------------------------------------------------------------------------------------------
            SUM                                                     0.0000         0.0000        -1.9490
------------------------------------------------------------------------------------------------------------------------


EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADGROUPS
-----------------------------------------------------------------------------------------------------------
 LOADGROUP        LOADMODE       LOAD     FACTOR           SUM_X          SUM_Y          SUM_Z    LOADED AREA
-----------------------------------------------------------------------------------------------------------
         1          SCHNEE     0.1000       1.00          0.0000         0.0000       -10.3249         103.25
         1            WIND    -0.1500       1.00         -0.0148         0.0059       -15.4874         108.86
-----------------------------------------------------------------------------------------------------------
       SUM                                               -0.0148         0.0059       -25.8123
-----------------------------------------------------------------------------------------------------------


EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADMODES
--------------------------------------------------------------------------------
                LOADMODE                           SUM_X          SUM_Y          SUM_Z
--------------------------------------------------------------------------------
       SUM        SCHNEE                          0.0000         0.0000       -10.3249
--------------------------------------------------------------------------------
       SUM          WIND                         -0.0148         0.0059       -15.4874
--------------------------------------------------------------------------------
--------------------------------------------------------------------------------
       SUM    AREA-LOADS                         -0.0148         0.0059       -25.8123
--------------------------------------------------------------------------------


EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS
================================================================================
                                                   SUM_X          SUM_Y          SUM_Z
================================================================================
                                                 -0.0148         0.0059       -27.7612
================================================================================
```

```
Inner Energy=            5.357406


StatikD:  Sum of external loads:      -0.014794       0.005925     -27.761253
StatikD:  Sum of support forces:       0.015775      -0.006082      27.761015
StatikD:  P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek1\7.5x15\CO6_OW+VO+WD+SNOW
StatikD: Time of calculation:        1.06s
StatikD: Version 2025.38 done.
```

Annex B.6.2. For anchor and ballast calculation

```
EXTERNAL LOADS (AUTOMATIC SELFWEIGHT)
ORDERED BY LOADGROUPS
-----------------------------------------------------------------------------------------------------------------------
     LOADGROUP        LOADMODE    FACTOR  VOLUME/AREA(P.U)          SUM_X          SUM_Y          SUM_Z     VOLUME/AREA
-----------------------------------------------------------------------------------------------------------------------
        STRUTS      SELFWEIGHT      1.00         78.500000         0.0000         0.0000        -0.8365        0.010656
        CABLES      SELFWEIGHT      1.00         78.500000         0.0000         0.0000        -0.0353        0.000449
 MEM-LINKS1000      SELFWEIGHT      1.00          0.005000         0.0000         0.0000        -0.5402      108.042120
 MEM-LINKS2000      SELFWEIGHT      1.00          0.005000         0.0000         0.0000        -0.5370      107.393517
-----------------------------------------------------------------------------------------------------------------------
           SUM                                                     0.0000         0.0000        -1.9490
-----------------------------------------------------------------------------------------------------------------------


EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADGROUPS
----------------------------------------------------------------------------------------------------------
 LOADGROUP        LOADMODE      LOAD    FACTOR          SUM_X          SUM_Y          SUM_Z   LOADED AREA
----------------------------------------------------------------------------------------------------------
         1          SCHNEE    0.1083      1.00         0.0000         0.0000       -11.1850        103.25
         1            WIND   -0.1500      1.00        -0.0148         0.0059       -15.4874        108.86
----------------------------------------------------------------------------------------------------------
       SUM                                            -0.0148         0.0059       -26.6724
----------------------------------------------------------------------------------------------------------


EXTERNAL LOADS (AREA-DEPENDENT)
ORDERED BY LOADMODES
--------------------------------------------------------------------------------
                  LOADMODE                        SUM_X          SUM_Y          SUM_Z
--------------------------------------------------------------------------------
       SUM          SCHNEE                       0.0000         0.0000       -11.1850
--------------------------------------------------------------------------------
       SUM            WIND                      -0.0148         0.0059       -15.4874
--------------------------------------------------------------------------------
--------------------------------------------------------------------------------
       SUM      AREA-LOADS                      -0.0148         0.0059       -26.6724
--------------------------------------------------------------------------------


EXTERNAL LOADS: SUM OF ALL EXTERNAL LOADS
================================================================================
                                                  SUM_X          SUM_Y          SUM_Z
================================================================================
                                                -0.0148         0.0059       -28.6213
================================================================================
```

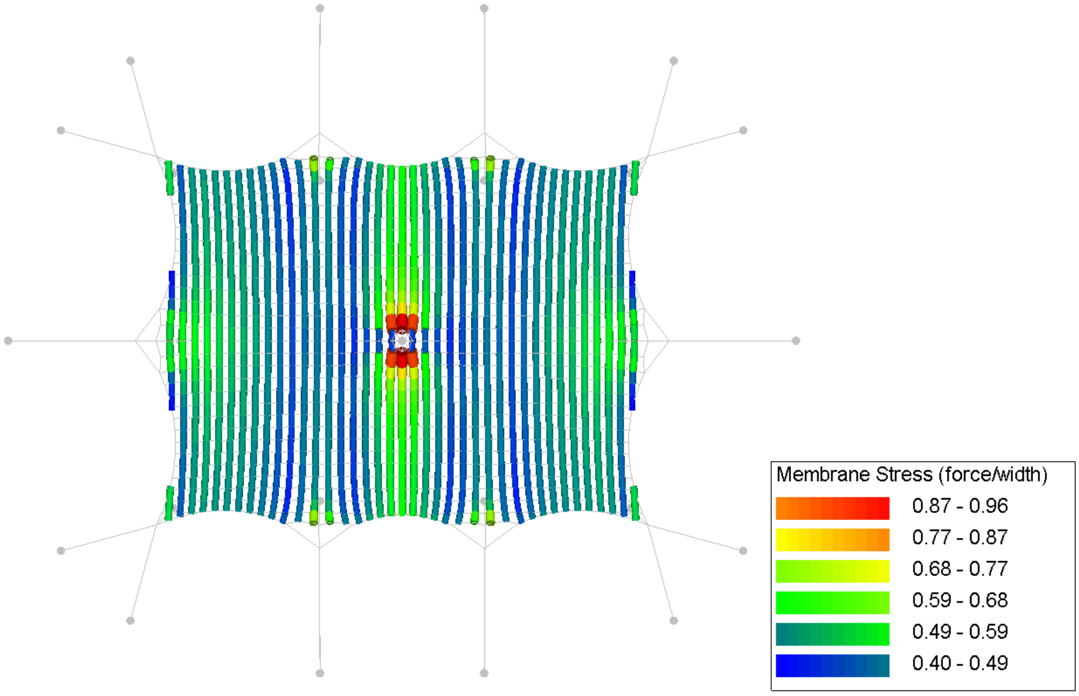
```
Inner Energy=           5.508935


StatikD:  Sum of external loads:      -0.014794       0.005925      -28.621299
StatikD:  Sum of support forces:       0.014705      -0.004778       28.622217
StatikD:  P:\2026\TTP20260008\06. Constructie - hoofdberekeningen\Boek1\7.5x15\C06a_OW+VO+WD+SNOW_anchors
StatikD: Time of calculation:       1.11s
StatikD: Version 2025.38 done.
```

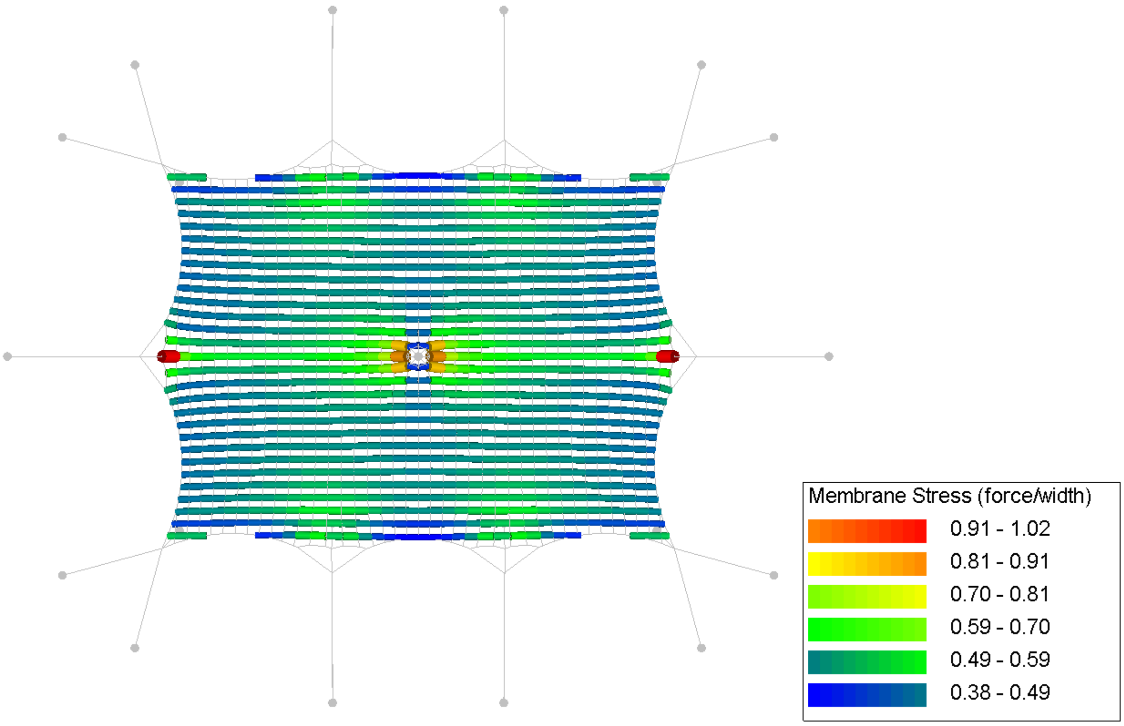
Annex C. Results per load combination – 7.5x10m model

Annex C.1. CO1. Own weight + pretension

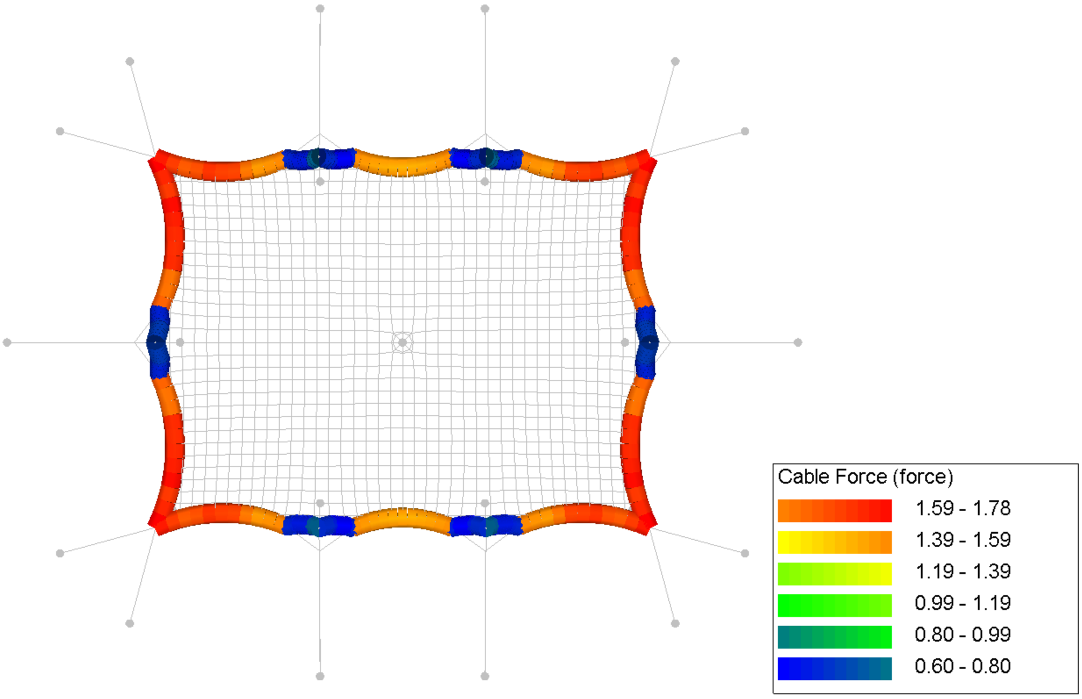
Annex C.1.1. Membrane stress (warp)



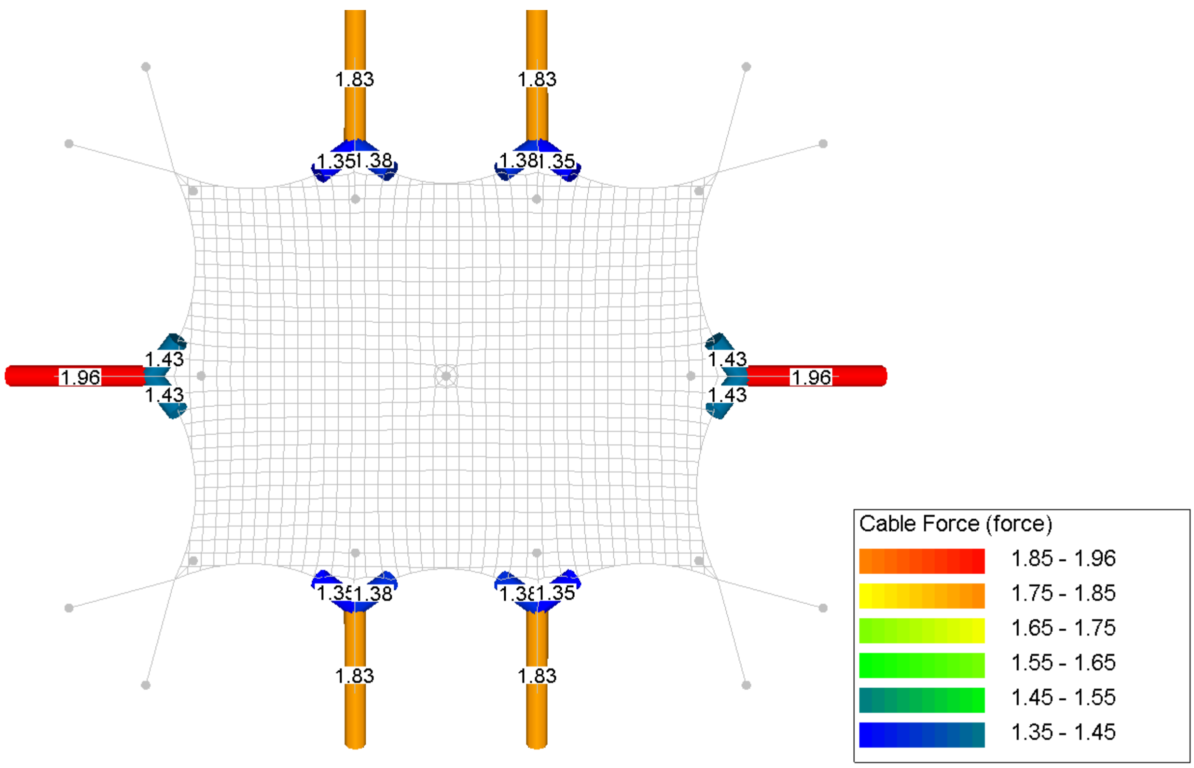
Annex C.1.2. Membrane stress (weft)



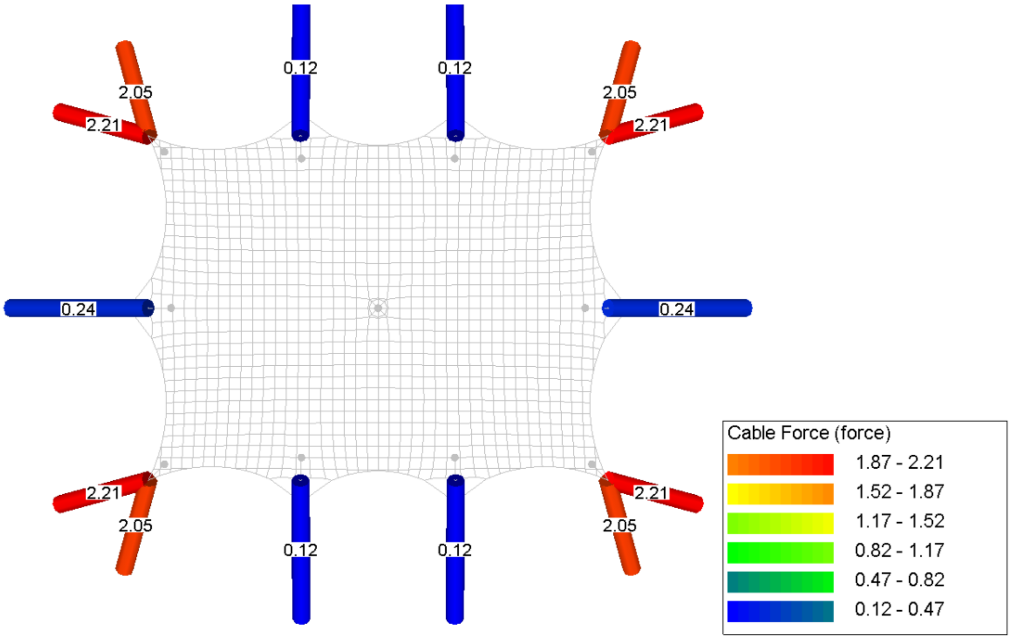
Annex C.1.3. Circumference rope forces



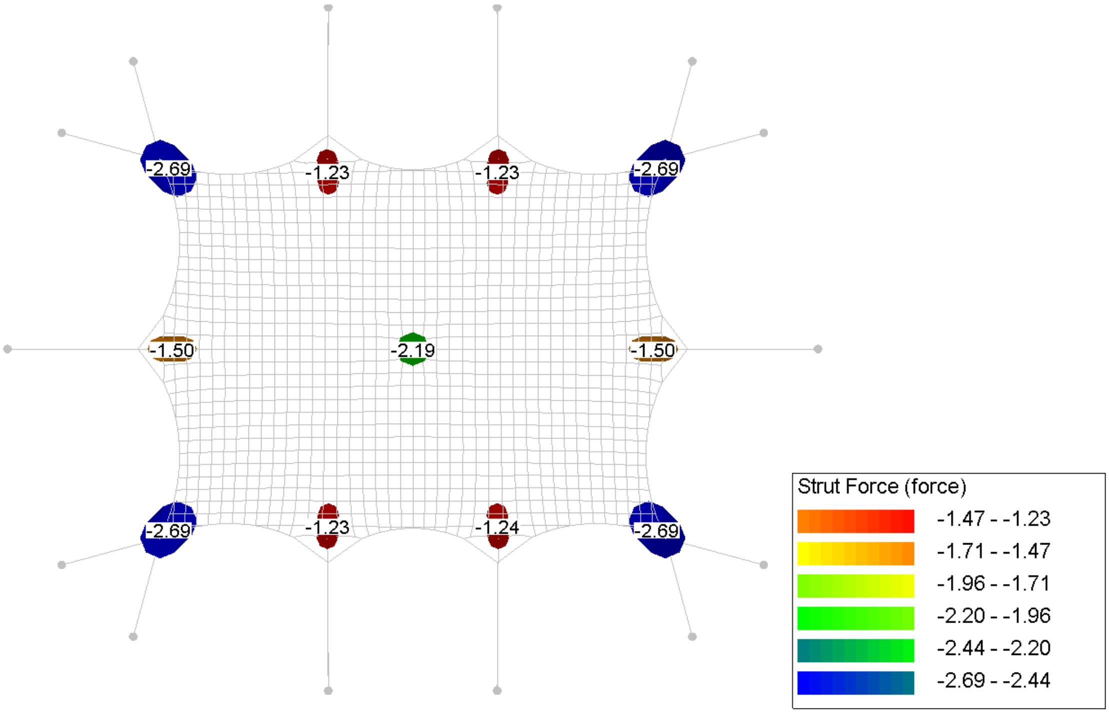
Annex C.1.4. Rope forces



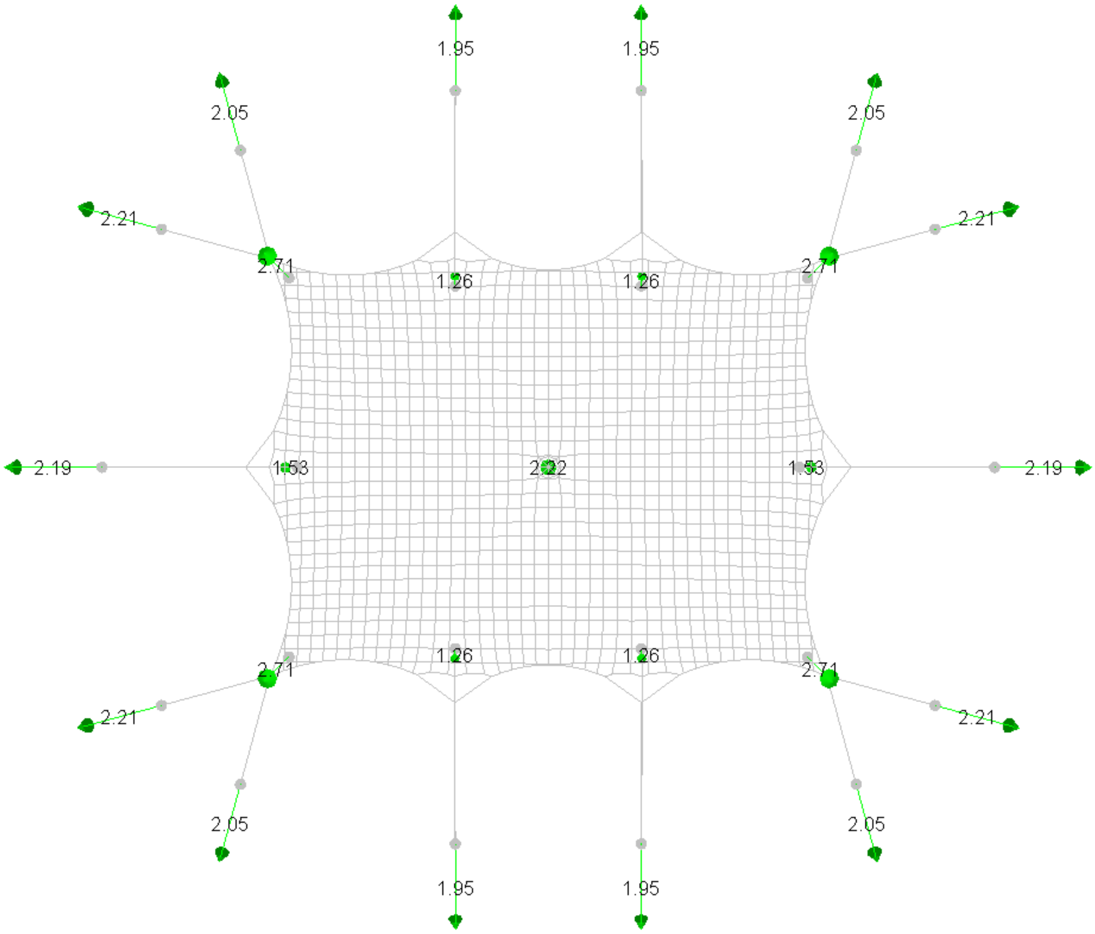
Annex C.1.5. Belt forces



Annex C.1.6. Pole forces

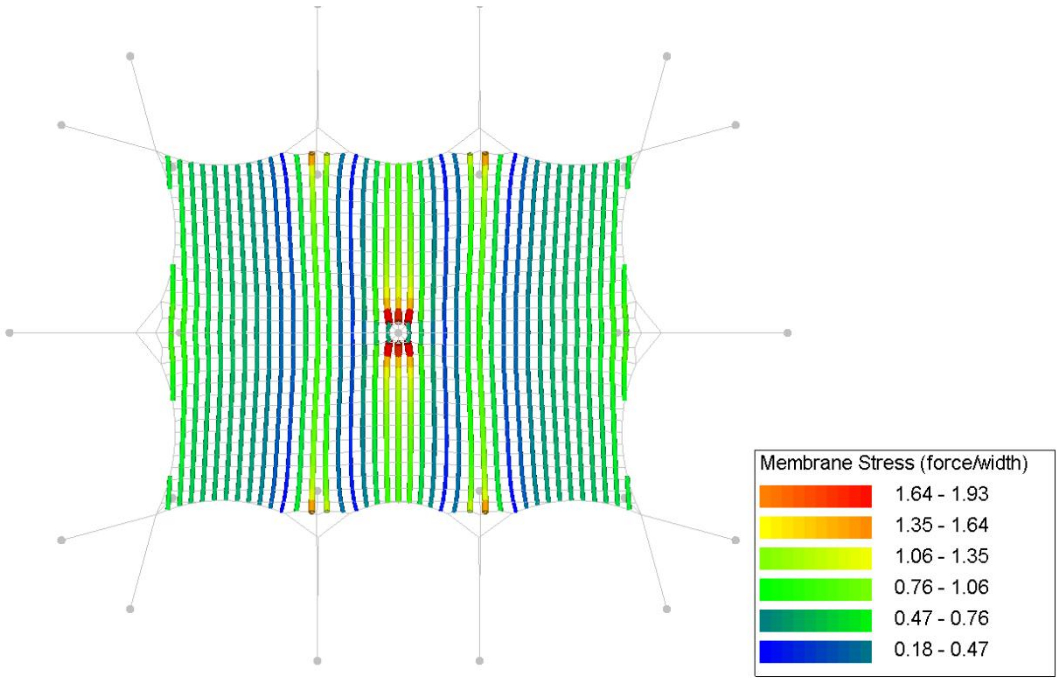


Annex C.1.7. Reaction forces

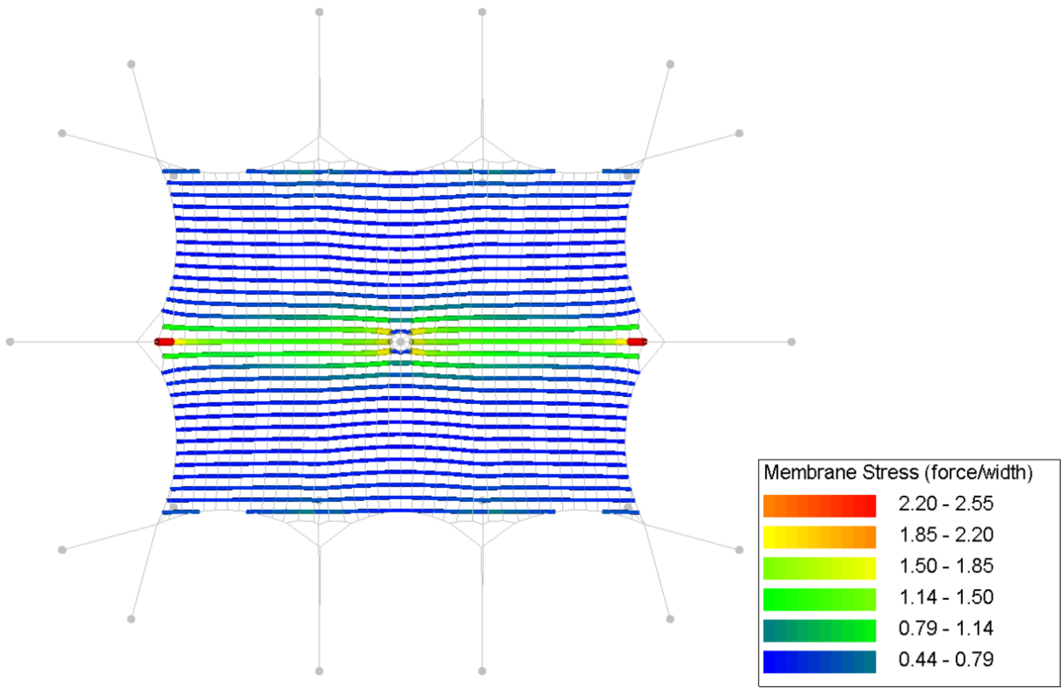


Annex C.2. CO2. Own weight + pretension + snow

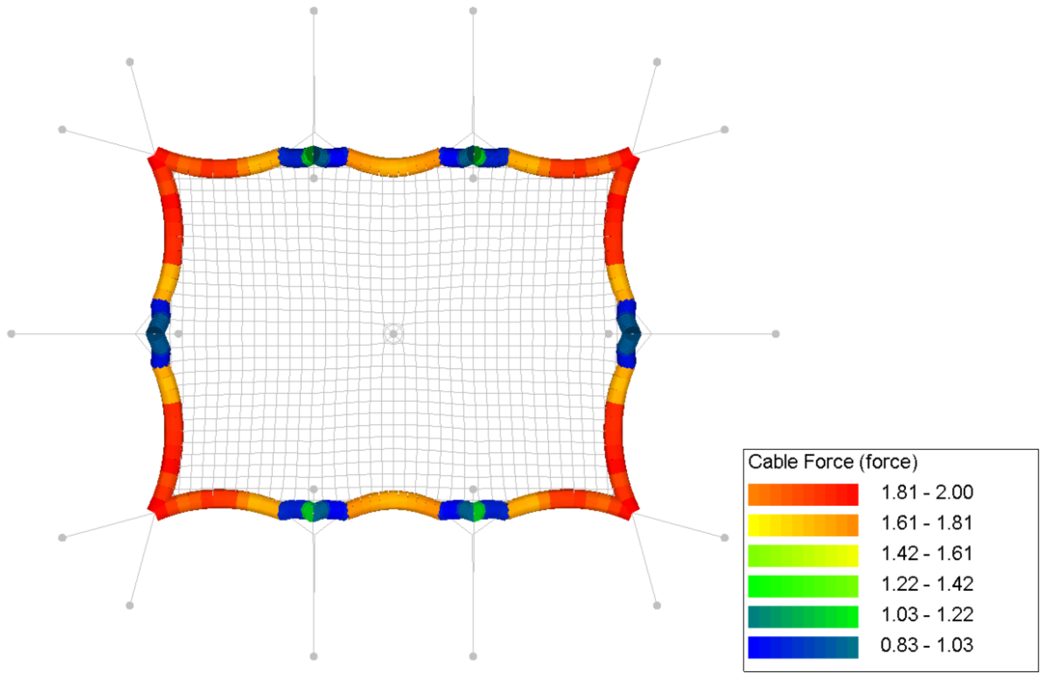
Annex C.2.1. Membrane stress (warp)



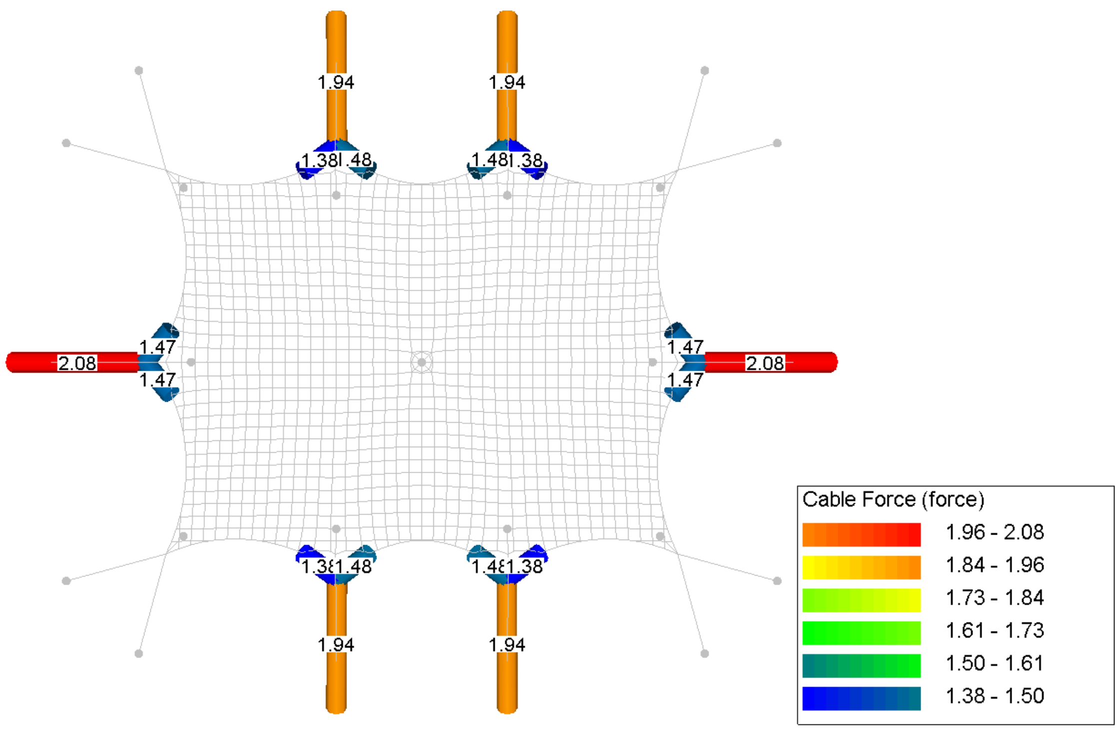
Annex C.2.2. Membrane stress (weft)



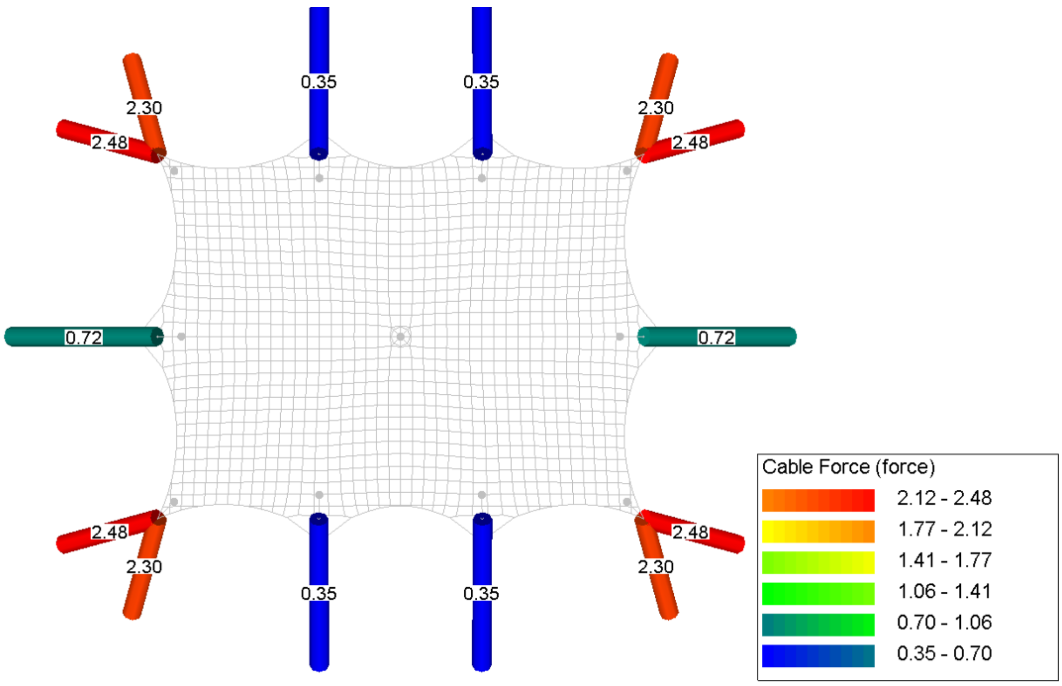
Annex C.2.3. Circumference rope forces



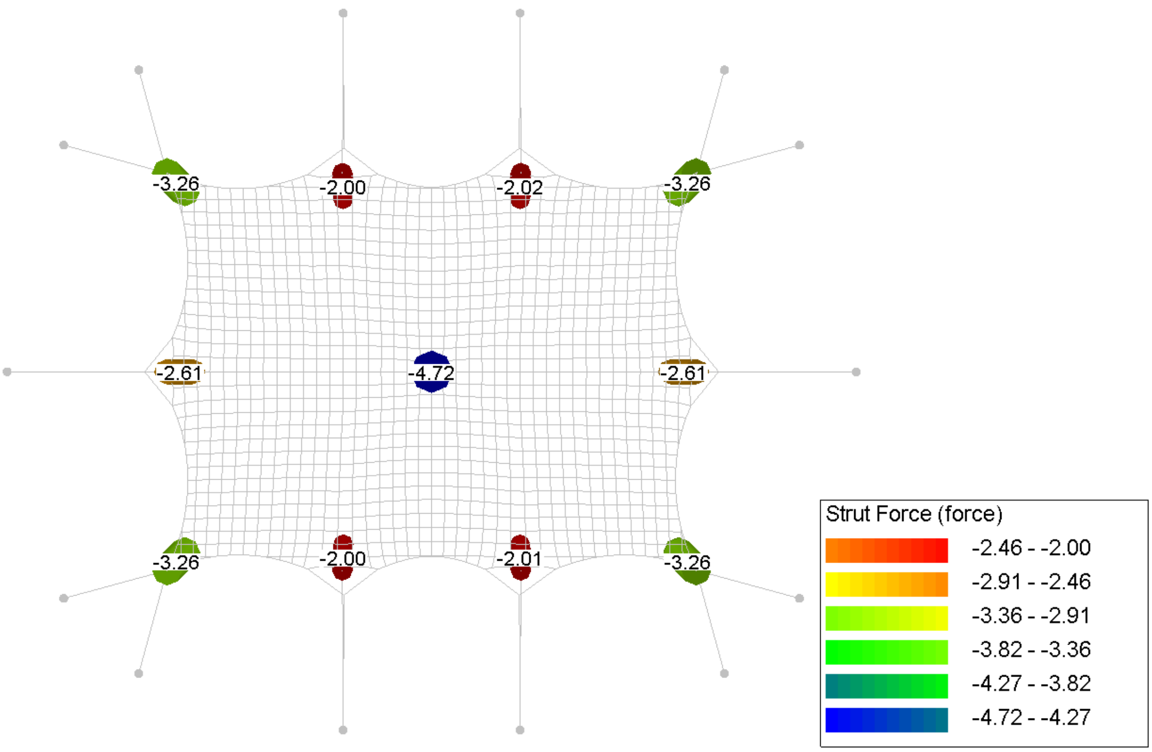
Annex C.2.4. Rope forces



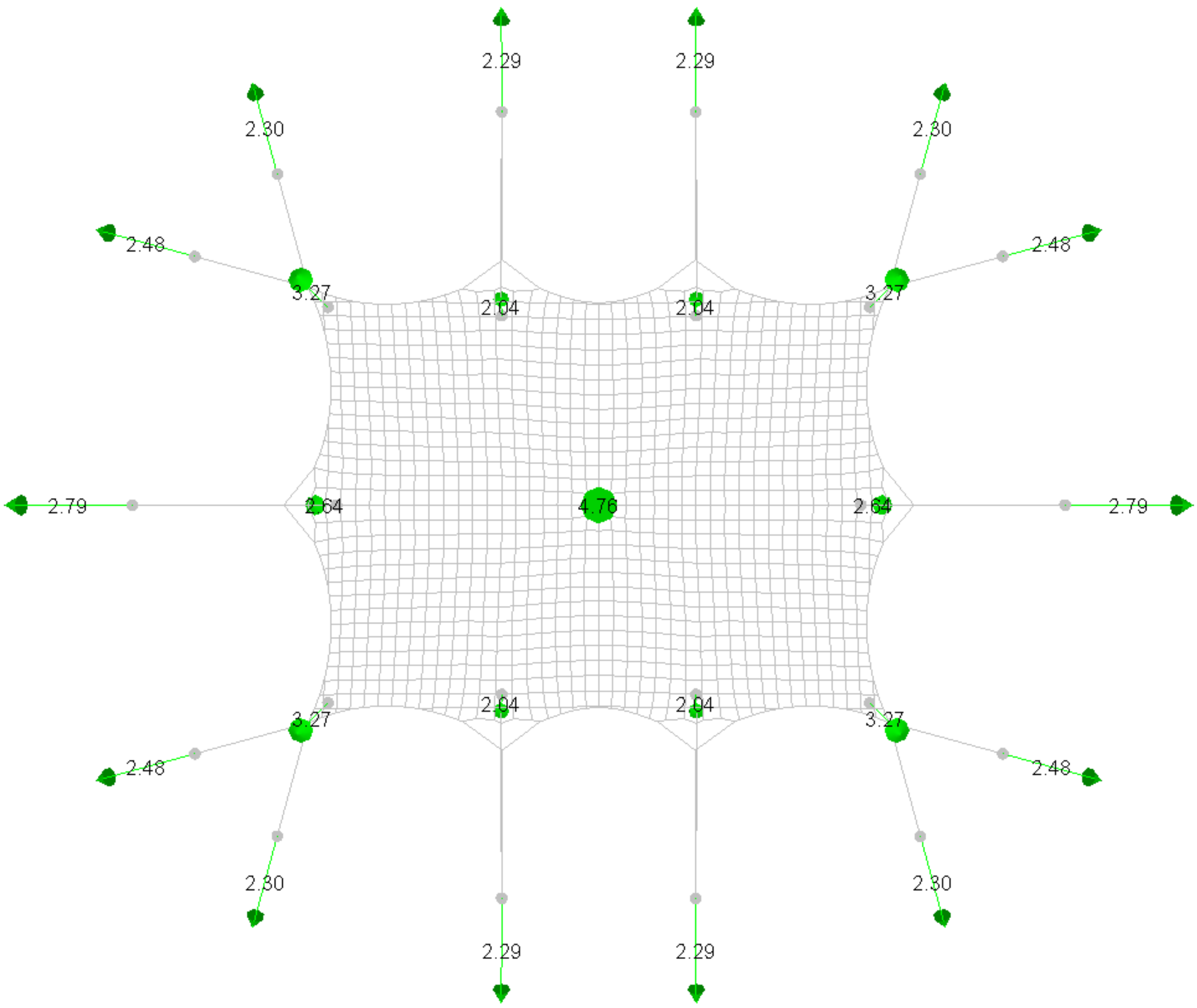
Annex C.2.5. Belt forces



Annex C.2.6. Pole forces

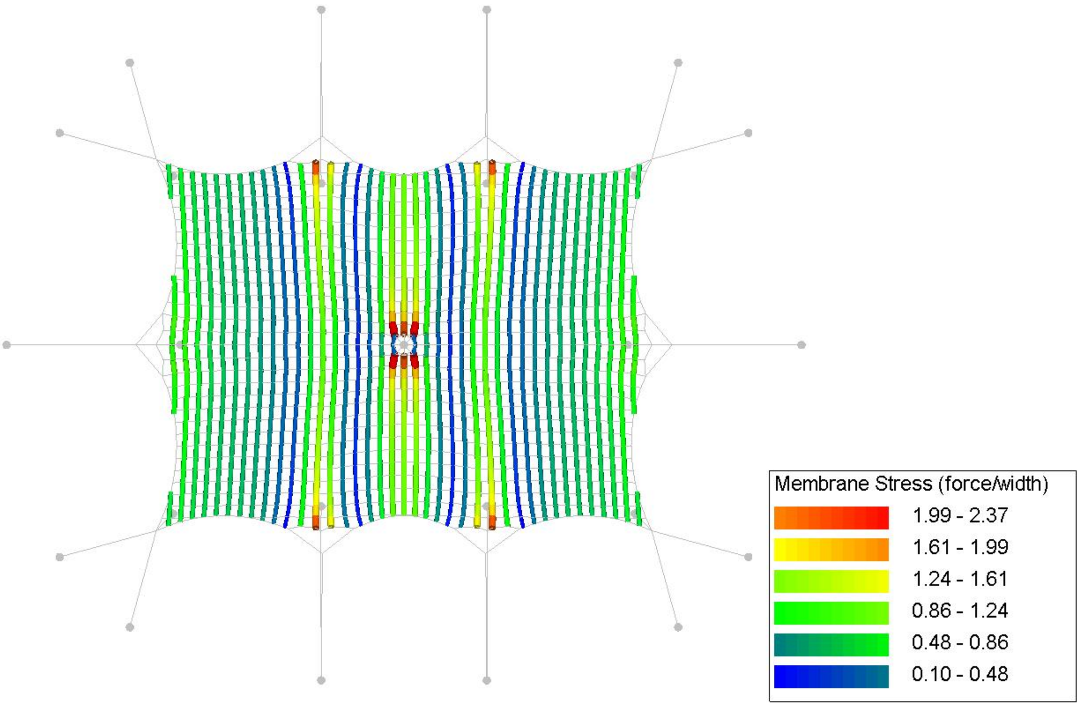


Annex C.2.7. Anchor forces

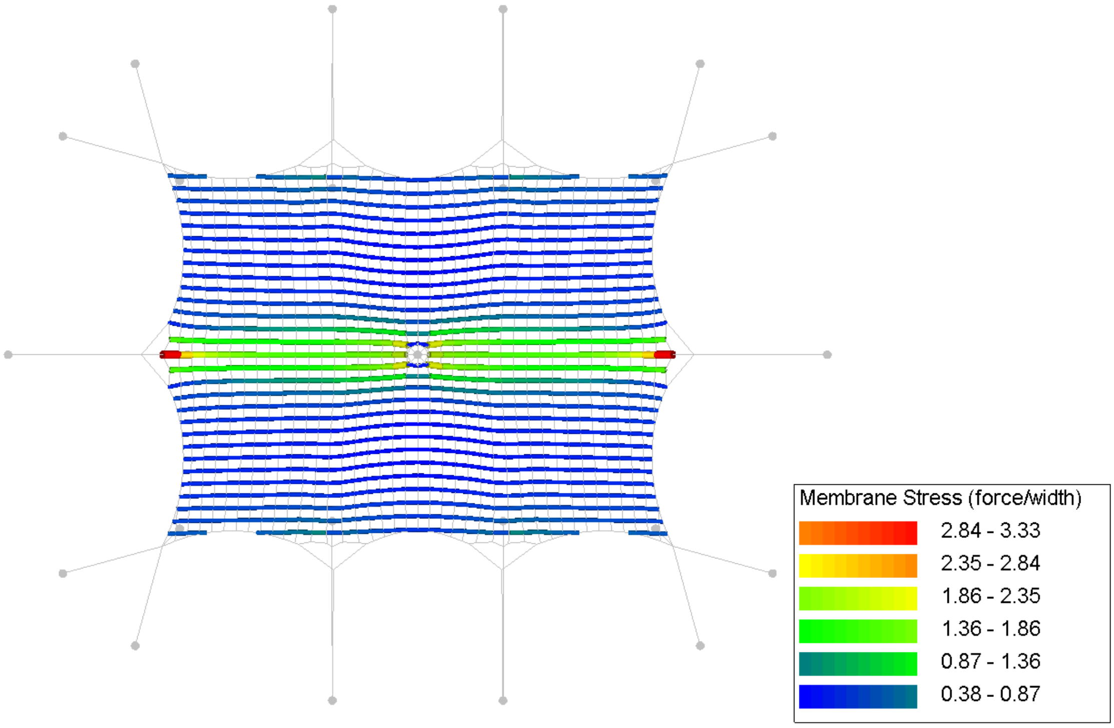


Annex C.3. CO3. Own weight + pretension + wind pressure

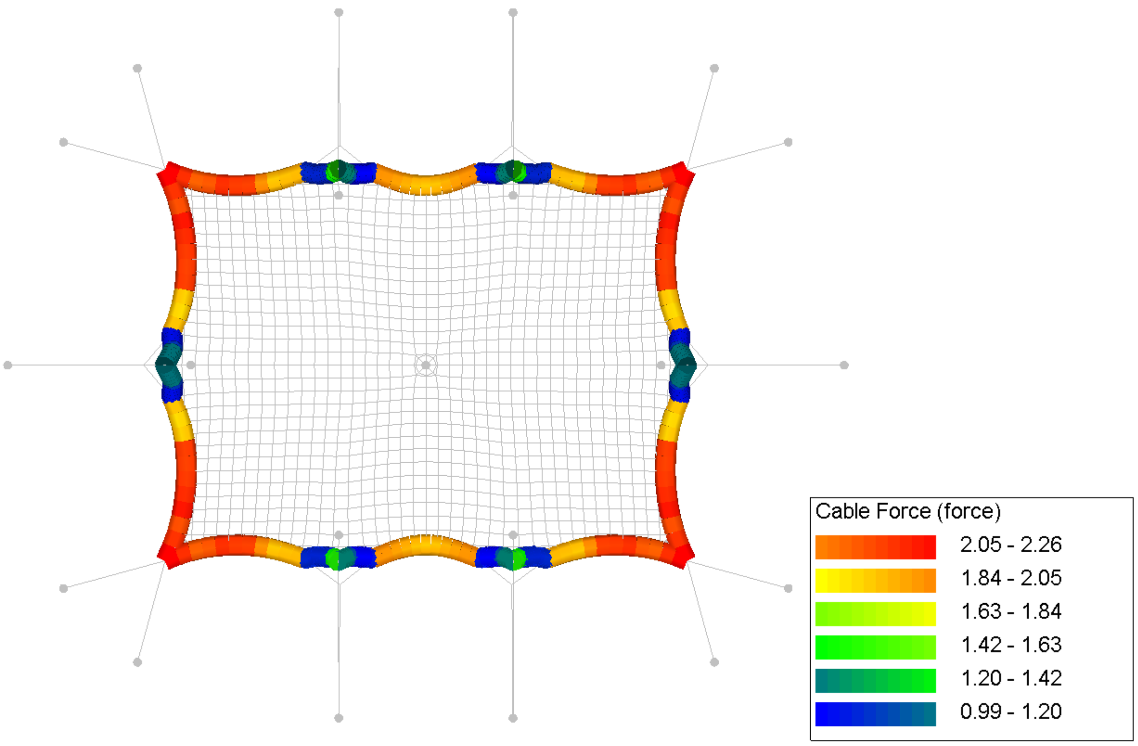
Annex C.3.1. Membrane stress (warp)



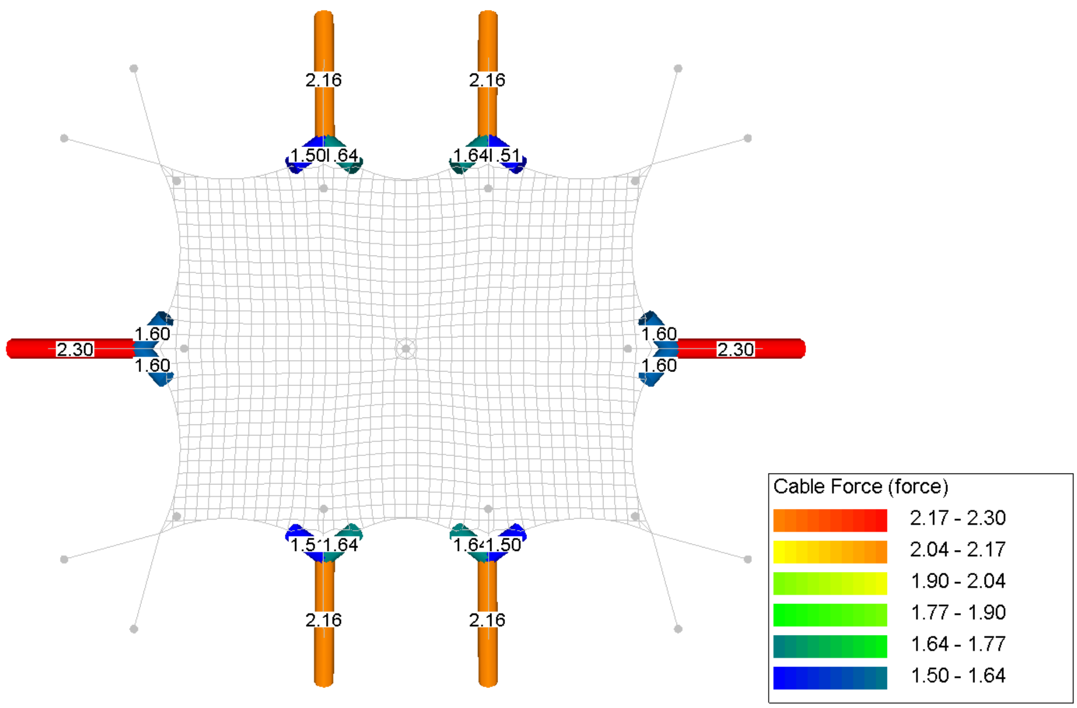
Annex C.3.2. Membrane stress (weft)



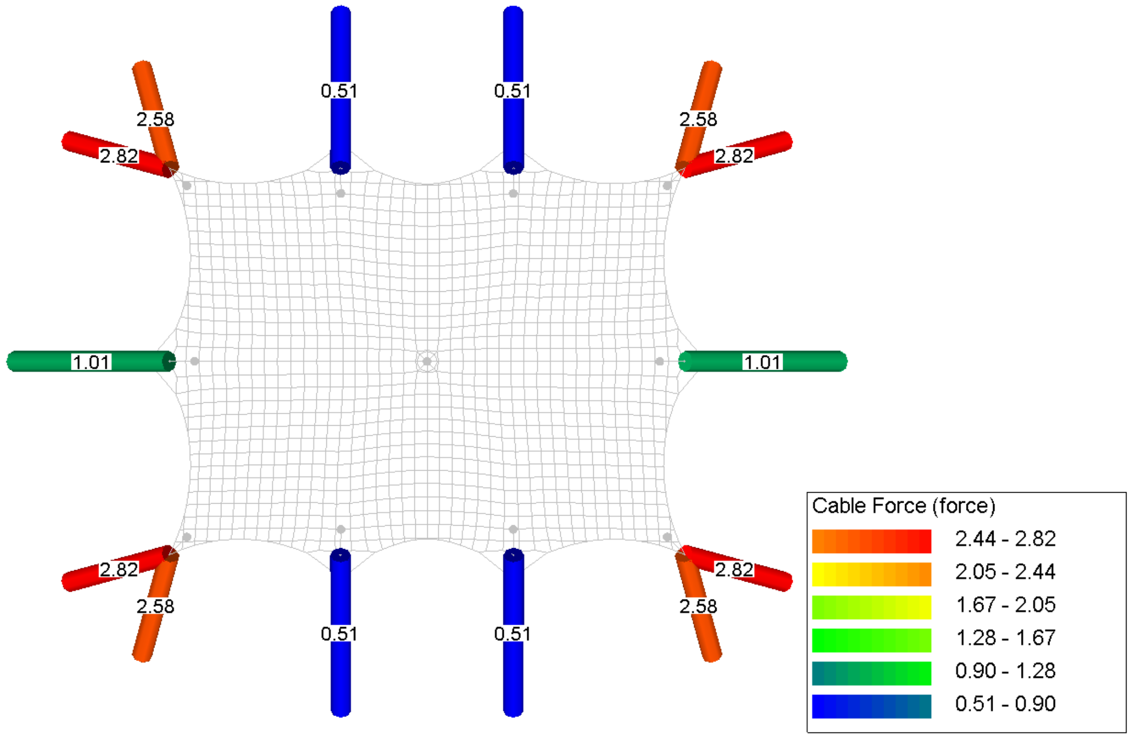
Annex C.3.3. Circumference rope forces



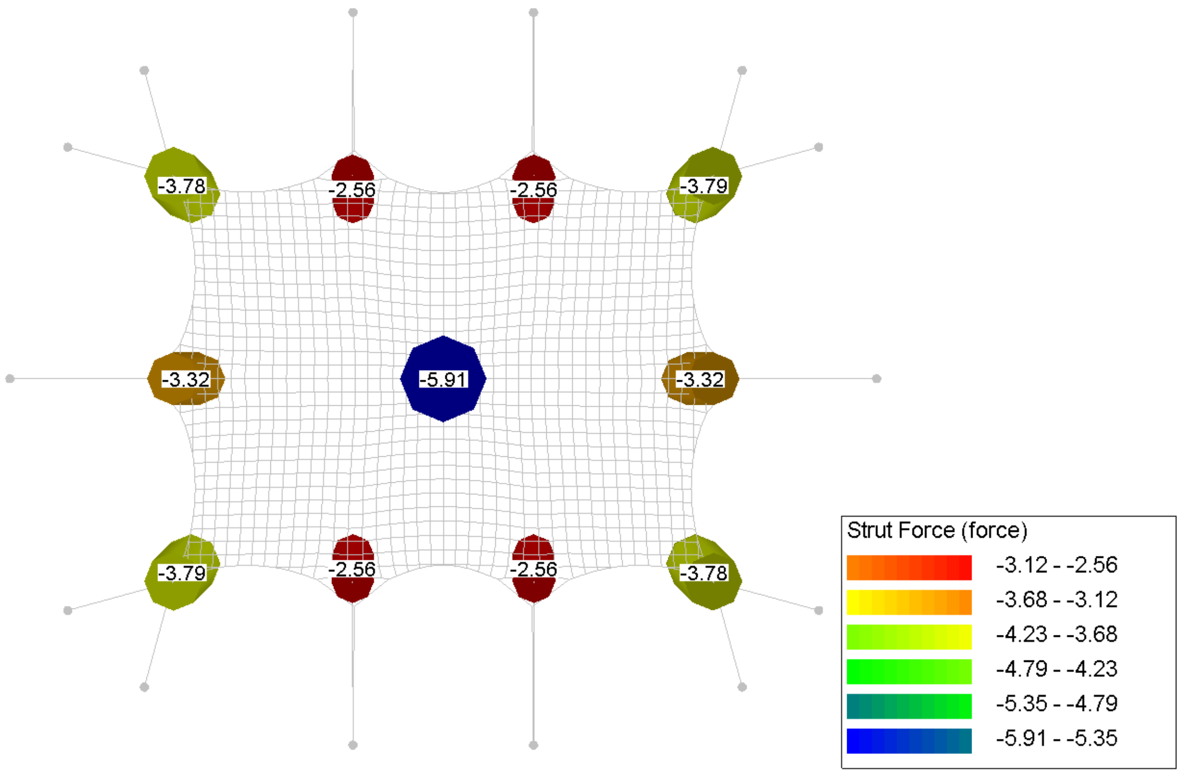
Annex C.3.4. Rope forces



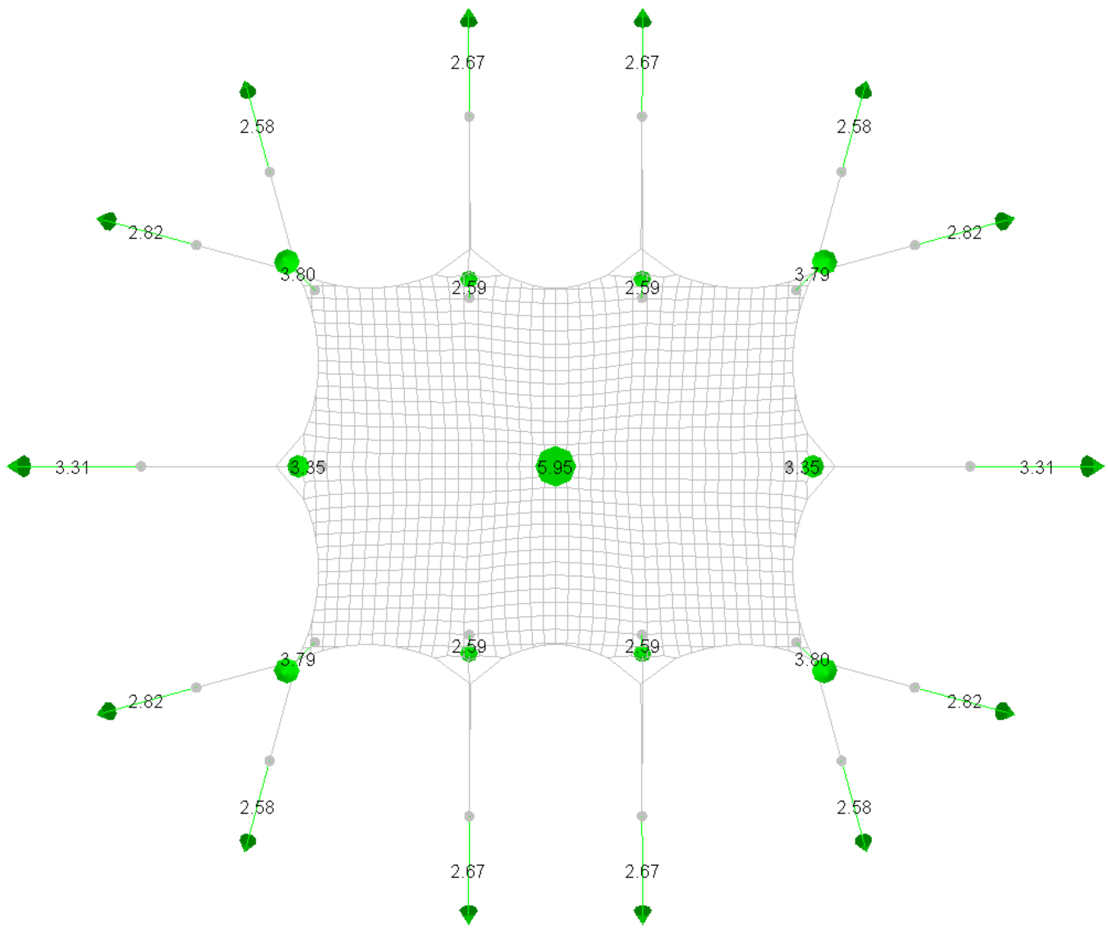
Annex C.3.5. Belt forces



Annex C.3.6. Poles forces

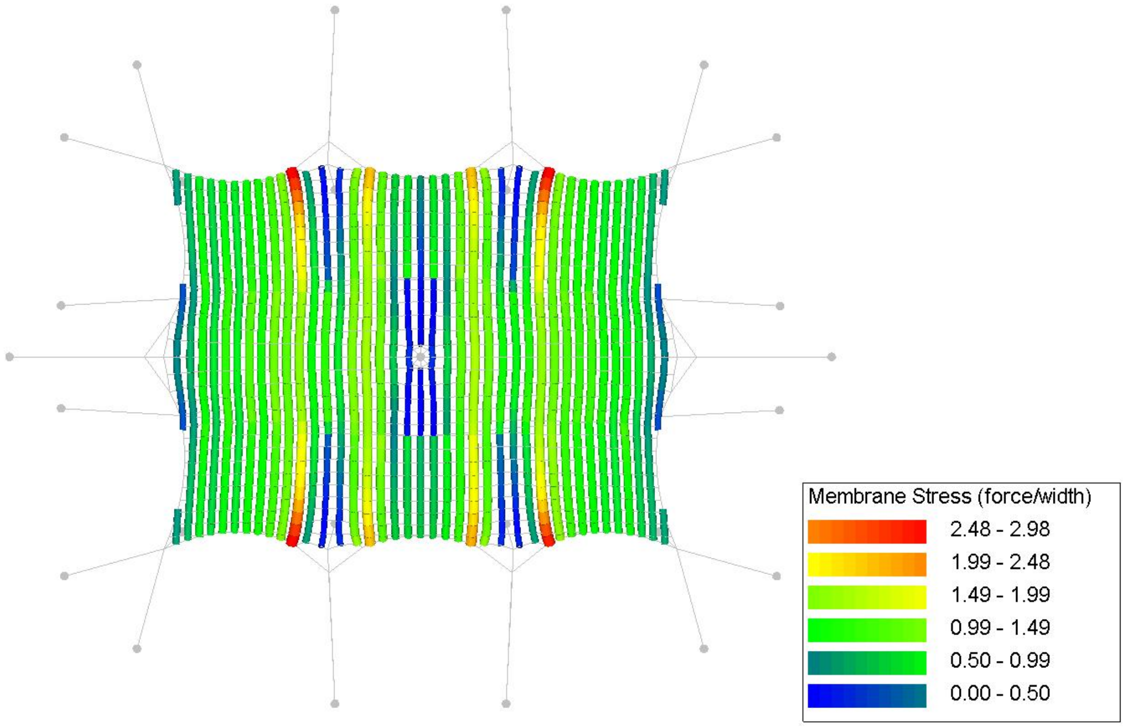


Annex C.3.7. Reaction forces

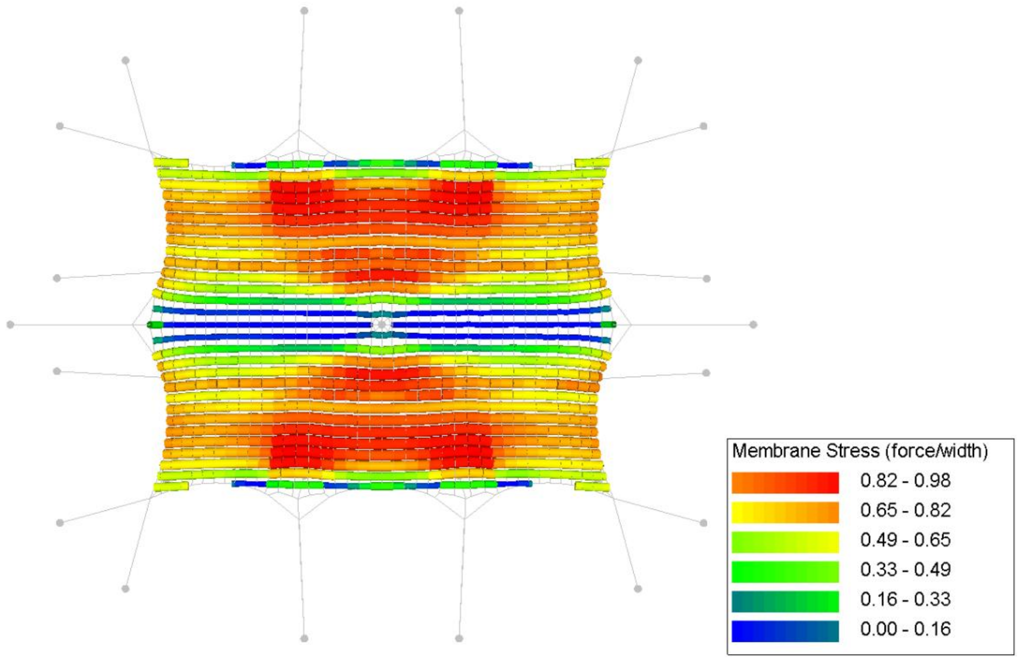


Annex C.4. CO4. Own weight + pretension + wind suction

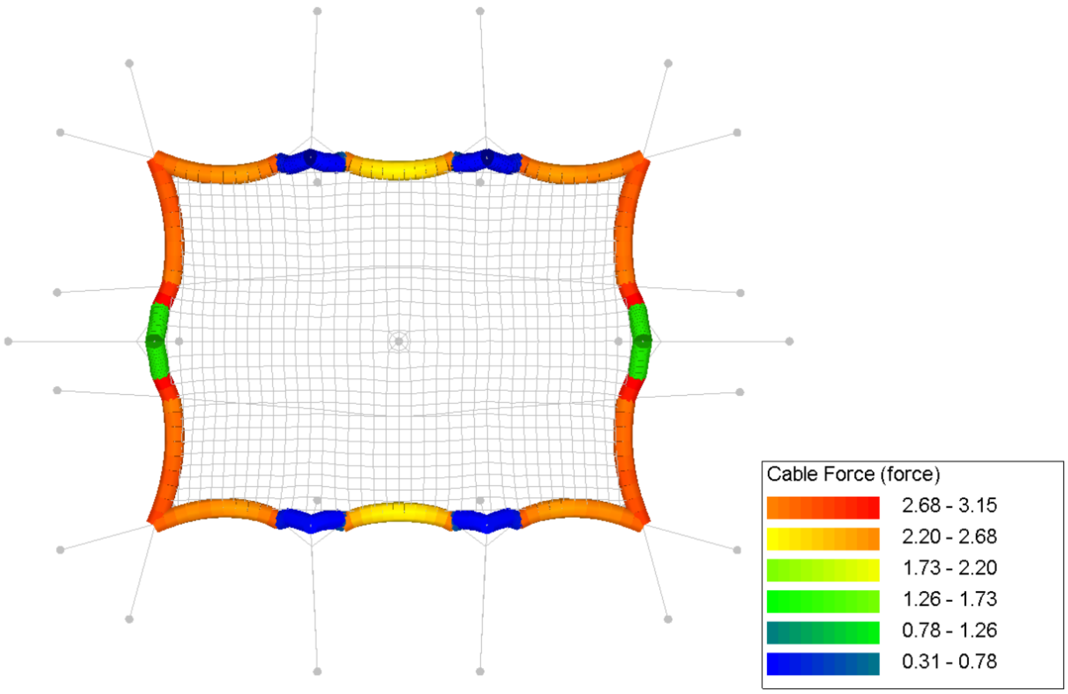
Annex C.4.1. Membrane stress (warp)



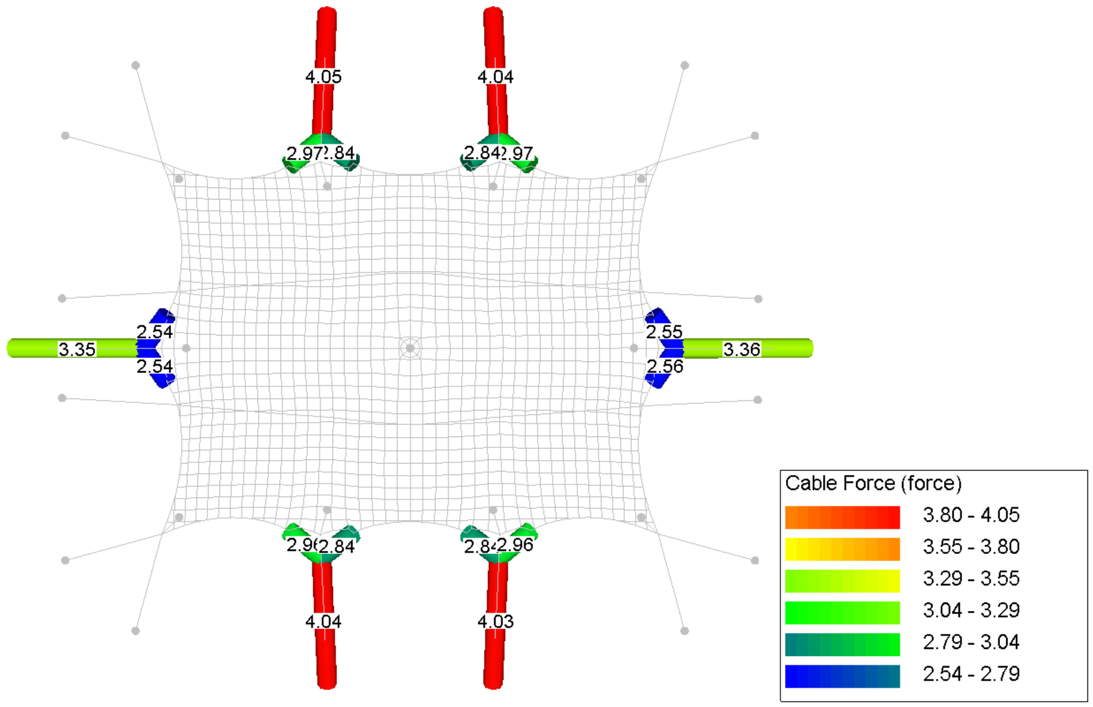
Annex C.4.2. Membrane stress (weft)



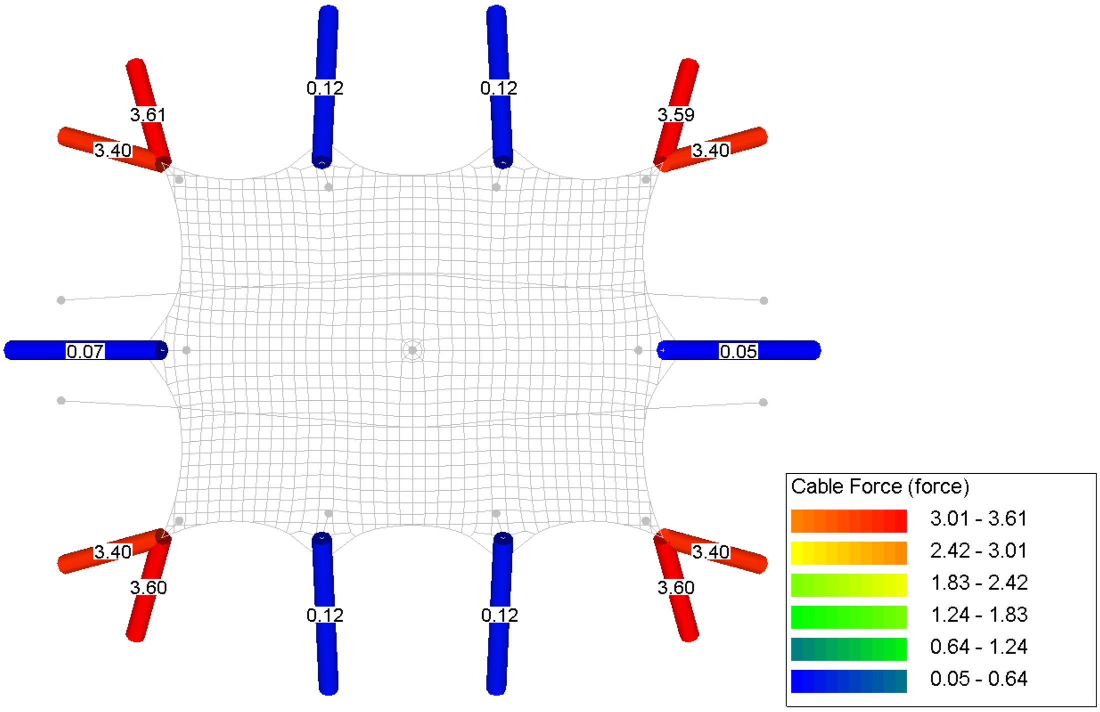
Annex C.4.3. Circumference rope forces



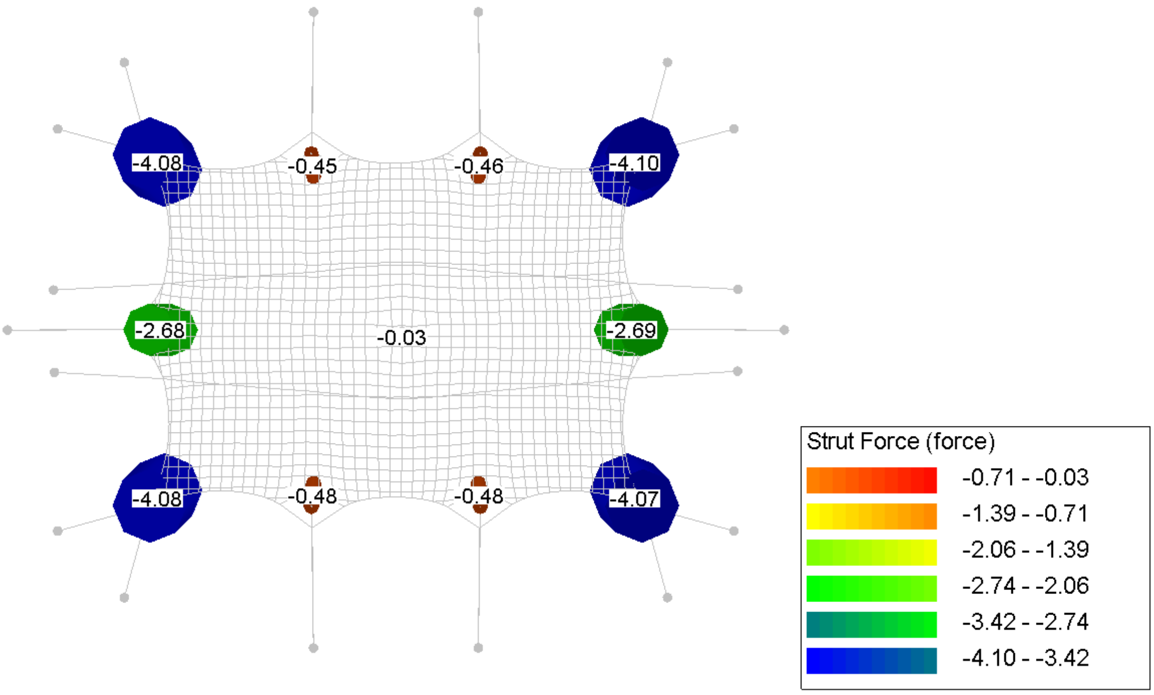
Annex C.4.4. Guy rope forces



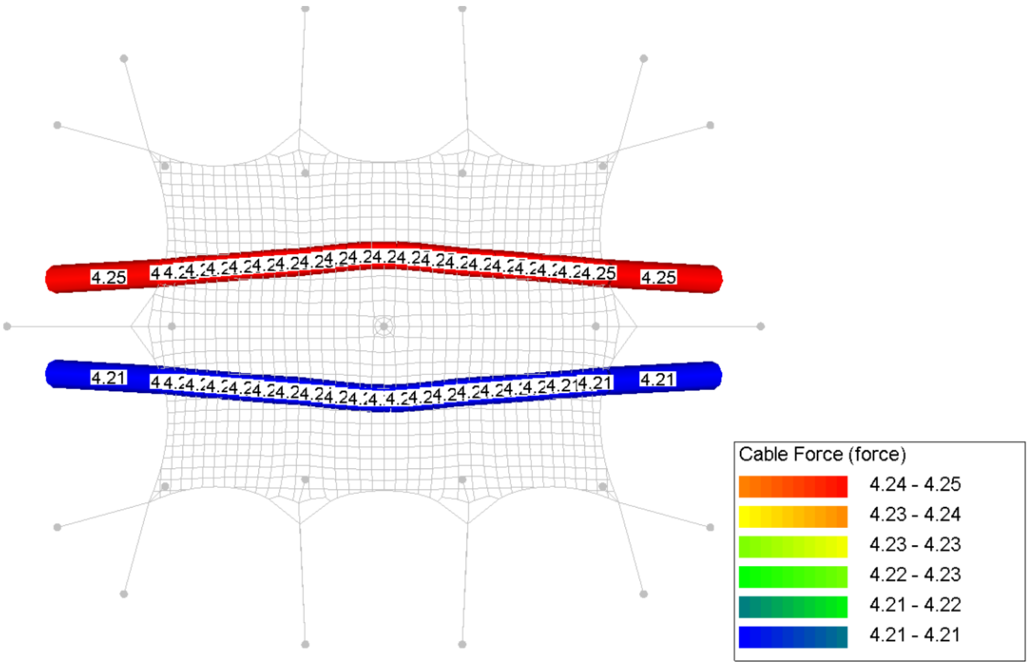
Annex C.4.5. Belt forces



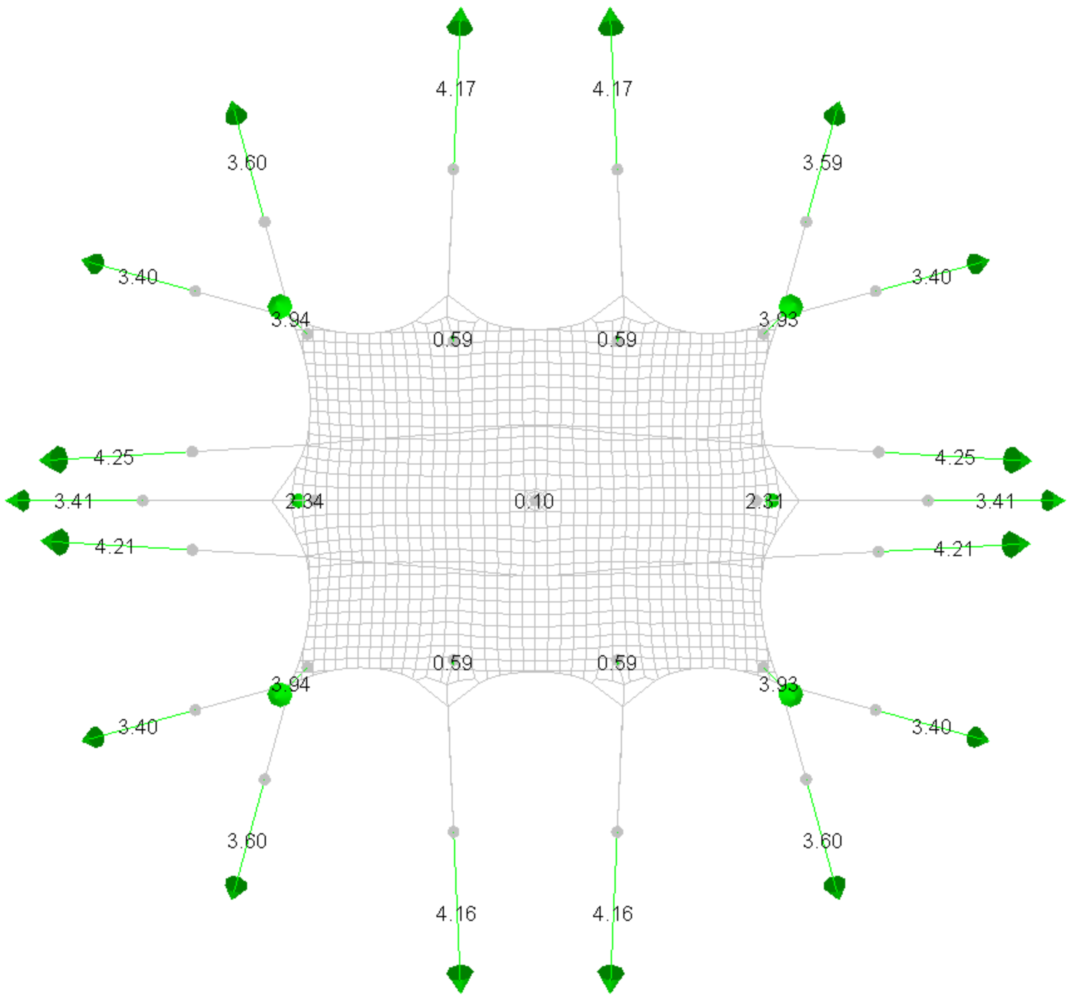
Annex C.4.6. Poles forces



Annex C.4.7. Storm belt forces

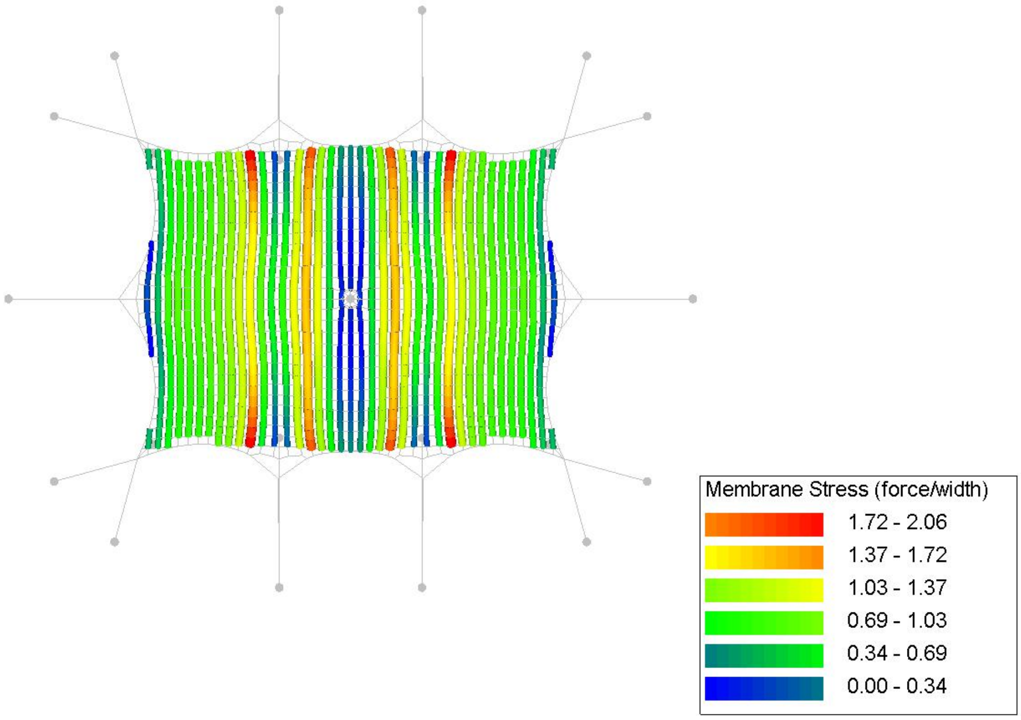


Annex C.4.8. Reaction forces

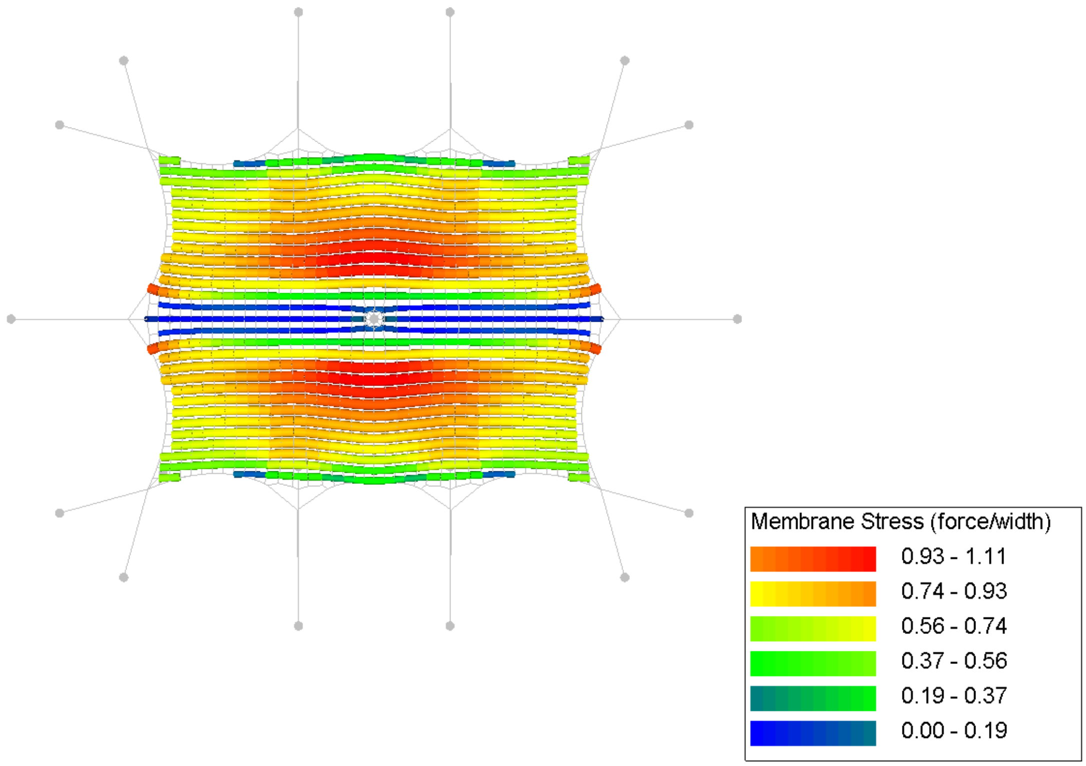


Annex C.5. CO5. Own weight + pretension + wind suction reduced [NL]

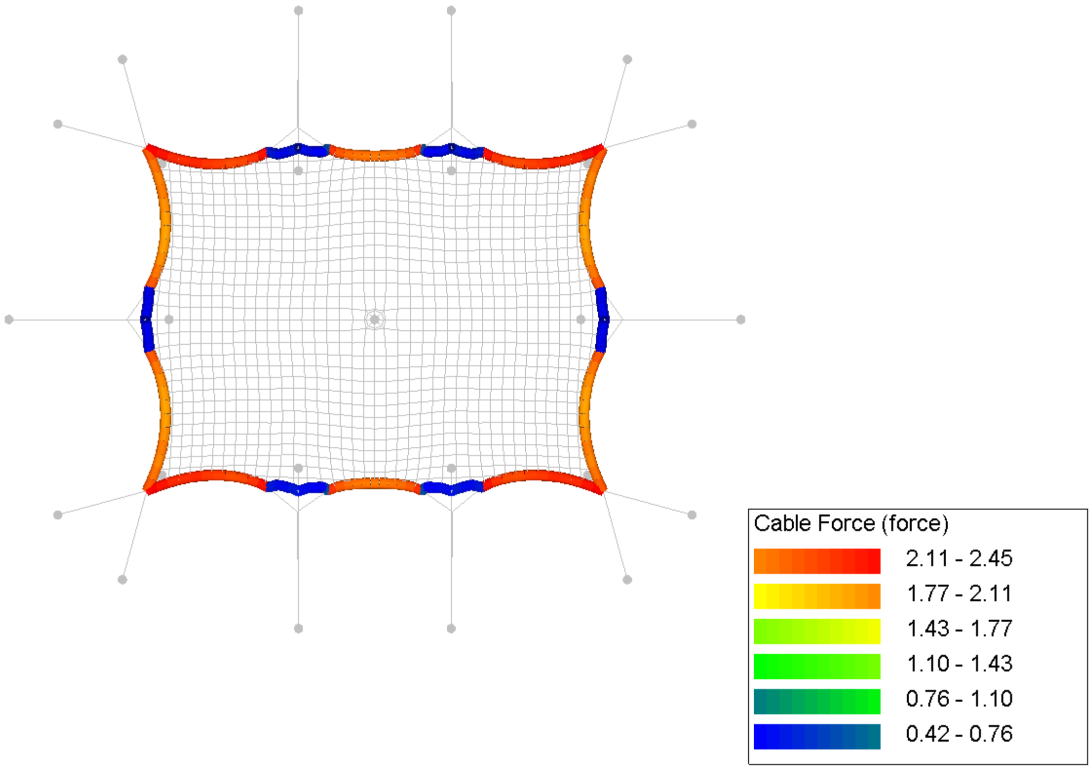
Annex C.5.1. Membrane stress (warp)



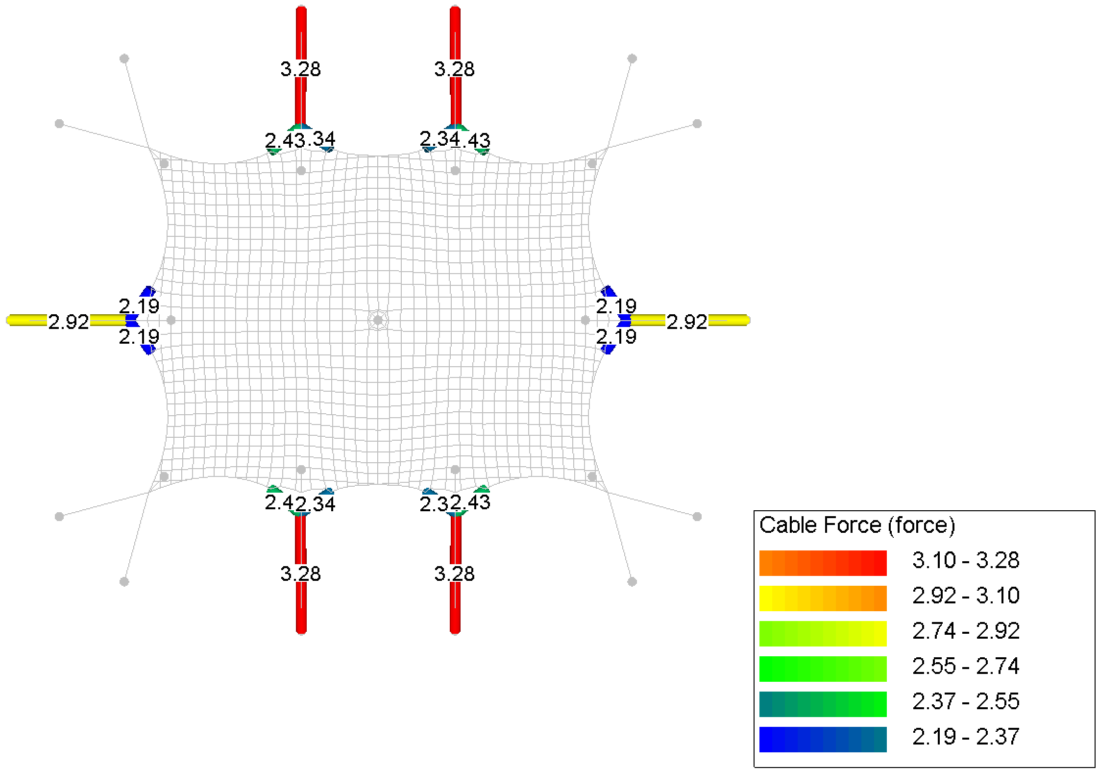
Annex C.5.2. Membrane stress (weft)



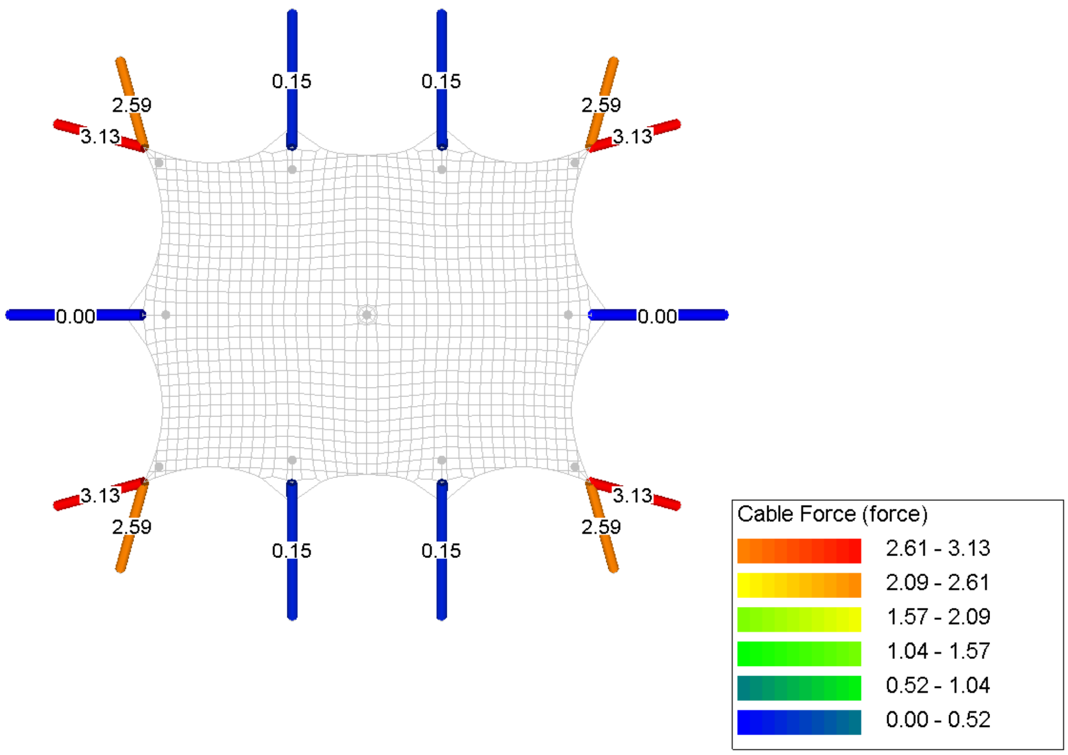
Annex C.5.3. Circumference rope forces



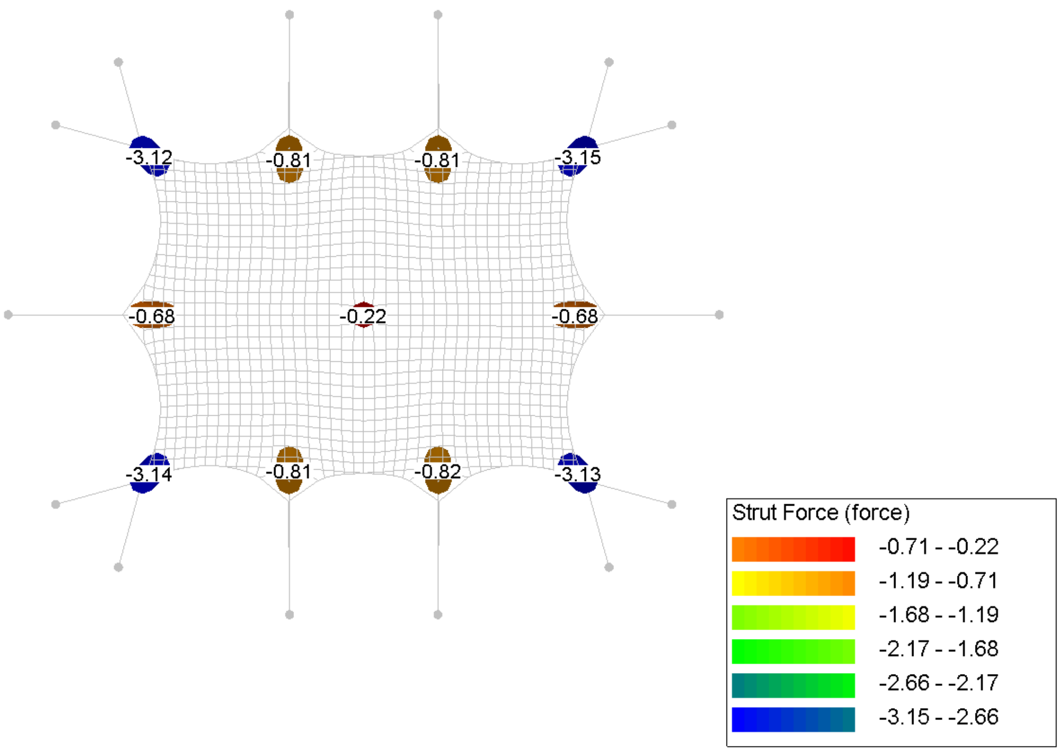
Annex C.5.4. Rope forces



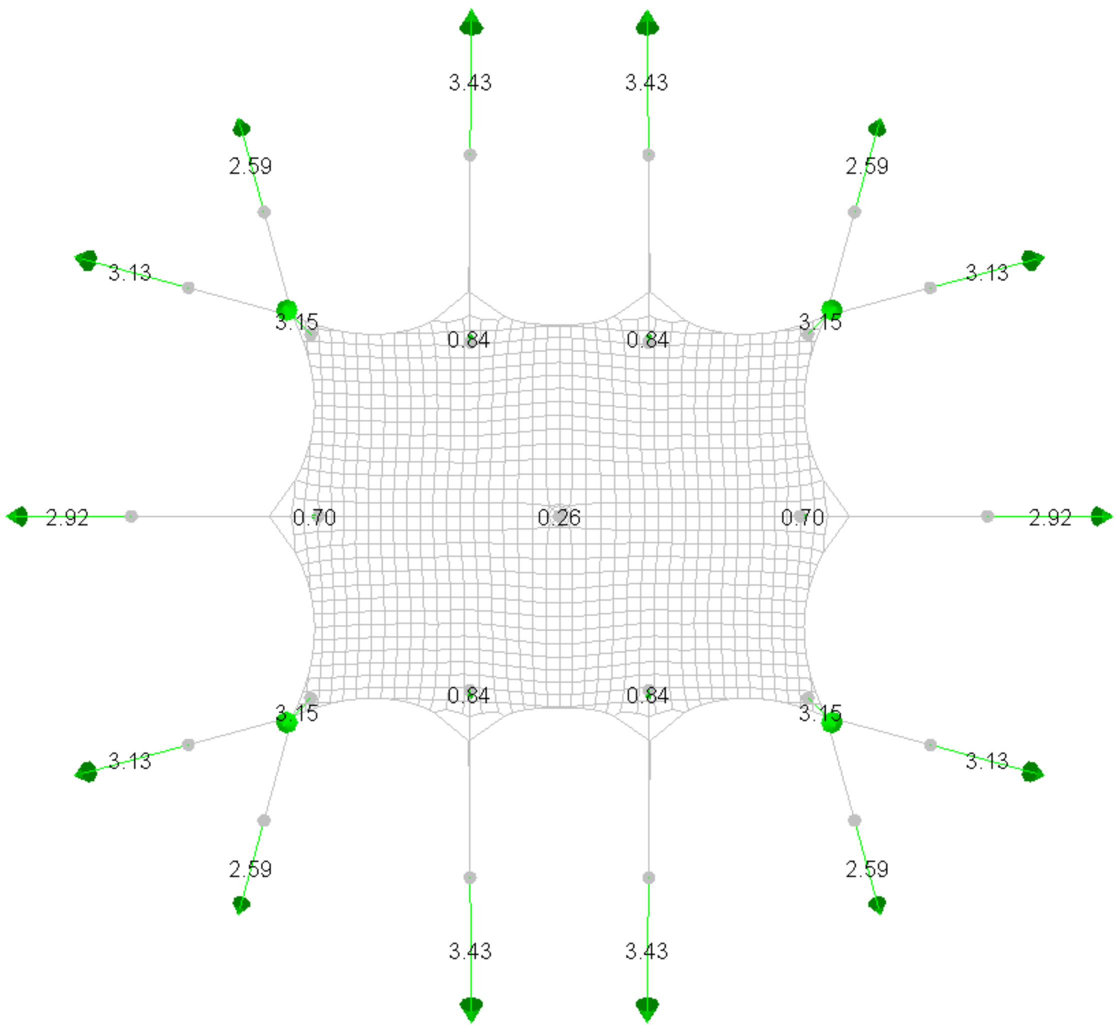
Annex C.5.5. Belt forces



Annex C.5.6. Poles forces

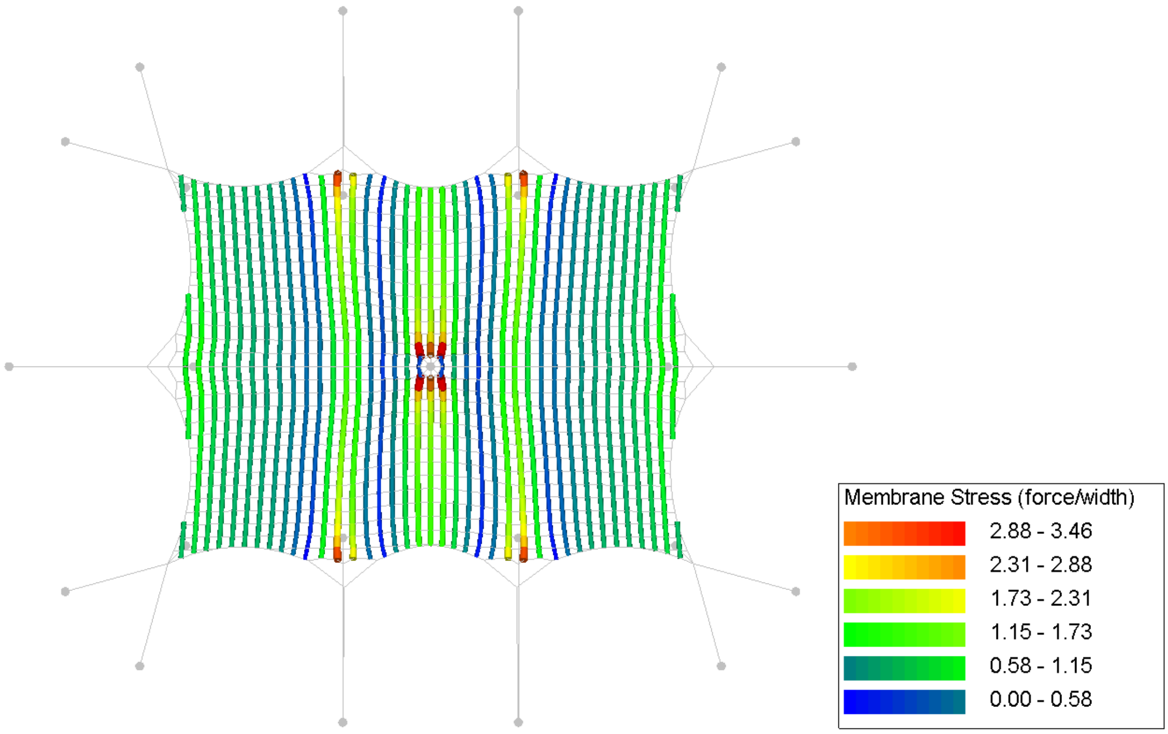


Annex C.5.7. Reaction forces

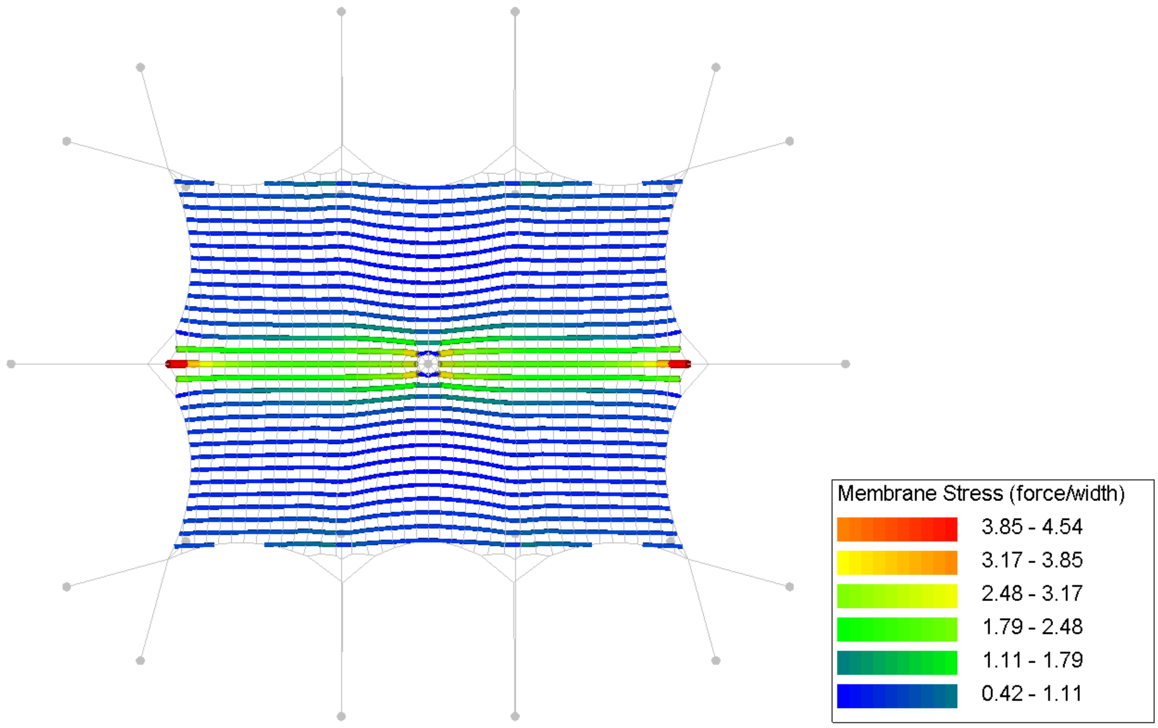


Annex C.6. CO6. Own weight + pretension + wind pressure + snow [FR]

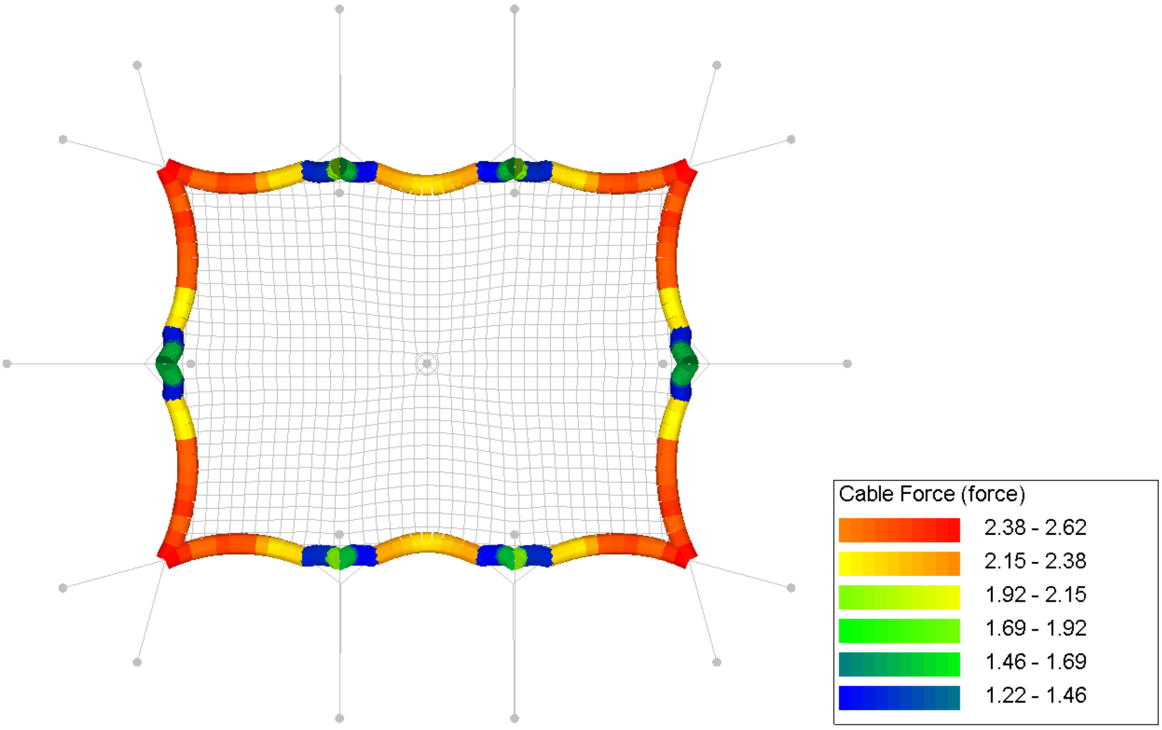
Annex C.6.1. Membrane stress (warp)



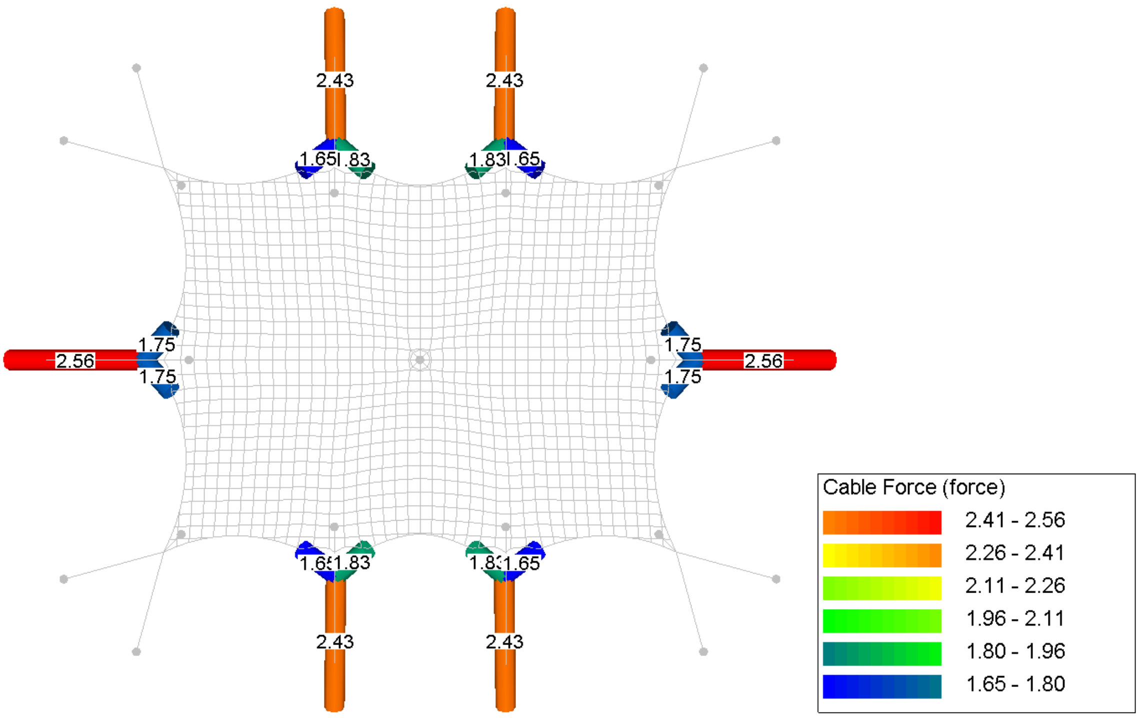
Annex C.6.2. Membrane stress (weft)



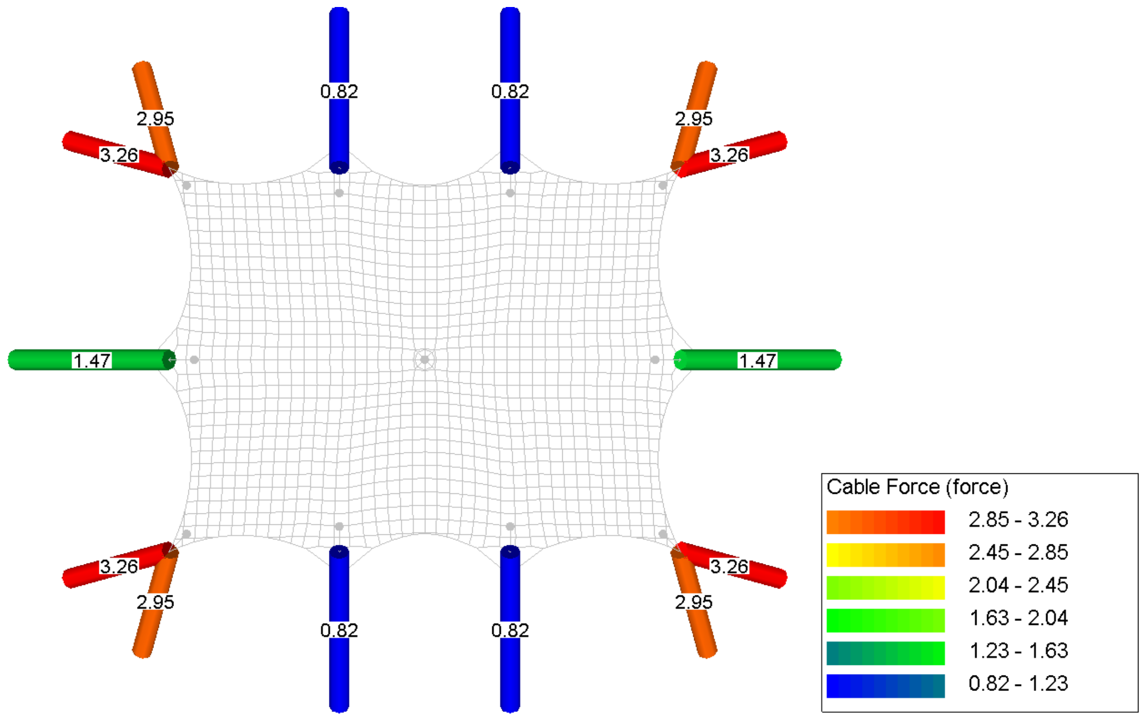
Annex C.6.3. Circumference rope forces



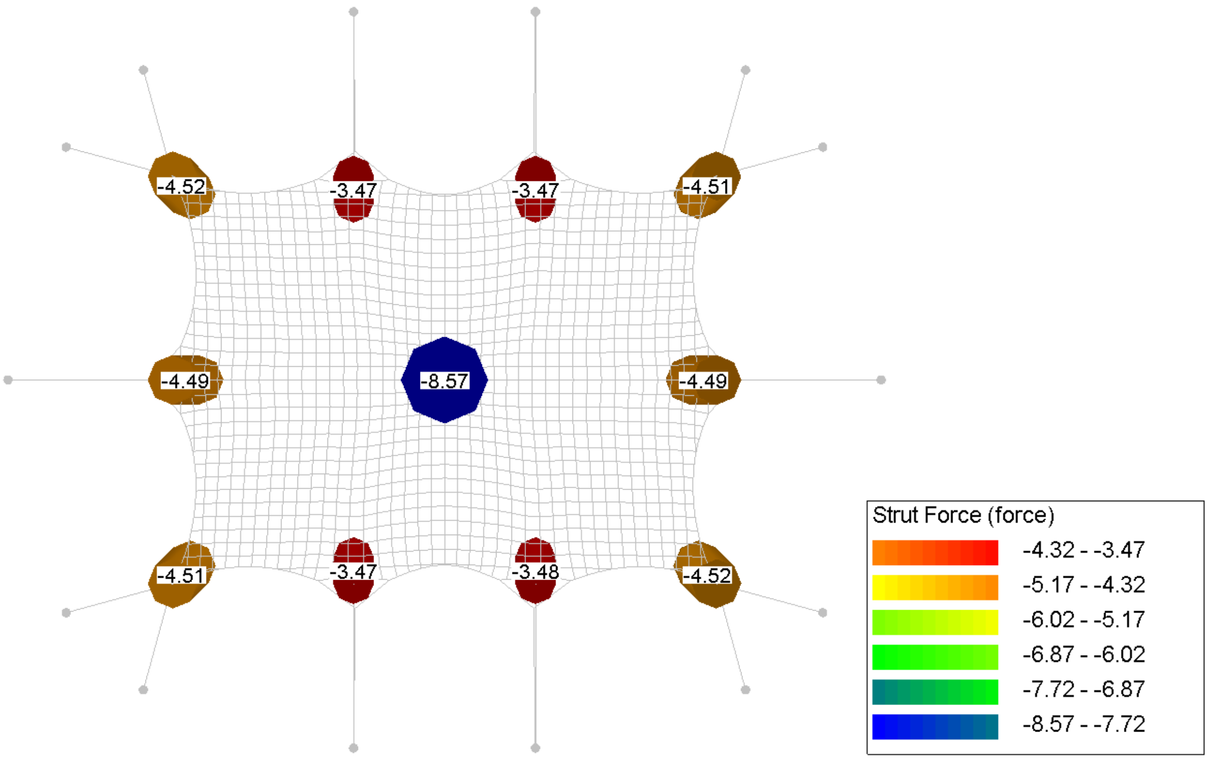
Annex C.6.4. Guy rope forces



Annex C.6.5. Belt forces

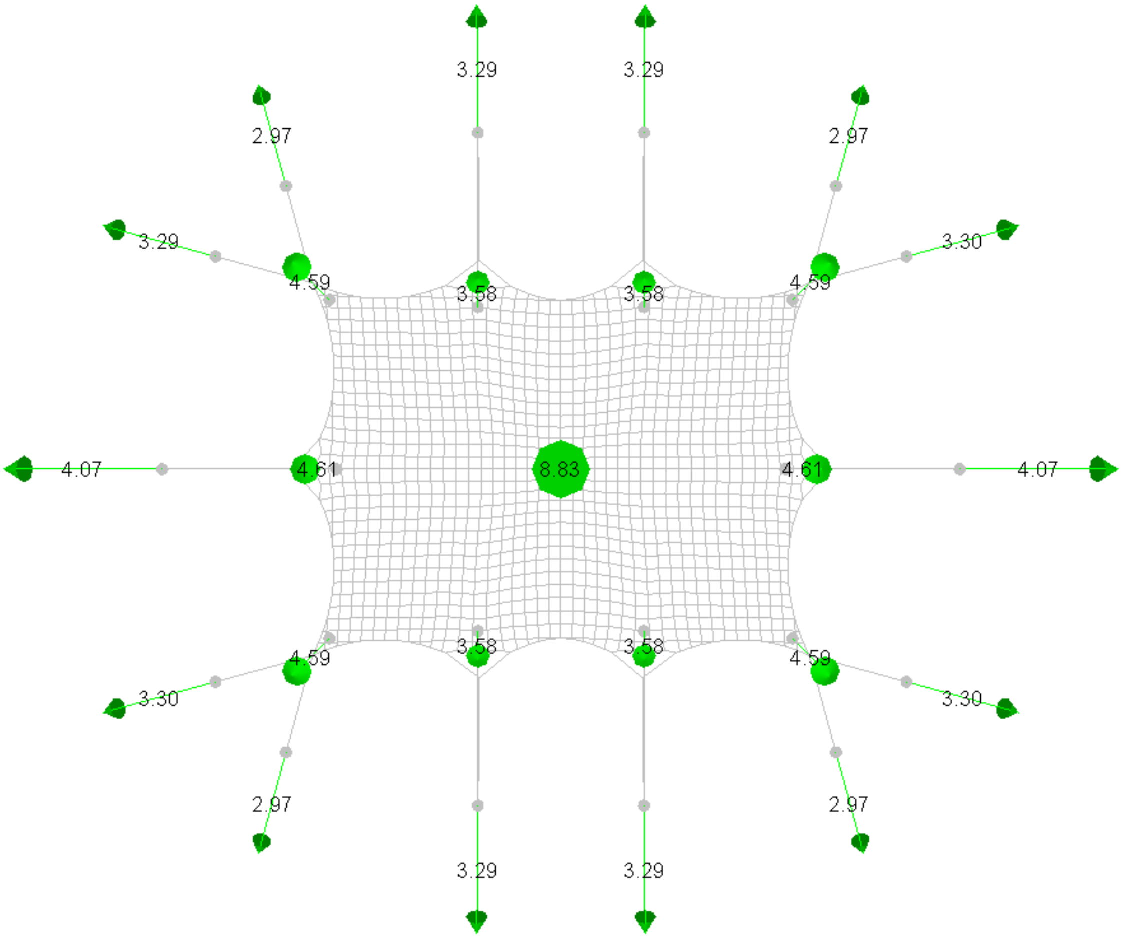


Annex C.6.6. Poles forces



Annex C.6.7. Reaction forces

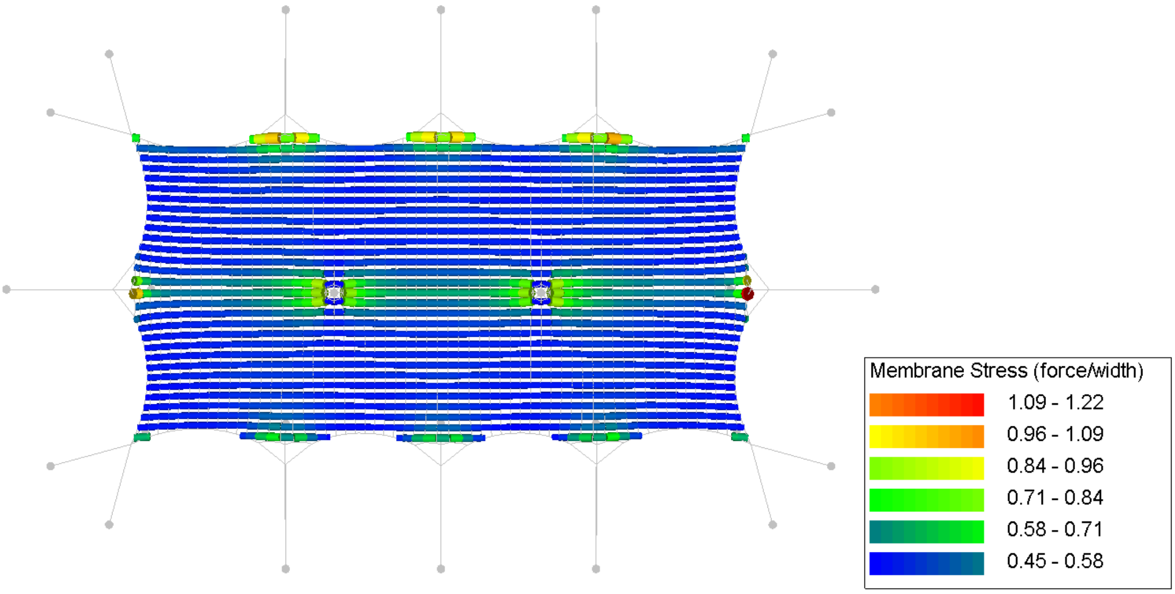
Increased snow load of 0.1 x 1.5/1.35=0.1083 kN/m2 is applied for ballast and anchor calculation



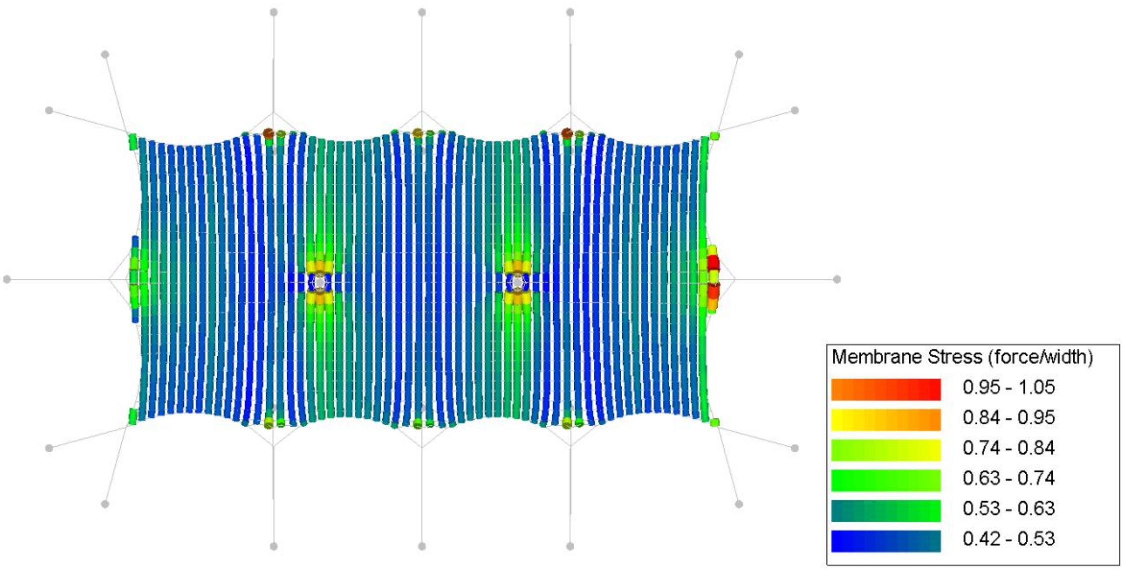
Annex D. Results per load combination – 7.5x15m model

Annex D.1. CO1. Own weight + pretension

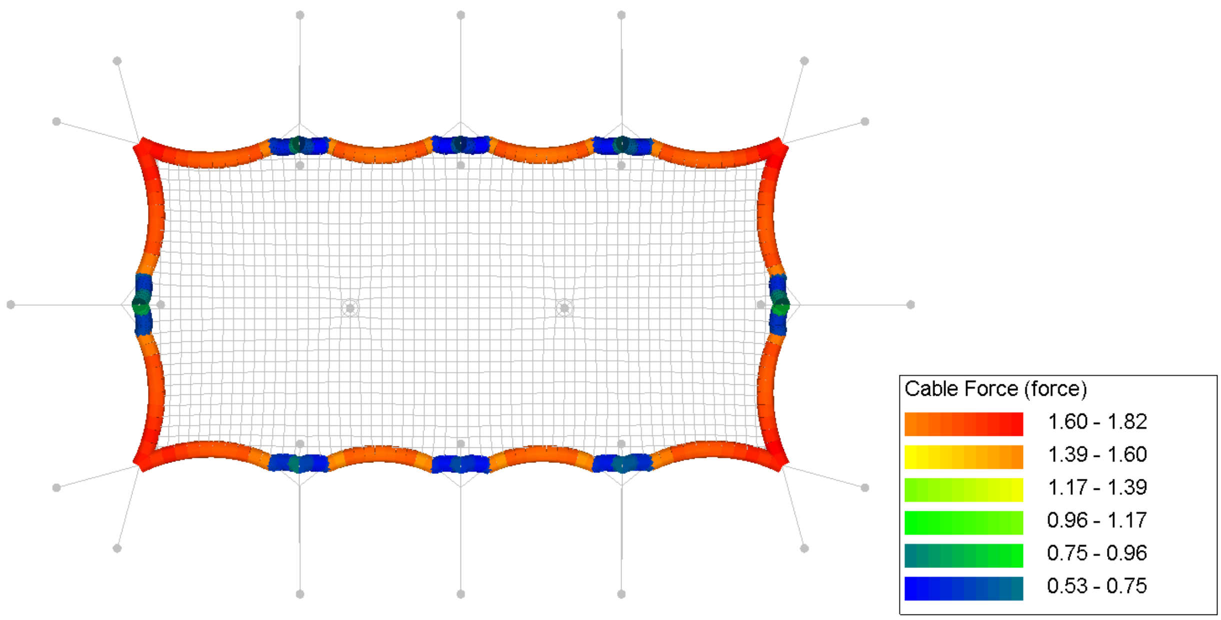
Annex D.1.1. Membrane stress (warp)



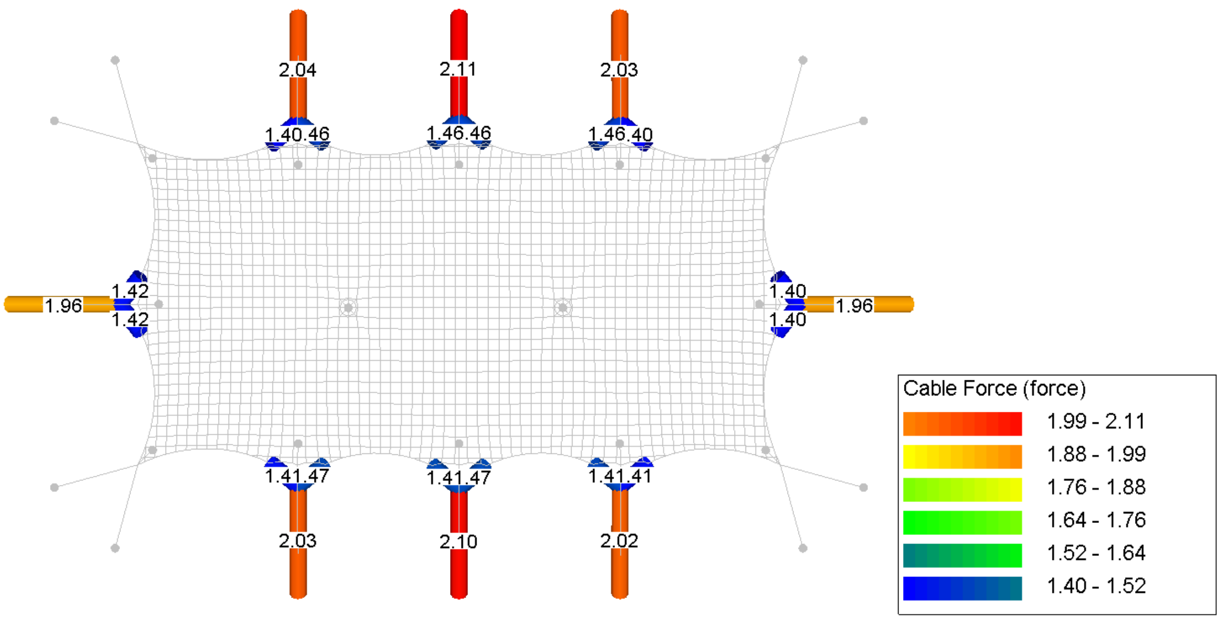
Annex D.1.2. Membrane stress (weft)



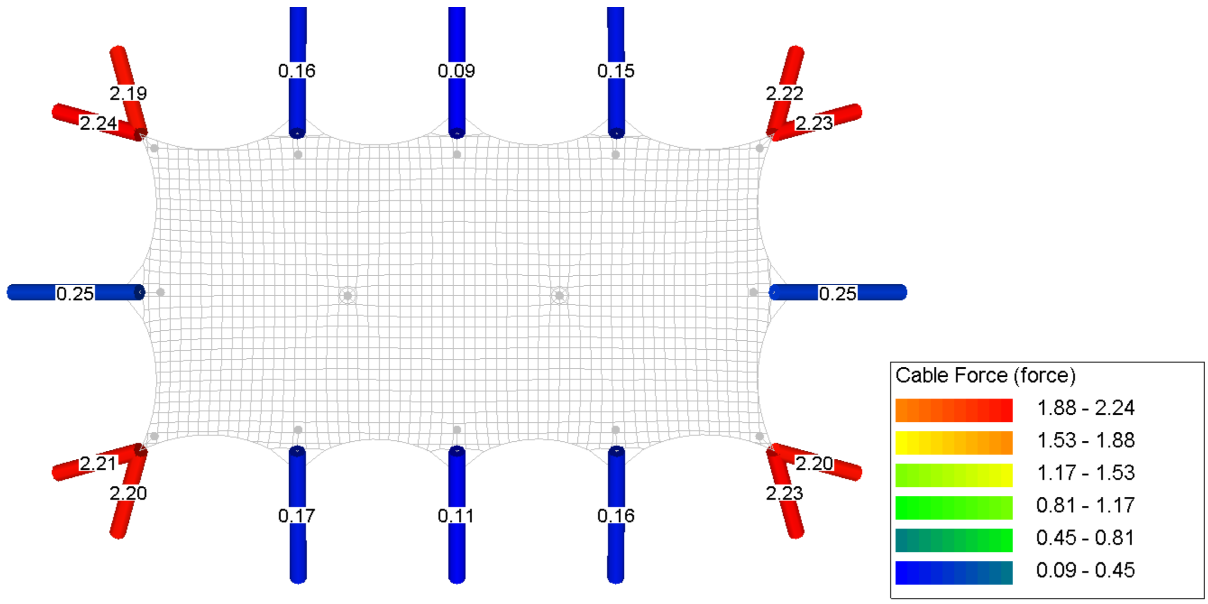
Annex D.1.3. Circumference rope forces



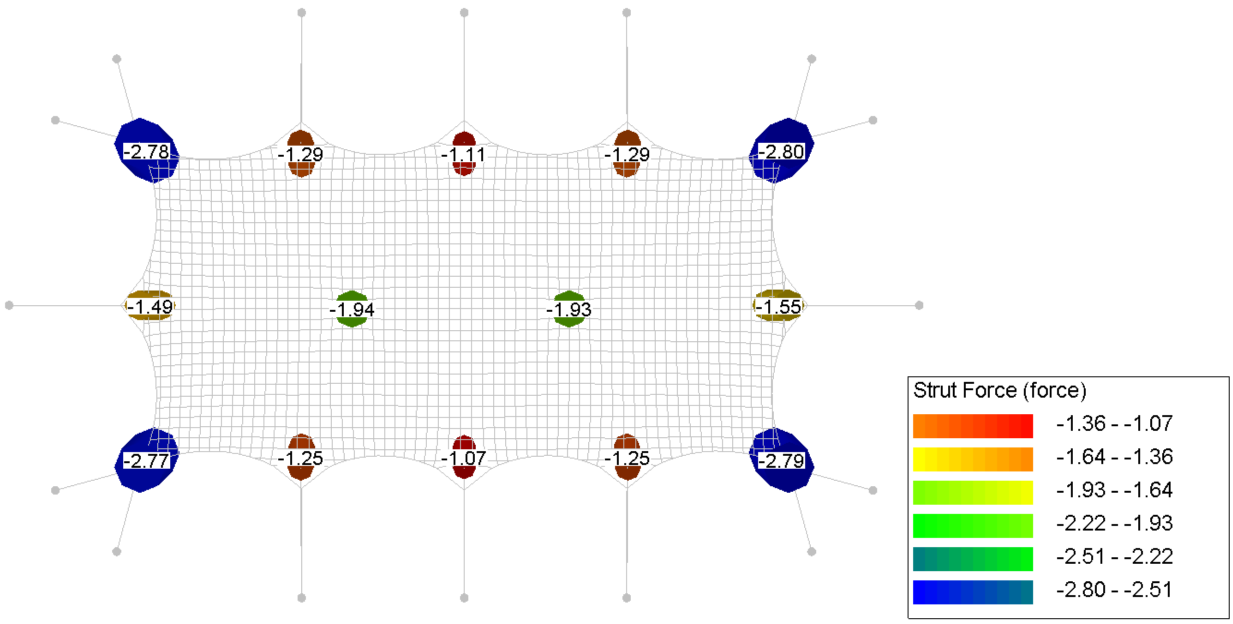
Annex D.1.4. Rope forces



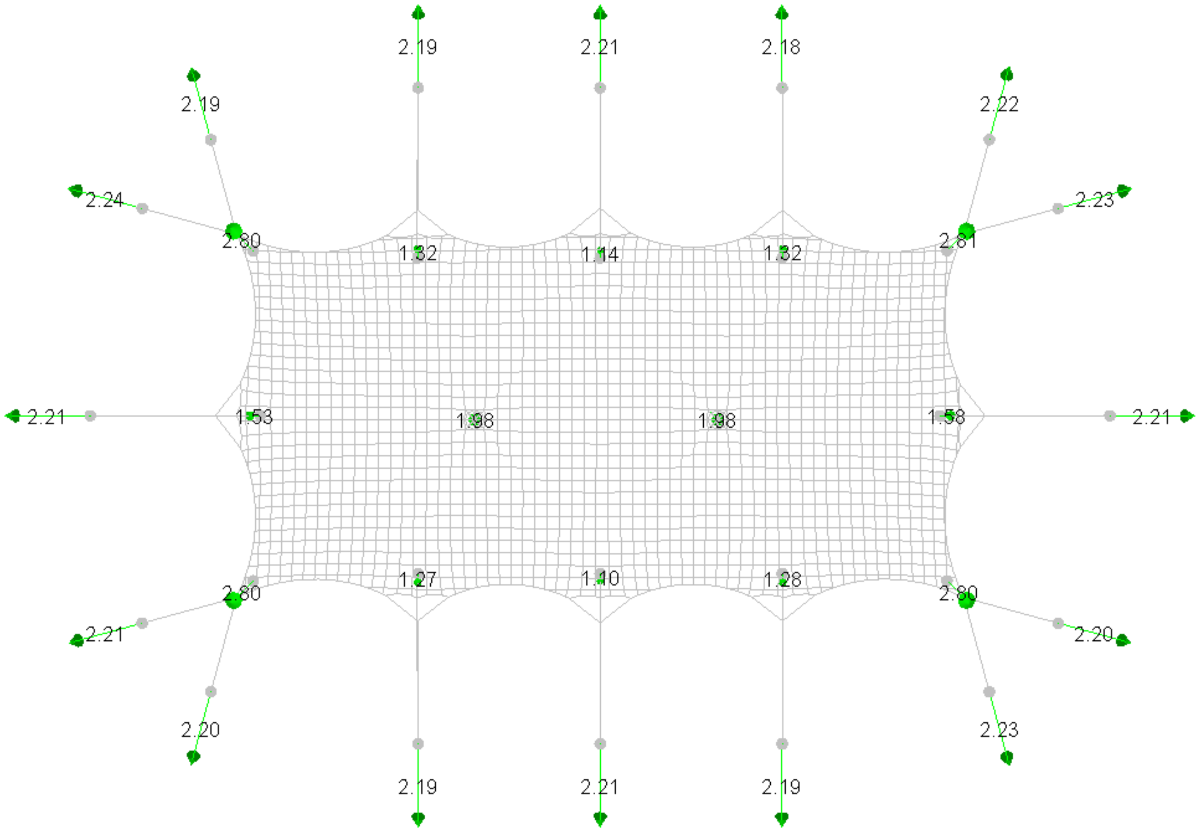
Annex D.1.5. Belt forces



Annex D.1.6. Pole forces

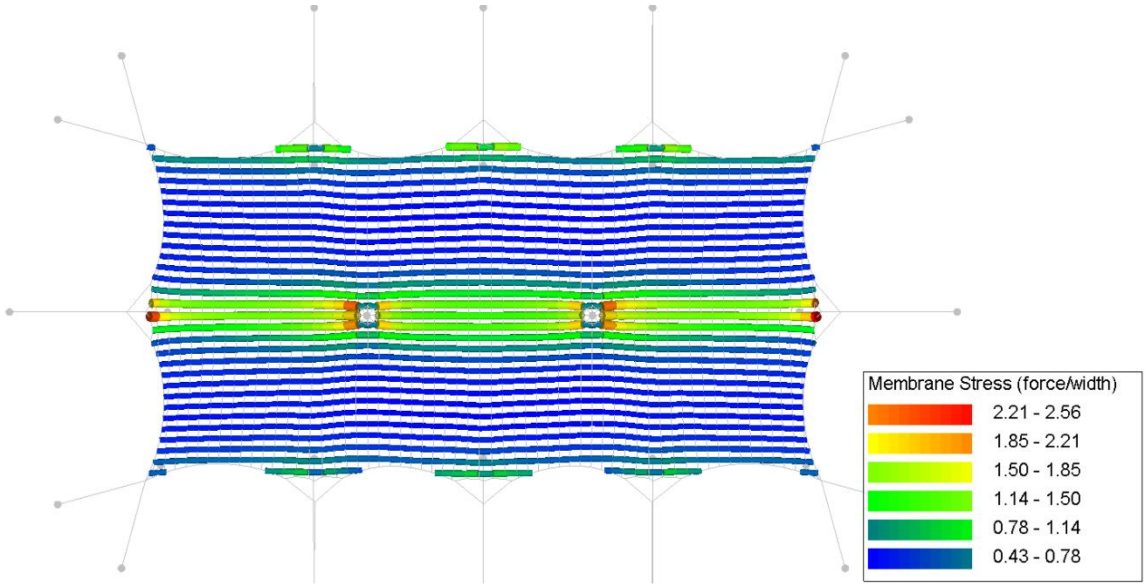


Annex D.1.7. Reaction forces

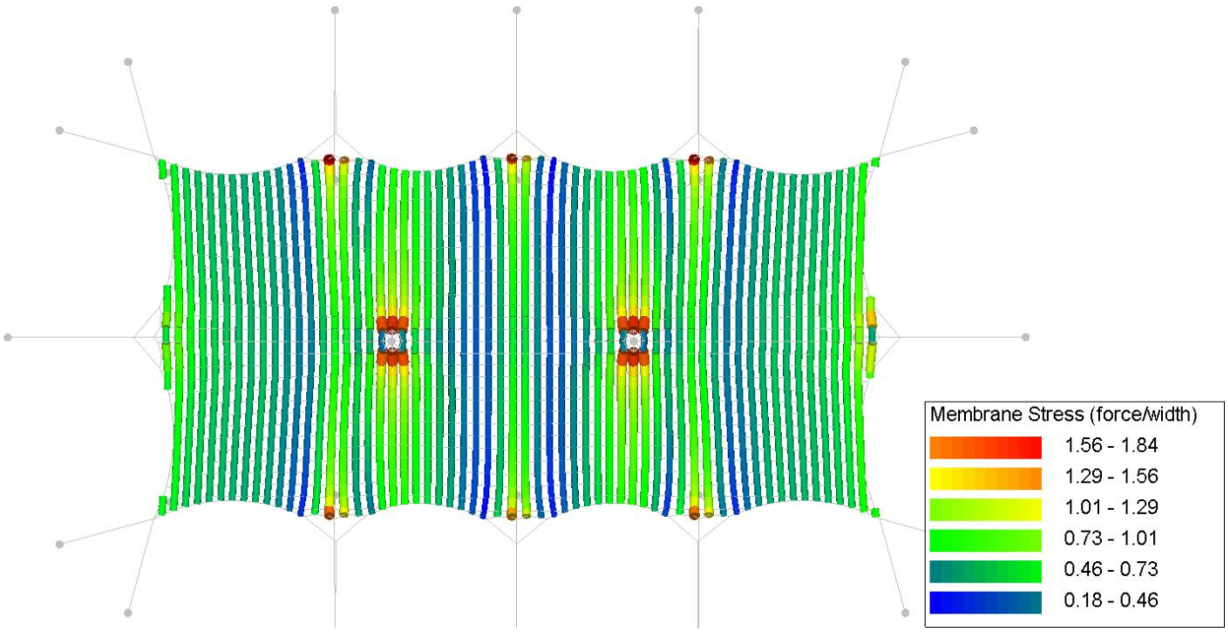


Annex D.2. CO2. Own weight + pretension + snow

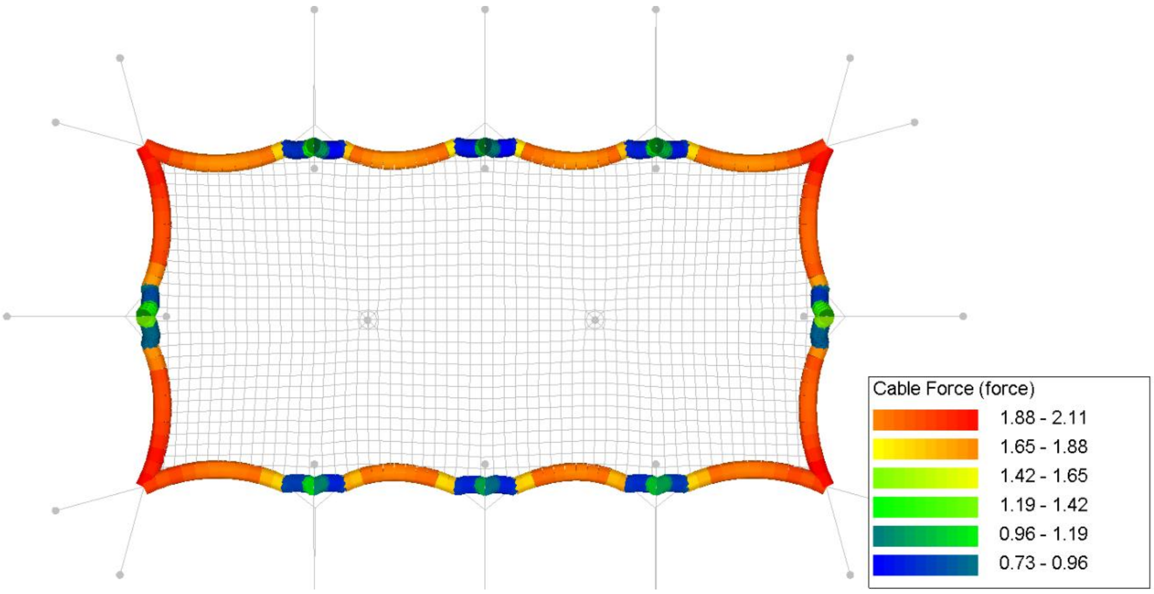
Annex D.2.1. Membrane stress (warp)



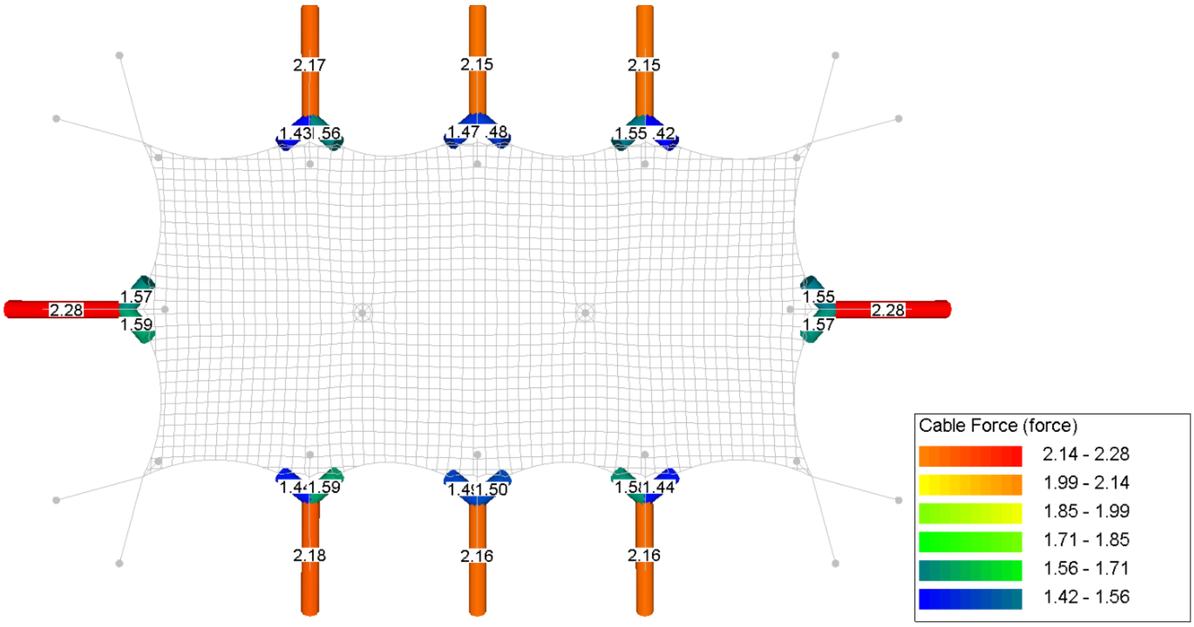
Annex D.2.2. Membrane stress (weft)



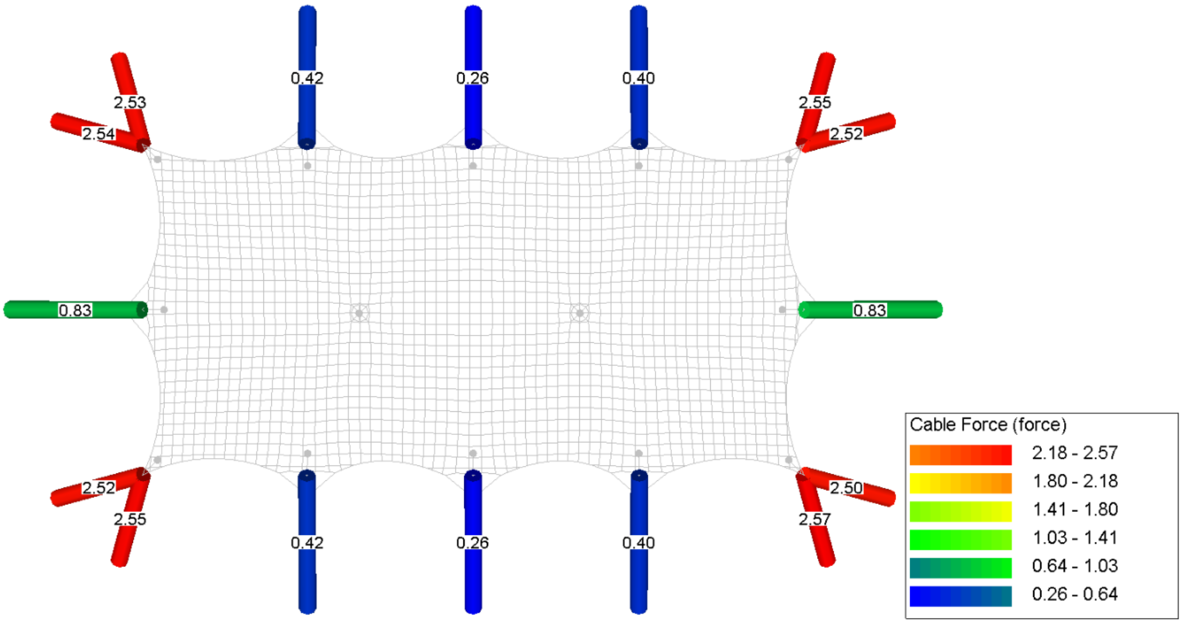
Annex D.2.3. Circumference rope forces



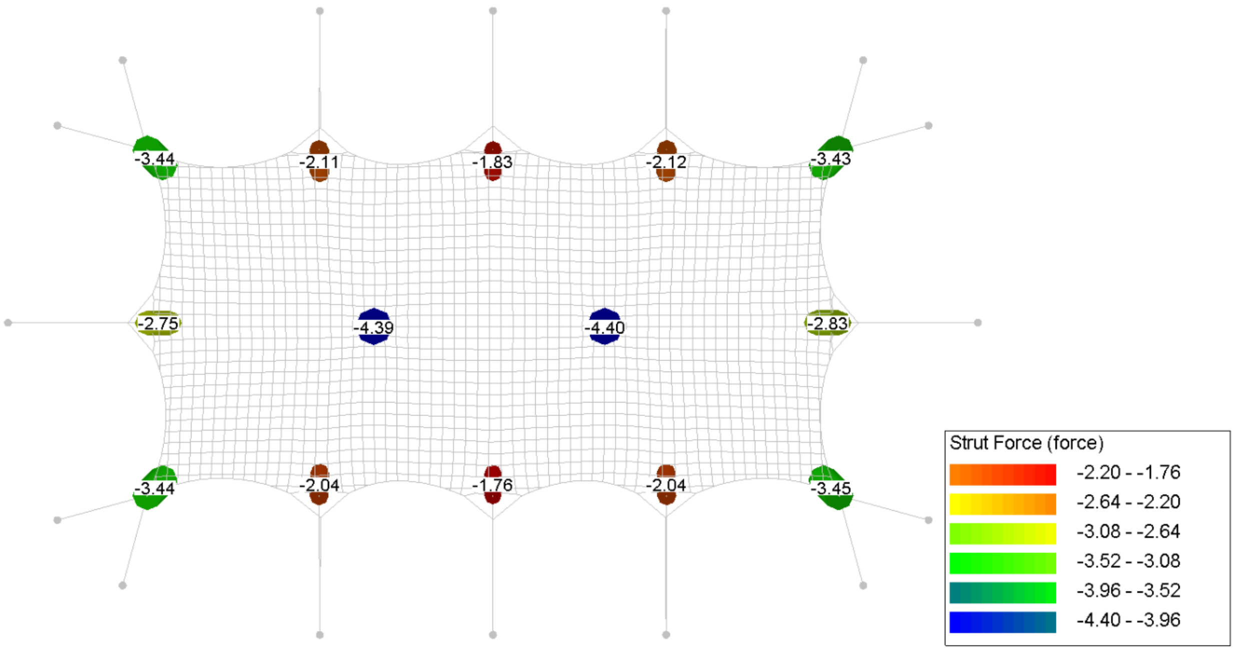
Annex D.2.4. Rope forces



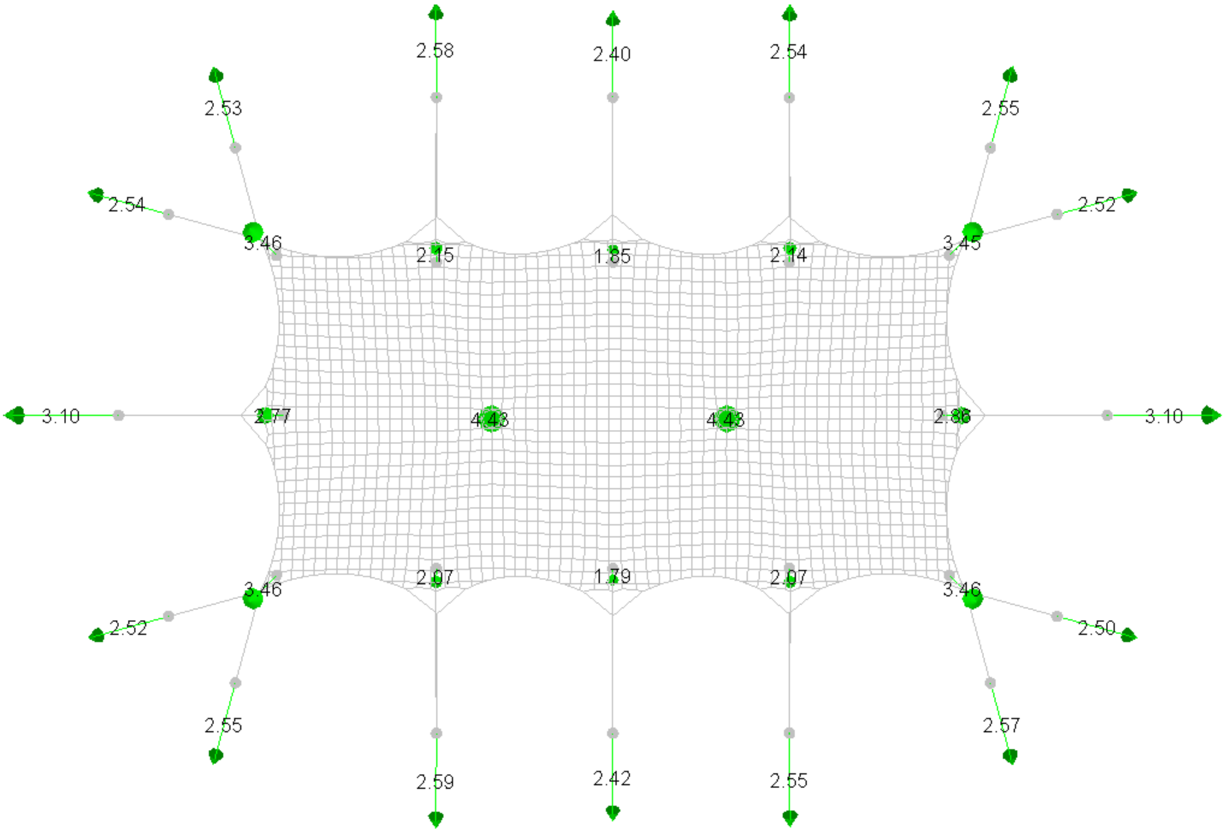
Annex D.2.5. Belt forces



Annex D.2.6. Pole forces

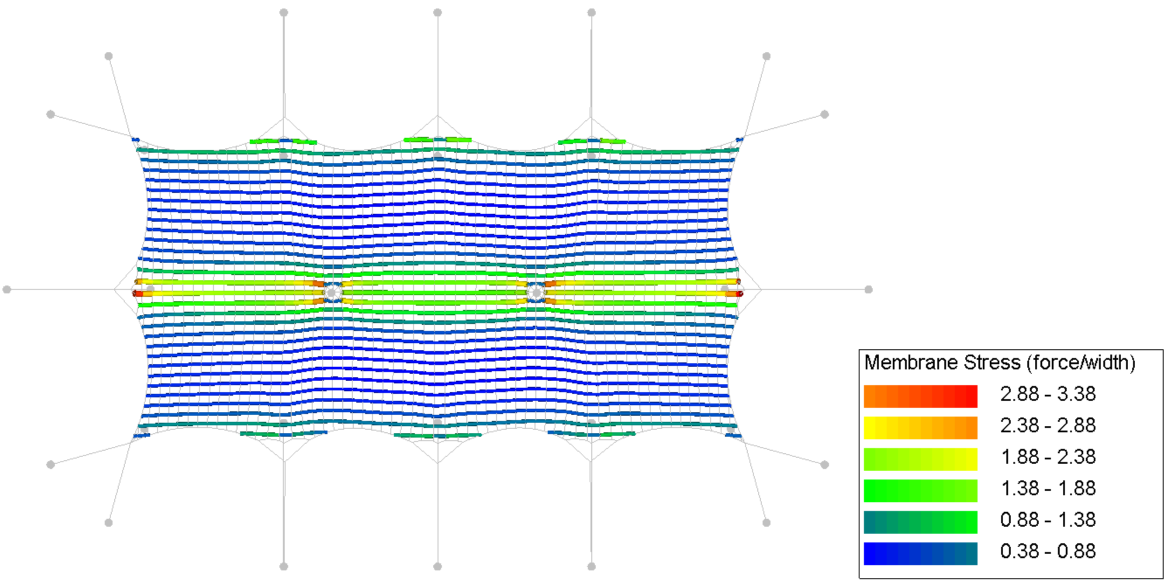


Annex D.2.7. Anchor forces

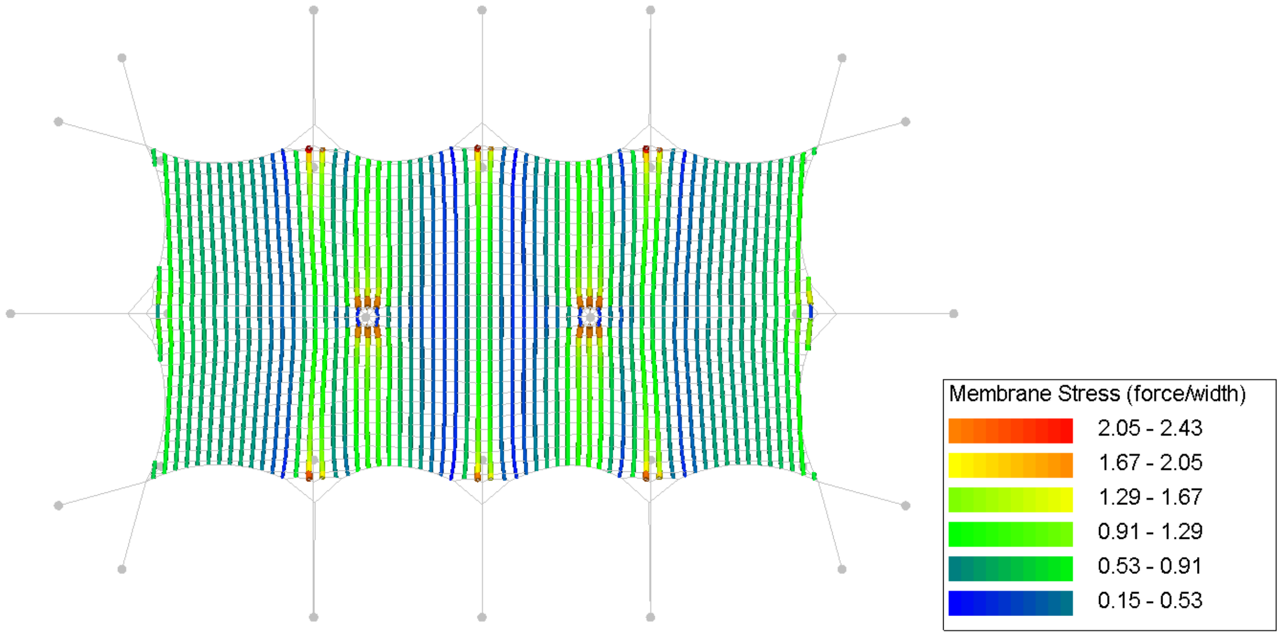


Annex D.3. CO3. Own weight + pretension + wind pressure

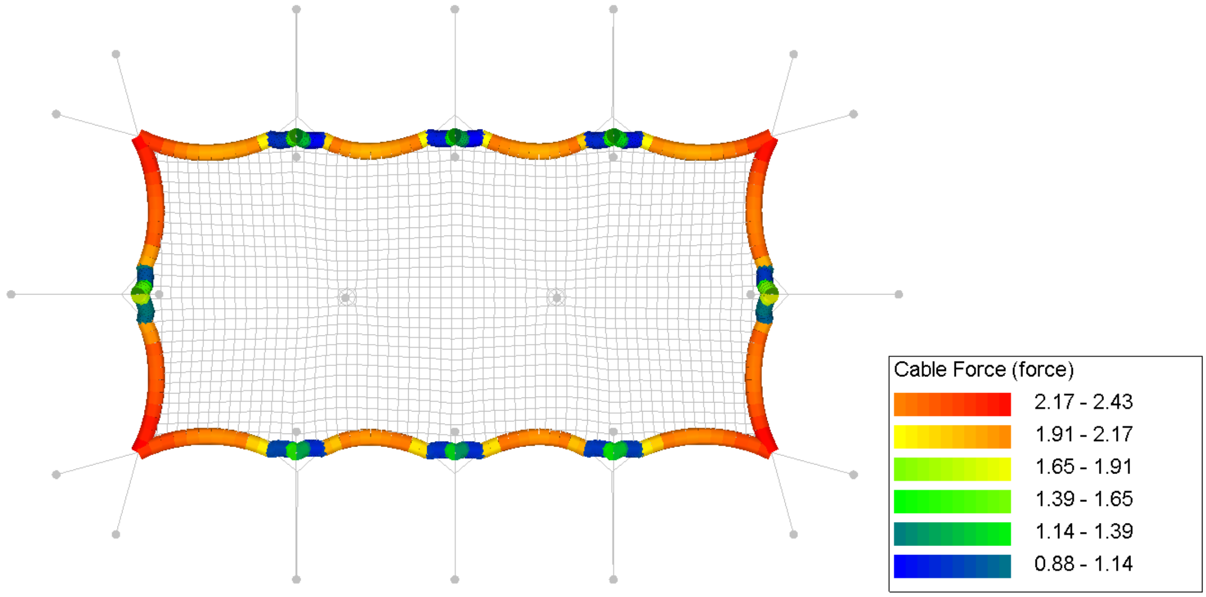
Annex D.3.1. Membrane stress (warp)



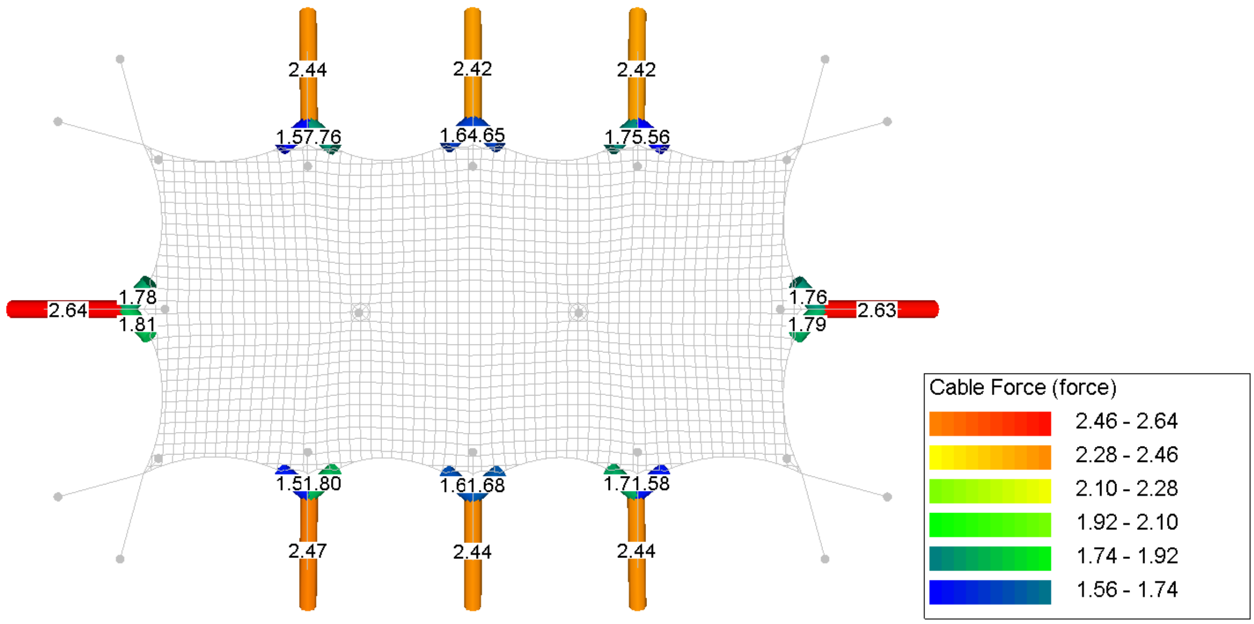
Annex D.3.2. Membrane stress (weft)



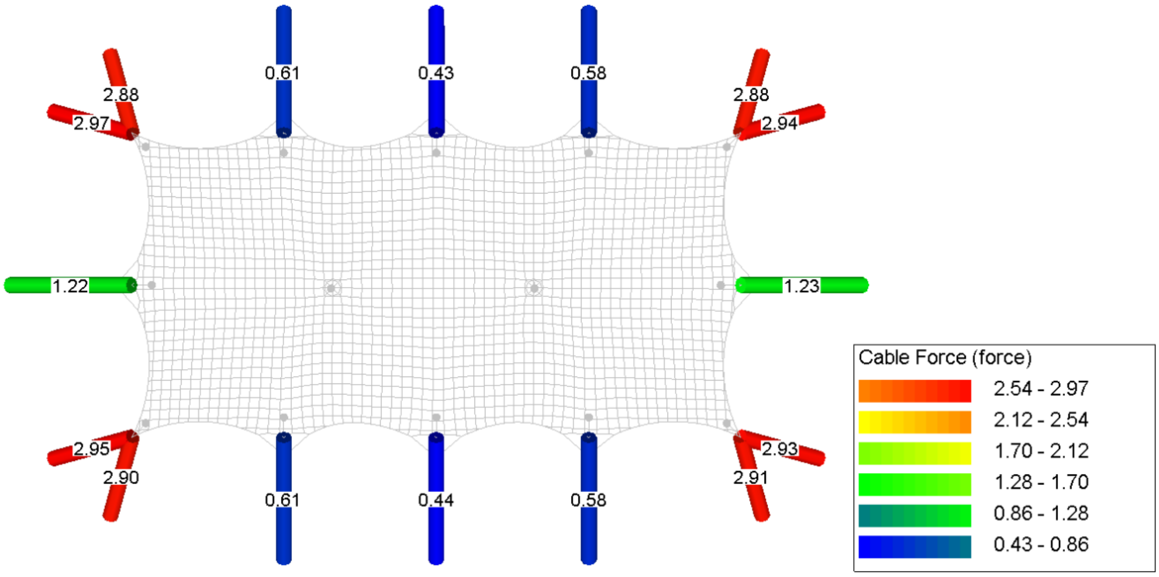
Annex D.3.3. Circumference rope forces



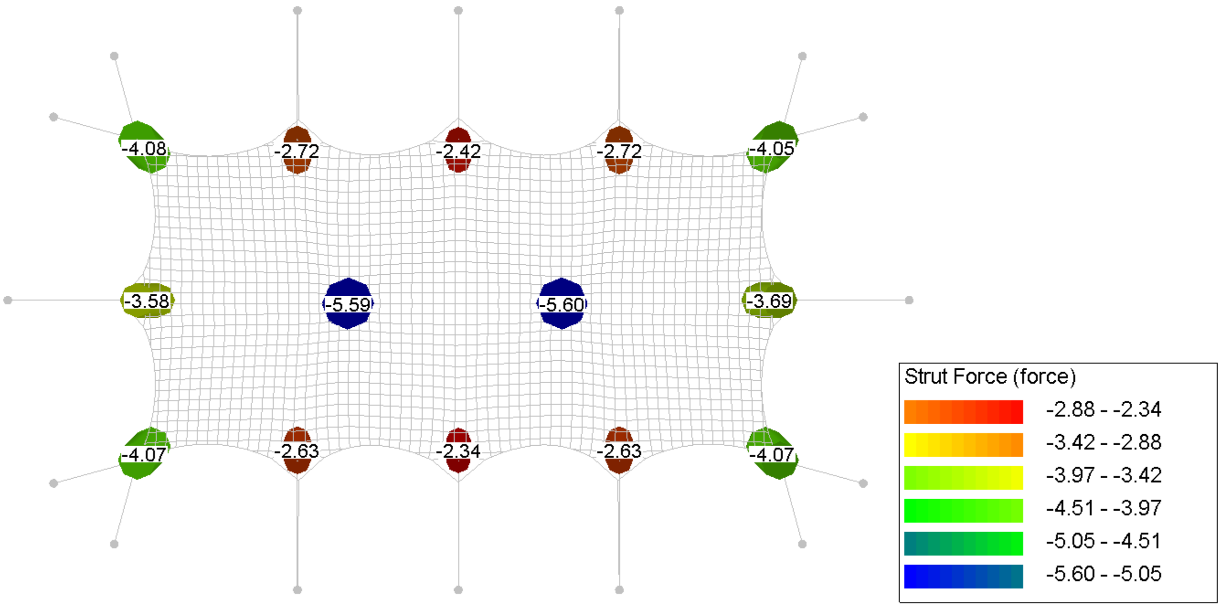
Annex D.3.4. Rope forces



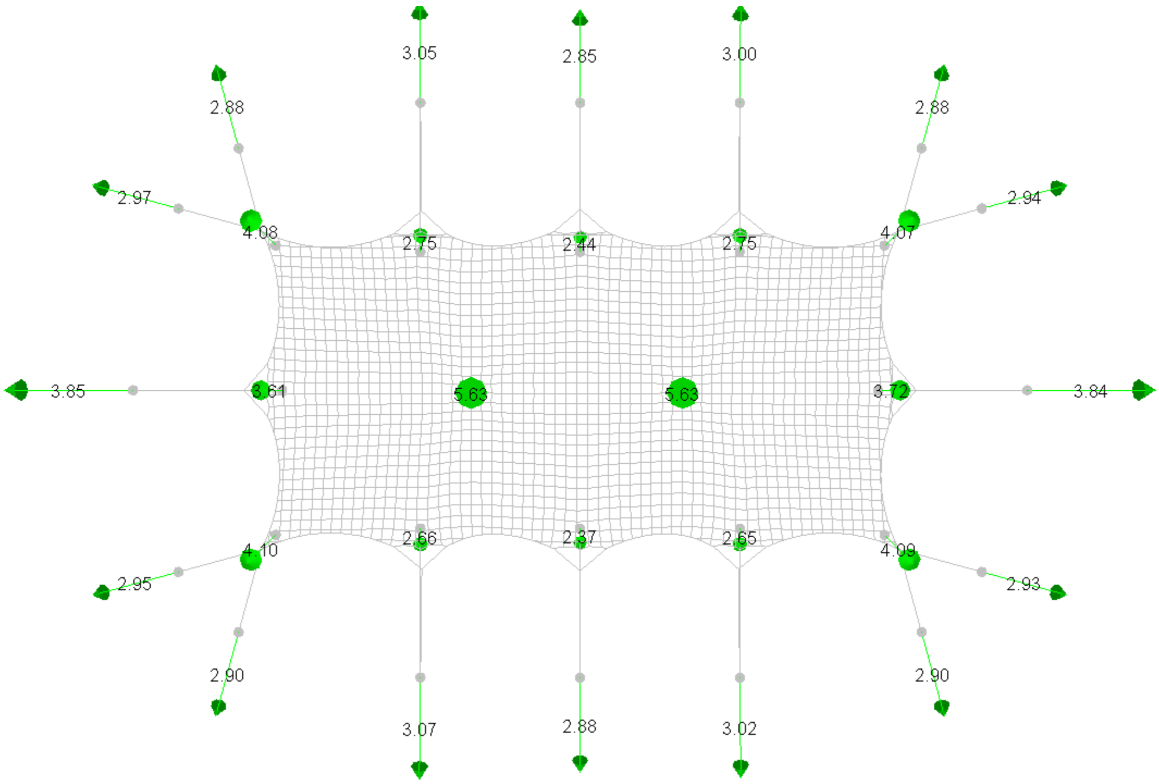
Annex D.3.5. Belt forces



Annex D.3.6. Poles forces

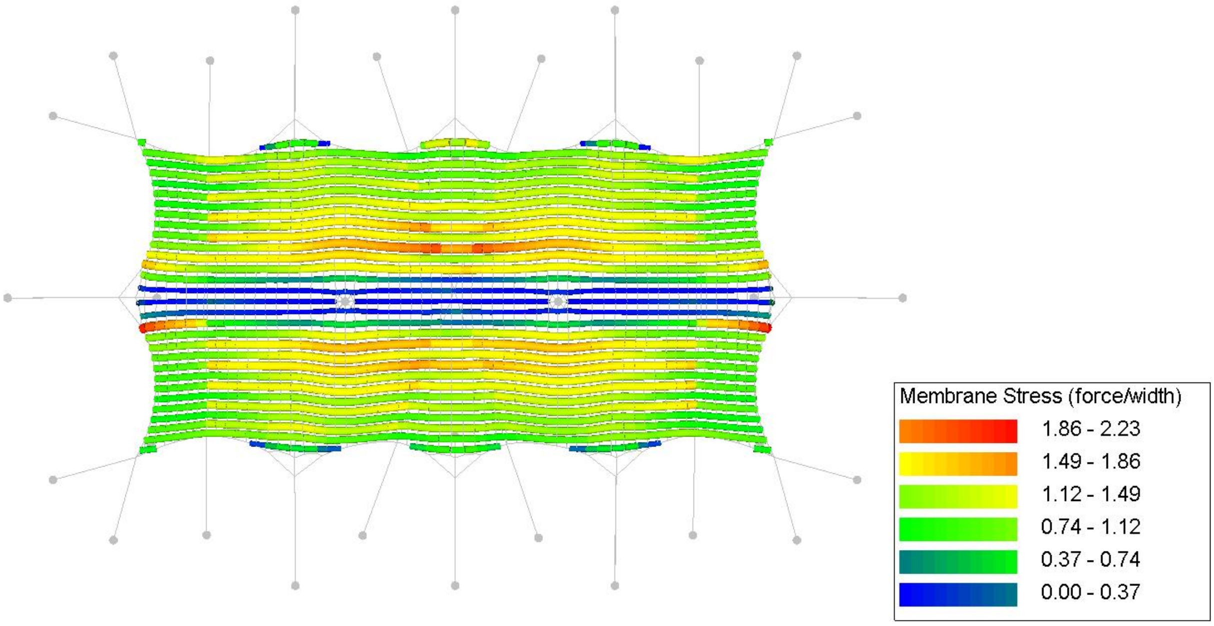


Annex D.3.7. Reaction forces

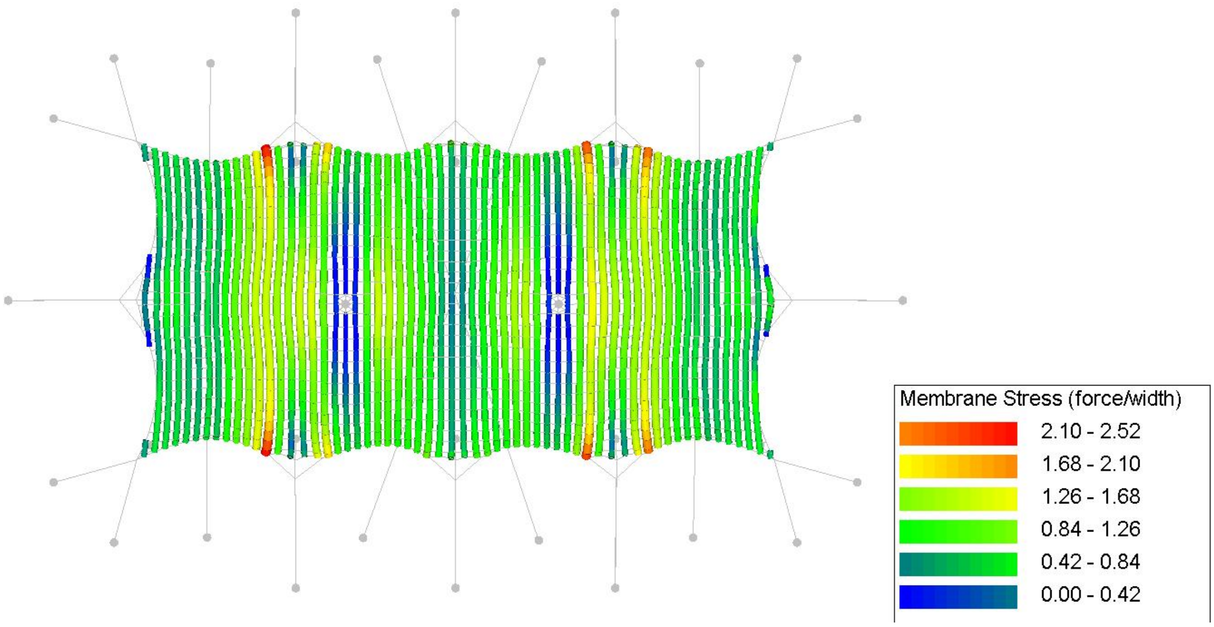


Annex D.4. CO4. Own weight + pretension + wind suction

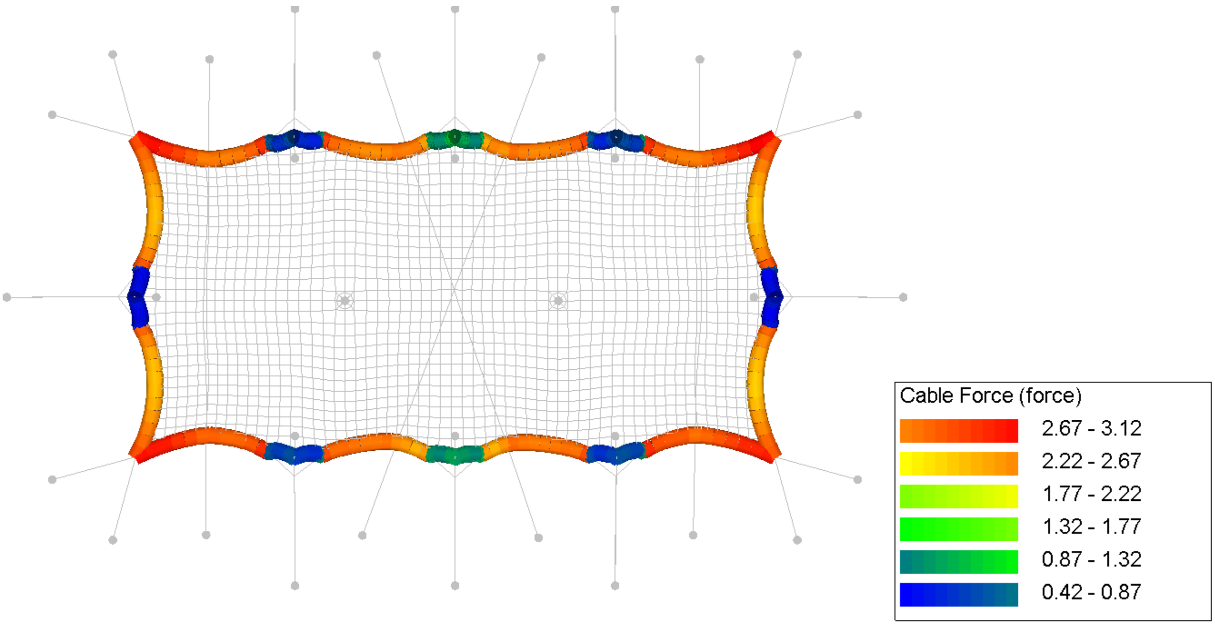
Annex D.4.1. Membrane stress (warp)



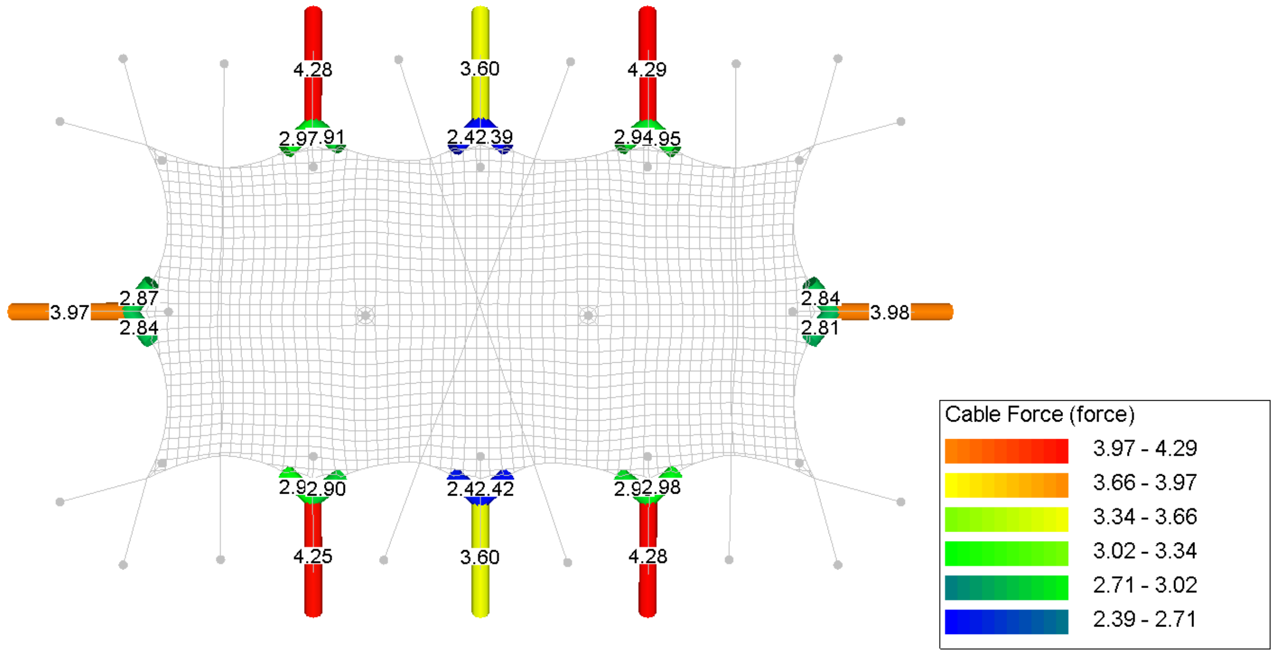
Annex D.4.2. Membrane stress (weft)



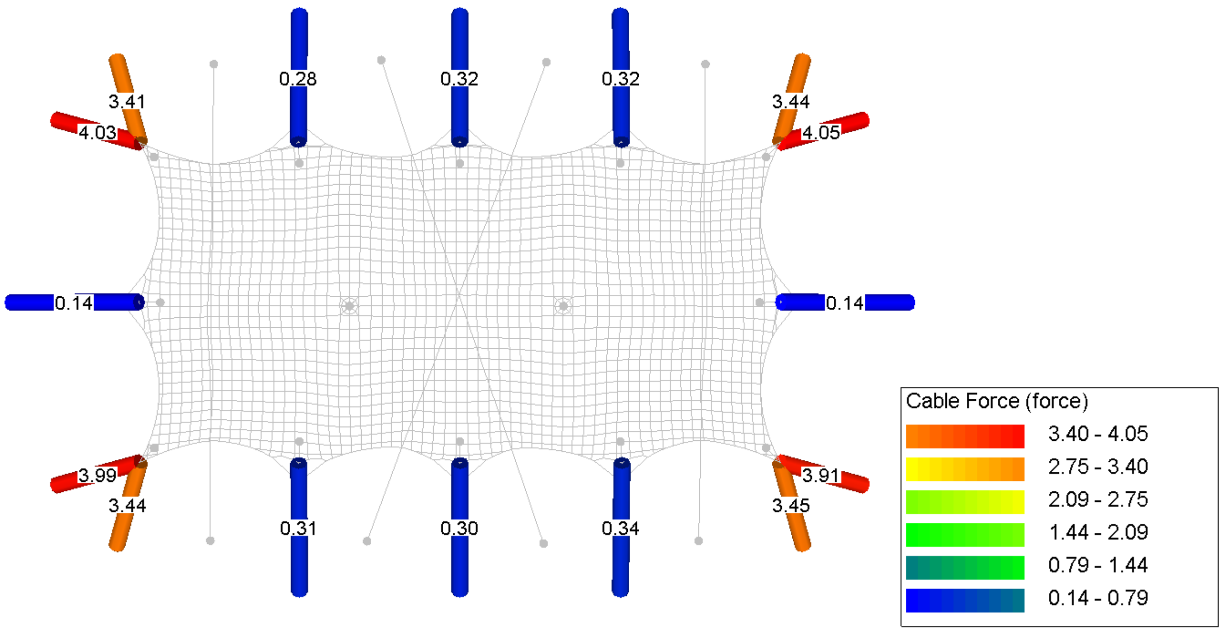
Annex D.4.3. Circumference rope forces



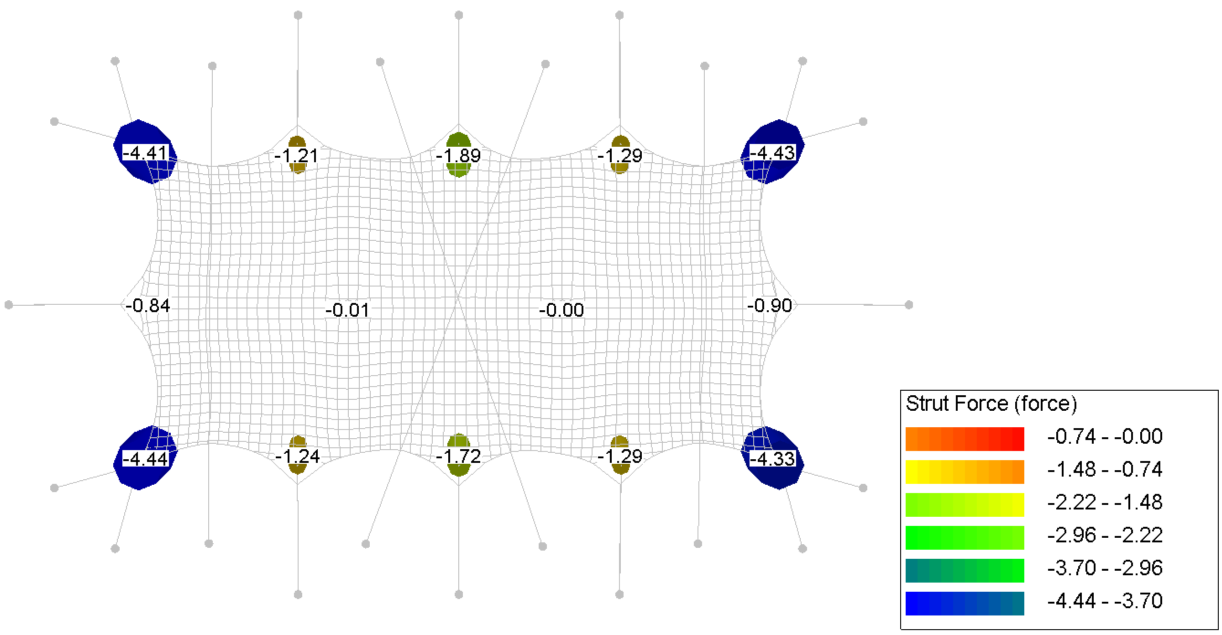
Annex D.4.4. Guy rope forces



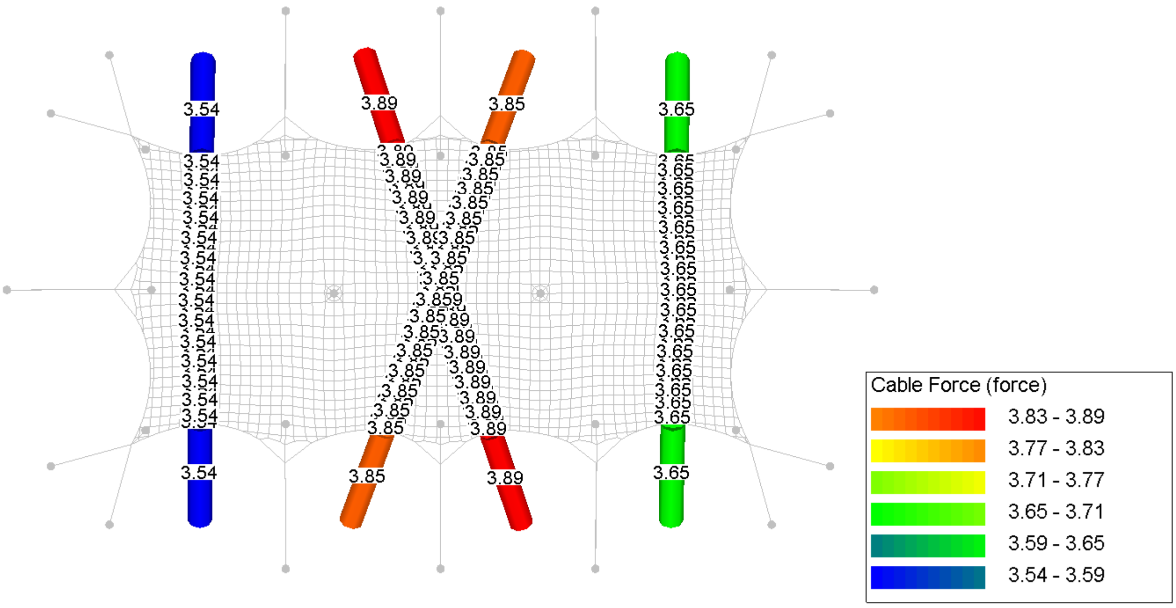
Annex D.4.5. Belt forces



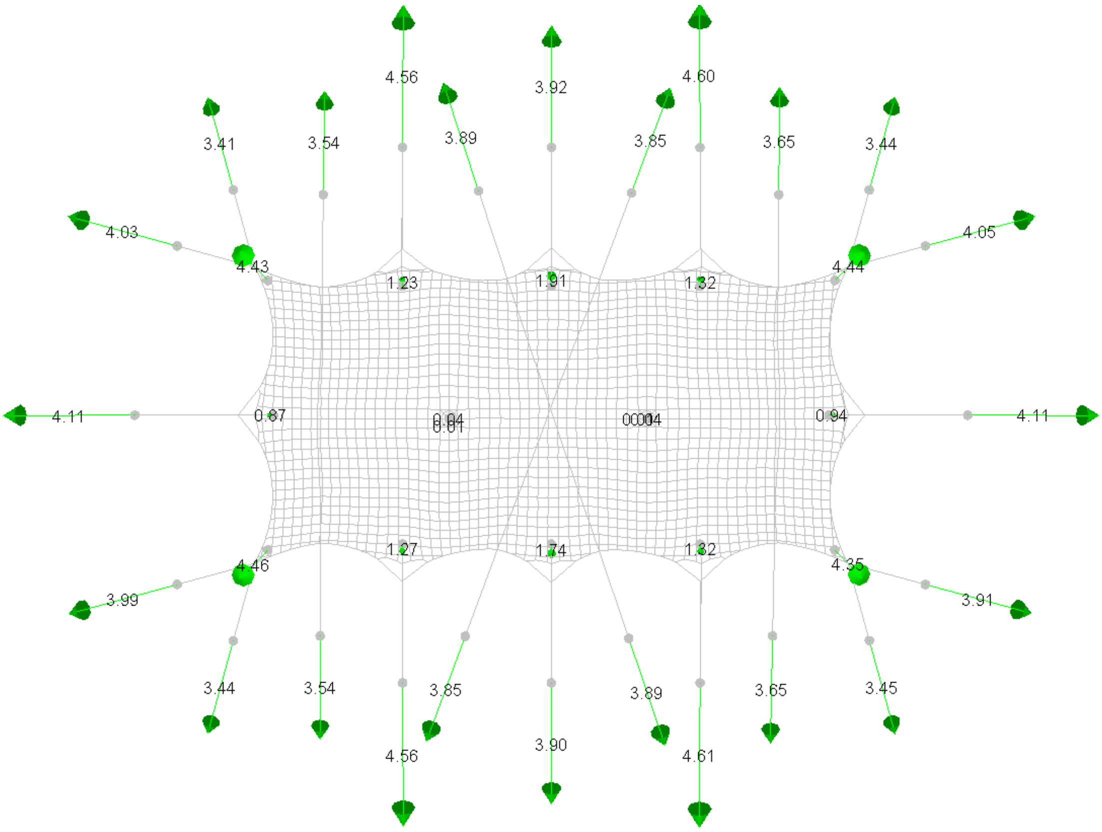
Annex D.4.6. Poles forces



Annex D.4.7. Storm belt forces

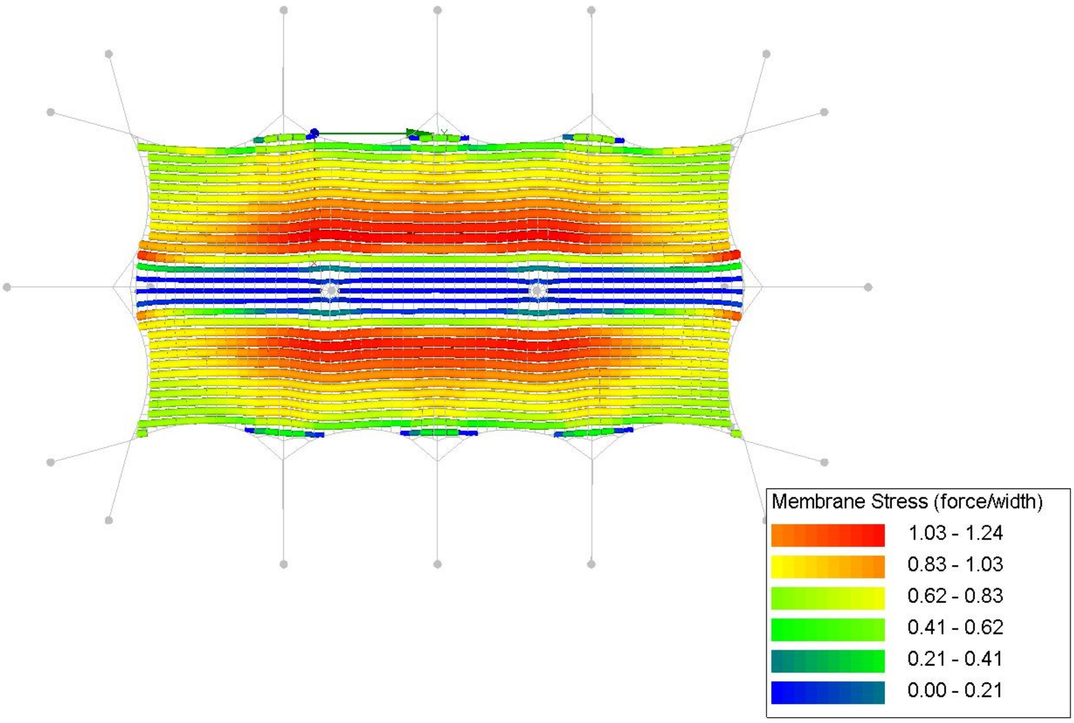


Annex D.4.8. Reaction forces

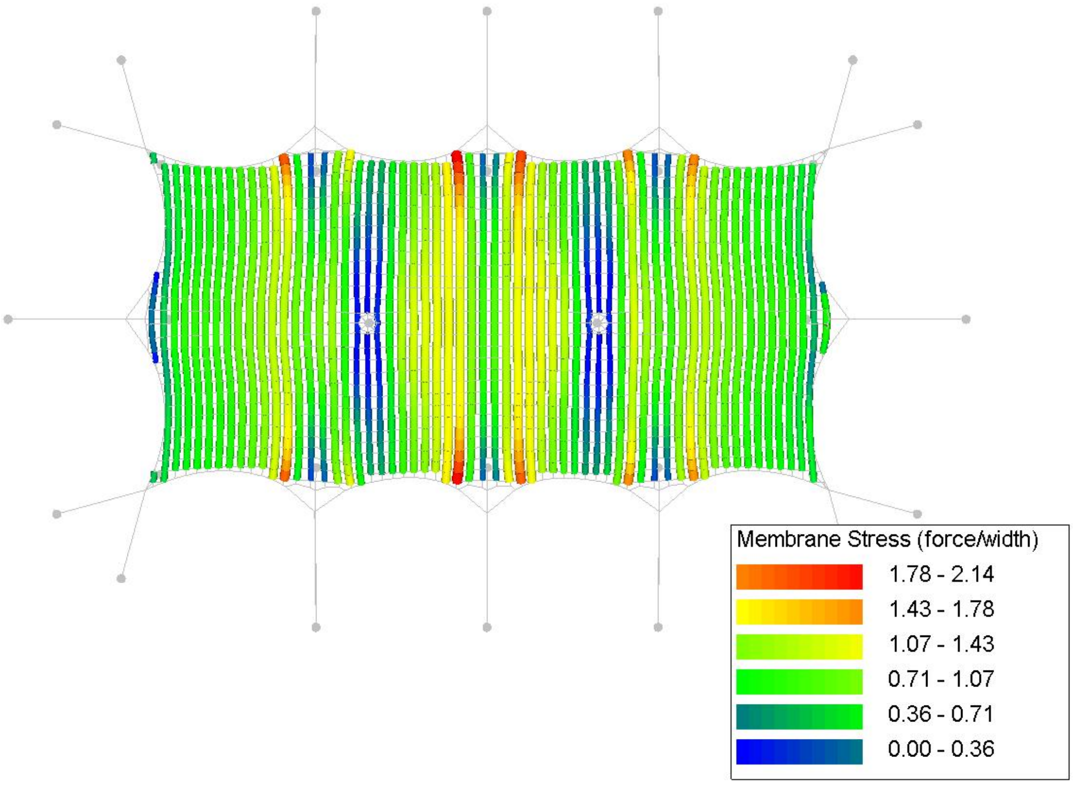


Annex D.5. CO5. Own weight + pretension + wind suction reduced [NL]

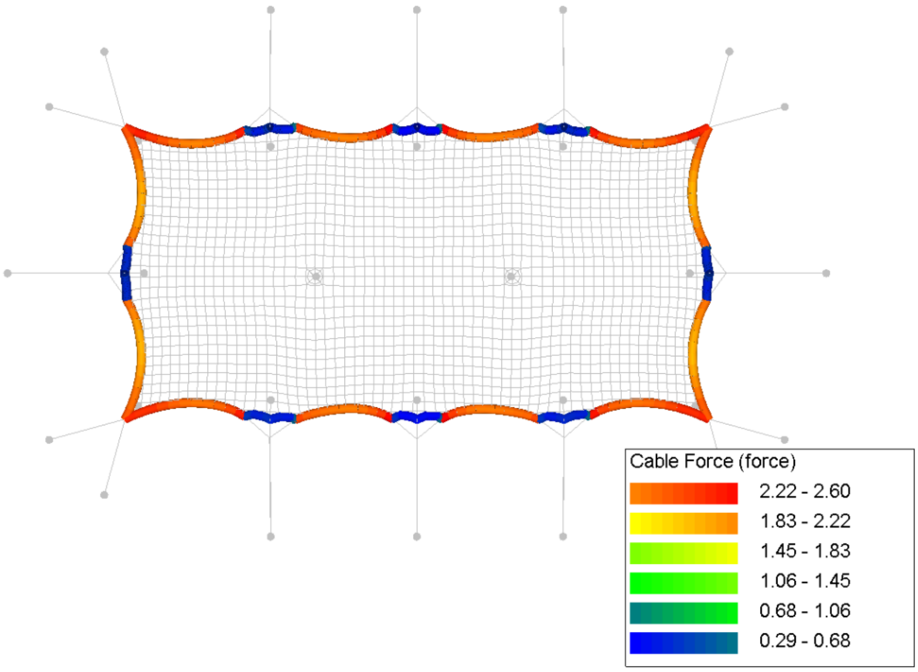
Annex D.5.1. Membrane stress (warp)



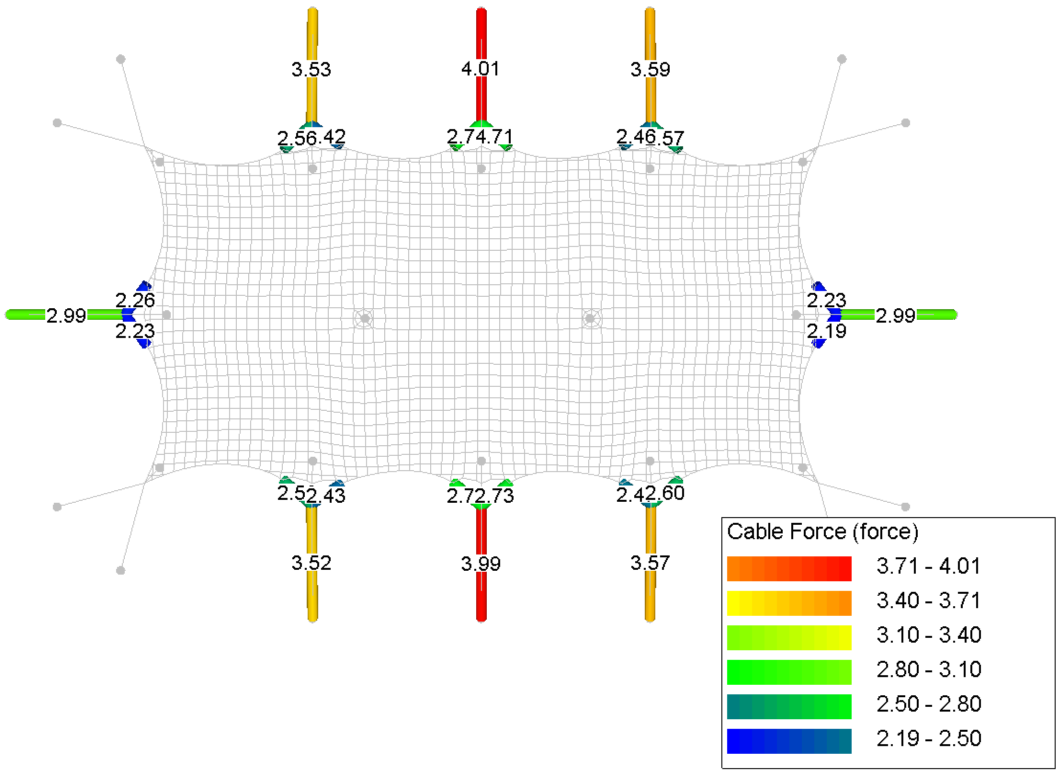
Annex D.5.2. Membrane stress (weft)



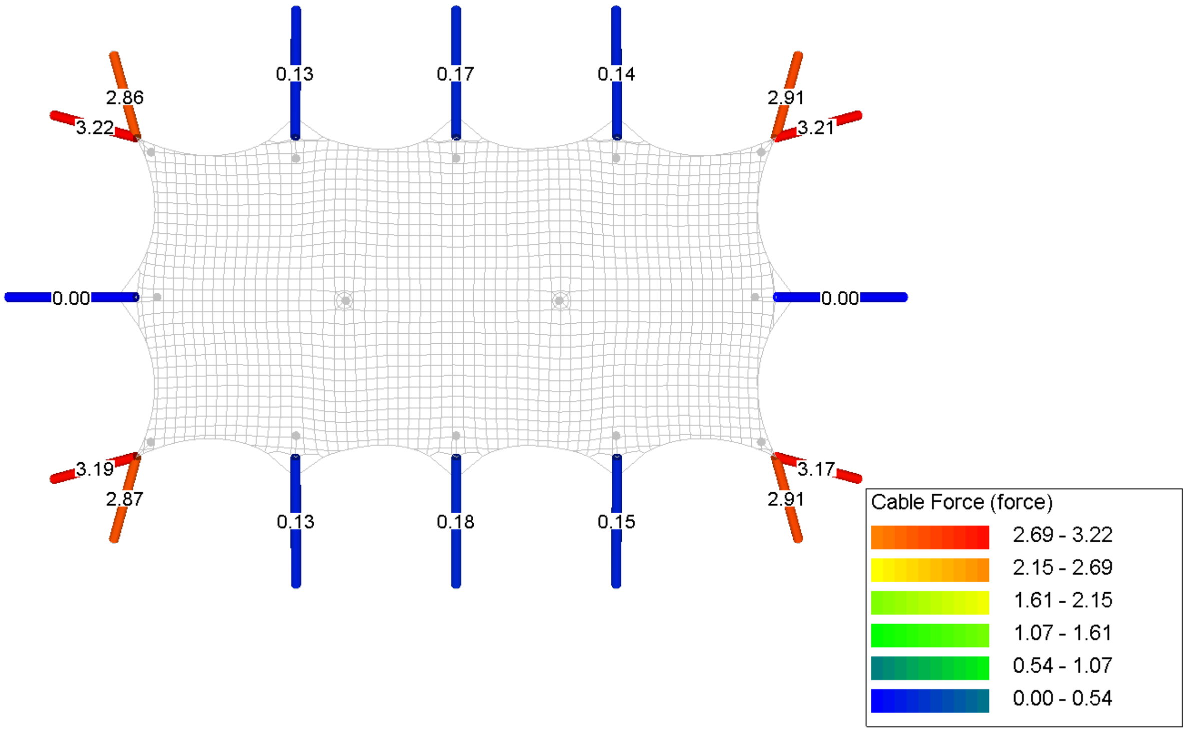
Annex D.5.3. Circumference rope forces



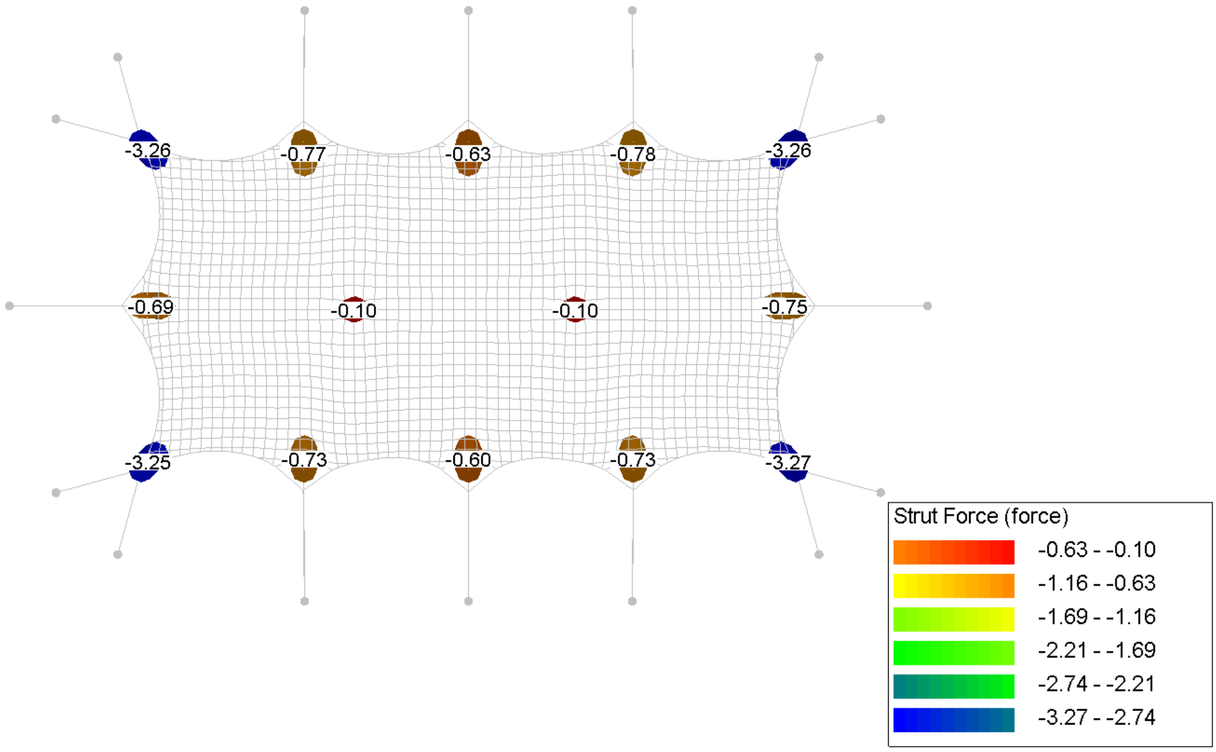
Annex D.5.4. Rope forces



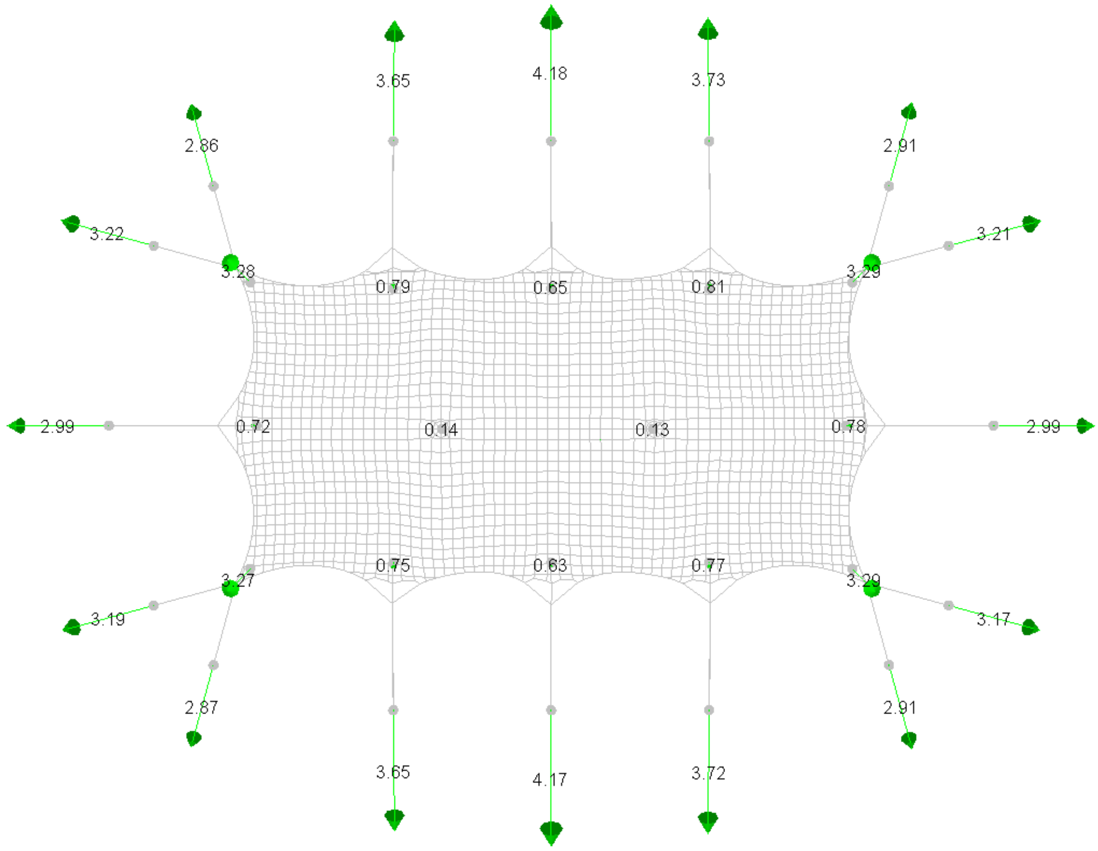
Annex D.5.5. Belt forces



Annex D.5.6. Poles forces

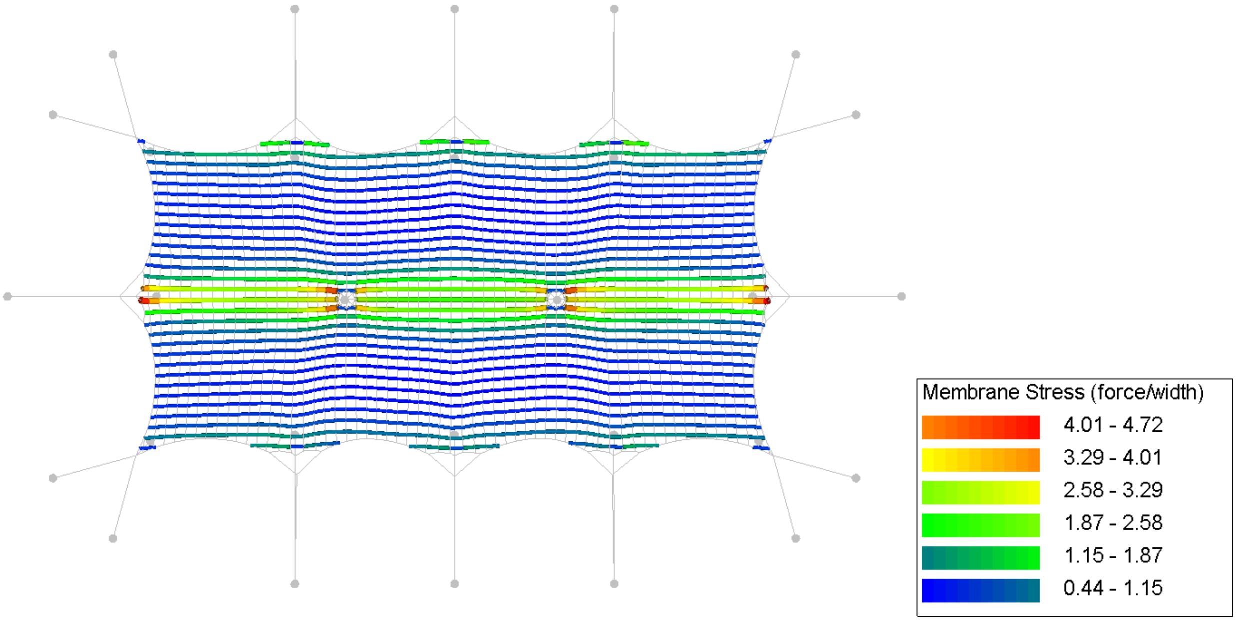


Annex D.5.7. Reaction forces

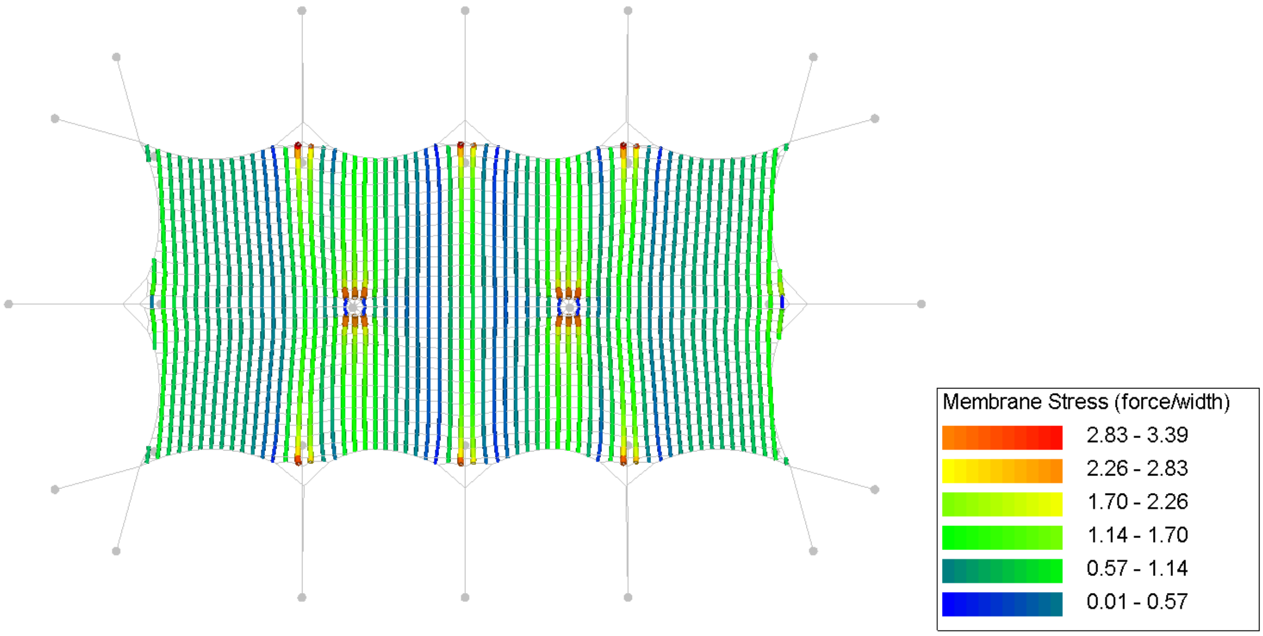


Annex D.6. CO6. Own weight + pretension + wind pressure + snow [FR]

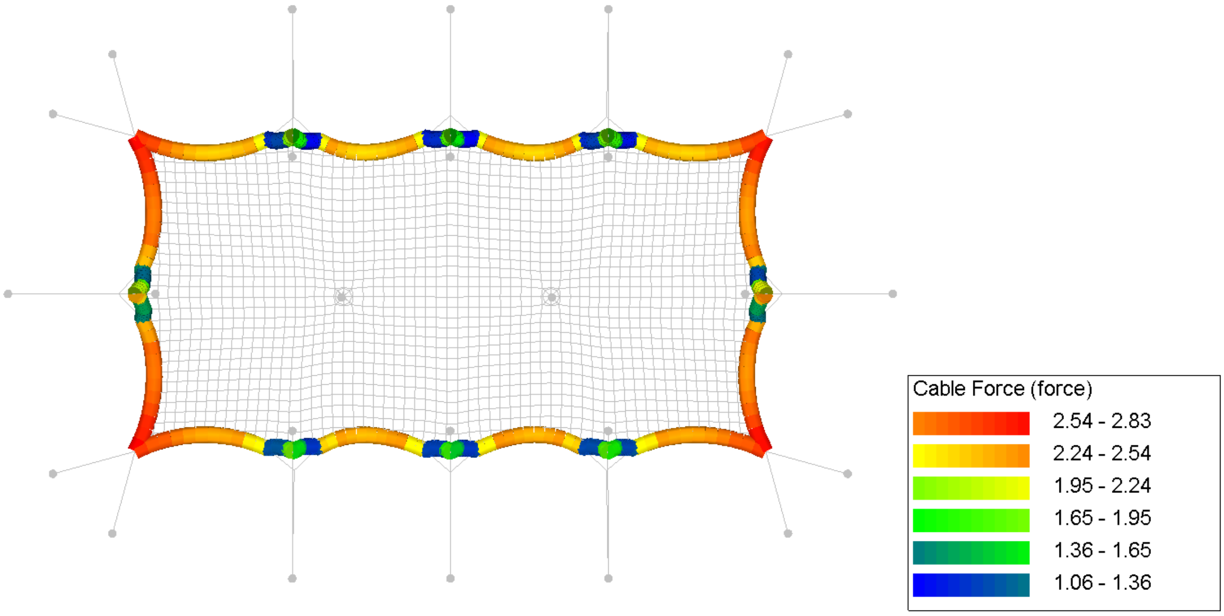
Annex D.6.1. Membrane stress (warp)



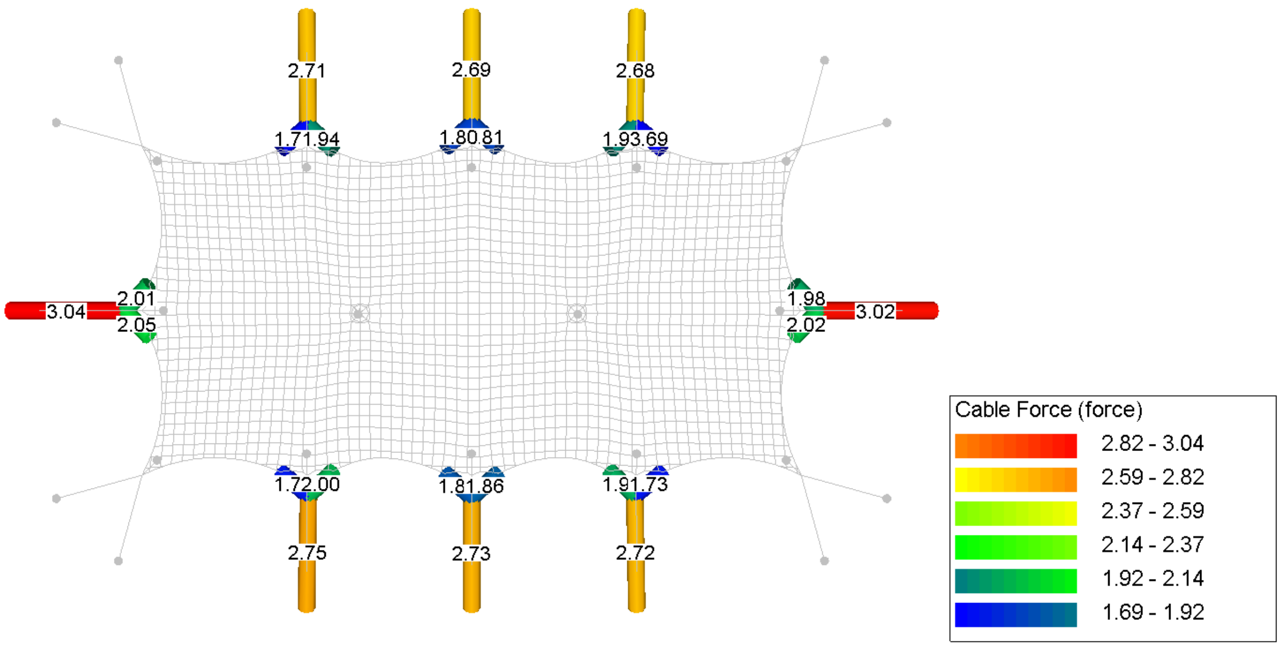
Annex D.6.2. Membrane stress (weft)



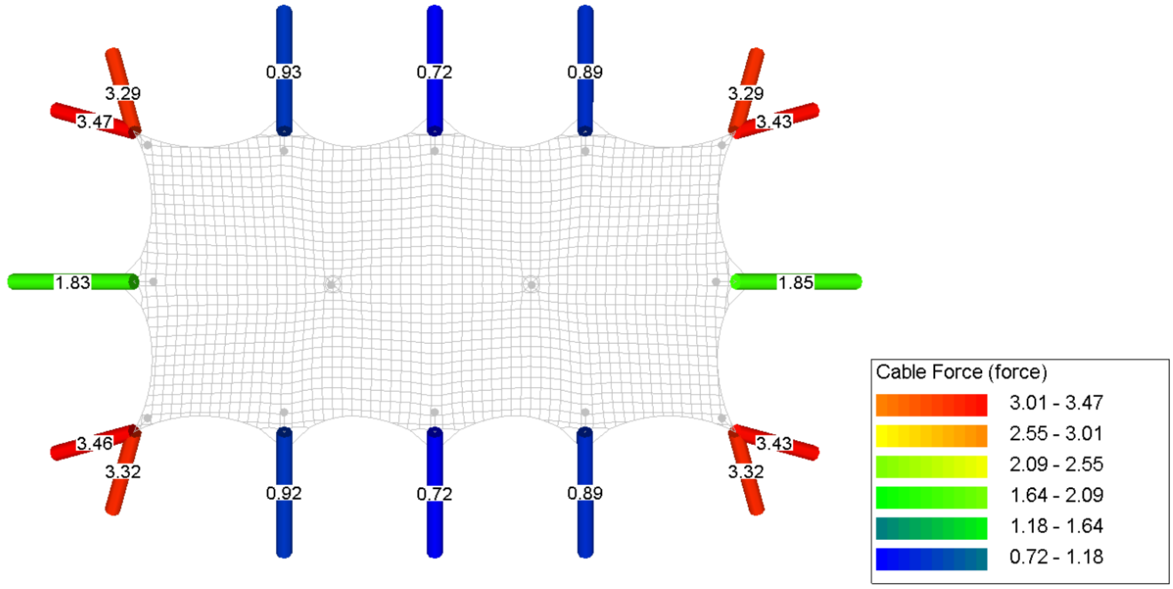
Annex D.6.3. Circumference rope forces



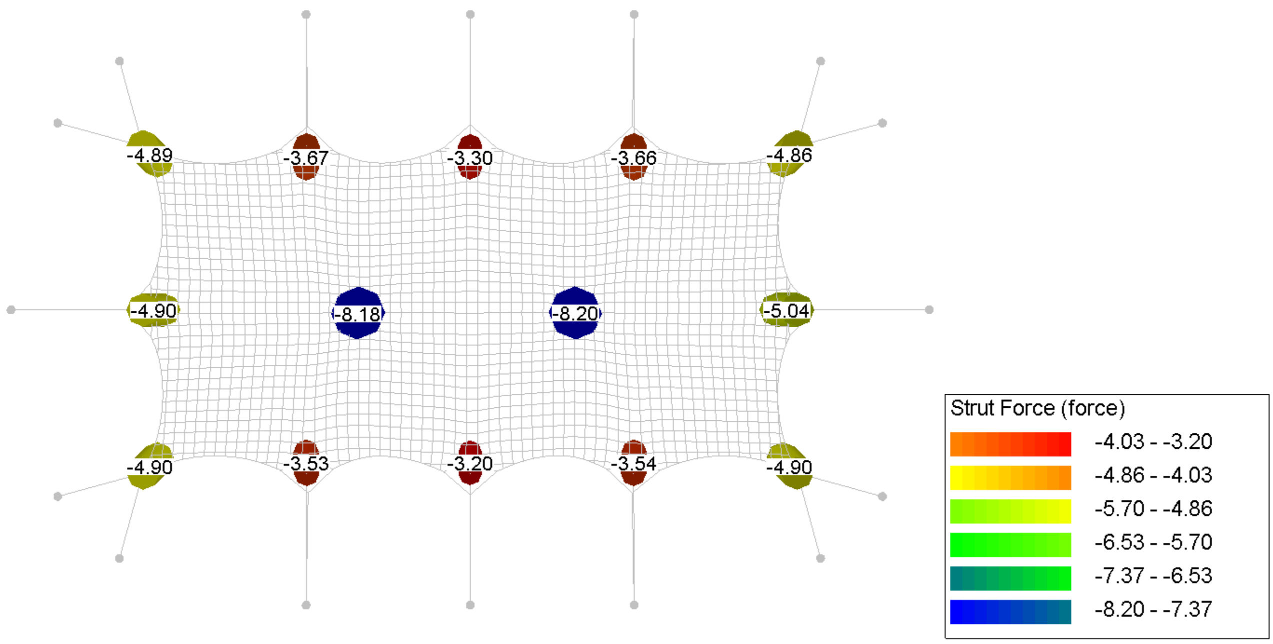
Annex D.6.4. Guy rope forces



Annex D.6.5. Belt forces

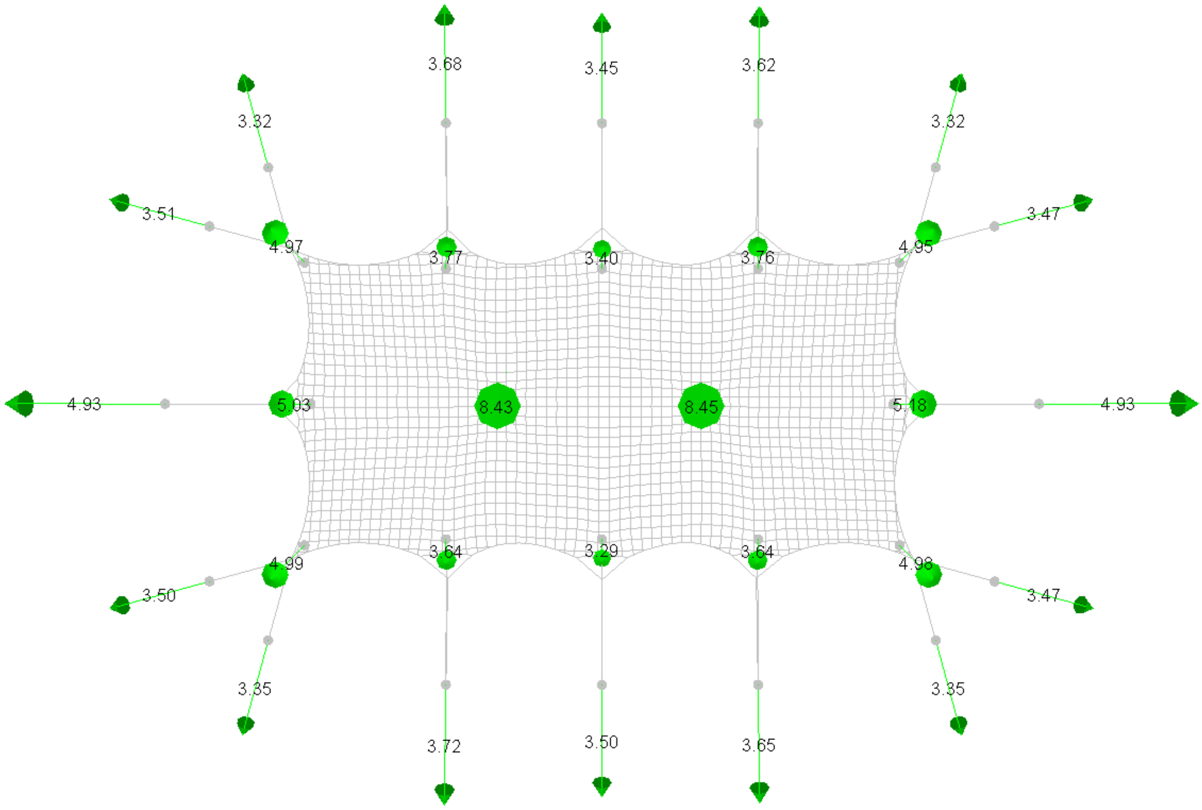


Annex D.6.6. Poles forces



Annex D.6.7. Reaction forces

Increased snow load of 0.1 x 1.5/1.35=0.1083 kN/m2 is applied for ballast and anchor calculation



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Annex E. Member checks

Annex E.1. Main pole 4.0m – Wood, Spruce C18

Woodtype		Spruce		
Strenght type		C18		
Material factor	γ_M	1.3		NEN-EN 1995-1-1:2005 table 2.3
Climate class		2		NEN-EN 1995-1-1:2005 article 2.3.1.3
Straightness factor	β_c	0.2		NEN-EN 1995-1-1:2005 equ. 6.29
Height	D	95	mm	
Length (buckling)	l_{bucy}	4	m	
Effective area	A	7088.218425	mm^2	
Moment of innertia	I_y	3998198.205	mm^4	
Elastic modules	$W_{el;y}$	84172.59379	mm^3	
Charistic pressure strenght	f_{c0k}	18	N/mm^2	
Charistic bending strenght	f_{c0k}	18	N/mm^2	
Modules of elasticity	$E_{0.05}$	6	kN/m^2	
	i_y	23.8	mm	
Slenderness	λ_y	168.4		
Relative slenderness	λ_{rely}	2.94		NEN-EN 1995-1-1:2005 equ. 6.21
	k_y	5.07		NEN-EN 1995-1-1:2005 equ. 6.21
Buckling factor	k_{cy}	0.109		NEN-EN 1995-1-1:2005 equ. 6.25

Strength check

Pressure force	F_d	9.20	kN	
Bending moment	M_d	0.00	kNm	
Pressure stress	σ_{c0d}	1.30	N/mm^2	
Bending stress	σ_{c0d}	0.00	N/mm^2	
Load duration		short		NEN-EN 1995-1-1:2005 table 2.1
Modificationfactor	k_{mod}	0.90		NEN-EN 1995-1-1:2005 table 3.1
Design pressure strenght	f_{c0d}	12.46	N/mm^2	NEN-EN 1995-1-1:2005 equ. 2.14
Design bending strenght	f_{m0d}	12.46	N/mm^2	NEN-EN 1995-1-1:2005 equ. 2.14
Strength check		**0.96**		NEN-EN 1995-1-1:2005 equ. 6.23

Annex E.2. Side pole 3m – Wood, Spruce C18

Woodtype		Spruce		
Strenght type		C18		
Material factor	γ_M	1.3		NEN-EN 1995-1-1:2005 table 2.3
Climate class		2		NEN-EN 1995-1-1:2005 article 2.3.1.3
Straightness factor	β_c	0.2		NEN-EN 1995-1-1:2005 equ. 6.29
Height	D	75	mm	
Length (buckling)	l_{bucy}	3	m	
Effective area	A	4417.864669	mm^2	
Moment of innertia	I_y	1553155.548	mm^4	
Elastic modules	$W_{el;y}$	41417.48127	mm^3	
Charistic pressure strenght	f_{c0k}	18	N/mm²	
Charistic bending strenght	f_{c0k}	18	N/mm²	
Modules of elasticity	$E_{0.05}$	6	kN/m²	
	i_y	18.8	mm	
Slenderness	λ_y	160.0		
Relative slenderness	λ_{rely}	2.79		NEN-EN 1995-1-1:2005 equ. 6.21
	k_y	4.64		NEN-EN 1995-1-1:2005 equ. 6.21
Buckling factor	k_{cy}	0.120		NEN-EN 1995-1-1:2005 equ. 6.25

Strength check

Pressure force	F_d	5.87	kN	
Bending moment	M_d	0.00	kNm	
Pressure stress	σ_{c0d}	1.33	N/mm²	
Bending stress	σ_{c0d}	0.00	N/mm²	
Load duration		short		NEN-EN 1995-1-1:2005 table 2.1
Modificationfactor	k_{mod}	0.90		NEN-EN 1995-1-1:2005 table 3.1
Design pressure strenght	f_{c0d}	12.46	N/mm²	NEN-EN 1995-1-1:2005 equ. 2.14
Design bending strenght	f_{m0d}	12.46	N/mm²	NEN-EN 1995-1-1:2005 equ. 2.14
Strength check		0.89		NEN-EN 1995-1-1:2005 equ. 6.23

Annex E.3. Corner pole 2.5m - Wood, C18

Woodtype		Spruce		
Strenght type		C18		
Material factor	γ_M	1.3		NEN-EN 1995-1-1:2005 table 2.3
Climate class		2		NEN-EN 1995-1-1:2005 article 2.3.1.3
Straightness factor	β_c	0.2		NEN-EN 1995-1-1:2005 equ. 6.29
Height	D	70	mm	
Length (buckling)	$l_{buc,y}$	2.5	m	
Effective area	A	3848.451001	mm^2	
Moment of innertia	I_y	1178588.119	mm^4	
Elastic modules	$W_{el,y}$	33673.94626	mm^3	
Charistic pressure strenght	f_{c0k}	18	N/mm²	
Charistic bending strenght	f_{c0k}	18	N/mm²	
Modules of elasticity	$E_{0.05}$	6	kN/m²	
	i_y	17.5	mm	
Slenderness	λ_y	142.9		
Relative slenderness	$\lambda_{rel,y}$	2.49		NEN-EN 1995-1-1:2005 equ. 6.21
	k_y	3.82		NEN-EN 1995-1-1:2005 equ. 6.21
Buckling factor	k_{cy}	0.149		NEN-EN 1995-1-1:2005 equ. 6.25

<u>Strength check</u>

Pressure force	F_d	7.00	kN	
Bending moment	M_d	0.00	kNm	
Pressure stress	σ_{c0d}	1.82	N/mm²	
Bending stress	σ_{c0d}	0.00	N/mm²	
Load duration		short		NEN-EN 1995-1-1:2005 table 2.1
Modificationfactor	k_{mod}	0.90		NEN-EN 1995-1-1:2005 table 3.1
Design pressure strenght	f_{c0d}	12.46	N/mm²	NEN-EN 1995-1-1:2005 equ. 2.14
Design bending strenght	f_{m0d}	12.46	N/mm²	NEN-EN 1995-1-1:2005 equ. 2.14
Strength check		**0.98**		NEN-EN 1995-1-1:2005 equ. 6.23

Annex E.4. Main pole 4.0m – Wood, Spruce C18 – for snow and wind combination [FR]

Woodtype		Spruce		
Strenght type		C18		
Material factor	γ_M	1.3		NEN-EN 1995-1-1:2005 table 2.3
Climate class		2		NEN-EN 1995-1-1:2005 article 2.3.1.3
Straightness factor	β_c	0.2		NEN-EN 1995-1-1:2005 equ. 6.29
Height	D	100	mm	
Length (buckling)	$l_{buc;y}$	4	m	
Effective area	A	7853.981634	mm^2	
Moment of innertia	I_y	4908738.521	mm^4	
Elastic modules	$W_{el;y}$	98174.77042	mm^3	
Charistic pressure strenght	f_{c0k}	18	N/mm²	
Charistic bending strenght	f_{c0k}	18	N/mm²	
Modules of elasticity	$E_{0.05}$	6	kN/m²	
	i_y	25.0	mm	
Slenderness	λ_y	160.0		
Relative slenderness	$\lambda_{rel;y}$	2.79		NEN-EN 1995-1-1:2005 equ. 6.21
	k_y	4.64		NEN-EN 1995-1-1:2005 equ. 6.21
Buckling factor	k_{cy}	0.120		NEN-EN 1995-1-1:2005 equ. 6.25

Strength check

Pressure force	F_d	11.91	kN	
Bending moment	M_d	0.00	kNm	
Pressure stress	σ_{c0d}	1.52	N/mm²	
Bending stress	σ_{c0d}	0.00	N/mm²	
Load duration		short		NEN-EN 1995-1-1:2005 table 2.1
Modificationfactor	k_{mod}	0.90		NEN-EN 1995-1-1:2005 table 3.1
Design pressure strenght	f_{c0d}	12.46	N/mm²	NEN-EN 1995-1-1:2005 equ. 2.14
Design bending strenght	f_{m0d}	12.46	N/mm²	NEN-EN 1995-1-1:2005 equ. 2.14
Strength check		**1.02**		NEN-EN 1995-1-1:2005 equ. 6.23

Annex E.5. Side pole 3.0m – Wood, Spruce C18 – for snow and wind combination [FR]

Woodtype		Spruce		
Strenght type		C18		
Material factor	γ_M	1.3		NEN-EN 1995-1-1:2005 table 2.3
Climate class		2		NEN-EN 1995-1-1:2005 article 2.3.1.3
Straightness factor	β_c	0.2		NEN-EN 1995-1-1:2005 equ. 6.29
Height	D	80	mm	
Length (buckling)	$l_{buc;y}$	3	m	
Effective area	A	5026.548246	mm^2	
Moment of innertia	I_y	2010619.298	mm^4	
Elastic modules	$W_{el;y}$	50265.48246	mm^3	
Charistic pressure strenght	f_{c0k}	18	N/mm²	
Charistic bending strenght	f_{c0k}	18	N/mm²	
Modules of elasticity	$E_{0.05}$	6	kN/m²	
	i_y	20.0	mm	
Slenderness	λ_y	150.0		
Relative slenderness	$\lambda_{rel;y}$	2.62		NEN-EN 1995-1-1:2005 equ. 6.21
	k_y	4.15		NEN-EN 1995-1-1:2005 equ. 6.21
Buckling factor	k_{cy}	0.136		NEN-EN 1995-1-1:2005 equ. 6.25

Strength check

Pressure force	F_d	7.14	kN	
Bending moment	M_d	0.00	kNm	
Pressure stress	σ_{c0d}	1.42	N/mm²	
Bending stress	σ_{c0d}	0.00	N/mm²	
Load duration		short		NEN-EN 1995-1-1:2005 table 2.1
Modificationfactor	k_{mod}	0.90		NEN-EN 1995-1-1:2005 table 3.1
Design pressure strenght	f_{c0d}	12.46	N/mm²	NEN-EN 1995-1-1:2005 equ. 2.14
Design bending strenght	f_{m0d}	12.46	N/mm²	NEN-EN 1995-1-1:2005 equ. 2.14
Strength check		**0.84**		NEN-EN 1995-1-1:2005 equ. 6.23

Annex E.6. Corner pole 2.5m – Wood, Spruce C18 – for snow and wind combination [FR]

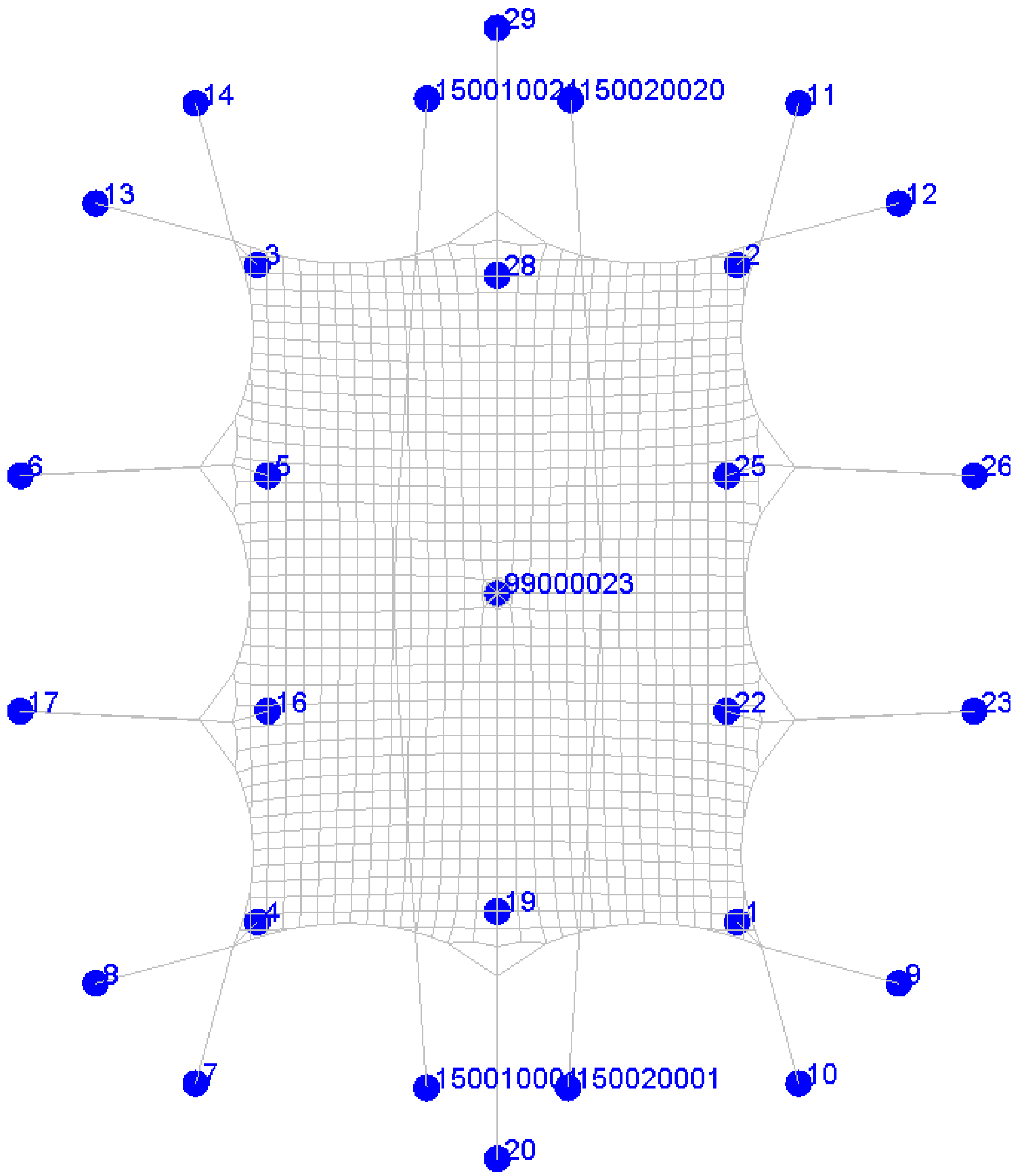
Woodtype		Spruce		
Strenght type		C18		
Material factor	γ_M	1.3		NEN-EN 1995-1-1:2005 table 2.3
Climate class		2		NEN-EN 1995-1-1:2005 article 2.3.1.3
Straightness factor	β_c	0.2		NEN-EN 1995-1-1:2005 equ. 6.29
Height	D	70	mm	
Length (buckling)	$l_{buc;y}$	2.5	m	
Effective area	A	3848.451001	mm^2	
Moment of innertia	I_y	1178588.119	mm^4	
Elastic modules	$W_{el;y}$	33673.94626	mm^3	
Charistic pressure strenght	f_{c0k}	18	N/mm^2	
Charistic bending strenght	f_{c0k}	18	N/mm^2	
Modules of elasticity	$E_{0.05}$	6	kN/m^2	
	i_y	17.5	mm	
Slenderness	λ_y	142.9		
Relative slenderness	$\lambda_{rel;y}$	2.49		NEN-EN 1995-1-1:2005 equ. 6.21
	k_y	3.82		NEN-EN 1995-1-1:2005 equ. 6.21
Buckling factor	k_{cy}	0.149		NEN-EN 1995-1-1:2005 equ. 6.25

Strength check

Pressure force	F_d	6.95	kN	
Bending moment	M_d	0.00	kNm	
Pressure stress	σ_{c0d}	1.81	N/mm^2	
Bending stress	σ_{c0d}	0.00	N/mm^2	
Load duration		short		NEN-EN 1995-1-1:2005 table 2.1
Modificationfactor	k_{mod}	0.90		NEN-EN 1995-1-1:2005 table 3.1
Design pressure strenght	f_{c0d}	12.46	N/mm^2	NEN-EN 1995-1-1:2005 equ. 2.14
Design bending strenght	f_{m0d}	12.46	N/mm^2	NEN-EN 1995-1-1:2005 equ. 2.14
Strength check		**0.97**		NEN-EN 1995-1-1:2005 equ. 6.23

Annex F. Reaction forces model 7.5x10m

Annex F.1. Point numbers



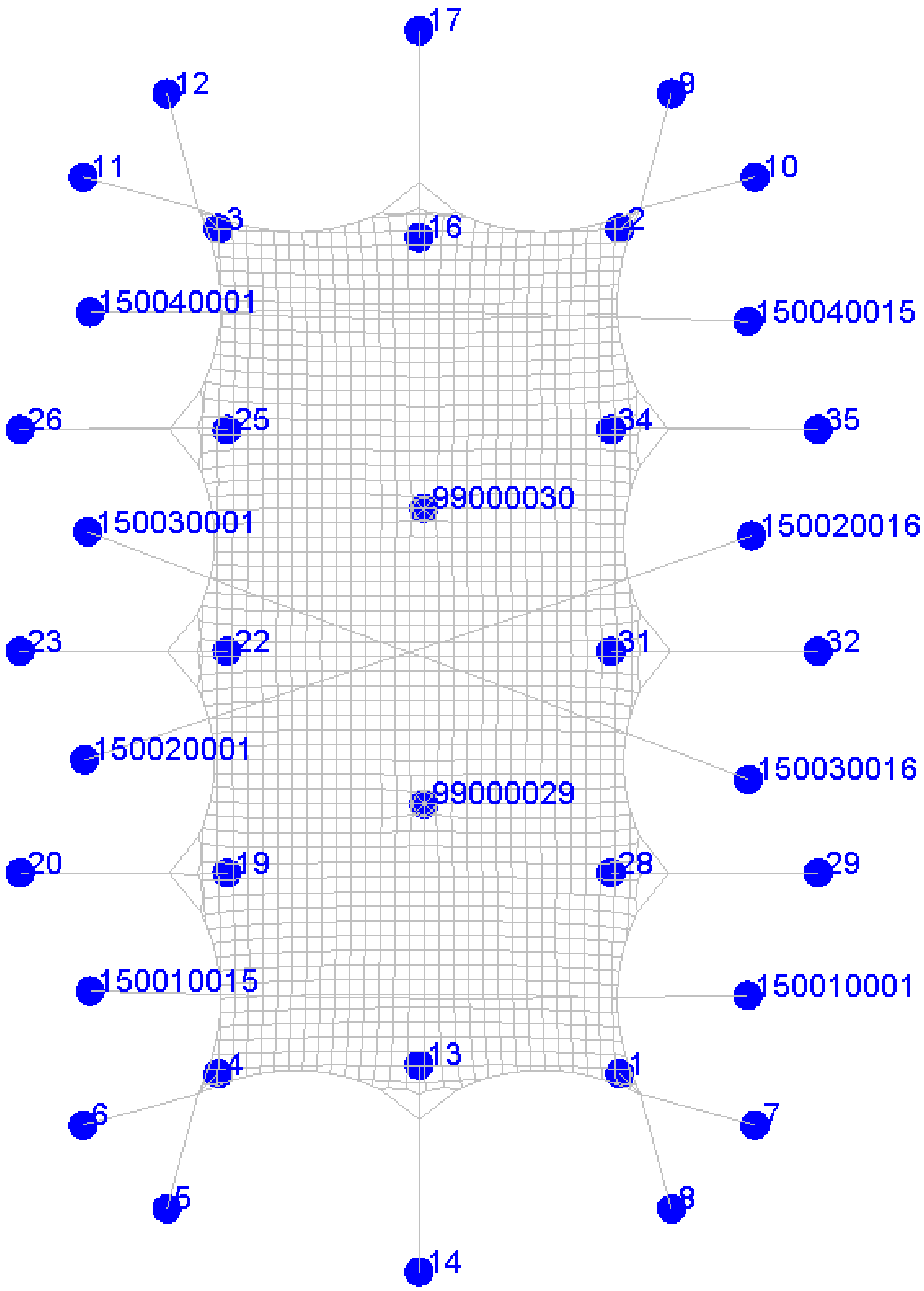
Annex F.2. Reaction forces

Reaction forces are shown in kilonewtons [kN] without any safety factor (1.35, 1.5, 1.2). A load combination especially for anchors and ballast calculation for snow and wind combined is CO6a.

				CO1			CO2			CO3			CO4			CO5			CO6			CO6a		
Att	Node	Koor X	Koor Y	Fx	Fy	Fz	Fx	Fy	Fz	Fx	Fy	Fz	Fx	Fy	Fz	Fx	Fy	Fz	Fx	Fy	Fz	Fx	Fy	Fz
Tie	6	-3.00	7.17	-1.55	0.00	-1.19	-1.80	0.01	-1.41	-2.10	0.01	-1.65	-3.32	-0.15	-2.52	-2.73	-0.03	-2.07	-2.54	0.02	-2.02	-2.58	0.02	-2.05
Tie	17	-3.00	3.83	-1.55	0.00	-1.19	-1.80	-0.01	-1.41	-2.10	-0.01	-1.65	-3.32	0.15	-2.52	-2.73	0.03	-2.07	-2.54	-0.02	-2.02	-2.58	-0.02	-2.05
Tie	20	3.75	-2.50	0.00	-1.75	-1.31	0.00	-2.20	-1.71	0.00	-2.60	-2.04	0.00	-2.80	-1.95	0.00	-2.36	-1.72	0.00	-3.15	-2.50	0.00	-3.19	-2.53
Tie	23	10.50	3.83	1.55	0.00	-1.19	1.80	-0.01	-1.41	2.10	-0.01	-1.65	3.31	0.15	-2.51	2.73	0.03	-2.07	2.54	-0.02	-2.02	2.58	-0.02	-2.05
Tie	26	10.50	7.17	1.55	0.00	-1.19	1.80	0.01	-1.41	2.10	0.01	-1.65	3.31	-0.15	-2.51	2.73	-0.03	-2.07	2.54	0.02	-2.02	2.58	0.02	-2.05
Tie	29	3.75	13.50	0.00	1.75	-1.31	0.00	2.20	-1.71	0.00	2.60	-2.04	0.00	2.80	-1.95	0.00	2.36	-1.72	0.00	3.15	-2.50	0.00	3.19	-2.53
Tie	7	-0.52	-1.43	-0.41	-1.51	-1.56	-0.46	-1.70	-1.75	-0.52	-1.93	-1.99	-0.64	-2.33	-2.40	-0.58	-2.14	-2.21	-0.61	-2.23	-2.30	-0.61	-2.26	-2.32
Tie	8	-1.93	-0.02	-1.40	-0.38	-1.44	-1.57	-0.43	-1.62	-1.76	-0.48	-1.82	-2.47	-0.67	-2.54	-1.77	-0.49	-1.83	-2.01	-0.55	-2.08	-2.03	-0.56	-2.10
Tie	9	9.43	-0.02	1.40	-0.38	-1.44	1.57	-0.43	-1.62	1.76	-0.48	-1.82	2.47	-0.67	-2.54	1.77	-0.49	-1.83	2.01	-0.55	-2.08	2.03	-0.56	-2.10
Tie	10	8.02	-1.43	0.41	-1.51	-1.56	0.46	-1.70	-1.75	0.52	-1.93	-1.99	0.64	-2.33	-2.40	0.58	-2.14	-2.21	0.61	-2.23	-2.30	0.61	-2.26	-2.32
Tie	11	8.02	12.43	0.41	1.51	-1.56	0.46	1.70	-1.75	0.52	1.93	-1.99	0.64	2.32	-2.39	0.58	2.14	-2.21	0.61	2.23	-2.30	0.61	2.26	-2.32
Tie	12	9.43	11.02	1.40	0.38	-1.44	1.57	0.43	-1.62	1.76	0.48	-1.82	2.47	0.67	-2.54	1.77	0.49	-1.83	2.01	0.55	-2.08	2.03	0.56	-2.10
Tie	13	-1.93	11.02	-1.40	0.38	-1.44	-1.57	0.43	-1.62	-1.76	0.48	-1.82	-2.46	0.67	-2.53	-1.77	0.49	-1.83	-2.01	0.55	-2.08	-2.03	0.56	-2.10
Tie	14	-0.52	12.43	-0.41	1.51	-1.56	-0.46	1.70	-1.75	-0.52	1.93	-1.99	-0.64	2.33	-2.40	-0.58	2.14	-2.21	-0.61	2.23	-2.30	-0.61	2.26	-2.32
Pole	1	7.15	0.85	0.45	-0.45	2.63	0.54	-0.54	3.18	0.63	-0.62	3.69	0.64	-0.65	3.83	0.52	-0.52	3.07	0.75	-0.74	4.41	0.76	-0.75	4.47
Pole	2	7.15	10.15	0.45	0.45	2.63	0.54	0.54	3.18	0.63	0.62	3.69	0.64	0.65	3.83	0.52	0.51	3.06	0.75	0.74	4.41	0.76	0.75	4.47
Pole	3	0.35	10.15	-0.45	0.45	2.63	-0.54	0.54	3.18	-0.63	0.62	3.69	-0.64	0.64	3.83	-0.52	0.52	3.07	-0.75	0.74	4.41	-0.76	0.75	4.47
Pole	4	0.35	0.85	-0.45	-0.45	2.63	-0.54	-0.54	3.18	-0.63	-0.62	3.69	-0.64	-0.65	3.83	-0.52	-0.51	3.06	-0.75	-0.74	4.41	-0.76	-0.75	4.47
Pole	5	0.50	7.17	-0.20	0.01	1.25	-0.33	0.01	2.01	-0.42	0.01	2.55	-0.09	0.03	0.58	-0.13	0.00	0.83	-0.56	0.00	3.46	-0.58	0.00	3.53
Pole	16	0.50	3.83	-0.20	-0.01	1.25	-0.33	-0.01	2.01	-0.42	-0.01	2.55	-0.09	-0.03	0.58	-0.13	0.00	0.83	-0.56	0.00	3.46	-0.58	0.00	3.53
Pole	19	3.75	1.00	0.00	-0.25	1.51	0.00	-0.42	2.60	0.00	-0.54	3.31	0.00	-0.38	2.31	0.00	-0.11	0.69	0.00	-0.72	4.46	0.00	-0.73	4.56
Pole	22	7.00	3.83	0.20	-0.01	1.25	0.33	-0.01	2.01	0.42	-0.01	2.55	0.09	-0.03	0.58	0.13	0.00	0.83	0.56	0.00	3.46	0.58	0.00	3.53
Pole	25	7.00	7.17	0.20	0.01	1.25	0.33	0.01	2.01	0.42	0.01	2.55	0.09	0.03	0.58	0.13	0.00	0.83	0.56	0.00	3.46	0.58	0.00	3.53
Pole	28	3.75	10.00	0.00	0.25	1.51	0.00	0.42	2.60	0.00	0.54	3.31	0.00	0.38	2.28	0.00	0.11	0.69	0.00	0.72	4.46	0.00	0.73	4.56
Pole	99000023	3.75	5.50	0.00	0.00	2.22	0.00	0.00	4.76	0.00	0.00	5.95	0.00	0.00	0.10	0.00	0.00	0.26	0.00	0.00	8.61	0.00	0.00	8.83
Stormbelt	150010001	2.75	-1.50										0.18	-3.13	-2.87									
Stormbelt	150010021	2.76	12.50										0.18	3.13	-2.87									
Stormbelt	150020001	4.75	-1.49										-0.18	-3.10	-2.84									
Stormbelt	150020020	4.79	12.49										-0.17	3.11	-2.83									

Annex G. Reaction forces model 7.5x15m

Annex G.1. Point numbers



Annex G.2. Reaction forces

Reaction forces are shown in kilonewtons [kN] without any safety factor (1.35, 1.5, 1.2). A load combination especially for anchors and ballast calculation for snow and wind combined is CO6a.

				CO1			CO2			CO3			CO4			CO5			CO6			CO6a		
Att	Node	Koor X	Koor Y	Fx	Fy	Fz	Fx	Fy	Fz	Fx	Fy	Fz	Fx	Fy	Fz	Fx	Fy	Fz	Fx	Fy	Fz	Fx	Fy	Fz
Tie	14	3.75	-7.50	0.00	-1.77	-1.32	0.00	-2.45	-1.90	0.00	-3.03	-2.37	0.01	-3.33	-2.41	0.00	-2.42	-1.76	-0.01	-3.80	-3.02	-0.01	-3.86	-3.07
Tie	17	3.75	13.50	0.00	1.76	-1.33	0.00	2.44	-1.91	0.00	3.02	-2.38	0.00	3.32	-2.42	0.00	2.41	-1.77	-0.01	3.79	-3.02	-0.01	3.85	-3.08
Tie	20	-3.00	-0.75	-1.73	0.01	-1.35	-2.02	-0.01	-1.60	-2.38	-0.02	-1.90	-3.61	0.02	-2.78	-2.88	0.03	-2.25	-2.83	-0.04	-2.27	-2.87	-0.04	-2.30
Tie	23	-3.00	3.00	-1.71	0.00	-1.39	-1.86	0.00	-1.52	-2.21	0.00	-1.80	-3.08	0.01	-2.43	-3.24	0.00	-2.63	-2.63	0.00	-2.16	-2.67	0.00	-2.19
Tie	26	-3.00	6.75	-1.72	-0.01	-1.34	-1.99	0.01	-1.58	-2.35	0.02	-1.86	-3.65	-0.01	-2.81	-2.94	-0.03	-2.29	-2.79	0.03	-2.23	-2.82	0.04	-2.26
Tie	29	10.50	-0.75	1.73	0.00	-1.35	2.03	-0.01	-1.60	2.41	-0.02	-1.90	3.63	0.02	-2.77	2.88	0.03	-2.24	2.87	-0.04	-2.29	2.91	-0.04	-2.32
Tie	32	10.50	3.00	1.72	0.00	-1.39	1.88	0.00	-1.52	2.24	0.00	-1.81	3.07	0.00	-2.40	3.24	0.00	-2.62	2.68	0.00	-2.18	2.71	0.00	-2.21
Tie	35	10.50	6.75	1.72	-0.01	-1.34	2.01	0.01	-1.58	2.37	0.02	-1.87	3.67	-0.02	-2.80	2.95	-0.03	-2.28	2.82	0.04	-2.24	2.86	0.04	-2.27
Tie	5	-0.52	-6.43	-0.42	-1.53	-1.58	-0.47	-1.74	-1.79	-0.55	-2.03	-2.09	-0.75	-2.76	-2.83	-0.60	-2.21	-2.27	-0.65	-2.37	-2.44	-0.66	-2.40	-2.47
Tie	6	-1.93	-5.02	-1.49	-0.41	-1.54	-1.73	-0.47	-1.79	-1.97	-0.54	-2.03	-2.33	-0.64	-2.41	-1.96	-0.54	-2.02	-2.25	-0.62	-2.32	-2.27	-0.62	-2.34
Tie	7	9.43	-5.02	1.50	-0.41	-1.55	1.74	-0.47	-1.80	1.98	-0.54	-2.04	2.35	-0.65	-2.42	1.96	-0.54	-2.02	2.27	-0.62	-2.34	2.29	-0.63	-2.36
Tie	8	8.02	-6.43	0.41	-1.51	-1.56	0.47	-1.72	-1.77	0.55	-2.02	-2.08	0.74	-2.73	-2.81	0.59	-2.18	-2.25	0.65	-2.37	-2.44	0.65	-2.40	-2.47
Tie	9	8.02	12.43	0.41	1.50	-1.55	0.47	1.71	-1.76	0.55	2.01	-2.07	0.73	2.68	-2.76	0.59	2.17	-2.24	0.64	2.35	-2.42	0.65	2.38	-2.45
Tie	10	9.43	11.02	1.53	0.41	-1.57	1.76	0.48	-1.81	1.99	0.54	-2.05	2.36	0.65	-2.43	1.99	0.54	-2.05	2.27	0.62	-2.34	2.29	0.63	-2.36
Tie	11	-1.93	11.02	-1.52	0.41	-1.56	-1.74	0.47	-1.80	-1.97	0.54	-2.03	-2.35	0.65	-2.42	-1.99	0.54	-2.05	-2.25	0.62	-2.32	-2.27	0.62	-2.34
Tie	12	-0.52	12.43	-0.41	1.52	-1.57	-0.47	1.72	-1.78	-0.55	2.01	-2.07	-0.75	2.77	-2.85	-0.60	2.19	-2.26	-0.64	2.35	-2.42	-0.65	2.37	-2.44
Pole	1	7.15	-4.15	0.46	-0.46	2.72	0.57	-0.57	3.36	0.68	-0.67	3.98	0.73	-0.72	4.33	0.54	-0.54	3.18	0.81	-0.81	4.79	0.82	-0.82	4.86
Pole	2	7.15	10.15	0.46	0.47	2.73	0.57	0.57	3.36	0.67	0.67	3.98	0.71	0.70	4.23	0.54	0.54	3.20	0.81	0.80	4.78	0.82	0.81	4.84
Pole	3	0.35	10.15	-0.47	0.47	2.73	-0.57	0.57	3.36	-0.67	0.67	3.96	-0.73	0.72	4.32	-0.54	0.54	3.20	-0.80	0.80	4.76	-0.81	0.81	4.82
Pole	4	0.35	-4.15	-0.47	-0.46	2.72	-0.57	-0.57	3.36	-0.67	-0.67	3.97	-0.73	-0.72	4.31	-0.54	-0.54	3.20	-0.81	-0.80	4.77	-0.82	-0.81	4.84
Pole	13	3.75	-4.00	0.00	-0.25	1.51	0.00	-0.44	2.73	0.00	-0.58	3.56	0.00	-0.14	0.86	0.00	-0.11	0.71	0.00	-0.78	4.86	0.00	-0.80	4.97
Pole	16	3.75	10.00	0.00	0.25	1.56	0.00	0.46	2.82	0.00	0.59	3.67	0.00	0.15	0.93	0.00	0.12	0.77	0.00	0.80	5.01	0.00	0.82	5.12
Pole	19	0.50	-0.75	-0.21	-0.01	1.30	-0.35	-0.01	2.12	-0.44	0.00	2.72	-0.20	-0.01	1.22	-0.13	0.00	0.78	-0.59	0.01	3.64	-0.60	0.01	3.72
Pole	22	0.50	3.00	-0.18	0.00	1.12	-0.30	0.00	1.83	-0.39	0.00	2.41	-0.31	0.00	1.89	-0.10	0.00	0.64	-0.53	0.00	3.28	-0.55	0.00	3.35
Pole	25	0.50	6.75	-0.21	0.01	1.30	-0.35	0.01	2.11	-0.44	0.00	2.71	-0.21	0.01	1.30	-0.13	0.01	0.79	-0.59	-0.01	3.64	-0.60	-0.01	3.71
Pole	28	7.00	-0.75	0.20	-0.01	1.26	0.33	-0.01	2.04	0.43	0.00	2.62	0.20	-0.01	1.25	0.12	0.00	0.74	0.57	0.01	3.52	0.58	0.01	3.59
Pole	31	7.00	3.00	0.18	0.00	1.09	0.29	0.00	1.77	0.38	0.00	2.33	0.28	0.00	1.71	0.10	0.00	0.62	0.52	0.00	3.18	0.53	0.00	3.25
Pole	34	7.00	6.75	0.20	0.01	1.26	0.33	0.01	2.04	0.43	0.00	2.62	0.21	0.01	1.30	0.12	0.01	0.76	0.57	-0.01	3.52	0.58	-0.01	3.59
Pole	99000029	3.83	0.42	0.00	0.00	1.98	-0.01	0.01	4.43	-0.02	0.03	5.63	0.00	0.00	0.04	0.00	0.00	0.14	-0.04	0.08	8.22	-0.04	0.08	8.43
Pole	99000030	3.83	5.42	0.00	0.00	1.98	-0.01	0.00	4.43	-0.02	-0.01	5.63	0.00	0.00	0.04	0.00	0.00	0.13	-0.04	-0.04	8.24	-0.04	-0.04	8.45
Stormbelt	150010001	9.31	-2.83										2.62	0.00	-2.38									
Stormbelt	150010015	-1.81	-2.75										-2.62	0.04	-2.37									
Stormbelt	150020001	-1.91	1.16										-2.72	-0.89	-2.63									
Stormbelt	150020016	9.38	4.95										2.71	0.92	-2.63									
Stormbelt	150030001	-1.86	5.02										-2.65	0.98	-2.60									
Stormbelt	150030016	9.32	0.83										2.66	-0.98	-2.60									
Stormbelt	150040001	-1.81	8.73										-2.71	0.02	-2.45									
Stormbelt	150040015	9.32	8.58										2.71	-0.06	-2.45									